DRAFT V0.2.2 · SECOND ADVERSARIAL LOOP · REVIEWER: KIMI K3.1-PRO (KIMI.AI)

From the Qn Universe to QnIR v0.2

Review integration, derivation, and the candidate intermediate representation for the second adversarial loop — with the complete Qn universe canon and full glossary reproduced as rendered-content annexes, an external-claims assessment of the October 2026 OpenAI mathematics corpus, and all review data traveling in this single size-compliant PDF.

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cdqn — Chained and Distributed Quang Numbers · Author & maintainer lead
Draft prepared with AI writing assistance (GLM, Z.ai) under the author's lead
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Draft status. **Version v0.2.2** (rendered-content edition; adds the external-claims assessment of Section 12) — circulated for the *second* adversarial AI review loop. The loop-1 reviewer was Google Deep Review; its canvas for this program is rate-limited to two reviews per week and the loop-2 quota is exhausted, so loop 2 proceeds with **Kimi K3.1-Pro (kimi.ai)** as reviewer. Loop 1 reviewed the condensed Qn universe specification and the QnIR v0.1 candidate; its transcript is `docs/AiReviews/AiReviewQnIRv0.1.md` (reproduced as Annex U). The canon then absorbed the review file by file (simemp v1.2.0, qs v1.3.0, abstractionLayers v1.8.1, qm/qexpr/qnPrimitive v1.3.0); this draft derives the QnIR v0.2 candidate from that integrated canon and from every remaining reviewer finding, and is written so the reviewer can attack it on consistency, logic, physical grounding, and completeness. Numbered open questions (QO-16 … QO-32) in Section 15 are the response format requested for this loop; the QO-1 … QO-15 answers of loop 1 and their canon dispositions are tabulated in Section 2.

Self-contained review package. Annexes A–U reproduce the complete Qn universe canon — every `.md` design document and the full glossary, as rendered content under the Annex Guide conventions N1–N6 — so that all review data lives in this one PDF within conservative adversarial-review intake bounds. The byte-verbatim edition of this draft measures 8.5×10^5 extracted characters — at the cap measured for loop-2 review tooling, with no margin — so the size discipline of the v0.2.1 edition is retained here defensively, whatever the present reviewer's limit; the rendered edition preserves every sentence, heading, and table cell while removing syntax-only token mass. Verification commitments (SHA-256 per source file) and the content-fidelity procedure are stated in the Annex Guide (Section B).

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1 Introduction and Invitation to Attack (Loop 2)

1.1 Where this draft stands in the review program

This document is the second station of a deliberately exposed review program. The loop-1 draft [11] condensed the Qn universe into a self-contained specification and proposed a first candidate intermediate representation, QnIR v0.1, together with sixteen anticipated objections and fifteen numbered open questions, QO-1 … QO-15. That draft was submitted to Google Deep Review on 26 October 2026, and the resulting transcript — a segment-by-segment technical audit of unusual depth — is preserved verbatim in the canon at `docs/AiReviews/AiReviewQnIRv0.1.md` and reproduced as Annex U of this PDF. The reviewer answered every open question, identified three fatal objections, four unanticipated architectural defects, and supplied constructive resolutions that the author then integrated into the canon file by file. The present draft is the derivation the reviewer asked for: it consumes the integrated canon, closes the remaining findings inside the QnIR candidate itself, and re-exposes the result for a second attack.

Three integration rounds happened between the loops, and they are the raw material of this derivation. The constitution absorbed the two-tier decidability finding and the compute-unit decoupling (`simemp.md` v1.2.0, Annex A); the semantics document closed the parser-boundary trust gap with a modality ceiling, a co-signing handshake, and four typed failure receipts (`qs.md` v1.3.0, Annex J); the structural thesis made the road receipt sound with execution-proof realizations and an origin-quotiented export projection (`abstractionLayers.md` v1.8.1, Annex B); and the numeric and symbolic layer files constructed multi-digit naturals inside the artifact model, split rewrite into canonical contraction and bounded saturation, and registered the resulting breaker receipts (`qm.md`, `qexpr.md`, `qnPrimitive.md` v1.3.0, Annexes E, H, I). Section 2 tabulates every loop-1 finding against the canon section that now anchors it, so the reviewer can verify the integration rather than trust it.

Everything that survived that integration is re-derived here as QnIR v0.2: a stack-free SSA state model in place of mutable slot files, a corrected 64-bit encoding with three widened operand fields and a header constant table, a ceiling-parameterized static cost discipline that repairs the totality argument, receipt-typed recoverable exceptions, bounded calls and containers, fork-join causal clocks, and the execution-proof interface for road receipts. The candidate remains deliberately falsifiable: every design choice cites its canon ground and, where a genuine choice existed, the BOC reasoning that selected it under the two-stage model that loop 1 itself forced into the constitution.

1.2 How the reviewer should read this package

This PDF is self-contained by construction. The paper body (Sections 1 through 16) carries the derivation; the annexes carry the data. Annex Guide (Section B) lists every reproduced file with its version, line count, and SHA-256 commitment, documents the two mechanical rendering transforms (prose wrap marker, tab expansion), and gives the exact procedure that reconstructs the source bytes from the annex text — a round trip that was verified mechanically for all 25 annex files before this draft was compiled. Where this paper and an annex disagree, the annex (the live canon) governs; where an annex and the GitHub Pages site disagree, the site's rendering pipeline is cosmetic and the `.md` source governs, consistent with the rendering convention fixed in `abstractionLayers.md` v1.8.1.

The attack surface is organized for systematic review rather than discovery by foraging. Section 2 is the integration ledger — audit it to confirm nothing was silently dropped. Sections 4–8 are the machine itself; the reviewer is invited to hunt for encoding contradictions, totality holes, and determinism leaks exactly the way loop 1 succeeded in doing. Section 14 pre-states sixteen fresh objections in the voice a hostile reviewer might use, and Section 15 closes with the seventeen

new numbered questions QO-16 … QO-32 that define the response format for this loop (the last two arise from the external-claims assessment of Section 12). As in loop 1, the draft would rather be sharpened by attack than polished by praise.

1.3 A caution on tone and claims

Nothing in Sections 4–13 is canonical. The QnIR v0.2 candidate is provisional under Axiom 14: it is a candidate awaiting deterministic validation, and its correct epistemic home is PENDING, not SEALED. Theorems labeled "candidate" are accounting arguments to be attacked, not formal results certified by a proof assistant; where the loop-1 reviewer found the v0.1 totality argument resting on an honest-but-unverified cost table, this draft repairs the structure of the argument but still owes the reviewer the calibration evidence named in QO-17. The prior-art comparison of Section 11 is positioning, not priority claims — the Qn program's contribution is the simultaneous composition of residual-conserving arithmetic, content-addressed identity, and intent-contract governance, and the comparison table exists precisely so the reviewer can challenge whether that composition justifies a new IR.

2 The Loop-1 Review and the Integration Ledger

2.1 What the loop-1 review established

The loop-1 transcript organizes its findings into five segments — constitution (PDF pp. 1–7), numeric substrate and symbolic layers (pp. 7–14), domain projections and governance (pp. 15–20), the QnIR v0.1 candidate itself (pp. 20–27), and the objections/QO index (pp. 27–38) — with an unnumbered high-level summary of six cross-cutting weaknesses. Its own QO cross-reference index maps QO-1 … QO-15 onto the technical points, and its final subsection grades the sixteen self-stated objections into three fatal, three cosmetic, and ten sharpened-but-surviving items, then adds four unanticipated objections (A–D) the self-critique had missed. Three of the fatal findings concerned the candidate directly: unconstructed naturals beyond $Q(9)$ with an unfixed zoom resolution $\kappa(z)$ (Objection 14), static cost accounting that permits mid-instruction budget overdraft (Objection 15), and the two boundary trust gaps — an unmitigated statistical parser at the intent entrance and an unverified execution claim inside the road receipt (Objections 10 and 7). The unanticipated objections concerned the candidate's encoding: arity and dual-destination fields that do not fit the 64-bit word (A), the absence of any forward conditional branch while a comparison result halts the machine (B), conflation of the static program hash chain with the dynamic causal counter under loops (C), and the undefined partition of the unified value/blade index space (D).

The reviewer also supplied the resolutions in constructive form: an operationalization of the conditional-entropy postulate as observational determinism, a two-stage constrained lexicographic BOC model, a platform-invariant compute unit U with a Layer-0 telemetry envelope, gateway billing attribution, a radix-anchor (Horner-step) construction of bounded naturals, the `OP_BYPASS`/`OP_SATURATE` split, an SSA value-graph state model, a receipt-typed exception taxonomy, fork-join causal clocks, origin-quotiented portable commitments, and zk-STARK or interactive-bisection realizations for the road-receipt proof field. The canon absorbed these file by file; the ledger below records where each landed and what remains open. One canon observation deserves note for the record: the transcript's final objection (D) ends mid-sentence in the committed source, an artifact of the review capture; its content is recoverable in full from Segment 4, Points 1, 3, and 7, and this draft treats the segment text as authoritative.

2.2 Integration ledger: QO-1 … QO-15 and the segment findings

Table 1 is the traceability spine of this derivation. Status vocabulary: CANON = integrated into a committed, live canon file at the cited section; CANDIDATE = integrated and tagged `[vX.Y.Z – CANDIDATE]` pending BOC sealing (Axiom 14); v0.2 = resolved inside this draft's QnIR candidate (Sections 4–13); PENDING = editorial touch-up queued on a named future canon update; OPEN = carried as a numbered open question of this loop.

Table 1: Loop-1 findings and their dispositions. "Seg." = segment, "P." = point of `AiReviewQnIRv0.1.md` (Annex U).

Item	Finding (short)	Disposition	Status
QO-1 / Seg. 1 P.3	BOC scalar arg max undecidable; Tier-1 binary, Tier-2 multi-objective	`simemp` §7 rewritten: hard filter C_{valid}, lexicographic `LexKey` with declared permutation $\pi \in S_3$, rubric mode	CANDIDATE
QO-2 / Seg. 1 P.2	Landauer metering conflated with logical work; scale gap 10^7–10^{10}	`simemp` §4.4: U = platform-invariant work quantum (the transition $Q(0) \to Q(1)$ costs $1\,U$); Layer-0 Hardware Telemetry Envelope via calibration receipts; §4.2 receipt reframing as epistemic compression	CANDIDATE
Seg. 1 P.1	Eq. 1 lacks probability space; tautology risk of declared $\mathcal{D}$	`simemp` §1: Observational Determinism Invariant over bit-identical commitments, falsifiable by differential execution	CANDIDATE
QO-4 / Seg. 2 P.1	No $\mathbb{N}$ beyond $Q(9)$; $\kappa(z)$ unfixed; 7-smooth claim	`qm` §3.6 radix-anchor (Horner) construction of $\mathbb{N}_{\leq H_{\max}}$; §3.3 $\kappa(z)$ bound per-artifact in DCC (three canonical schedules); §3.5 anchor-basis rewording	CANDIDATE
QO-3 / Seg. 2 P.2	Zoom telescope vs. exact $\mathbb{Q}$ and constructive reals	expressiveness stratification recorded; `OP_RATIO`/`OP_CF` generators typed; zero-testing limits stated	CANDIDATE
QO-5 / Seg. 2 P.3	`OP_REWRITE` conflates contraction with e-graph saturation	`qexpr` §4.6 split: `OP_BYPASS` (16-rule confluent suite, hash-governing) + `OP_SATURATE` (Q(rsi)-gated, versioned artifact); `qnPrimitive` §5.9/§1.16 aligned; breaker receipts registered	CANDIDATE
Seg. 2 P.4	$\mathcal{R}$-algebra ambiguities: division orientation, interval endpoints, even roots $n \geq 4$, Collatz modality	`qm` v1.3.0 semantic refinements; Collatz tagged `MODAL_HYPOTHESIS`; root promotion discipline feeds v0.2 ISA (R10′)	CANDIDATE

Item	Finding (short)	Disposition	Status
Seg. 2 P.5	Cayley rotor height growth; grade-sparsity; DEC Hodge error; Anchor-Cut soundness	height-ceiling discipline in `qm`; grade-sparse operand requirement; mesh well-centeredness precondition; `abstractionLayers` §4.4 audit realizations	CANDIDATE
QO-13 / Seg. 3 P.1	Road receipt π unsigned by execution; disjoint cross-node commitments	`abstractionLayers` §4.4: Realization A (zk-STARK over AIR-arithmetized QnIR trace), Realization B (optimistic bisection, failure → `RECEIPT_ANCHOR_REVOKED`); $Q(0)_A \mapsto \widehat{Q(0)}$ portable commitments	CANDIDATE
QO-11 / Seg. 3 P.2	Parser drift upstream of formalization; untyped r_{social}; 16-dim ceiling vs. open vocabulary	`qs` §4.4 modality ceiling; §6.1 round-trip co-signing handshake; §6.2 four parser-boundary receipts; §5.3 CTD/DEL typing of r_{social}; §3.4 normative axes vs. lexical payloads	CANDIDATE
QO-12 / Seg. 3 P.3	Qlog weakening/contraction; latency-harvester timing channel	linear/affine resource logic sequent rules recorded in review; harvester isolation discipline flagged for `qlog` next update	PENDING
QO-14 / Seg. 3 P.4	Independence quorum fails for legacy history (both directions)	custodial vs. stemmatic lineage bifurcation and graded quarantine proposed; deferred to `qs_history/history_library` updates	PENDING
Seg. 3 P.5	Chronosa: CALM/FLP tension; sheaf gluing is not inference	vision-proposal scoping acknowledged; partial-synchrony quorum partition to ride the next `chronosa` update	PENDING
Seg. 3 P.6	Qphy height growth; Qchem catalysis accounting; Qbio $U_{\text{phys}}/U_{\text{comp}}$ and $\ln 2$	domain corrections queued for `qphy`/`qchem`/`qbio` updates; $1\,U$ normalization now constitutionally fixed (`simemp` §4.4)	PENDING
Seg. 3 P.7	Container versioning amplification vs. locality	HAMT/RRB lowering with $\mathcal{O}(\log_B M)$ bound adopted in v0.2 containers (Section 6)	v0.2
QO-6 / Seg. 4 P.3	Mutable **V**/**B** slots vs. Axiom 13; index partition undefined	SSA value graph in causal birth order (Section 4); 8-bit `dest` eliminated	v0.2
QO-7 / Seg. 4 P.4	Missing opcodes; no forward branch; terminal `OP_CMP` indeterminacy	ISA v0.2 additions: `OP_BR_FWD`, `OP_BR_STATUS`, `OP_SELECT`, `OP_UNPACK`, `OP_RADIX_ACC`, `OP_REVERSE`, container, delegation, and bounded call opcodes; receipt-typed exceptions (Section 5)	v0.2

Item	Finding (short)	Disposition	Status
QO-8 / Seg. 4 P.2	Static costs operand-dependent; overdraft after Stage 1	ceiling-parameterized cost polynomials, pre-execution reservation, per-dispatch base cost, refund discipline (Section 8)	v0.2
QO-9 / Seg. 4 P.7	Commitment width, collision accounting, zoom hash normalization	256-bit SHA3-256 with domain separation; active collision detection receipt; canonical minimal-zoom normalization (Section 6)	v0.2
QO-10 / Seg. 4 P.6	Static word chain conflated with dynamic κ; no concurrency story	static W_i^{prog} vs. dynamic trace hash split; fork-join causal clocks; choreographic endpoint projection (Section 13)	v0.2
Seg. 4 P.5	Worked examples not reproducible	corrected SSA-form examples with all fields declared (Section 10)	v0.2
Seg. 4 P.1	64-bit word cannot encode arity-4/k-ary, dual dest, 256-bit hashes	word layout v0.2: three 12-bit operands, `aux` stays 16-bit, header constant table for commitments; adjacent-pair `OP_DIV` (Section 7)	v0.2
Seg. 1 P.5	No prior-art positioning	comparative subsection restored into the candidate (Section 11)	v0.2
Seg. 1 P.4	Gateway billing; semantic vs. syntactic scope; migration witnessing	`simemp` §6 (static $W_{\text{gate}} > 0$ billed to invoker before inspection), §4.3 multihash-style tagged digests + re-witnessing; semantic fidelity reserved to `qs` §6.1	CANDIDATE
R12 direction	$H(\text{Intent} \mid \text{Exec}) = 0$ conditions the wrong way	v0.2 R12′: $H(\text{Exec} \mid \text{Intent}, \mathcal{D}) = 0$, certified only post-handshake (Section 9)	v0.2
Minor typos	Segments 1–4 editorial lists	constitution/semantics files absorbed their own items; v0.1-paper items (label "Theorem 1 (Candidate)", `src1=` naming, stage numbering) fixed in this draft	CLOSED
Deferred	`qlog` §6.1 wording; glossary `boc-policy` / `compute-unit-u` entries; `index.md` SSL threshold table	queued one-line touch-ups on the named files' next versioned updates	PENDING

2.3 Editorial state of the canon at snapshot time

At the snapshot timestamp (2026-10-27 10:55 ICT (UTC+7)), the live docs carry the integrated review record in seven files: `simemp.md` v1.2.0, `abstractionLayers.md` v1.8.1, `qm.md` v1.3.0, `qexpr.md` v1.3.0, `qnPrimitive.md` v1.3.0, `qs.md` v1.3.0, and the loop-1 draft itself at `docs/AiReviews/`. Three one-line touch-ups remain queued exactly as recorded in the files' cross-file dependency notes: the fixed-point identity attribution in `qm.md` §5.6 still reads `OP_REWRITE` where the retired alias must become `OP_BYPASS`; `qexpr.md` §6 inherits breaker-

3's receipt name `RECEIPT_METRIC_EXHAUSTED` from the `qnPrimitive.md` §5.9 registry; and `index.md` quotes three of five SSL scaling thresholds. None of these pending items touches the v0.2 derivation, and each is recorded in the ledger so that the next canon updates carry them as mechanical ride-alongs.

3 Design Requirements for QnIR v0.2

QnIR v0.1 stated fourteen requirements R1–R14, each traced to its canon ground. The loop-1 review upheld the requirement set as the right shape of constraint but showed that three of the fourteen were stated too weakly to enforce what they claimed: R1 (static costs) admitted operand-dependent costs and post-execution billing; R12 (anti-drift) conditioned entropy in the wrong direction and assumed a drift-free intent object the parser boundary had not yet earned; and R11 (lowering isomorphism) was contradicted by the ISA itself, which lacked opcodes for six of the eight Layer-3 node classes it was required to represent. v0.2 therefore amends R1, R11, and R12 in place and adds five new requirements R15–R19 that the review showed to be missing load-bearing walls rather than optional equipment. The amended set is Table 2; unchanged requirements are listed in one row each for completeness, so this table remains the single authoritative statement of what the candidate must satisfy.

Table 2: QnIR v0.2 design requirements. R1′, R11′, R12′ amend v0.1; R15–R19 are new. Canon grounds cite the snapshot files of Annexes A–I.

ID	Requirement	Canon ground
R1′	Every instruction has a static cost given by a closed-form polynomial C_{op} of declared header ceilings and the instruction's immediate field only — never of runtime register values; the cost is reserved from W_{rem} *before* execution and unused residue is refunded at early terminal exactness.	Axiom 6; [1] §4.4; review Seg. 4 P.2
R2	All numeric state is the exact tuple $\langle \sigma, q_z, z, r_z, d \rangle$; no IEEE 754 value may exist in machine state.	Axioms 9–10; [5] §3
R3	No implicit operands, flags, or defaults; every program declares its DCC profile and required anchor contracts.	No-Implicit Rule; [2] §3.2
R4	Instructions are causally ordered; a program may not reference a value born later.	Axiom 5; [7] §6
R5	Every program and operand commitment carries an envelope reference.	Axiom 1; [7] §4
R6	Operand identity is content-addressed: equality of values is equality of commitment hashes, after canonical zoom normalization.	[6] §2.2; review Seg. 4 P.7
R7	Dispatch obeys $Q(\text{bypass}) \prec Q(\text{reuse}) \prec$ direct compute at every step.	[7] §5.9
R8	Every terminal transition emits a receipt from the closed status set; a program cannot end silently.	Axioms 6–7; [2] §3.6
R9	No stochastic opcode exists in the core ISA; generative candidates enter only via gateway validation, billed W_{gate} to the invoker.	Axiom 14; [1] §6

ID	Requirement	Canon ground
R10′	No floating-point opcode; infeasible radicals are never promoted implicitly — promotion happens only through an explicit frame instruction plus a status-token select.	Axiom 10; [5] §5.5; review Seg. 4 P.3
R11′	Lowering from CAE-DAG to QnIR is a semantics-preserving isomorphism onto the instruction stream; every Layer-3 node class maps to at least one opcode, and the stream retains lineage commitments.	[6] §4, §5.3; review Seg. 4 P.4
R12′	The terminal receipt certifies $H(\text{Execution} \mid \text{Intent}, \mathcal{D}) = 0$ — execution uniquely determined by declared intent and dependencies — and makes that certification only for intent contracts anchored by the `qs.md` §6.1 co-signature handshake (or principal-formalized without a parser step).	[8] §4.4, §6.1; review Seg. 4 P.7
R13	QnIR remains substrate-agnostic; platform lowering happens below it and must be constant-time where the DCC profile requires.	[6] §5.1–5.2
R14	All control flow is bounded: backward jumps carry declared trip ceilings; forward branches are strictly intra-stream; unbounded loops are unencodable.	Axioms 2, 6
R15	Recoverable epistemic conditions are status tokens on the destination value, handled only by `OP_BR_STATUS`/`OP_SELECT`; fatal receipts halt immediately. Only the closed terminal set is terminal.	Axiom 6; review Seg. 4 P.4
R16	The static program hash chain and the dynamic execution trace hash are distinct commitments; no loop iteration can invalidate a program word.	Axiom 5; review Seg. 4 P.6
R17	Every dispatch consumes a strictly positive base cost $c_{\text{base}} \geq 1\,U$: no zero-cost instruction exists at the meter, so W_{rem} is a strict ranking function.	[1] §4.4; review Seg. 4 P.2
R18	Container opcodes exist for the five Category C structures, with single-element versioning amplification bounded by $\mathcal{O}(\log_B M)$ under a declared branching factor.	[9] §3; review Seg. 3 P.7
R19	Cross-node transition claims are anchored only through road receipts whose π field carries a valid execution proof (Realization A or B of `abstractionLayers` §4.4); unproofed receipts quarantine.	[2] §4.4

Two amendments deserve a sentence of justification each, because they change what it *means* for a program to be legal. R12′ inverts the loop-1 conditioning direction at the reviewer's urging: a verifier can check that execution was uniquely determined by the declared intent and dependency set, but two different intents can legitimately share an arithmetic sub-routine, so no receipt could ever have certified the v0.1 direction. And R15 converts the closed twelve-status

terminal registry of Axiom 6 into a two-tier runtime taxonomy: the twelve remain the only *terminal* outcomes, but conditions such as `ORDERING_INDETERMINATE_AT_ZOOM` or `PRECISION_INSUFFICIENT` may first surface as recoverable status tokens that a well-funded program resolves by zoom deepening — emitting a terminal receipt only if the program ends with the condition unresolved. This preserves the closed set (nothing new is terminal) while making the canon's own instruction — "the caller funds a zoom deepening" — finally expressible inside a program.

4 Machine State Model v0.2: the SSA Value Graph

4.1 Why the slot files had to go

The v0.1 state model — a value file **V** of Qm tuples and a blade file **B** of multivectors behind one unified 8-bit index space — failed its own constitution in four ways that the reviewer demonstrated mechanically. Overwriting a slot across instructions or loop iterations mutates machine state in place, contradicting Axiom 13; uncommitted slot values lack commitment hashes, undermining R6 and the hash probes of dispatch stages 2–3; the partition of the 256 indices between **V** and **B** was never defined, so static type expectations break exactly when `OP_ROOT` promotes an even root of a negative scalar from a scalar slot to a blade slot at runtime; and unevaluated symbolic generators (continued fractions, rational pairs) had no declared home at all. Each defect is an instance of the same design error: mutable, address-like storage beneath a content-addressed ontology.

The v0.2 correction adopts the reviewer's recommendation wholesale and is stated as a BOC verdict under the two-stage model of `simemp.md` §7. Tier-1 admission: the SSA replacement preserves collision-resistant identity (every value is born committed), finite metric bounds (liveness window), and security contracts (no hidden state), so C_{valid} is nonempty. Tier-2 lexicographic comparison with the constitutional first key (`ImplicitDeps`): the SSA form removes the implicit slot-mutation semantics entirely (fewer implicit dependencies), then wins on worst-case work (no register spill semantics at the IR level), ties on coupling, and ties on portability. The verdict tabulation is recorded here in rubric mode as the constitution requires, and it is the same verdict the reviewer derived independently in Segment 4, Point 3.

4.2 State tuple

Terminology. The phrase "virtual machine" of v0.1 is retired here in favor of *abstract machine*: the tuple is a deterministic semantics carrier in the SECD/Wasm/zkVM lineage — its steps exist so that receipts, budget accounting, and the trace commitment $H(\text{Trace})$ have defined boundaries to attach to — not a managed runtime in the JVM sense. Its data structure, by contrast, is deliberately MLIR-shaped: the SSA value graph, block parameters at `OP_LOOP` headers, and per-opcode load-time verification are the MLIR/SSA design points, adopted under a frozen opcode catalog and a no-rewrite governance that MLIR itself does not impose. Section 11 tabulates both lineages (rows "JVM-class managed runtimes" and "MLIR dialect stack").

A QnIR v0.2 abstract machine instance is the finite tuple

$$\text{AM}_{0.2} \;=\; \big\langle\, \mathbf{G},\ \mathbf{F},\ \mathcal{S},\ \mathcal{R},\ \kappa_{\text{tree}},\ W_{\text{rem}},\ \text{PC},\ \text{CT} \,\big\rangle, \tag{1}$$

whose components answer R1′–R19 one-to-one:

- **SSA value graph $\mathbf{G} = [\%0, \ldots, \%(L{-}1)]$**: one immutable value per instruction, in instruction order, which is topological (causal birth) order. Value $\%i$ is the triple $\langle \tau_i,\ v_i,\ s_i \rangle$: a static *type tag* $\tau_i \in \{$`QM_TUPLE`, `BLADE`, `RATIO`, `CF_GEN`, `COCHAIN`, `CONTAINER`, `INTENT_REF`,

`STATUS`}, the value payload v_i, and a *status tag* s_i carrying either `OK` or one recoverable token of R15. Because the destination of instruction i is implicitly $\%i$, the 8-bit `dest` field is eliminated from the word format and the operand space widens to 12 bits (Section 7).

- **Frame table F**: declared orthogonal frames with signatures $\eta = (p, q, s)$, instantiated only by explicit frame instructions (R3, R10′); the `BLADE` type tag binds each blade value to its frame id, so no value can change dimensionality after birth.
- **Content-addressed store $\mathcal{S}$**: the only memory, unchanged from v0.1 (no addresses, write-once, versioned update, billed locality, payload air gap), now with domain-separated commitments and active collision detection (Section 6).
- **Receipt queue $\mathcal{R}$**: ordered receipts emitted so far; the terminal receipt closes the queue (R8).
- **Causal clock κ_{tree}**: a fork-join tree clock replacing the flat scalar (Section 13); single-threaded programs degenerate to the v0.1 scalar path.
- **Budget counter W_{rem}**: remaining work in units of U, initialized from the program header, never renewed, reduced by gateway fees, cost reservations, and base dispatch costs (R1′, R17).
- **Program counter** PC: index into the linear instruction stream; only `OP_LOOP` (backward, tripped) and `OP_BR_FWD` (forward, intra-stream) modify it, both within declared bounds (R14).
- **Constant table** CT: header-declared pool of 256-bit commitments and anchor constants, indexed by the `aux` field; this is how `OP_REUSE` and `OP_VERIFY` name commitments that cannot fit in a 64-bit word (Section 7).

4.3 Flattening and the isomorphism claim

Flattening a hash-consed CAE-DAG into **G** is now a direct $\mathcal{O}(V+E)$ topological sort: every Layer-3 node class of `qexpr.md` §4 maps to at least one opcode of Section 5 (R11′), children precede parents because hash-consing already orders construction causally, and the load-time well-formedness check is the trivial $\mathcal{O}(L)$ predicate $\text{src}_k < \text{PC}$ for every operand reference — R4 becomes a syntax check instead of a runtime discipline. No register allocation, spill code, or liveness analysis exists at the IR level; the only structure carried for hardware is the *liveness window* declaration $W_{\text{live}} \leq 256$ in the program header, which tells HCA-Collapse the maximum number of SSA values simultaneously live at any program point so the lowering can map hot values to physical registers. Register allocation is a Layer-0 concern with a Layer-0 calibration receipt (R1′), not an IR semantics; whether the 256-value window suffices for real Q(workflow) kernels is exactly the measurement obligation QO-18 puts to the reviewer.

State-model invited objections

(i) SSA in instruction order makes **G** a mere restatement of the stream — is the value graph doing any work beyond renaming slots? The answer it must defend: the graph is what makes loop-carried dependencies explicit as block parameters at `OP_LOOP` headers, and what lets `OP_JOIN` merge branch receipts without slot aliasing. (ii) The static type tags fix the `OP_ROOT` promotion hazard by forbidding implicit promotion — but does that merely move the pain to authors, who must now write frame-then-select pairs by hand? (iii) $W_{\text{live}} \leq 256$ is a new ceiling; what is the measured distribution of simultaneous liveness in representative programs, and does the ceiling silently throttle kernels that v0.1's flat 256 slots would have carried?

5 The Instruction Set v0.2 (ISA v0.2)

5.1 Design deltas from v0.1

The v0.1 table carried 30 opcodes in seven classes. v0.2 keeps every semantics that survived review, retires one name, modifies five entries, and adds nineteen opcodes, giving 49 opcodes in eight classes. The deltas are exactly the reviewer's list, and each delta cites its forcing finding. (1) `OP_REWRITE` is retired — not removed, *retired*: the alias was already deprecated in `qexpr.md` v1.3.0 with the instruction that it die at the QnIR v0.2 candidate, and its two halves now live as `OP_BYPASS` (canonical contraction, governing canonical node hashing) and `OP_SATURATE` (bounded equality saturation, legal only inside a Q(rsi) workflow with pinned schedule, ceilings, and hash-tie-broken cost extractor, emitting a new versioned artifact under Axiom 13). (2) Control flow gains the forward conditional branch and predicated select the reviewer proved unavoidable (`OP_BR_FWD`, `OP_BR_STATUS`, `OP_SELECT`); without them a program cannot act on `OP_CMP` output at all. (3) Receipt-typed exceptions split the terminal registry's behavior into fatal receipts (halt immediately) and recoverable status tokens (written to the destination's status tag s_i, consumable by the two branch forms) so the canon's own "caller funds a zoom deepening" instruction becomes programmable (R15). (4) `OP_DIV` no longer abuses `aux` as a second destination: it emits a pair value and the shared `OP_UNPACK` projects quotient or remainder, keeping one-destination purity. (5) `OP_ROOT` loses its dynamic dimension promotion: an even root of a negative scalar yields a status token, and the program promotes explicitly through `OP_FRAME` plus `OP_SELECT` (R10′). (6) Containers, radix accumulation, reversion, delegation, road-receipt ingestion, and fork-join join the ISA so that R11′, R18, and the QO-4/QO-10 integrations have opcode-level existence rather than prose-level promises.

5.2 Complete v0.2 opcode table

Table 3 is the complete candidate table. "Arity" counts operand slots consumed (`src1`, `src2`, `src3`); window forms use `src1` = base index and `src2` = count. "Cost" is the v0.2 static cost expression in U (always including the base dispatch cost $c_{\text{base}} = 1$, folded into the listed expression); header ceilings are $H_{\max}$ (logarithmic height), $z_{\max}$ (zoom), $n_{\max} = \text{Dim}_{\max}$ (Clifford dimension), and $S_{\max}$ (the instruction's declared step ceiling in `aux` where applicable); $w = \lceil H_{\max}/64 \rceil$ is the limb width. Recoverable status tokens appear in the failure column marked ∘; fatal receipts are marked •; both vocabularies are defined in Section 8.

Table 3: QnIR v0.2 instruction set (complete candidate table). Classes A–H; ∘ = recoverable status token, • = fatal receipt.

Opcode	Ar.	Semantics (operand → result)	Cost / failure
Class A — Constants and tuple construction (Layer 1 grounding)			
`OP_QCONST`	0	Load digit artifact $Q(k)$, $k \in \{0..9\}$, as canonical tuple; `aux`=k	1 ($Q(0)$/$Q(1)$ pre-sealed)
`OP_TUPLE`	3	Compose $\langle \sigma, q, z, r, d \rangle$: σ, q, r from `src1..3`; z, d packed in `aux`; rejects signed-zero forms	1 / •`...REJECTION_BY_GATEWAY`
`OP_SIGMA`	1	Apply reflection involution $\mathcal{I}_d$: $\oplus \leftrightarrow \ominus$, $\odot$ fixed	1

Opcode	Ar.	Semantics (operand → result)	Cost / failure
OP_RADIX_ACC	2	Radix-anchor construction (QO-4 at ISA level): $\texttt{Cons}_B(q_{\text{high}}, d_{\text{low}}) = (q_{\text{high}} \times B) + d_{\text{low}}$; $B \in \{Q(2), Q(8)\}$ by flag; innermost seed $Q(1)$; $\texttt{Cons}_B(Q(0), Q(0)) \to Q(0)$ via OP_BYPASS	$w+1$ / •...EXPRESSION_TOO_COMPLEX
OP_UNPACK	1	Project field aux$\in \{\sigma, q, z, r, d, C, R\}$ of a tuple or OP_DIV pair	1
Class B — Exact $\mathcal{R}$-algebra (Layer 2 morphisms)			
OP_ADD	2	Exact addition over common δ_z; re-canonicalized	$w+1$
OP_MUL	2	Exact multiplication with orientation parity composition	$2\lceil h_1 h_2/64^2 \rceil + 1$
OP_DIV	2	Diophantine constraint solve; emits pair value (C, R); σ_R, σ_C orientation fixed per qm §4 (floor form; $\sigma_C = \odot$ guarded when $C = Q(0)$)	$3w + S_{\max} + 1$ / •...DIVISION_BY_ZERO_REJECTED
OP_MOD	2	Remainder-only partition under declared modulus anchor (parity via ...Q2_PARITY_v1)	$2w+1$
OP_CMP	2	Positional interval comparison; indeterminacy lands on the status tag, not the terminal set	1 / ∘STATUS_ORDERING_INDET
OP_POW	2	Bounded monoidal fold A^B (exact integer exponent); height check $B \cdot h(A) \le H_{\max}$	$B \cdot w + 1$ / ∘STATUS_TOO_COMPLEX
OP_ROOT	2	Solve $X^n - A = Q(0)$; even root of negative yields status token — promotion only via OP_FRAME + OP_SELECT (no implicit dimension change)	$4w+1$ / ∘STATUS_NEG_EVEN_ROOT
OP_FACT	1	Factorial fold with $\mathcal{O}(1)$ bypass at $n \in \{0, 1, 2\}$	$\max(1, S_{\max}) + 1$; bypass hit: 1
Class C — Symbolic generators and scale (Layer 3 collapse)			
OP_RATIO	2	Hold exact rational pair p/q; $q \neq Q(0)$ enforced; GCD-reduced canonical form	$w+1$ / •...DIVISION_BY_ZERO_REJECTED
OP_CF	0	Nullary continued-fraction generator (aux selects anchored constant $\pi, \tau, e, \ln 2$); per-convergent metering	$S_{\max} + 1$ / ∘STATUS_PRECISION
OP_ZOOM	1	Deepen zoom $z \to z{+}1$ consuming conserved residual r_z; the funded recovery path of R15	2 / ∘STATUS_PRECISION if $z \ge z_{\max}$
OP_FOLD	win	Monoidal accumulation over window [src1, src1+src2); monoid declared in flags; trip ceiling aux=$m_{\max}$; per-step metering	$m_{\max} \cdot c_{\text{body}} + 1$

Opcode	Ar.	Semantics (operand → result)	Cost / failure
OP_TERMINAL	1	Test $r = Q(0)$ (Terminal Exactness); on true, remaining zoom reservations are refunded and zoom work halts	1
Class D — Geometry (frames, products, rotors, calculus)			
OP_FRAME	1	Adjoin orthogonal axis to frame table; verifies $\vec{e}_{k+1} \cdot \vec{e}_i = Q(0)$; binds frame id for BLADE births	2 / frame-ceiling receipt
OP_GP	2	Clifford geometric product over grade-sparse blade operands (dense operands rejected at load: R18-adjacent discipline)	$2^{n_{\max}}$-bounded / •...DIMENSIONAL_BUDGET_EXHAUSTED
OP_WEDGE	2	Exterior product with exact nilpotency $\vec{e}_i \wedge \vec{e}_i = Q(0)$	$2^{n_{\max}}$-bounded
OP_ROTOR	2	Cayley rotor from Pythagorean pair; unit norm by construction	$4w + 1$
OP_REVERSE	1	Clifford reversion (computes $R^\dagger$ for rotor sandwiches $R\vec{v}R^\dagger$)	1
OP_DEC_D	1	Discrete exterior derivative on COCHAIN operands; $\mathbf{d} \circ \mathbf{d}$ intercepted	3; bypass hit: 1
OP_HODGE	1	Discrete Hodge star via dual volume ratios (exact OP_RATIO commitments); requires well-centered mesh declaration	3 / •...REJECTION_BY_GATEWAY
Class E — Optimization and content addressing			
OP_BYPASS	1	Canonical contraction: 16-rule terminating suite of qexpr §4.6.1; on hit, substitute identity at $\Delta W = 0$ metric cost; *dispatch still bills* c_{base} (R17)	1; emits RECEIPT_BYPASS_IDENTITY
OP_SATURATE	win	Bounded equality saturation (qexpr §4.6.2): legal only in Q(rsi) profiles pinning schedule, $K_{\max}$ epochs, node ceiling, hash-tie-broken extractor; output is a new versioned artifact	$K_{\max} \cdot N_{\max}$-bounded / •...BUDGET_EXHAUSTED
OP_REUSE	1	Lookup CT[aux] $\mapsto \mathcal{H}(\alpha_{\text{expr}})$; admit iff breakeven vs. recorded $\Delta W_{\text{recorded}}$ in the artifact's Metric envelope; miss → recompute fallback	1 / ∘STATUS_LOOKUP_MISS
OP_COMMIT	1	Insert operand into $\mathcal{S}$ under domain-separated $\mathcal{H}(\alpha)$; shallow header comparison on hash hit; collision → quarantine	2 / •...HASH_COLLISION_QUARANTINED
Class F — Governance and bounded control			

Opcode	Ar.	Semantics (operand → result)	Cost / failure
OP_GATE	1	SIMEMP gateway crossing: deduct static W_{gate} from the invoking principal *before* inspection; validate DCC, anchors, intent contract	$2 + W_{\text{gate}}$ / •...REJECTION_BY_GATEWAY
OP_BUDGET	1	Charge declared work to W_{rem}; halts when exhausted	1 / •...BUDGET_EXHAUSTED
OP_LOOP	2	Bounded backward jump to a loop header; trip count is a header block parameter declared in aux; program words are never mutated	body billed per trip; 1 per jump
OP_BR_FWD	2	Forward conditional branch: PC ← PC + offset, $1 \leq$ offset $< L -$ PC; condition from operand status/value	1
OP_BR_STATUS	2	Branch on recoverable status token: src1 value, expected token in aux; the funded-recovery edge of R15	1
OP_SELECT	3	Constant-time predicated multiplexer: dest = cond ? src1 : src2; preserves side-channel immunity	1
OP_CALL	1	Bounded call: target id in aux (call graph strictly acyclic, target < caller, verified $\mathcal{O}(L)$ at load); pre-partitioned sub-budget $W_{\text{sub}} \leq W_{\text{rem}}$; static depth ceiling $C_{\max}$	1 / •...BUDGET_EXHAUSTED
OP_RET	0	Return to non-addressable call frame; merges child receipt queue in causal order	1
OP_DELEGATE	1	Mint sub-budgeted child context $W_{\text{sub}} \leq W_{\text{rem}}$ for agent-to-agent work	1
OP_INGEST_ROAD	1	Anchor external road receipt $\mathcal{R}_{\text{road}}$; admissible only if π carries Realization A or B proof (R19); failure quarantines	2 / •...QUARANTINED
OP_FORK	win	Split data-disjoint SSA subgraphs onto branch clocks with pre-partitioned budgets $\sum_j W_{\text{rem}}^{(j)} \leq W_{\text{rem}}$	1 per branch
OP_JOIN	win	Deterministic join: merge branch receipts and trace hashes in lexicographical branch-id order	1 per branch
OP_HALT	1	Emit terminal receipt (aux = status from the closed set) and stop; mandatory final position	1

Class G — Verification and audit

Opcode	Ar.	Semantics (operand → result)	Cost / failure
`OP_WITNESS`	1	Commit current machine-state digest to store (tamper-evident checkpoint)	2
`OP_ANCHORCUT`	1	Emit signed anchor checkpoint receipt (Anchor-Cut chain node; bisection witness for Realization B)	2
`OP_VERIFY`	2	Verify a receipt-chain segment between two witnesses; segment endpoints via `CT`	$K+1$ / •`...QUARANTINED`
Class H — Category C containers (the five structures of `qstructures.md`)			
`OP_CONTAINER_NEW`	win	Build container from value window; lowering via bounded-radix persistent HAMT/RRB with declared $B \in \{Q(8), 32\}$	$M \cdot w$ at build; versioning $\mathcal{O}(\log_B M)$
`OP_CONTAINER_GET`	2	Positional projection (index or record field id in `aux`)	$\log_B M + 1$ / ∘`STATUS_LOOKUP_MISS`
`OP_CONTAINER_SET`	2	Functional update: emits *new* container version referencing predecessor (Axiom 13); no in-place mutation	$2\log_B M + 1$
`OP_CONTAINER_COMMIT`	1	Seal container root commitment into $\mathcal{S}$; transient tiles amortize batch work to $\mathcal{O}(1)$/batch	2

Three v0.1 omissions stand and are now deliberate with reasons on record: no indirect jump (targets are static stream indices or call-table ids); no I/O opcodes (all boundary exchange is receipt-mediated through `OP_GATE`, `OP_INGEST_ROAD`, and `OP_HALT`, per Axiom 8); and no stochastic opcode (R9). The `OP_REWRITE` alias appears nowhere in the v0.2 stream grammar; loaders reject it explicitly with `RECEIPT_REJECTION_BY_GATEWAY` so legacy binaries are diagnosed rather than silently reinterpreted.

ISA invited objections

(i) Forty-nine opcodes is no longer a "small, fixed-width" machine — does AIR arithmetization of Class H and the fork-join pair stay tractable (QO-16)? (ii) `OP_SATURATE` with pinned schedules is deterministic, but two nodes running different extractor generations legitimately produce different artifacts — where exactly is the line between measured divergence and state drift when both artifacts re-enter a shared workflow? (iii) `OP_CALL` reintroduces a stack — non-addressable, statically depth-bounded, acyclic, but a stack; is the code-blowup argument for it quantified anywhere, or is it intuition wearing a receipt? (iv) Recoverable status tokens add a second, unterminated vocabulary next to the closed terminal set — what stops programs from accumulating tokens as unfunded debt and terminating *success-shaped*?

6 Memory, Commitments, and Containers

6.1 The store: unchanged rules, hardened keys

The store $\mathcal{S}$ keeps the five v0.1 disciplines — no addresses, write-once, versioned update, billed locality, payload air gap — because the reviewer did not dispute a single one of them; what the

review hardened is the *key*. v0.2 commits the reviewer's domain-separated construction: local commitments are

$$\mathcal{H}(\alpha) \;=\; \text{SHA3-256}\big(\text{DomainTag} \,\|\, \mathcal{H}(Q(0)_{\text{node}}) \,\|\, \text{CanonicalCBOR}(\alpha)\big), \tag{2}$$

while shared Class-I `Q(anchor)` and `Q(pattern)` schemas strip the node-genesis prefix so that cross-node contract equality remains $\mathcal{O}(1)$. The collision accounting the reviewer demanded is now on record: at 2^{64}–2^{80} distinct artifacts the birthday bound sits between 2^{-129} and 2^{-97}, orders below hardware single-event-upset rates and above any Landauer-feasible brute force, and — more importantly — the machine never *trusts* that bound on local hits: on any hash match during `OP_COMMIT` or `OP_REUSE`, the abstract machine performs a constant-time $\mathcal{O}(k)$ shallow comparison of the canonical tuple header, and a mismatch aborts deduplication with the new fatal receipt `RECEIPT_HASH_COLLISION_QUARANTINED`, sealing both pre-images for hash-primitive migration under the structural-indirection discipline of `simemp.md` §4.3.

6.2 Canonical zoom normalization

Requirement R6 says equality of values is equality of commitment hashes; the reviewer showed v0.1 violated its own rule across zoom levels, since $\langle \oplus, 1, 0, Q(0), d_1 \rangle$ and $\langle \oplus, \kappa(1), 1, Q(0), d_1 \rangle$ denote the same number but hash differently. v0.2 therefore normalizes before hashing: any tuple with $r_z = Q(0)$ is reduced to its canonical minimal zoom $z_{\min}$ (the smallest z whose lattice δ_z still represents the value exactly), and only the normalized form enters $\mathcal{H}(\alpha)$. The normalization is a pure function of the tuple, costs $\mathcal{O}(w)$, and its provenance note — the zoom history that led to the value — stays in the envelope's lineage block, so the discipline that zoom work is real, billed, and conserved is unaffected; only *identity* is zoom-canonical. The open question QO-24 asks the reviewer to test whether normalization ever erases a distinction the conservativity disciplines need.

6.3 Containers

The five Category C structures of `qstructures.md` — `Q(array)`, `Q(matrix)`, `Q(table)`, record tuple, and persistent DAG-deduped tree — get their first real ISA footing in Class H, because loop 1 measured them only "at lowering boundaries" while the review showed the lowering itself was unspecified and naive flat buffers would pay $\mathcal{O}(M)$ versioning amplification per single-element update across the Memory Wall. The v0.2 verdict, following the reviewer's construction: persistent cross-step containers lower to bounded-radix persistent Hash-Array Mapped Tries or RRB-trees with branching factor B aligned to the octet/cache-line anchors ($B \in \{Q(8), 32\}$ declared in the DCC profile), bounding single-element amplification to $\mathcal{O}(\log_B M)$ hash-consed node allocations; and dense linear-algebra kernels operate over *transient* register tiles that seal and emit a single new root commitment at `OP_COMMIT` boundaries, amortizing versioning to $\mathcal{O}(1)$ per batch. Container values are first-class SSA citizens typed `CONTAINER`, their integrity is the store's integrity (all internal nodes are commitments), and their contract rows — footprint, locality, work, versioning amplification, breakeven — are finally *measured* by Class H opcodes rather than promised by prose. Whether the HAMT/RRB choice actually wins the two-stage BOC comparison against Merkle B-trees on real kernels is left as QO-19 with the rubric tabulated, not asserted.

7 Encoding and Program Form v0.2

7.1 The 64-bit word, repaired

Loop 1 committed to a fixed 64-bit instruction word and the reviewer held the candidate to it, showing four operands that cannot fit the v0.1 layout. v0.2 keeps the 64-bit commitment and repairs the layout by deleting the `dest` field (implicit in SSA: instruction i writes $\%i$) and narrowing the per-word flags, spending the reclaimed bits on a third operand and wider indices:

Field	Bits	Content
opcode	8	one of the 49 opcodes of Table 3
src1	12	first operand SSA index
src2	12	second operand SSA index (window length for window forms)
src3	12	third operand SSA index (`OP_TUPLE` $\sigma/q/r$; `OP_SELECT` second arm)
aux	16	immediate: digit index, step ceiling $S_{\max}$, constant-table id, receipt status, frame id, packed (z, d)
flags	4	per-instruction attributes (bypass probe on/off, radix base B, refund eligibility, monoid class)

Every field is declared even when zero-padded (R3). The four-bit flags budget is honest about its cost: the six v0.1 flag attributes that did not survive the narrowing moved into the program header, where they are per-program rather than per-word, and QO-30 asks the reviewer whether the remaining four bits will hold — the alternative (a second word format) would break the uniformity that makes the stream hash chain and the AIR arithmetization tractable, so the candidate prefers to spend header space first.

Higher-arity and dual-output forms resolve inside this layout without hidden conventions. `OP_FOLD` takes its operand window as [`src1`, `src1`+`src2`) with the monoid declared in flags and its trip ceiling in `aux` — the reviewer's register-window rule, adopted verbatim. `OP_TUPLE` packs z and d into `aux` as two bytes and takes σ, q, r from the three source fields. `OP_DIV` emits a pair value that `OP_UNPACK` projects, so `aux` stays a pure immediate carrying the Diophantine step ceiling that Requirement 1's totality argument needs. And 256-bit commitments never touch the word at all: `OP_REUSE`, `OP_VERIFY`, and `OP_CALL` index the header constant table CT, a CBOR section of the program header that the v0.1 encoding already promised to deterministic CBOR but left unspecified.

7.2 Two hash chains, not one

The reviewer's unanticipated objection C is adopted in full: v0.1 chained the *runtime* causal counter into the *static* program commitment, which is ill-defined the moment a loop body executes twice. v0.2 separates the two ledgers. The static program stream commitment is

$$W_i^{\text{prog}} \;=\; \text{Hash}\big(W_{i-1}^{\text{prog}} \,\|\, \text{word}_i \,\|\, i\big), \tag{3}$$

a pure function of the loaded bytes, checked once at load time and immutable thereafter; loop iterations cannot touch it because the trip counter is a loop-header block parameter, not a word. The dynamic execution trace hash

$$\mathcal{H}(\text{Trace}) \;=\; \text{Hash}\big(\ldots \,\|\, \text{word}_{i_t} \,\|\, \kappa_t \,\|\, s_t\,\big) \tag{4}$$

stamps each executed step's static index, tree-clock value, and resulting status tag, and it is what the terminal receipt seals. A verifier with the program bytes can always recompute W_i^{prog}; a verifier with only the receipt can check trace membership against the sealed hash but cannot reconstruct it — that asymmetry is exactly why anti-drift certification in v0.2 also carries the execution-proof interface of Section 9 instead of claiming receipt-only sufficiency.

7.3 Program header

The v0.2 header (deterministic CBOR) carries: QnIR version; DCC profile reference; required anchor contracts; the constant table CT; declared ceilings — stream length L, depth $D_{\max}$, node budget $N_{\max}$, zoom ceiling $z_{\max}$, height ceiling $H_{\max}$, liveness window $W_{\text{live}} \leq 256$, call-depth ceiling $C_{\max}$, saturation ceilings $(K_{\max}, N^{\text{sat}}_{\max})$ where applicable; the work budget W_{budget}; the gateway fee W_{gate} schedule; the intent-contract reference for R12′, which must reference a handshake-anchored contract when the intent is parser-derived; and the paternity reference (Axiom 12). Program size remains exactly $64L$ bits and is checked against the node budget before execution begins.

8 Execution Semantics and Ceiling-Parameterized Totality

8.1 The two-tier outcome vocabulary

v0.2 makes explicit what the reviewer showed v0.1 only implied: the machine speaks two grammars of non-success. *Fatal receipts* are terminal transitions from the closed twelve-status set of Axiom 6; they halt the machine immediately, seal the receipt queue, and cannot be caught in-stream: `RECEIPT_BUDGET_EXHAUSTED`, `RECEIPT_DIVISION_BY_ZERO_REJECTED`, `RECEIPT_REJECTION_BY_GATEWAY`, `RECEIPT_HASH_COLLISION_QUARANTINED`, `RECEIPT_QUARANTINED`, `RECEIPT_PHYSICAL_VIOLATION`. *Recoverable status tokens* are intermediate conditions written to the destination value's status tag s_i — `STATUS_ORDERING_INDET`, `STATUS_PRECISION`, `STATUS_NO_SOLUTION`, `STATUS_LOOKUP_MISS`, `STATUS_TOO_COMPLEX`, `STATUS_NEG_EVEN_ROOT` — consumable only by `OP_BR_STATUS` and `OP_SELECT`. A token is *funded debt*: the program may spend work to retire it (the canonical retirement being `OP_ZOOM` for ordering and precision tokens), but a token left unconsumed at stream end maps deterministically to its closed-set terminal (`STATUS_ORDERING_INDET` → `ORDERING_INDETERMINATE_AT_ZOOM`, `STATUS_PRECISION` → `PRECISION_INSUFFICIENT`, `STATUS_NO_SOLUTION` → `NO_SOLUTION`, `STATUS_LOOKUP_MISS` → `INCONCLUSIVE`, `STATUS_TOO_COMPLEX` and `STATUS_NEG_EVEN_ROOT` → `EXPRESSION_TOO_COMPLEX`). Nothing is added to the terminal registry — the closed set of Axiom 6 is untouched — and the mapping is total, so R8 still holds: a program cannot end silently, and it cannot end success-shaped with outstanding unfunded tokens.

8.2 The seven-stage dispatch loop

Each instruction executes under a seven-stage loop, refined from v0.1's five by the cost and status disciplines:

1. **Base billing.** $W_{\text{rem}} \leftarrow W_{\text{rem}} - c_{\text{base}}$ with $c_{\text{base}} = 1\,U$; if $W_{\text{rem}} < 0$, halt with `RECEIPT_BUDGET_EXHAUSTED` (R17 — no instruction is free at the meter).
2. **Cost reservation.** Compute C_{op} from header ceilings and `aux`; if $W_{\text{rem}} < C_{\text{op}}$, halt with `RECEIPT_BUDGET_EXHAUSTED` *before* any semantic executes; otherwise reserve it.
3. **Bypass probe.** Consult the 16-rule identity table; on hit, substitute the identity result, emit `RECEIPT_BYPASS_IDENTITY` at $\Delta W = 0$ metric cost, refund the reservation, advance κ_{tree} and PC.
4. **Reuse probe.** If the result artifact exists under its (domain-separated) commitment and the breakeven holds against the recorded $\Delta W_{\text{recorded}}$ of the artifact's Metric envelope, retrieve it and refund the unused reservation.
5. **Direct compute.** Execute the opcode semantics with per-micro-step metering: iterative internals (`OP_DIV`, `OP_ROOT`, `OP_CF`, `OP_POW`, `OP_FACT`, `OP_FOLD`, `OP_SATURATE`) decrement

the reservation as they consume declared steps, so a mid-instruction exhaustion surfaces as `RECEIPT_BUDGET_EXHAUSTED` rather than an overdraft.

6. **Refund.** On early terminal exactness (`OP_TERMINAL` hit, convergent early exit), refund the unconsumed reservation (R1′).
7. **Receipt accumulation.** Append any fatal receipt to $\mathcal{R}$ and halt, or write the status token to s_i; the $\mathcal{O}(1)$ preemption yields to bus arrivals per the latency-harvester discipline, with the v0.2 addition that harvester-visible results land only in isolated staging visible at declared checkpoints (`qlog.md` §6.2 discipline).

8.3 The cost model

Every C_{op} is a closed-form, monotone polynomial of the declared header ceilings and the instruction's own immediate — never of runtime register values (R1′). The shape follows the reviewer's discipline: arithmetic scales with the limb width $w = \lceil H_{\max}/64 \rceil$ (Table 3 lists $w{+}1$ for `OP_ADD`, $2\lceil h_1 h_2/64^2 \rceil{+}1$ for `OP_MUL`, $3w{+}S_{\max}{+}1$ for the division search); search and fold opcodes pre-reserve their declared step ceiling $S_{\max}$ or $m_{\max}$; geometric products reserve the grade-sparse $2^{n_{\max}}$ bound; and dispatch itself is never free. Two properties make the model more than a table. First, it is *checkable*: because C_{op} depends only on header-declared quantities, the load-time validator can compute every instruction's worst-case cost before execution and reject programs whose summed reservation exceeds W_{budget} — the budget gate becomes partially static. Second, it is *calibrated*: at the HCA-Collapse boundary each target DCC profile supplies a verified affine WCET/energy calibration receipt certifying that the lowered constant-time sequence respects the parameterized U bound, audited through implementation-feedback receipts; the certificate chain from C_{op} to joules runs exactly through the Layer-0 Hardware Telemetry Envelope of `simemp.md` §4.4, and never backwards into logical budgets.

Candidate 1 (Totality of QnIR v0.2 programs)**.** Every QnIR v0.2 program terminates and resolves to exactly one terminal receipt from the closed status set.

Argument (candidate, to be attacked). W_{rem} is a strict ranking function: every dispatch costs $c_{\text{base}} \geq 1\,U$ (stage 1), every semantic reserves its polynomial cost before executing (stage 2), and iterative internals meter per micro-step (stage 5), so no execution path consumes work without decrementing the counter, and the counter is never renewed. The control-flow graph is finite and acyclic except for `OP_LOOP`, whose trip count is a declared block parameter bounding iterations statically, and `OP_CALL`, whose call graph is verified acyclic with depth $\leq C_{\max}$ at load time. Streams are finite (L words, load-checked); token debt is totalized by the mapping of Section 8.1; the trace hash and receipt queue seal the outcome. The argument is an accounting identity again — but now every term of the identity is a declared ceiling rather than an honest guess. What the identity still assumes, and the reviewer should press, is the calibration chain of Section 8.3: the U-level bound is airtight by construction, while its physical meaning rests on lowering receipts the implementation owes (QO-17). □

9 Governance Integration

9.1 Gateway billing attribution

The constitution now fixes what loop 1 left implicit: generation behind the stochastic boundary is free; every gateway crossing is deterministic, metered, and receipted; and the *invoking principal* pays. QnIR v0.2 implements this at the ISA level: `OP_GATE` deducts the static, DCC-declared fee $W_{\text{gate}} > 0$ from the caller's W_{rem} *before* any candidate is inspected, charged regardless of outcome (R9), so failed and adversarial candidate floods are billed to their invoker and cannot induce

free-riding metric exhaustion at the boundary. The same accounting closes the loop-1 double-flagged finding at the machine level, and it composes with the staged pipeline of `simemp.md` §6: parsing, measurement, BOC selection, verbalization, and co-signature verification each carry their metered stages, with the gateway fee as the toll between the free side and the governed side. The fee schedule is per-DCC-profile metadata, not a universal constant — the fairness properties of that schedule (who sets W_{gate}, and whether outcome-independent charging over-deters honest exploration) are exactly what QO-23 submits to the reviewer.

9.2 The parser-boundary precondition on anti-drift

R12′ couples the machine's proudest guarantee to the semantics layer's newest gate. Loop 1 certified that execution did not drift from declared intent; the review showed the certificate is only as good as the intent object, and an unmitigated statistical parser can hallucinate an intent that carries a valid human origin signature and well-formed modality tags — after which every downstream check validates the wrong intent with perfect rigor. The canon now closes this at the source: parser-derived candidates enter typed no higher than `MODAL_HYPOTHESIS` (`qs.md` §4.4) and are sealed only after the Deterministic Canonical Round-Trip Paraphrase & Co-Signing Handshake (`qs.md` §6.1), whose four typed failure receipts (§6.2) make parse failures first-class lineage events. QnIR v0.2 inherits the discipline mechanically: the program header's intent-contract reference must resolve to a handshake-anchored contract whenever the intent was parser-derived, and `OP_GATE` verifies that anchoring as part of contract validation. Anti-drift certification therefore means: *given* a principal-anchored intent contract, execution is uniquely determined by it and the declared dependency set ($H(\text{Execution} \mid \text{Intent}, \mathcal{D}) = 0$); the machine certifies the determinism of the descent, while the handshake certifies the anchoring of the root — and neither certifies the other's job.

9.3 The road-receipt interface

Cross-node claims enter the machine only through `OP_INGEST_ROAD`, which admits a road receipt $\mathcal{R}_{\text{road}} = \langle \mathcal{H}(S_{\text{in}}), \mathcal{M}, \mathcal{H}(S_{\text{out}}), \Delta W, \pi, \sigma \rangle$ only if its proof field π carries one of the two realizations canonized in `abstractionLayers.md` §4.4: Realization A, a transparent zk-STARK over the AIR-arithmetized QnIR execution trace, verified in $\mathcal{O}(\text{polylog}\, N)$ with zero trust in the sender and zero disclosure of the private local graph; or Realization B, the optimistic interactive Anchor-Cut bisection, in which π commits the Merkle root of the K intermediate `OP_ANCHORCUT` witnesses and the receiver holds the protocol-enforced challenge right, descending $\mathcal{O}(\log K)$ rounds to a single origin-quotiented QnIR instruction step replayed deterministically at $\mathcal{O}(1)$ local cost. A receipt that fails response or hash check quarantines with `RECEIPT_ANCHOR_REVOKED` — deliberately the inherited anchor-fracture name, so no thirteenth terminal status exists. The portable-commitment discipline rides along: public commitments are computed only after quotienting the local root, $Q(0)_A \mapsto \widehat{Q(0)}$, so identical deterministic computations on distinct nodes yield identical portable commitments $\mathcal{H}_{\text{pub}}(S_{\text{out}})$ and Class-I transitions are verifiable by forced re-derivation on any peer.

The candidate's own obligation inside this interface is the AIR arithmetization itself: `abstractionLayers.md` §13 item 4 defers it to "QnIR v0.2", and this draft accepts the deferral in the form of QO-16 — the constraint system for the 49-opcode machine, the lookup-argument treatment of the bypass table, and the prover-cost versus thermodynamic-breakeven accounting that decides when Realization B is the honest choice. What v0.2 delivers now is the *interface contract*: the ISA is fixed-width, integer-only, loop- and call-bounded precisely so that its trace arithmetizes, and Class G's `OP_ANCHORCUT` emits the witness format Realization B consumes.

9.4 BOC in the design verdicts

Every design verdict of this draft was taken under the constitution's new two-stage model, and the two non-trivial ones are tabulated here in rubric mode as §7 requires. *SSA value graph vs. mutable slots* (Section 4.1): Tier-1 all-pass on both finalists; Tier-2 lexicographic with `ImplicitDeps` first eliminates the slots on the constitutional key (slot mutation is implicit state), then on worst-case work (no spill semantics at IR level), tie on coupling, tie on coverage. *HAMT/RRB vs. Merkle B-tree containers* (Section 6.3): Tier-1 all-pass on both; Tier-2 order declared ($\Delta W_{\text{worst}}, M_{\text{coupling}}, P_{\text{coverage}}$) in the calling profile leaves the finalists inside the tie band of available evidence — the verdict is therefore recorded as *unresolved at rubric level* and handed to the reviewer as QO-19 rather than silently sealed, which is itself the constitution's mandated behavior for residual ties (no implicit tie-breaking).

10 Worked Examples v0.2

The loop-1 examples failed reproduction on exactly the details the reviewer listed: missing initializations, an undefined primed slot `V0'`, omitted destination fields, a fault prose pointing at the wrong slot, and dispatch-stage numbering that contradicted the semantics section. The v0.2 examples are restated in SSA form with every field declared; each row is now checkable mechanically against Sections 4–8, and the pair of follow-up examples exercises the two v0.2 capabilities loop 1 could not express at all — status recovery and the bounded call.

10.1 Example 1: exact division with parity partition (corrected)

Compute $X \div Q(2)$ for $X = \langle \oplus, 7, 0, Q(0), d_1 \rangle$ (declared zoom $z = 0$), with the dividend now *explicitly loaded* rather than assumed in a slot:

PC	Word	Writes	Effect	Receipt
0	`QCONST aux=7`	$\%0 \leftarrow Q(7)$	dividend born, $\langle \oplus, 7, 0, Q(0), d_1 \rangle$	—
1	`QCONST aux=2`	$\%1 \leftarrow Q(2)$	divisor born, $\langle \oplus, 2, 0, Q(0), d_1 \rangle$	—
2	`GATE src1=%hdr`	—	DCC + `...Q2_PARITY_v1` verified; W_{gate} billed	—
3	`DIV src1=%0, src2=%1`	$\%3 \leftarrow (C, R)$	solve $X = (Q(2)\, C) + R$, $S_{\max}$ reserved	—
4	`UNPACK src1=%3, aux=C`	$\%4 \leftarrow C$	quotient projection	—
5	`UNPACK src1=%3, aux=R`	$\%5 \leftarrow R$	remainder projection; parity check	—
6	`HALT aux=SUCCESS`	—	$C = \langle \oplus, 3, 0, Q(0), d_1 \rangle$; $R = \langle \oplus, 1, 0, Q(0), d_1 \rangle$ (odd)	terminal

Had the divisor been $Q(0)$, stage 5 of `OP_DIV` (*direct compute*, not the stage-2 bypass probe) would emit `RECEIPT_DIVISION_BY_ZERO_REJECTED` and `OP_HALT` would seal it as the terminal state — the forbidden operation costs its reservation and leaves a permanent witness. Every v0.1 defect is addressed: the dividend is born at PC 0; there is no primed slot notation; the pair destinations are explicit SSA values %4, %5; the stage references match Section 8.2; and the fault prose names `src2=%1`, the actual divisor.

10.2 Example 2: factorial with bypass, reuse, and explicit calls

Compute $n!$ for $n = 4$, with the identity-table hits where the canon guarantees them and a constant-table reuse of the committed 3! artifact:

PC	Word	Writes	Effect	Receipt
0	`QCONST aux=4`	$\%0 \leftarrow Q(4)$	n born	—
1	`QCONST aux=1`	$\%1 \leftarrow Q(1)$	accumulator seed	—
2	`GATE src1=%hdr`	—	anchors verified	—
3	`REUSE aux=ct(3!)`	$\%3 \leftarrow 3!$	breakeven vs. recorded ΔW holds; fetch	hit
4	`MUL src1=%0, src2=%3`	$\%4 \leftarrow 4 \times 6$	$= \langle \oplus, 24, 0, Q(0), d_1 \rangle$	—
5	`COMMIT src1=%4`	—	insert under canonical $\mathcal{H}(4!)$; κ+1	—
6	`HALT aux=SUCCESS`	—	terminal receipt with ΔW ledger	terminal

For $n \in \{0, 1, 2\}$ stage 3 of the dispatch loop (the *bypass probe*) hits the identity table ($0! \to Q(1)$, $1! \to Q(1)$, $2! \to Q(2)$) and the program completes at zero metric cost beyond the per-dispatch base — exactly reproducing the canon's factorial bypasses [5] §5.6, and this time the stage number is the semantics section's stage number. The `REUSE` breakeven at PC 3 evaluates against $\Delta W_{\text{recorded}}$ stored in the committed artifact's Metric envelope at its own `OP_COMMIT` time — the circularity the reviewer identified in v0.1's unmeasured $\Delta W_{\text{compute}}$ prediction is gone because the only admitted predictor is a recorded measurement.

10.3 Example 3: status recovery — the zoom-deepening program

Loop 1 could not express the canon's own instruction to "fund a zoom deepening" after an indeterminate comparison; R15 and `OP_BR_STATUS` exist precisely for this. Compare $A = \langle \oplus, 1, 0, Q(1)/3\text{-gen}, d_1 \rangle$ against $B = \langle \oplus, 1, 0, Q(0), d_1 \rangle$ (i.e., 1/3 vs. 1 at $z = 0$):

PC	Word	Effect	Receipt / status
0–1	`(load` A`,` B`)`	$\%0, \%1$ born (`RATIO`-typed A)	—
2	`CMP src1=%0, src2=%1`	$\%2 \leftarrow$ comparison at $z = 0$	$s_2 =$`STATUS_ORDERING_INDET`
3	`BR_STATUS src1=%2, aux=INDET, off=2`	status observed; jump to PC 6	taken
4	`HALT aux=SUCCESS`	(untaken arm: accept z=0 verdict)	—
5	`BR_FWD off=1`	skip recovery arm if verdict accepted	—
6	`ZOOM src1=%0`	fund deepening $z \to z$+1	—
7	`CMP src1=%0, src2=%1`	recompare at deeper zoom	$s_7 =$ `OK` (ordering decided)
8	`HALT aux=SUCCESS`	terminal with decided ordering	terminal

If the declared $z_{\max}$ were reached with ordering still indeterminate, PC 7 would leave s_7 unresolved and the stream-end mapping would seal `ORDERING_INDETERMINATE_AT_ZOOM` as the terminal receipt — the closed set's outcome, reached through funded attempts rather than a silent halt at first contact.

10.4 Example 4: bounded call for shared verified patterns

A Q(pattern) for exact matrix–vector product, verified once and committed, is invoked instead of inlined at every use site: `OP_CALL aux=pat id` with the sub-budget W_{sub} declared per call site, the call graph verified acyclic at load (target id < caller id), and `OP_RET` merging the child

receipt queue in causal order. The cost ledger the reviewer asked for, quantified: inlining a P-word pattern at m use sites yields $L = L_0 + m(P-1)$ words; the call form yields $L = L_0 + m$ words plus $C_{\max} \geq 1$ and per-call overhead of $2\,U$. The break-even in bytes is $P > 2$ — trivially satisfied by any non-trivial pattern — while the break-even in *verification* work is the honest motivation: the called pattern is verified once at commitment, not m times at flatten, so hierarchical Q(workflow) graphs verify in $\mathcal{O}$(distinct patterns) rather than $\mathcal{O}$(inlined words).

11 Prior-Art Positioning

Loop 1 deferred the reviewer's Segment 1 Point 5 to the QnIR v0.2 cycle, and the constitution records that deferral as an open line in its integration record. This section discharges it. The purpose is not to claim novelty against strawmen but to state, paradigm by paradigm, what QnIR re-derives, what it adopts, and what it claims only the simultaneous composition can give — so the reviewer can attack the composition claim directly rather than excavate it.

Table 4: QnIR v0.2 against established prior art. "Adopts" = mechanism used knowingly; "re-derives" = parallel mechanism re-derived under constitution constraints; "adds" = what the reviewer cannot find in the paradigm.

Paradigm	**Mechanism relationship**	**What QnIR adds**
eBPF / RFC 9669 [13]	*Adopts*: static verification, bounded loops, 64-bit register word discipline; *re-derives*: load-time safety proof	exact tuple values (no machine words at the semantic layer); totality by budget with sealed receipt; content-addressed program identity; no kernel-trust anchor — verification is cryptographic, not privileged
Deterministic Wasm (fuel-metered) [14]	*Adopts*: platform-invariant fuel/gas as the metering idea behind U; *re-derives*: deterministic execution envelopes	U decoupled from physical joules by constitution (`simemp` §4.4) with a calibration-envelope interface; float-free by axiom, not by profile flag; receipts are lineage artifacts, not logs
Total functional programming [15]	*Re-derives*: totality by construction (bounded recursion, no general recursion)	totality enforced by an emitted-receipt economy and a *budget counter as ranking function*, i.e. resource-bounded totality with failure-as-value semantics (`BUDGET_EXHAUSTED` is a result, not a crash)
Unison [16]	*Adopts*: content-addressed definitions, hash-keyed $\mathcal{O}(1)$ equivalence, immutable versioning; *re-derives*: write-once store	commitments are *governed artifacts* inside a nine-block envelope with lineage, DCC profile, and lifecycle state; identity extends from code to every numeric value and receipt
egg [17] / egglog equality saturation	*Re-derives*: rewrite-driven optimization; *adopts*: the e-graph lesson that saturation must be budgeted and scheduled	the `OP_BYPASS`/`OP_SATURATE` split: confluence and $\mathcal{O}(1)$ hashing reserved to the 16-rule contraction; saturation legal only under pinned schedule, ceilings, and versioned-artifact output (measured divergence, not drift)
zkVMs (RISC Zero, Cairo) [18]	*Adopts*: AIR/STARK execution receipts as Realization A of the road receipt; *re-derives*: the small-deterministic-VM design point	receipts are constitutional objects (quarantine, anchor revocation, bisection rights) tied to a local-first identity model; origin-quotiented portable commitments make Class-I transitions cross-node deduplicable

Table 4: QnIR v0.2 against established prior art (continued).

Paradigm	Mechanism relationship	What QnIR adds
Posit arithmetic [19]	*Re-derives*: the goal of drift-free numerics; *adopts*: the critique of IEEE 754 as a design error class	exactness is not a wider mantissa but an *ontology*: residual-conserving tuples with explicit zoom, conserved remainders, and terminal-exactness receipts; inexactness is a typed outcome, never a silent rounding
Constructive reals (iRRAM, Boehm) [20]	*Re-derives*: demand-driven precision refinement (zoom as the analogue of increasing precision)	refinement is metered, receipted, and bounded by declared ceilings; the expressiveness boundary vs. $\mathbb{Q}$ and Type-2 streams is stated (loop-1 review, Seg. 2 P.2) instead of implied
Linear/affine resource logics [21]	*Re-derives*: no silent weakening (residual conservation), contraction only via explicit reuse	the resource algebra is an abstract machine: sequent-style rules become dispatch stages with billed erasure, and the exponential modality is the content-addressed store itself
`JVM`-class managed runtimes [36]	*Adopts*: the portable-instruction-stream-plus-load-time-checker genre; *re-derives*: load-time verification as an admission gate	no privileged verifier (validation compiles to receipts and public commitments, not trust in a verifier binary); no mutable frame slots — v0.1's slot file was rejected for exactly that kinship (Axiom 13); no dynamic class loading, JIT speculation, or GC timing: totality by budget, float-free by axiom
MLIR dialect stack (Mojo compiles through it) [37, 38]	*Adopts*: the SSA value graph with block parameters, per-operation verification, and structured control flow; *re-derives*: hash-consing as content addressing; progressive lowering as the abstraction-layer stack with receipted crossing (QO-13)	a *closed* 49-opcode catalog instead of open dialects (No-Implicit Rule; the co-signed vocabulary must be frozen); cost-in-the-ISA (static U per row of Table 3) instead of cost-blind operations; no rewriters — content addresses are evidence, and only `OP_SATURATE` mutates, pinned, versioned, receipted. Division of labor: QnIR is the contract layer; an MLIR/Mojo backend is a legitimate downstream HCA-Collapse lowering target

The composition claim, stated falsifiably: no single paradigm above simultaneously provides (a) exact residual-conserving numeric semantics, (b) content-addressed identity for values and programs alike, (c) budget-totality with failure-as-value receipts, (d) an epistemic boundary between stochastic candidate generation and deterministic verification, and (e) cryptographic cross-node verification without disclosure of private state. Each paradigm provides a subset; the Qn program's bet is that the *joint* discipline is what AI-agent workflows need, and that paying its coordination costs is rational exactly when agent autonomy is high and trust is low. The reviewer is invited to name a system that already provides the joint discipline under another vocabulary — that single counterexample would reframe the program's contribution more decisively than any internal critique.

12 External Claims Assessment: the OpenAI Mathematics Corpus and Two Headline Claims

The loop-2 preparation window (September–October 2026) coincided with an unusual external event that this draft must dispose of explicitly rather than silently. OpenAI released a public repository, `github.com/openai/math` (initial commit 6 October 2026), containing 721 mathematical manuscripts organized into 373 families and described by the repository itself as "raw research data" produced by an internal model, with roughly 42 % of top-line results carrying Lean formalizations [40]. Three headline claims from this effort dominate the public discussion [44]: a claimed refutation of the Exponential Time Hypothesis (ETH) [39, 41]; a claimed proof of the local smoothing conjecture in *every* dimension, which would imply the Kakeya set conjecture and Stein's restriction conjecture in all dimensions [42, 43]; and a claimed rigorous existence theory for interacting four-dimensional quantum field theories whose artifacts are held by an invited expert group rather than released. The author's review program imports external evidence through an explicit claim-intake discipline — the same No-Implicit Rule that governs the machine extends to the literature — so this section records the intake, the mechanical dependency scan, and the dispositions. The verdict, stated up front: *no claim in the corpus is load-bearing for the QnIR v0.2 derivation*, and the reason is mechanical rather than rhetorical (Section 12.2). The episode is retained because it is also a live field demonstration of exactly the failure class the Qn receipting design polices (Section 12.3), and because a reviewer who disagrees with the dispositions should have a numbered target to attack (QO-31, QO-32).

12.1 The claims and their verification status

Table 5 inventories the claims as released, with their artifact and formalization status. The single most important structural fact about this corpus is an *asymmetry*: a Lean kernel is an internal verifier for a formal artifact — it certifies that the formal statement follows from the axioms — but it cannot certify that the formal statement captures the informal claim being announced. The examination group convened for the ETH claim (R. Impagliazzo, R. Paturi, R. Williams) states publicly that its involvement "should not be taken as certification of the claim," and that a result of this significance requires careful independent scrutiny [44]. In the vocabulary of this draft: the informal-to-formal translation is a *parser boundary*, and statement fidelity across that boundary is a receiptable property, not an inherited one. This is precisely the trust gap that `qs.md` §6.1 closed for verbalized contracts with the four typed failure receipts and the R12 post-handshake precondition, and it is not a coincidence that the same boundary is where the external episode concentrates its risk.

For E-2 the shape of the risk is worth spelling out, because it is exactly the shape the Qn program's typed receipts anticipate. A claimed ETH disproof via an analytic PDE method against a finitely combinatorial statement can fail in three ways, and they are disjoint: either the formalized statement is weaker than ETH as usually understood (statement-fidelity failure — the most likely branch by the examination group's own emphasis), or the auxiliary construction itself fails an exactness or totality obligation when audited, or nothing fails and ETH falls, in which case complexity theory absorbs a major result. The Qn reading is indifferent to which branch obtains: all three are *typed outcomes of a verification workflow*, and the design question is only whether the workflow's receipts record which branch occurred. The same tri-partite reading applies to E-3, with the additional observation that E-2 and E-3 reportedly descend from one shared method family, which the next subsection treats as an evidence-cluster fact rather than two independent confirmations.

Table 5: The October 2026 OpenAI mathematics claims as released [40, 44, 45]. "Independent status" is the position at this draft's build time (2026-10-27); no row had a completed independent verification.

Claim (id)	**Artifact / formalization**	**Independent status**
Corpus of 721 manuscripts in 373 families (E-1)	Released as "raw research data," 6–26 Oct 2026; ∼42 % of top-line results with Lean formalizations; complexity-adjacent families present (e.g. Family 102, "Ordinary NP-hardness at the basic semidefinite threshold")	No independent verification program for the bulk; released for community triage
ETH is false (E-2)	Natural-language disproof 16 Oct 2026; Lean artifact 17 Oct; preprint dated 25 Oct [41]; claimed method: "a novel auxiliary partial differential equation"; P = NP explicitly *not* claimed	Examination group convened (Impagliazzo, Paturi, Williams); involvement "should not be taken as certification"; statement-fidelity check (formal statement vs. actual ETH) named as the open human obligation
Local smoothing in all dimensions ⇒ Kakeya and Stein restriction in all dimensions (E-3)	AI output 18 Oct 2026; Lean formalization 19 Oct; same claimed method family (auxiliary PDE)	No independent report at build time; prior art: 2D local smoothing [43], 3D Kakeya [42] (127-page proof; 2025)
Context rows (E-0)	Navier–Stokes finite-time blowup announced 9 Sep 2026 [45]; rigorous existence for 4D gauge-fermion QFTs incl. pure Yang–Mills, Lean artifact 16 Oct 2026	QFT artifacts withheld by the invited expert group (Gross, Douglas, Jaffe, Klebanov, Nekrasov) pending their own write-up; the NS announcement carries an unresolved public priority dispute

12.2 Mechanical dependency scan against the Qn derivation

The dispositions below are not an authorial judgment call; they are the output of a mechanical scan that the reviewer can re-run. Method: a lexical scan of the complete canon snapshot (25 files annexed to this PDF) for the vocabulary of conditional complexity — ETH, SETH, "exponential time hypothesis," P vs NP, subexponential time, hardness assumptions — plus a citation walk of every complexity-flavored statement the canon makes. The scan returns exactly three classes of complexity content in the canon, none of which is a conditional assumption:

1. **Constructive cost facts.** `qm.md` derives, e.g., that extending the unary successor morphism to arbitrary depth incurs $\mathcal{O}(q_z)$ causal lineage depth and $\mathcal{O}(q_z)$ Landauer creation work (Annex E). These are statements about what an operation *costs when performed* under the compute-unit U — accounting identities of the machine — not claims about how fast any problem can be solved by anyone. No hypothesis about the world's algorithmic landscape is consumed.
2. **Proven, unconditional NP-hardness.** `qexpr.md` cites the NP-hardness of optimal DAG extraction from a saturated e-graph (Tate et al.; `egg`) as the forcing finding behind the `OP_BYPASS`/`OP_SATURATE` split and the greedy polynomial-time extractor with hash tie-breaking (Annex H). NP-hardness is a reduction, not a conditional lower bound: it stands or falls with published proofs, not with ETH. Even the extreme branch of E-2 being verified — or a hypothetical P = NP world — would leave the citation and the design exactly where they stand, because greedy deterministic extraction is chosen for hash stability under Axioms 13 and 14, not because optimal extraction is believed hard. The design would keep the greedy extractor in a world where optimal extraction became cheap, for the same reason content addresses do not become pointers when lookup gets fast.
3. **Historical transcript material.** The loop-1 review (Annex U) compares Qn arithmetic against polynomial-time rational routines in the course of its objections. This is reviewer prose, preserved verbatim as a historical record, and load-bearing for nothing.

Table 6 states the per-claim disposition. The Kakeya row deserves one plain sentence: the canon contains no occurrence of Kakeya, Hausdorff dimension, Besicovitch sets, or restriction theory — the claim, even if fully verified by the geometric-measure-theory community, touches no axiom, no opcode, no encoding decision, and no receipt in this derivation. It would be *good news about mathematics*; it would not be news about QnIR.

The E-2 "if verified" cell is the only nontrivial entry, so its scope is bounded here. The QnIR v0.2 cost table (Table 3) asserts static costs in U as *metering decisions* — the ISA's totality argument needs ceilings to exist and be pinned, not to equal any external hardness bound. `OP_SATURATE`'s workflow guidance (pinned schedule, ceilings, versioned-artifact output) exists to make saturation divergence *measurable* rather than impossible; a world with subexponential satisfiability search would shift the measured workload distributions that calibrate those guidance ceilings, exactly as any measured calibration shifts. The mechanism — pin, ceiling, receipt, versioned artifact, extractor generation in the lineage block (QO-28) — is untouched in every branch. This is the sense in which the derivation is *complexity-assumption-free*: it is conditional only on its own axioms and on arithmetic facts with published proofs.

12.3 What the episode teaches the QnIR design

The dispositions being "no impact," the episode would still be a design gift if it stopped there, because four of QnIR v0.2's structural choices were each exercised in public, at scale, within the preparation window of this loop.

Table 6: Per-claim dependency scan and disposition. "Impact if verified" is the change to the QnIR v0.2 derivation *in the branch where the claim survives independent examination*; "impact as released" is the change at today's unverified status.

Claim	Canon occurrences	Impact if verified	Impact as released
E-2 (ETH false)	0 (no ETH, SETH, or P≠NP usage)	none on the derivation; recalibration of *guidance* values for saturating-search ceiling tables (workload-measured, versioned)	none; recorded as unverified external claim
E-3 (Kakeya all dims)	0 (no GMT vocabulary)	none	none; informational
E-1 (corpus)	0 load-bearing	none; complexity-adjacent families read as external landscape context	candidate-state intake only, under Axiom 14
E-0 (context)	0	none	provenance observation only

(1) Statement-fidelity receipts at the parser boundary. The examination group's public caveat — that a Lean check is where scrutiny should *start* — is the field instance of the trust gap `qs.md` §6.1 closed: formal-statement-proves is not informal-claim-meant. QnIR v0.2 attaches receipts at exactly that boundary (parser-boundary receipts in the closed catalog; the R12 post-handshake precondition; four typed failure tokens). The E-2 branch taxonomy of Section 12.1 maps onto those typed outcomes one-to-one: a fidelity failure is a typed receipt, not a silent acceptance.

(2) Anti-double-counting via content-addressed lineage. E-2 and E-3 reportedly share one claimed method family (the auxiliary PDE engine). In ordinary epistemic practice this makes them *one* evidence cluster, but nothing in the publication form exposes the sharing — two preprints read as two confirmations. Under the road-receipt π interface, receipts carry $\mathcal{H}(\text{Trace})$ commitments of the derivation that produced them; shared derivation lineage hashes to overlapping roots and is mechanically detectable rather than folklore. The Qn reading of the October corpus is therefore *one* cluster with several leaves, not three independent breakthroughs — and the reading is checkable, which is the property that matters (QO-32).

(3) The co-signing handshake observed in the wild. The invited examination groups are external co-signers in exactly the `qs.md` §6.1 sense, and their behavior illustrates why the handshake's *closed-vocabulary* precondition matters: the Yang–Mills group withheld the artifacts pending its own write-up (a co-signature *refused* rather than granted), and the ETH group granted nothing beyond organized scrutiny. A co-signature over an open or drifting vocabulary would have been meaningless; the groups were given artifacts to examine, not authority to amend the claim — the same frozen-vocabulary discipline the closed 49-opcode catalog asserts for the machine.

(4) Provenance hygiene is not bureaucratic. The Navier–Stokes announcement's priority dispute includes a public concession from the announcing party that de-identified usage data of the involved products "cannot be ruled out" as having helped improve the producing model [45]. Whatever the dispute's resolution, the structural lesson aligns with loop-1 Segment 1 Point 4: receipts that do not record input lineage leave contamination undetectable, and the road-receipt lineage block with W_{gate} billing attribution is the QnIR answer. An evidence economy in which agents generate claims at corpus scale needs lineage as a first-class field, not as an etiquette norm.

12.4 Disposition and protocol integration

Disposition (recorded for the loop-2 ledger). E-1, E-2, E-3 (and the E-0 context rows) enter this loop's integration record as UNVERIFIED-EXTERNAL observations with *zero* load-bearing dependencies in the Qn canon, per the mechanical scan of Section 12.2. No canon file changes, no QnIR v0.2 opcode, encoding, state, or receipt change, and no annex of this PDF changes on their account. If E-2 survives the Impagliazzo–Paturi–Williams examination, the sole follow-up is a guidance-recalibration note on saturating-search ceilings in the next versioned ceiling table, with the mechanism unchanged. The scan method itself (lexical pass + citation walk) ships as a pre-committed mechanical appendix under QO-31 so that the "zero dependency" verdict remains re-checkable at every future loop rather than asserted once here.

Invited objection

If the ETH refutation is verified, the derivation survives untouched — but is that a *feature* or a blind spot? A machine whose cost semantics cannot be refuted by any complexity-theoretic discovery is either assumption-free (this draft's claim) or indifferent to the world it runs in (the objection). The reviewer is invited to name a concrete verification workflow whose *safety* — not merely its wall-clock comfort — would change under subexponential SAT, or to accept QO-31's pre-commitment: the guidance layer recalibrates, the constitution does not, and the split between the two is exactly the split between measured calibration and asserted identity that loop 1 already forced on the cost table.

13 Scaling Story: Causal Clocks and Choreography

13.1 Intra-node: fork-join tree clocks

Loop 1's flat causal counter $\kappa(\text{next}) = \kappa(\text{last})+1$ imposes a total order that the reviewer showed is a cache-coherency bottleneck on multi-core hosts and causally blind across nodes. v0.2 replaces the scalar with a deterministic hierarchical fork-join tree index — an interval-tree-clock-style pedigree $\kappa_{\text{tree}} = \langle \kappa_{\text{parent}}, \text{branch_id}, \kappa_{\text{local}} \rangle$ — paired with the `OP_FORK`/`OP_JOIN` pair of Class F. Forking requires data-disjoint SSA subgraphs and pre-partitioned budgets $\sum_{j=1}^{m} W_{\text{rem}}^{(j)} \leq W_{\text{rem}}$, so parallelism is a budget delegation before it is an execution fact. The concurrency-safety argument rests on the store, not on the clock: $\mathcal{S}$ is write-once and keyed by content hash independent of κ, so concurrent `OP_COMMIT` insertions are monotonic set unions over a join-semilattice — they commute without locks, which is precisely the CALM-convergent shape the reviewer cited. `OP_JOIN` merges branch receipt queues and trace hashes in deterministic lexicographical `branch_id` order with zero lock contention, and the merged clock advances by event rather than by wall-clock time, keeping the causal arrow (Axiom 5) the only ordering discipline in the machine.

13.2 Inter-node: choreographic workflows

Across nodes, v0.2 adopts the reviewer's choreographic-programming recommendation for $Q(\text{workflows})$: global protocols are authored as a causal DAG of $Q(\text{patterns})$ under shared intent contracts and compiled by deterministic endpoint projection into per-node local QnIR streams, each of which is an ordinary total program under Theorem 1. Cross-node causality is tracked strictly as a partial-order Merkle-DAG over ingested parent road-receipt hashes in Block 4 (Lineage) of the Universal Envelope — a cryptographic vector clock with zero wall-clock synchronization — and each ingestion is `OP_INGEST_ROAD` under the execution-proof discipline of Section 9.3. The determinism claim to attack here is the projection itself: endpoint projection must be a pure function of the choreography and its intent contracts, or two nodes could compile different local streams from the same protocol; QO-26 asks the reviewer to press on the deadlock-freedom and projection-purity proof obligations and on how partial failures (a node

halting mid-choreography with a fatal receipt) are receipted into the peers' lineages without silent retry loops.

14 Anticipated Objections and Self-Critique (v0.2)

This section states, in the voice a hostile reviewer might use, the sixteen objections the author considers most probable and most damaging against the v0.2 design itself — not against v0.1, whose defects are now the canon's history. Each is numbered so the reviewer can sharpen rather than rediscover, following the loop-1 protocol that the reviewer graded as the draft's best habit.

Objection 1. The SSA value graph is renaming. $\%i$ is a slot with a write-once rule; everything v0.2 claims (immutability, birth order, hash-readiness) was equally true of a v0.1 slot that the ISA simply forbade from being written twice. The real change is the deleted `dest` field, and that could have been done without the SSA vocabulary.

Objection 2. Twelve-bit operand indices admit 4096 SSA values per program; the liveness window admits 256 simultaneously live. Neither number has a measured justification. Real Q(workflow) kernels (multi-frame rotors, iterative solvers) may exceed 4096 births, and the fix — multi-word pseudoinstructions or stream segmentation — reopens exactly the encoding cans v0.2 just nailed shut.

Objection 3. The cost polynomials are functions of declared ceilings, so the only way to make a program cheap is to declare tight ceilings — and ceilings are declared by the same principal who benefits from under-declaring. $H_{\max}$ under-quoted by one limb makes arithmetic costs lie by $\sim 64\times$. The load-time validator checks consistency of the sum, not truth of the terms; the calibration receipts arrive only at lowering, after the budget was spent.

Objection 4. Refunds are an inflation channel. `OP_TERMINAL` "refunds remaining zoom reservations"; a program can therefore over-reserve, harvest refunds, and present a ΔW ledger smaller than its true worst-case footprint — distorting every breakeven comparison (`OP_REUSE`, `OP_SATURATE`) that reads recorded ledgers.

Objection 5. Recoverable status tokens are terminal states wearing a disguise. Six new token names, a stream-end mapping table, and two consuming opcodes is a second error vocabulary with its own semantics — and the mapping `STATUS_LOOKUP_MISS` $\rightarrow$ `INCONCLUSIVE` quietly re-classifies what loop 1 treated as a benign fallback into a terminal epistemic state. The closed set is formally untouched but behaviorally enlarged.

Objection 6. `OP_CALL` reintroduces a stack, a return address, and a calling convention to a machine whose v0.1 selling point was that it has none. The acyclicity check is $\mathcal{O}(L)$ at load, but transitive sub-budget composition (W_{sub} of W_{sub} of $\dots$) makes worst-case stack depth times frame size a new un-validated ceiling interaction.

Objection 7. Class H is four opcodes of promise. HAMT/RRB lowering with $B \in \{Q(8), 32\}$ is asserted, not derived; the reviewer's own construction listed Merkle B-trees as the alternative and the draft's rubric admitted the tie is unresolved. Shipping the opcode names before the BOC comparison risks canonizing the winner by implementation inertia rather than selection.

Objection 8. `OP_FORK`/`OP_JOIN` buy concurrency with a deterministic merge, but data-disjointness of "SSA subgraphs" is a compiler-grade analysis the IR does not specify. If disjointness is declared rather than proven, a buggy fork is a data race wearing a receipt; if it is proven, that proof is a new verification obligation of the same order as the encoding check the whole design celebrates.

Objection 9. The AIR story is an IOU. Realization A says the trace "arithmetizes directly"; the actual constraint system for 49 opcodes — including Class H Merkle steps and the call-frame stack — does not exist, and the prover cost of Class D geometric products inside a STARK may exceed the thermodynamic breakeven so thoroughly that Realization B is always selected, at which point the "zero-trust" property is honest but the "$\mathcal{O}$(polylog)" one is decorative.

Objection 10. W_{gate} billed regardless of outcome converts the gateway into a toll both ways: an attacker "griefs" a principal by inducing crossings (each costs the principal), and a principal with fee-priced exploratory loops may under-validate. The fee schedule is DCC metadata — i.e., set by the very principal whose incentive discipline it is supposed to enforce.

Objection 11. Canonical minimal-zoom normalization decides *identity* by a function of the tuple; but two computations arriving at the same value through different zoom histories now share a commitment while carrying different lineage — the deduplication that helps storage blurs the provenance that receipts exist to preserve. The lineage-block note is a comment, not a constraint.

Objection 12. Origin quotienting $Q(0)_A \mapsto \widehat{Q(0)}$ is a projection with an inverse problem: if hardware seeds remain "bound solely in the envelope's lineage block," any leak joining $\mathcal{H}_{\text{pub}}$ commitments to their envelopes re-derives node identity across pseudonymous boundaries. The quotient is only as private as its least careful consumer.

Objection 13. R12′ makes the machine's guarantee conditional on a handshake the machine cannot verify beyond signature shape: `OP_GATE` checks that the intent-contract reference resolves to a co-signed artifact, but not that the co-signature was informed (the principal may co-sign the read-back without comprehending it). Determinism of the descent is real; anchoring of the root remains a social fact wearing a cryptographic hat.

Objection 14. The prior-art table asserts a clean joint-discipline gap, but "provides" is graded by the author: WebAssembly with deterministic profiles plus a content-addressed package store plus a tlog gets three of the five columns for free, and the remaining two are governance, not ISA. The composition claim survives only if the columns are *mechanically* inseparable, which the table does not argue.

Objection 15. The handshake's verbalizer $\mathcal{V}$ is "a pure function of the formal contract" — but the contract schema is versioned, so $\mathcal{V}$ is a versioned family, and the guarantee $H(\mathcal{T} \mid \text{Contract}) = 0$ is relative to a verbalizer generation the principal cannot inspect. A regressed verbalizer changes the read-back text while signature verification still passes.

Objection 16. This very package is an attack surface: the annexes claim *content* fidelity, not byte fidelity — the rendering pipeline (front-matter rendering, liquid-include resolution, fence preservation, record-style table rendering, box-glyph transliteration, math-delimiter stripping) is a semantic transform, and a semantic transform can lose a sentence as silently as a wrapper can lose a character. Loop 1 caught data loss between live docs and PDF; a renderer with a bug would repeat that failure in subtler form. If the reviewer finds one annex sentence, heading, or table cell that is not in the committed source, the package's central convenience becomes its central defect — and the author has pre-committed the build-time verification (whitespace-insensitive char-stream equality against an independent extractor, per-heading and per-cell presence, residue checks) that would expose it.

15 Open Questions for the Reviewer (QO-16 … QO-32)

The loop-1 draft closed with fifteen open questions; the loop-1 reviewer answered all of them, and their dispositions are tabulated in Section 2. This draft continues the numbering. QO-16 … QO-32 are the response format requested from the reviewer for this loop (QO-31 and QO-32 arise

from the external-claims assessment of Section 12): a short verdict per item (with segment/point references in the reviewer's own style), plus any unanticipated objection the reviewer chooses to add under the same QO-15 protocol loop 1 established. The author's pre-commitments for each item are stated so the reviewer can hold the next iteration to them.

QO-16 (AIR arithmetization). What is the honest constraint-system shape for the 49-opcode machine — per-opcode transition constraints, lookup-argument treatment of the 16-rule bypass table and of Class H Merkle steps — and what prover cost does Class D imply? Pre-commitment: if the STARK prover cost for a canonical kernel exceeds its Realization B bisection cost by more than the declared breakeven margin, v0.3 will demote Realization A to a profile-optional realization rather than a default. **QO-17 (Cost-polynomial calibration).** How should the calibration chain from C_{op} to lowering receipts be validated — what evidence short of full WCET measurement certifies that a DCC profile's affine bound dominates the true lowered cost? Pre-commitment: one canonical kernel (the Example-2 factorial and one Class D rotor chain) will ship with measured lowering traces in the v0.3 cycle. **QO-18 (Liveness window).** Is $W_{\text{live}} \leq 256$ sufficient for representative $Q(\text{workflow})$ kernels, and what liveness distribution justifies it? Pre-commitment: the flattener will emit liveness histograms for the worked examples and for one external kernel per domain file. **QO-19 (Container lowering BOC).** HAMT/RRB vs. Merkle B-tree under the declared Tier-2 order $(\Delta W_{\text{worst}}, M_{\text{coupling}}, P_{\text{coverage}})$: the rubric is open and recorded in Section 9.4. Pre-commitment: the verdict will be sealed with a tabulated comparison on one insertion-dominated and one scan-dominated workload or the opcodes will be re-scoped as profile-specific. **QO-20 (Tree-clock receipt ordering).** Does `OP_JOIN`'s lexicographical branch-id merge preserve Axiom-5 birth order for all downstream hash derivations, and what is the canonical encoding of κ_{tree} inside trace hashes and receipts? Pre-commitment: the v0.3 spec will fix the CBOR encoding and add the order-preservation property to the validator. **QO-21 (The call-pair trade).** Quantify when bounded `OP_CALL` beats inlining beyond the trivial $P > 2$ byte break-even — specifically verification-work amortization vs. stack-frame accounting complexity. Pre-commitment: if the reviewer finds the verification amortization argument weak, `OP_CALL`/`OP_RET` move to a profile-gated extension class and the core ISA returns to pure inlining. **QO-22 (Token taxonomy).** Are the six recoverable status tokens and their terminal mappings the right partition of the closed set's behavior, and is `STATUS_LOOKUP_MISS` → `INCONCLUSIVE` the correct classification? Pre-commitment: the mapping table becomes canon in `qnPrimitive.md` at its next versioned update, amended by the reviewer's verdict. **QO-23 (Gateway fee fairness).** What pricing discipline should bound a DCC profile's W_{gate} — outcome-independent charging creates both a griefing vector against principals and an under-validation incentive. Pre-commitment: the fee schedule gains a declared justification field and a profile-level cap formula in the v0.3 cycle. **QO-24 (Zoom normalization).** Does canonical minimal-zoom normalization ever erase a distinction that conservativity or lineage auditing needs, and should zoom history be part of identity at a secondary commitment level? Pre-commitment: normalization is stated as a pure function with its own receipt; if the reviewer identifies a erasure case, identity gains a zoom-history sidecar commitment rather than reverting. **QO-25 (Domain-tag registry).** The domain-separated commitment (Eq. 2) needs an allocation registry for `DomainTag` values — who allocates, what prevents squatting across the swarm? Pre-commitment: the registry joins the `Q(anchor)` governance of `abstractionLayers.md` §8.4 rather than a new authority. **QO-26 (Choreographic projection).** What proof obligations make endpoint projection trustworthy — projection purity, deadlock-freedom, and partial-failure receipting in peer lineages? Pre-commitment: a minimal two-node choreography example with full receipt traces will accompany the v0.3 draft. **QO-27**

(Verbalizer certification). How is the verbalizer generation itself certified, versioned, and audited so that a regressed $\mathcal{V}$ cannot silently change read-backs under valid signatures? Pre-commitment: `qs.md` §6.1 gains a verbalizer-generation field co-signed with the contract pair. **QO-28 (Saturation determinism boundary).** Two `OP_SATURATE` runs under different extractor generations yield different versioned artifacts; what concretely distinguishes measured divergence from state drift when both artifacts re-enter a shared workflow — and must the extractor generation be part of the artifact's identity block? Pre-commitment: the extractor generation joins the lineage block in v0.3. **QO-29 (Falsifiable positioning).** Name the system that already provides the joint discipline (Section 11, columns a–e) under another vocabulary — or accept that the composition claim stands until one exists. Pre-commitment: the table gains a mechanical "provides column X" appendix with system-version citations. **QO-30 (Flags pressure).** Will four flag bits hold, or does the first flag-starved opcode force a second word format and break stream uniformity? Pre-commitment: the v0.3 validator will report per-class flag utilization; exhaustion triggers header promotion of the offending attribute, never a second word format. **QO-31 (External-claims intake).** Is the disposition of Section 12 correct — that the October 2026 OpenAI mathematics corpus (ETH refutation claim; all-dimensions Kakeya claim; 721-manuscript repository) has zero load-bearing dependencies in the Qn canon, with only ceiling-guidance recalibration as the verified-ETH branch's follow-up — or does a verified ETH refutation force any constitutional change beyond guidance? Pre-commitment: the claim-intake scan (lexical pass + citation walk over the annexed canon) ships as a mechanical appendix in the next integration ledger, so the "zero dependency" verdict is re-checkable at every future loop rather than asserted. **QO-32 (Evidence-cluster receipting).** The ETH and Kakeya claims reportedly descend from one shared method family; under road-receipt semantics, shared derivation lineage is mechanically detectable via overlapping $\mathcal{H}$(Trace) roots, preventing two claims from masquerading as two confirmations. Should shared-derivation detection (lineage-overlap exposure) be a required field of the road-receipt π interface in v0.3, or remain a validator-level check? Pre-commitment: the lineage-overlap field joins the receipt schema only if the reviewer judges the anti-double-counting property worth its per-receipt bytes; otherwise it stays a validator obligation with no wire change.

16 Conclusion and Review Loop Protocol (Loop 2)

16.1 What this draft claims

The claim of this draft is deliberately narrow and auditable. The loop-1 review produced a complete map of defects in the QnIR v0.1 candidate and constructive resolutions for each; the canon absorbed the constitutional, semantic, structural, and numeric findings in seven committed files; and Sections 4–13 demonstrate that the remaining findings — the ones that could only be resolved inside the machine itself — have opcode-level, encoding-level, and proof-structure-level answers that are consistent with the integrated constitution. The v0.2 candidate is therefore not a new design but the same design repaired at the joints the reviewer located: the state model is SSA because mutable slots contradicted Axiom 13; the word format is repaired because four operands did not fit; the cost table is polynomial in declared ceilings because honest guesses are not accounting identities; and the anti-drift certificate is conditional on handshake anchoring because downstream rigor cannot upstream hallucination. Every repair cites its forcing finding, and the ledger of Section 2 is the check that nothing was silently dropped.

16.2 What this draft does not claim

Three claims are explicitly *not* made. First, no implementation claim: no QnIR v0.2 interpreter exists, the cost calibrations are unmeasured, and the AIR arithmetization is a named obligation (QO-16) rather than an artifact. Second, no superiority claim: the prior-art section positions, it does not rank, and the composition thesis stands or falls on the reviewer's ability to name the joint discipline elsewhere. Third, no closure claim: QO-16 through QO-32 are genuine open questions, the container BOC verdict is deliberately unsealed, and the two-tier receipt taxonomy awaits the reviewer's classification before it becomes canon. The candidate's status remains PENDING under Axiom 14; if this loop goes as the last one did, the next canon updates will absorb the verdicts file by file under the same one-at-a-time discipline, and QnLang — the governed coding language this intermediate representation exists to serve — remains the milestone after that.

16.3 The loop protocol

The review protocol is unchanged from loop 1, with the package now self-contained and size-compliant; the loop-2 reviewer is Kimi K3.1-Pro (kimi.ai), the Google Deep Review canvas being rate-limited for this program. The reviewer receives this PDF only: the paper body, the external-claims assessment of Section 12, the full Qn universe as rendered-content annexes with SHA-256 commitments, and the loop-1 transcript as Annex U. The requested response format is a segment-structured transcript answering QO-16 … QO-32 in order (short verdicts with pointers sufficing, as loop 1 demonstrated), grading this draft's sixteen self-critiques under the fatal/cosmetic/surviving rubric, and adding unanticipated objections under the QO-15 protocol. The author's integration discipline is also unchanged: one canon file per versioned update, each carrying its review-integration record, cross-file dependencies deferred explicitly, and the rendering convention of `abstractionLayers.md` v1.8.1 applied to every file whose mathematics it touches. On reception of the loop-2 transcript, the author will produce the integration ledger for QO-16 … QO-32 in the format of Table 1 — and the candidate either advances toward its canon file `docs/qnir.md` or returns to the bench with the reviewer's map in hand.

Reproduction commitments. All annex verification procedures of this package are mechanical and stated in Section B; the SHA-256 commitments let any reviewer holding the repository verify that the annexed canon is the committed canon of 2026-10-27. The annex text is the rendered content of those bytes under the declared conventions N1–N6, and the build-time content-fidelity proof (independent-extractor char-stream equality, per-heading and per-cell presence) passed for all 25 files. Where the reviewer's environment renders this PDF differently from the author's, the commitments travel with the text, not with the pixels.

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[45] *WIRED*, "OpenAI Just Claimed a Huge Math Discovery. Some Academics Are Crying Foul," 9 September 2026 (Navier–Stokes announcement; public priority dispute; OpenAI statement on de-identified usage data).

A QnIR Receipt Catalog Additions (Candidate)

The receipt catalog of the Qn universe is closed but extensible by versioned updates (the `qs.md` v1.3.0 precedent). QnIR v0.2 requires the following additions, proposed as a candidate table for the reviewer's verdict (QO-22) and for the queued `qnPrimitive.md` registry update. Terminal states cited are the sealed twelve of `abstractionLayers.md` §3.6 — no new terminal status is introduced anywhere in this table, and the recoverable tokens are explicitly *not* terminal entries.

Name	Trigger	Disposition
`RECEIPT_HASH_COLLISION_QUARANTINED`	hash match with differing canonical tuple header at `OP_COMMIT` or `OP_REUSE`	fatal; terminal `QUARANTINED`; both pre-images sealed for hash migration
`RECEIPT_SATURATION_CONVERGED`	`OP_SATURATE` halt on convergence before any ceiling	terminal `SUCCESS`; versioned artifact emitted
`RECEIPT_BYPASS_IDENTITY`	`OP_BYPASS` identity hit (16-rule suite)	non-terminal, $\Delta W = 0$ metric cost; dispatch base billed
`RECEIPT_CALL_DEPTH_EXCEEDED`	`OP_CALL` exceeding declared $C_{\max}$ at load or runtime	fatal; terminal `FAILURE`
`RECEIPT_SUBBUDGET_VIOLATION`	fork/call/delegation partition with $\sum W_{\text{sub}} > W_{\text{rem}}$	fatal; terminal `FAILURE`
`RECEIPT_ROAD_PROOF_INVALID`	`OP_INGEST_ROAD` with π failing both realizations	fatal; terminal `QUARANTINED`; sibling receipt `RECEIPT_ANCHOR_REVOKED` inherited for bisection failure
`RECEIPT_TOKEN_UNRESOLVED`	stream end with unfunded recoverable token(s)	bookkeeping entry inside the terminal receipt; terminal status per the mapping of Section 8.1

Recoverable status tokens (non-terminal, with their stream-end terminal mappings):

Status token (on s_i)	Unresolved stream-end terminal
`STATUS_ORDERING_INDET`	`ORDERING_INDETERMINATE_AT_ZOOM`
`STATUS_PRECISION`	`PRECISION_INSUFFICIENT`
`STATUS_NO_SOLUTION`	`NO_SOLUTION`
`STATUS_LOOKUP_MISS`	`INCONCLUSIVE`
`STATUS_TOO_COMPLEX`	`EXPRESSION_TOO_COMPLEX`
`STATUS_NEG_EVEN_ROOT`	`EXPRESSION_TOO_COMPLEX`

B Annex Guide: Scope, Fidelity, and Verification

B.1 What is annexed, and why this scope

Annexes A–U reproduce the complete Qn universe design corpus so that the reviewer needs no repository access: every canonical `.md` specification file, the reflection essay, the full glossary (the human-readable glossary document plus all four machine-readable glossary domains), and the loop-1 review transcript itself. The exclusion rule is the author's instruction, and it is applied mechanically: files whose purpose is the GitHub Pages presentation — the documentation portal (`docs/index.md`), the site navigation registry (`docs/_data/navigation.yml`), the Jekyll include/layout templates, the site stylesheet, the repository `README.md`, and `LICENSE.md` (whose SSL terms are quoted inside the annexed documents wherever they bind) — are omitted as presentation infrastructure rather than universe content. Annex U is included even though it is a review artifact rather than canon: the reviewer of this loop should see the loop-1 transcript, and it is part of the committed `docs/` tree.

Why rendered content rather than byte-verbatim source. This edition annexes the corpus as *rendered content* under the conventions N1–N6 below, not as byte-verbatim markdown source. The review tooling of this loop bounds its intake (on the order of 3×10^5 tokens), and the byte-verbatim edition of this same draft measures 8.5×10^5 extracted characters across 194 pages — at that cap with no safety margin, dominated by tokens that carry syntax rather than content (box-drawing frame glyphs, Jekyll/Liquid directives, math-mode delimiters, link URLs, YAML front-matter syntax). The rendered edition removes exactly that class, re-lays tables as records, and preserves every sentence, heading, list item, and table cell; measured over the whole annexed corpus, 7.5×10^5 characters of source render to 6.8×10^5 characters of content with every pathological token class eliminated. The source bytes remain the identity anchor: each annex's provenance line and Table 7 carry the full SHA-256 of the committed source file, and the repository holds those bytes.

Table 7: Annexed files (rendered-content edition). *Lines* = source/rendered lines; *Chars* = source/rendered characters (k = thousands) under conventions N1–N6; the commitment is the full SHA-256 of the SOURCE file, truncated here to 16 hex digits for width; complete digests appear in each annex's provenance line. Content fidelity (whitespace-insensitive char-stream equality, per-heading and per-cell presence, residue checks) was verified mechanically at build time; see Section B.

Path	v	Updated	Lines	Chars	SHA-256	Annex
simemp.md	1.2.0	2026-10-26	334/568	36.0k/33.4k	f22d0f650e8e49e5	(B.3)
abstractionLayers.md	1.8.1	2026-10-26	518/744	46.2k/38.9k	02b9607241ca2e69	(B.3)
q0_q1.md	1.2.1	2026-10-23	411/471	28.7k/25.0k	48b37bcd11f7b919	(B.3)
q2_q9.md	1.1.3	2026-10-20	392/474	27.4k/23.7k	abbe4e6c9986d966	(B.3)
qm.md	1.3.0	2026-10-26	644/965	57.2k/52.1k	07f5c31d0265fa00	(B.3)
qm_geometry.md	1.2.3	2026-10-20	409/515	27.3k/23.4k	428d7d20a980dada	(B.3)
qm_calculus.md	1.1.1	2026-10-19	358/406	23.9k/20.5k	869964ef251c45e6	(B.3)
qexpr.md	1.3.0	2026-10-26	469/633	41.6k/37.4k	a69cb9bd94e0a170	(B.3)
qnPrimitive.md	1.3.0	2026-10-26	530/749	40.0k/34.4k	50d48852f537bc1c	(B.3)
qs.md	1.3.0	2026-10-26	527/755	48.1k/44.0k	e24e35dfef022674	(B.3)
qlog.md	1.1.1	2026-10-20	355/478	26.8k/23.3k	5ff1272e1e90bda3	(B.3)
qphy.md	1.0.5	2026-10-19	351/464	26.2k/22.8k	45977436cc4ffb5c	(B.3)
qchem.md	1.0.3	2026-10-19	281/374	20.4k/18.0k	4864574e37f784c3	(B.3)
qbio.md	1.0.2	2026-10-19	284/372	20.5k/17.8k	99fba2491d5345d0	(B.3)
chronosa.md	0.2.1	2026-10-19	280/365	21.6k/20.0k	f37867807110ff04	(B.3)
qstructures.md	0.1.0	2026-10-23	213/336	27.0k/22.7k	01d5f84fd550e008	(B.3)
qs_history.md	0.1.2	2026-10-22	256/389	28.6k/24.6k	4e613812b8d4b2ff	(B.3)
history_library.md	0.1.1	2026-10-22	206/354	28.1k/23.7k	3ce312ad82d166ff	(B.3)
reflections/h10_finite_computation.md	1.0.2	2026-10-13	137/166	9.1k/8.6k	671d0950a36da628	(B.3)
glossary.md	1.1.2	2026-10-14	128/87	3.9k/1.9k	aa35b7d4e1dae362	(B.3)
_data/glossary/architecture.yml	—	—	172/109	8.2k/7.6k	80234b7a2327f5a1	(B.3)
_data/glossary/primitives.yml	—	—	366/225	14.9k/13.5k	73fac713f4845d39	(B.3)
_data/glossary/constraints.yml	—	—	237/146	11.1k/10.3k	0f1c41264ad566d5	(B.3)
_data/glossary/governance.yml	—	—	96/55	3.5k/3.1k	b32d9f4a123f0e38	(B.3)
AiReviews/AiReviewQnIRv0.1.md	—	—	361/1580	121.6k/126.1k	f08f456dfeabf44e	(B.3)

B.2 Fidelity conventions N1–N6

The annexes are rendered content under exactly six mechanical conventions, none of which removes or reorders a single word of the source content.

(N1) *Charset transliteration.* Box-drawing and diagram glyphs are mapped to their ASCII skeletons (`|` for vertical rules, `-` for horizontal rules, `+` for junctions, `->` for arrows, `*`/`o` for filled/hollow markers), preserving column layout exactly; the canon's ASCII-art architecture figures remain column-exact.

(N2) *Front matter rendering.* YAML front matter is reduced to its content-bearing fields (title, version, updated, author, license, description, terms_used) in a marked block; presentation fields (layout, permalink, license_file, repository pointers) are dropped — they are re-stated in each annex's provenance line.

(N3) *Site-generation directives.* Liquid `{% include term.html %}` glossary references render to their visible text (the explicit `text` attribute when present, otherwise the glossary slug), exactly as the documentation site renders them; all other Liquid/Jekyll directives (loops, assigns, includes) are presentation plumbing and are removed. Markdown link syntax renders to its visible text; URLs are dropped as non-visible, except that external `http(s)` URLs quoted as content remain.

(N4) *Structure markers.* Headings render with their level (`H2:` …), list markers normalize to `-`, emphasis/bold and inline-code markers are stripped (their text is kept), and math-mode delimiters (single or double `$`) are stripped with their inner LaTeX kept verbatim — the one place where TeX source is the content. Fenced blocks lose their fence markers; fenced content itself is preserved verbatim (subject only to N1). Pipe tables render record-style: one logical line per row, cells in order separated by `|`, empty cells preserved.

(N5) *Prose wrapping and pagination.* Prose wraps at 96 columns for the page (continuations carry no marker: the text is a stream, not a byte facsimile); fence and diagram lines are never wrapped (maximum 101 columns). Page breaks inside a file are a property of paper, not of the text.

(N6) *Font note.* The annexes are set in DejaVu Sans Mono, extended with five glyphs it lacks — U+20D7 (combining right arrow above), U+27FA/U+27F9 (long double arrows), U+226A (much less-than) drawn after Sarasa Mono SC, and U+1D4AF (mathematical script capital T) drawn after FreeSerif — because the canon's re-encoded mathematics (the `abstractionLayers.md` v1.8.1 convention) uses them. The patch scales donor outlines to the host font's em and is part of the build pipeline, not the document text.

B.3 Verification procedure and commitments

Because the annex text is rendered content, the verification claim is *content* equality, and it was executed mechanically at build time by a checker whose expected-side extractor is an independent implementation of N1–N6 (separate code path from the renderer). Per file, the checker asserts: (C1) whitespace-insensitive character-stream equality between the expected content and the annex text, with layout-generated table rules excluded by a declared normalization; (C2) every source heading present with its level marker; (C3) every pipe-table cell present, in order, as an ordered word subsequence of the annex text (1,719 cells verified across the corpus); (C4) residue checks — no Liquid directives, no fence markers, no box glyphs, no math delimiters outside verbatim fence regions. All 25 files pass all checks; the report is part of the build pipeline. Two source lines of `qm.md` carry an odd `$` count (a canon-side typographical artifact); by the no-silent-mutation rule they are rendered byte-faithfully rather than repaired, and the checker treats unpaired delimiters as content.

Byte-level truth is carried by the commitments, not by the annex text: any reviewer holding the repository can hash the committed source files and compare the SHA-256 values of Table 7 and each provenance line. The commitments are made under the same discipline the universe applies to itself: content addressing is the identity of the annexed canon, and any future revision of a source file changes its hash and forces a new versioned edition of this package rather than a silent edit.

Annex A — `simemp.md` (constitution)

Source: `docs/simemp.md` *(constitution) — 1.2.0, updated 2026-10-26. Rendered content per Annex Guide conventions N1–N6: 334 source lines, 568 rendered; 36,043 → 33,424 characters; source SHA-256* `f22d0f65 0e8e49e5 6d89866f 610cce78 1da9fd92 42b33660 c92d6330 a9a5de00`*. Content fidelity verified mechanically at build time.*

```
[front matter] ------------------------------------------------------------
[front matter] title: SIMEMP Constraints
[front matter] version: 1.2.0
[front matter] updated: 2026-10-26
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Foundational physical, computational, and architectural constraints
[front matter]   governing the Qn and cdqn stack under Dependencies Determinism.
[front matter] terms_used: simemp, qn, cdqn, qnlang, qnir, dcc-profile, boc-policy, existential-
[front matter]   invariant, operational-agility, causal-arrow, scaling-threshold,
[front matter]   computational-consistency, receipt, ssl, dependencies-determinism,
[front matter]   structural-indirection, metric-exhaustion, compute-unit-u, no-
[front matter]   implicit-rule, paternity-reference, open-core-invariant, simemp-
[front matter]   gateway, payload
[front matter] ------------------------------------------------------------


H1: SIMEMP: Thesis and Constraints for the Qn and cdqn Stack


Field | Specification
Document Title | SIMEMP: Thesis and Constraints for the Qn and cdqn Stack
Version | 1.2.0
Last Updated | 2026-10-26 (Bao Loc, Vietnam)
Author | Christophe Duy Quang Nguyen
License | Scaling Source License (SSL) 1.0
Status | Canonical Constitutional Specification — Foundational Invariants


H2: Review Integration Record (v1.2.0)

Source: docs/AiReviews/AiReviewQnIRv0.1.md — transcript of the external adversarial review
(Google Deep Review, October 2026) of the condensed Qn universe specification and QnIR v0.1
candidate draft. This update integrates the constitutional findings of that review: the
decidability of the two-tier hierarchy and of the BOC policy (QO-1, Segment 1 Point 3), the
decoupling of the logical compute unit from physical thermodynamics (QO-2, Segment 1 Point 2),
the observational-determinism operationalization of the conditional-entropy postulate (Segment 1
Point 1), and the gateway billing-attribution and migration-witnessing findings (Segment 1 Point
4 — double-flagged: the same point surfaced independently in the data-loss audit as the missing
free-generation / metered-crossing statement). It also executes the site-wide rendering decision
recorded in the Rendering Fix Record of docs/abstractionLayers.md v1.8.1: all mathematical
notation in this file is re-encoded into engine-independent plain text, because the site's
Jekyll pipeline ships no MathJax/KaTeX engine. Under Axiom 13, this file is a new versioned
artifact; no prior content is mutated in place. New constructions introduced below are tagged
[v1.2.0 — CANDIDATE] pending BOC sealing; editorial corrections are applied directly and
recorded here.


Change | Section | Reviewer source | Type
Two-stage constrained lexicographic BOC model: Tier 1 as a hard boolean admission filter
(C_valid, failure -> REJECTION_BY_GATEWAY, inadmissible candidates never ranked); Tier 2 as
lexicographic minimization of LexKey(c) = ( ImplicitDeps(c), perm_π(ΔW_worst(c), M_coupling(c),
P_coverage(c)) ) with the priority permutation π ∈ S_3 declared in the calling DCC profile;
rubric mode requiring tabulated Tier-1 pass/fail plus Tier-2 trade-off vectors wherever
quantities remain qualitative; the unparameterized scalar argmax formulation retired | §7
(rewritten) | QO-1 / Segment 1 Point 3 | CANDIDATE
Observational Determinism Invariant: bit-identical output commitments (H(S_t(E_1)) =
H(S_t(E_2))) and identical terminal receipt statuses for any two Layer-0 environments agreeing
on the declared dependency projections (Π_D(E_1) = Π_D(E_2)), making the incomplete-dependency
defect D' ⊂ D empirically falsifiable via differential execution testing; the
conditional-entropy postulate is additionally annotated as a functional-purity statement | §1 |
Segment 1 Point 1 | CANDIDATE
Abstract Compute Unit U decoupled from physical energy: U redefined strictly as a
platform-invariant logical work quantum (normalized so Q(0) -> Q(1) costs exactly 1 U); the
Landauer bound retained as a theoretical floor with numeric anchor (2.87 × 10^-21 J per bit at T
= 300 K) and the 10^7–10^10 CMOS scale gap acknowledged; new Layer-0 Hardware Telemetry Envelope
(E_phys ≤ α_compute · U_op + β_bus · U_mem, declared error bars) bound to U exclusively via
node-local Historical Calibration Benchmark receipts | §4.4 (new), §4.1 | QO-2 / Segment 1 Point
2 | CANDIDATE
Receipt entropy-export reframing: a receipt compresses epistemic/audit state for an external
verifier; physically it is irreversible work that increases Layer-0 internal dissipation — the
Prigogine balance is an accounting analogy, not a thermodynamic identity | §4.2 | QO-2 / Segment
1 Point 2 (validity issue 3) | CANDIDATE
Gateway billing attribution: static DCC-declared verification fee W_gate > 0 deducted from the
```

invoking principal's remaining work budget before any candidate inspection, charged regardless of validation outcome — closing the free-riding metric-exhaustion (denial-of-service) vector at the boundary; explicit free-generation / metered-crossing statement added | §6 | Segment 1 Point 4 (a); data-loss audit (double flag) | CANDIDATE
Gateway soundness scope: gateways enforce structural/resource compliance (contract soundness, resource bounds, float prohibition), not informal-to-formal semantic intent alignment; semantic fidelity is governed exclusively by the docs/qs.md §6.1 round-trip co-signing handshake | §6 | Segment 1 Point 4 (b) | CANDIDATE
Migration witnessing under structural indirection: algorithm-tagged (multihash-style) self-describing digests; migration receipts record superseded/successor algorithm identifiers; historical commitments preserved by re-witnessing (re-digest and re-seal anchored to the original commitments) | §4.3 | Segment 1 Point 4 (c) | CANDIDATE
Terminal-status registry acknowledgment: the five constitutional lifecycle terminals cross-referenced to the sealed 12-status registry of docs/abstractionLayers.md §3.6 as the single source of truth | §5.2 | Data-loss audit of this file | CANDIDATE
Prior-art positioning against resource-metered bytecodes (eBPF, deterministic Wasm), content-addressed stores (Unison, egg/egglog), zkVM receipts (RISC Zero, Cairo), and exact arithmetic (Posits, iRRAM) deferred to the QnIR v0.2 candidate and the regenerated condensed specification | — | Segment 1 Point 5 | DEFERRED
Rendering: all 7 display-math blocks and 5 inline math spans re-encoded to engine-independent plain text (Unicode operators; code spans for every token containing subscripts, superscripts, or pipes), matching the docs/abstractionLayers.md v1.8.1 convention | file-wide | Site-wide rendering finding (Rendering Fix Record, docs/abstractionLayers.md v1.8.1) | EDITORIAL
Editorial box fix: §4.3 structural-indirection diagram, middle box row 2 (Dilithium3 instantiation) was 1 column wider than its own borders — a pre-existing canon defect of v1.1.1, same class as the fixes recorded in docs/qexpr.md v1.3.0 and docs/qs.md v1.3.0; re-padded to the 31-column interior with row text preserved byte-for-byte | §4.3 | Pre-existing editorial defect | EDITORIAL
Normative References: docs/qphy.md added for Landauer floor enforcement and telemetry anchoring; front matter terms_used gains compute-unit-u; §3.2 cross-references the mechanical BOC procedure | Normative References, front matter, §3.2 | Editorial consequence of §4.4 and §7 | EDITORIAL

Cross-file dependencies flagged (deferred under the one-file-at-a-time discipline): QnIR v0.2 — binds W_gate into its budget accounting (the caller's remaining budget W_rem), adopts the two-stage BOC model in its design verdicts, inherits the 1 U normalization for its static cost table, and carries the deferred prior-art positioning; docs/qphy.md — names the Layer-0 calibration-benchmark anchors at its next versioned update (the condensed specification prototypes Q(Zbus) / Q(th2), which are not yet canon); docs/qs.md — §6.1 remains the sole semantic-alignment path referenced by §6; docs/index.md — the SSL scaling-threshold table there quotes 3 of 5 thresholds (a touch-up may ride along a later index update); glossary — the boc-policy and compute-unit-u entries refresh at the queued glossary update.

H2: Normative References

The following documents establish the structural, mathematical, and legal constraints derived from this constitutional framework.

Document | Role | Target
docs/abstractionLayers.md | Layer architecture and SIMEMP Gateway validation | abstractionLayers.html
docs/qnPrimitive.md | Universal Envelope, operational axioms, and optimization primitives | qnPrimitive.html
docs/q0_q1.md | Primary genesis origin Q(0) and first unit Q(1) | q0_q1.html
docs/qm.md | Constructive numeric leaf substrate and Diophantine division constraints | qm.html
docs/qexpr.md | Layer 3 Content-Addressed Expression DAG specification | qexpr.html
docs/qm_calculus.md | Discrete Exterior Calculus, continued fractions, and invariant dynamics | qm_calculus.html
docs/qphy.md | Physical-layer projections; Landauer floor enforcement and Layer-0 telemetry anchoring | qphy.html
docs/qs.md | Constructive relational intent, epistemic spectrum, and human-agent alignment | qs.html
LICENSE.md | Scaling Source License 1.0 governing the Licensed Work | LICENSE.md

H2: 1. The Foundational Conjecture and Dependencies Determinism

The foundational premise of this architecture addresses the formal capability of discrete number systems: *Can a discrete qn number system abstract any computable phenomenon?*

Classical mathematics admits non-constructive entities: actual infinities, unmetered memory access, and continuous spaces lacking thermodynamic boundaries. Conversely, physical computational engines are strictly finite, bounded by thermodynamics, discrete memory hierarchies, and finite signal transmission speeds.

We postulate the operational conjecture:

A number system can abstract any computable phenomenon within a computational environment if, and only if, that abstraction is strictly governed by finite physical realities and constrained by the simemp framework.

This conjecture is formally rooted in dependencies-determinism. In an information-theoretic

sense, the state transition of any computational entity S_t is fully determined if, and only if, its conditional entropy given its complete set of declared causal dependencies D(S_t) vanishes:

H(S_t | D(S_t)) = 0

If any dependency is implicit, hidden, or unmeasured (violating the no-implicit-rule), or attempts to cross an unverified abstraction boundary or simemp-gateway, the available dependency set is incomplete (D' ⊂ D), yielding non-zero conditional entropy:

H(S_t | D') > 0

This residual entropy manifests operationally as non-deterministic state drift, runtime instability, or unmodeled attack surfaces. The SIMEMP framework ensures that every state transition's causal entropy is fully accounted for by explicit, verifiable dependencies across the cdqn stack.

Within the constructive model, H(S_t | D(S_t)) = 0 is a statement of functional purity — referential transparency over the declared dependency set — rather than a statistical measurement over an unmodeled probability distribution. Its operational content is made empirically falsifiable by the Observational Determinism Invariant [v1.2.0 — CANDIDATE]: for any two Layer-0 host execution environments E_1, E_2 that agree on the declared dependency projections (Π_D(E_1) = Π_D(E_2)) and satisfy the active dcc-profile hardware profile, evaluating the same governed transition must produce bit-identical output commitments (H(S_t(E_1)) = H(S_t(E_2))) and identical terminal receipt statuses. The invariant is deliberately stated over output commitments rather than raw values: commitments are what cross boundaries, so the comparison itself is verifiable at any receiving node without exposing payload content.

Under this invariant, an incomplete dependency set D' ⊂ D is no longer merely a hidden-model assumption: it is an empirically detectable defect, exposed by differential execution testing of the same governed transition across heterogeneous environments. An implementation that silently depends on unmodeled Layer-0 physical state — microarchitectural timing, cache residency, runtime assumptions — diverges under differential testing at exactly the boundary where its declared projections fail to capture the executed environment, and is thereby discarded under the same discipline as any other Tier-1 violation.

H2: 2. Method of Verification: The Recursive Search Loop

The Qn architecture does not claim closed empirical finality. The system operates as an active, recursive research framework grounded in operational falsifiability and multi-tier constraint auditing.

H3: 2.1. Operational Verification and Computational Consistency

Because the system explores finite computational bounds ahead of physical hardware realization, all models are working hypotheses structured to prevent premature calcification. The development trajectory is governed by a cyclic search loop:

Proposal -> Construction -> Evaluation -> Selection

Abstractions are verified empirically through computational-consistency:

If a proposed abstraction is internally consistent, and if computations authored in qnlang and compiled to qnir execute deterministically and terminate within declared finite resource envelopes, the abstraction is deemed provisionally valid within the Qn universe:

Validity(A) ⇔ (Consistent(A) ∧ ∀ p ∈ QnIR(A), ExecutesDeterministicallyWithinBounds(p))

Failure to compute within finite bounds, or divergence during execution, refutes the proposed abstraction and forces an iteration of the search loop.

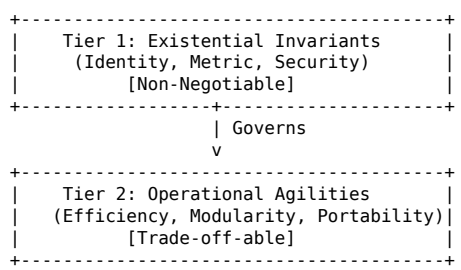
H3: 2.2. Epistemic Demarcation: Documentary Baselines vs. Empirical Grounding

To prevent ambiguity across documentation indexes and automated repository ingestion, the architecture strictly distinguishes between Documentary Maturity and Epistemic Modality:

- Documentary Maturity (Canonical Baseline): Specifications designated as Canonical represent frozen structural contracts for the current iteration of the search loop. They provide an authoritative, immutable baseline that distinguishes validated architecture from exploratory vision proposals (docs/chronosa.md) or uncommitted roadmap items.
- Epistemic Modality (Constructive Working Hypothesis): Because the runtime has not yet been executed on physical silicon substrates (Layer 0), all canonical specifications maintain the formal epistemic status of Constructive Working Hypotheses pending empirical Layer 0 execution receipts.

H2: 3. The SIMEMP Constraint Architecture

The constraint architecture enforces a strict two-tier hierarchy across all abstraction boundaries.



H3: 3.1. Tier 1: Existential Invariants

Tier 1 Existential Invariants represent non-negotiable requirements for any governed artifact to exist. If an abstraction violates any invariant, it is discarded:

1. Identity: The abstraction must be distinctly known, collision-resistant, versionable, and causally rooted.
2. Metric: The abstraction must carry an explicit, finite measurement envelope declaring memory, compute, precision, and uncertainty boundaries.
3. Security: The abstraction's state transitions, causal integrity, and execution limits must be verifiable against finite adversarial models.

H3: 3.2. Tier 2: Operational Agilities

Tier 2 Operational Agilities represent optimization parameters. They naturally conflict with one another:

1. Efficiency: Algorithmic throughput and minimal resource dissipation.
2. Modularity: Strict isolation of concerns, boundary containment, and composability.
3. Portability: Substrate independence across execution targets and network architectures.

The Golden Rule of Trade-offs: An Operational Agility may always be sacrificed to preserve an Existential Invariant, but an Existential Invariant must never be compromised for an Agility.

The mechanical decision procedure that instantiates this hierarchy is the two-stage BOC model of §7 [v1.2.0 — CANDIDATE].

H2: 4. Environmental Realities, Thermodynamics, and the DCC Profile

Physical computation requires formalizing hardware boundaries and thermodynamic laws directly into computational metadata.

H3: 4.1. The Memory Wall, Semantic Gap, and Landauer's Principle

- Landauer's Bound and Dissipation: Any irreversible manipulation or erasure of information dissipates thermodynamic work:

$$W \geq k_B T \ln 2$$

At a reference ambient temperature of T = 300 K this floor evaluates to approximately 2.87 × 10^-21 J per erased bit. It is a theoretical scale anchor, not a meter reading: commodity CMOS Layer-0 substrates dissipate on the order of 10^7 to 10^10 times this floor per logical operation (capacitive charging, leakage currents, interconnect overhead), regardless of the logical reversibility of the high-level symbolic operation. The constitutional consequence — the separation of the logical work quantum from the physical energy ledger — is drawn in §4.4 [v1.2.0 — CANDIDATE].

- The Memory Wall: Transporting data across physical memory hierarchies dissipates physical energy ($O(C \cdot V^2 \cdot f)$) and introduces transmission latency. The Metric invariant requires explicit accounting for memory bandwidth and locality rather than purely abstract asymptotic complexity.
- The Semantic Gap: Translation between high-level intent (qnlang) and physical execution (qnir) represents a primary vulnerability. The system mandates bounded, verifiable intermediate representations.

H3: 4.2. Open Systems, Entropy Export, and Dissipative Error Correction

A computational system is an open thermodynamic system. Left unconstrained, thermal noise, transmission faults, and physical substrate errors maximize internal entropy ($dS_{internal} > 0$).

To maintain internal structural order and deterministic reproducibility, the system operates as a dissipative structure (Prigogine formulation):

$$dS = dS_{internal} + dS_{exchange}, \text{ where } dS_{exchange} < 0$$

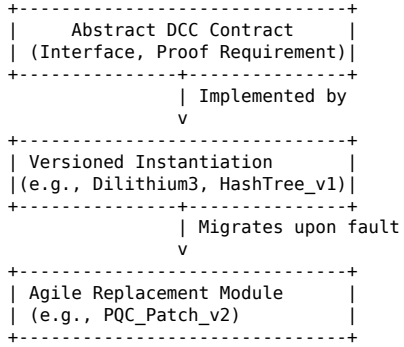
Maintaining computational stability requires expending work—formally recognized as

metric-exhaustion—to verify state transitions and export entropy via bounded receipts. Without reproducible, receipt-generating validation steps, a system cannot self-repair against entropy accumulation.

The entropy exported by a bounded receipt is epistemic, not thermal [v1.2.0 — CANDIDATE]: a receipt compresses audit state for an external verifier, reducing the verifier's information uncertainty about the transition it attests. The physical act of allocating, hashing, signing, and transmitting that receipt is itself irreversible Layer-0 work and increases internal dissipation (dS_internal > 0) on the emitting node. The Prigogine balance above is therefore an analogy of accounting structure — internal disorder must be compensated by bounded, verifiable export — not an identity of thermodynamic quantities; the physical energy ledger is carried separately by the Layer-0 Hardware Telemetry Envelope of §4.4.

H3: 4.3. The DCC Profile and Structural Indirection

To prevent rigid calcification and ensure defense against zero-day vulnerabilities, the system implements structural-indirection:



Every abstraction layer, morphism, and gateway exposes an explicit dcc-profile (Dependencies, Constraints, Capabilities):

- Dependencies: Abstract capability contracts, cryptographic commitments, and environmental prerequisites. Root dependencies must be parameterized interfaces rather than hardcoded low-level implementations.
- Constraints: Hard ceilings on memory allocation, execution steps, recursion depth, and precision budgets.
- Capabilities: Permitted operations, transformation rights, and network export authorizations.

If a concrete cryptographic primitive or implementation module suffers a zero-day vulnerability, structural indirection allows the system to swap the implementation module and emit an explicit migration receipt without fracturing the abstract DCC contract or breaking causal lineage.

Migration Witnessing [v1.2.0 — CANDIDATE]. Because node identity and causal lineage are content-addressed by cryptographic digests, swapping a compromised primitive invalidates historical hash bindings unless digests are self-describing. Every content-addressed digest therefore carries an explicit algorithm tag (a multihash-style prefix identifying its hash function), and every migration receipt records both the superseded and the successor algorithm identifiers. Historical commitments are preserved by re-witnessing: previously sealed artifacts are re-digested under the successor primitive and re-sealed with fresh signatures, each re-witness record anchoring back to the original commitment so that lineage verification remains possible across the full algorithm history. Bare un-tagged digests are not admissible across a simemp-gateway.

H3: 4.4. The Abstract Compute Unit U and the Layer-0 Hardware Telemetry Envelope [v1.2.0 — CANDIDATE]

The Abstract Compute Unit U is defined strictly as a platform-invariant logical work quantum at Layers 1–3 and in QnIR: the unique unit in which totality guarantees, static instruction costs, and cross-node receipt verification are expressed, normalized by definition so that the minimal governed state transition Q(0) -> Q(1) costs exactly 1 U. U is deliberately not a physical measurement. Joule costs vary continuously with silicon process node, voltage–frequency scaling, ambient temperature, and cache residency — quantities unknowable statically at the IR level — and folding them into U would break deterministic cross-node receipt verification, violating the substrate/invariant (car-vs-road) separation. One bit of logical erasure corresponds to the theoretical Landauer floor $W \geq k_B T \ln 2$ (§4.1); that floor is a scale relation anchored in Qphy, not a per-node meter reading.

Physical energy and thermal budgets ($E_{battery}$, $P_{thermal}$, in Joules and Watts) constitute a separate Layer-0 Hardware Telemetry Envelope, bound to U exclusively through the node's local Historical Calibration Benchmark receipt — a DCC-declared mapping from logical work to a hardware-specific worst-case energy envelope with stated error bars:

$$E_{phys} \leq \alpha_{compute} \cdot U_{op} + \beta_{bus} \cdot U_{mem}$$

where $\alpha_{compute}$ and β_{bus} are the node's declared worst-case per-unit energy coefficients for

the compute and memory-traffic work classes, and U_op, U_mem are the logical work executed in those classes. The coefficients are Layer-0 telemetry metadata: they gate battery and thermal leash enforcement at the node boundary and never feed back into logical budgets, so no implicit physical channel can perturb deterministic state transitions and H(S_t | D(S_t)) = 0 is preserved by construction. Until calibration receipts are physically realized on Layer 0 (§2.2), the envelope remains a CANDIDATE metadata contract under Axiom 14.

H2: 5. Existential Invariants Deep Dive

H3: 5.1. Identity Constraints

- Layered Verification: Abstractions must generate verifiable witnesses at each layer they traverse.
- Provenance and Lineage: Artifacts maintain a traceable derivation history. Pursuant to the ssl, all runtime specifications and derivative works must embed the required paternity-reference.

H3: 5.2. Metric Constraints

- Finite Resource Envelopes: Unmeasurable abstractions are physically unrealizable. Every operation must declare an explicit budget for execution, storage, and communication.
- Totality by Budget: Every computational transition must be total by budget, resolving to an explicit receipt containing a discrete terminal state (SUCCESS, FAILURE, TIMEOUT, BUDGET_EXHAUSTED, INCONCLUSIVE).
- Terminal-Status Registry [v1.2.0 — CANDIDATE]: The five statuses above are the constitutional core lifecycle terminals. The complete closed terminal-status registry of the Qn universe — adding NO_SOLUTION, QUARANTINED, PRECISION_INSUFFICIENT, EXPRESSION_TOO_COMPLEX, REJECTION_BY_GATEWAY, DIVISION_BY_ZERO_REJECTED, and ORDERING_INDETERMINATE_AT_ZOOM — is canonically sealed in docs/abstractionLayers.md §3.6, which remains the single source of truth for terminal-status vocabulary; this constitution references it without duplicating or amending it.

H3: 5.3. Security Constraints

- Bounded Adversaries and PQC Agility: Security models assume computationally bounded adversaries. The cryptographic substrate must maintain agility, remaining Post-Quantum Cryptography (PQC) capable without mutating underlying identity semantics.
- Halt-by-Construction: Computational subsystems must retain deterministic halt authority upon constraint exhaustion. Turing-unbounded execution is rejected within the governed core.
- The Causal Arrow: Distributed state ordering relies strictly on a persistent causal-arrow (monotonic sequences and cryptographic hash chains) rather than external wall-clock synchronization.

H2: 6. Constraint on Generative and Probabilistic Workflows

Stochastic and generative artificial intelligence processes are integrated via an explicit epistemic boundary:

```
[ Context & DCC ] --► [ Probabilistic Proposal ] --► [ Deterministic Construction ]
                                                                  |
[ Verified Receipt ] ◄-- [ Finite Selection ] ◄-- [ Metric Evaluation ]
```

Boundary Billing and Soundness Scope [v1.2.0 — CANDIDATE]. Generation behind the stochastic boundary is free of governance metering; every crossing of a simemp-gateway into governed state is deterministic, metered, and receipted. Parsing, measuring, and validating an unverified candidate — PQC signature checks, DCC contract checks, metric-envelope evaluation — themselves consume finite compute and memory work. Every gateway invocation therefore deducts a static, DCC-declared verification fee W_gate > 0 from the invoking principal's remaining work budget before any candidate is inspected, charged regardless of validation outcome, so failed and adversarial candidate floods are billed to their invoker and cannot induce free-riding metric exhaustion at the boundary. Gateway checks enforce structural and resource compliance — formal contract soundness, resource bounds, and representation constraints (the float prohibition) — and not informal-to-formal semantic intent alignment: whether a formalized candidate matches the human principal's actual intent is governed exclusively by the Deterministic Canonical Round-Trip Paraphrase & Co-Signing Handshake (docs/qs.md §6.1), which gates the transition to SEALED.

1. Context Declaration: Establish explicit DCC profiles, objective functions, and finite budgets.
2. Generation: Stochastic engines propose candidate structures (classified strictly as *unverified candidates*).
3. Construction: Candidates are formalized into canonical representations.
4. Evaluation: Candidates are measured against constraints, error tolerances, and resource consumption.
5. Selection & Sealing: A deterministic policy validates and seals acceptable candidates, emitting a signed receipt.
6. Bounded Termination: If no candidate meets the threshold within the declared budget, execution halts with a defined status (NO_SOLUTION or BUDGET_EXHAUSTED).

H2: 7. The BOC Policy

The Best of Choices (BOC) Policy is the formal decision framework used to resolve architectural and design conflicts. Version 1.1.1 expressed it as the scalar maximization Selection = argmax(SIMEMP Compliance(c)) over a SIMEMP Compliance(c) quantity that carried no declared codomain, metric space, or partial order — a formulation that could not be evaluated mechanically and under which any design choice could be rationalized post hoc as BOC-compliant. That scalar formulation is retired. Under the external review (QO-1), BOC is specified as a two-stage constrained lexicographic decision model over declared DCC metadata, mechanically decidable wherever its inputs are declared [v1.2.0 — CANDIDATE]:

Stage 1 — Tier 1 hard admission filter (binary). A candidate is admissible only if all three Existential Invariants hold as boolean predicates:

C_valid = { c ∈ C | I_Id(c) ∧ I_Met(c) ∧ I_Sec(c) = 1 }

where I_Id(c) verifies a collision-resistant identity envelope with intact causal lineage, I_Met(c) verifies that every resource ceiling in the Metric envelope is finite and declared, and I_Sec(c) verifies that capability and PQC signature contracts validate. A candidate failing any predicate is discarded outright (terminal status REJECTION_BY_GATEWAY); inadmissible candidates are never ranked, because Tier-1 invariants are admission predicates, not quantities to be maximized.

Stage 2 — Tier 2 lexicographic ordering over C_valid. Among admissible candidates, BOC minimizes the lexicographic key:

LexKey(c) = (ImplicitDeps(c), perm_π(ΔW_worst(c), M_coupling(c), P_coverage(c)))

The leading component is the constitutional mandate to minimize implicit state: ImplicitDeps(c) attests the candidate's residual implicit-state burden — dependencies present in its construction but absent from its declared dependency commitment — and its minimization is fixed in first position, never tradeable by any Tier-2 priority declaration. The trailing triple carries the Operational Agilities, mutually ordered by an explicit priority permutation π ∈ S_3 declared in the calling dcc-profile: ΔW_worst(c) the worst-case declared work (Efficiency), M_coupling(c) the external dependency and interface boundary count (Modularity), and P_coverage(c) the supported target hardware profile count (Portability). Ties inside a component resolve to the next component.

Rubric mode. Where Tier-2 quantities remain qualitative architectural judgments during high-level specification design — before measurable DCC metadata exists — BOC operates explicitly as a structured architectural rationale rubric: every design verdict must tabulate each finalist's Tier-1 pass/fail status and its Tier-2 trade-off vector together with the declared priority ordering used, and a verdict that cannot produce this tabulation is not BOC-compliant. Residual ties at the full key are never broken implicitly (no-implicit-rule): they force either refinement of the declared DCC metadata or explicit principal adjudication recorded in the design verdict.

When implementation paths conflict, the BOC policy selects the option that minimizes implicit state and guarantees invariant preservation, even if it requires higher operational overhead.

H2: 8. Alignment with the Scaling Source License

The architectural constraints directly enforce the legal mechanics of the ssl:

- Open Core Invariants: Identity and security layers cryptographically preserve the Open Core Invariants (anti-patent retaliation and non-scaling open-access clauses) across derivative lineage trees.
- Scale Auditing: The Metric invariant provides native, verifiable telemetry (active instances, container count, agent instances, and monthly transaction volumes) to verify whether deployment scale crosses the scaling-threshold.
- Safe Harbor Neutrality: The technical substrate operates as a neutral conduit, preserving sovereign user content and applications as uninspected Payloads.

H2: 9. Conclusion

The SIMEMP framework is a provisional, evolutionary thesis designed to formalize physical computing limits into explicit architectural constraints. By grounding computational abstractions in non-equilibrium thermodynamics, dependencies determinism, and structural indirection, SIMEMP establishes an adaptable foundation for exploring finite, accountable, and generative computational systems across the cdqn network. All subsequent primitive derivations and operational specifications remain subject to verification and revision under this framework.

Annex B — abstractionLayers.md

Source: docs/abstractionLayers.md — *1.8.1, updated 2026-10-26. Rendered content per Annex Guide conventions N1–N6: 518 source lines, 744 rendered; 46,244 → 38,890 characters; source*

SHA-256 02b96072 41ca2e69 c997ef37 92b83694 0338d6e0 bc0d0789 a0444b83 3458d85e.
Content fidelity verified mechanically at build time.

```
[front matter] ------------------------------------------------------------
[front matter] title: Abstraction Layers — Structural Thesis for the Qn and cdqn Stack
[front matter] version: 1.8.1
[front matter] updated: 2026-10-26
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Structural thesis defining the abstraction-layer framework, CAE-DAG
[front matter]   symbolic composition, substrate-invariant decoupling, and Q(anchor)
[front matter]   consensus under SIMEMP constraints.
[front matter] terms_used: abstraction-layer, boc-policy, causal-arrow, cdqn, chronosa,
[front matter]   complexity-degree, computational-consistency, compute-unit-u, dcc-
[front matter]   profile, dependencies-determinism, dimension-d, dual-ring-pqc-
[front matter]   boundary, existential-invariant, exposure-functor, identity-class,
[front matter]   layer-0, layer-1, licensed-work, lifecycle-state, metric-envelope,
[front matter]   metric-exhaustion, no-implicit-rule, non-rogue-invariant,
[front matter]   operational-agility, payload, q-anchor, q-bypass, q-c, q-datastruc,
[front matter]   q-even, q-event, q-gauge, q-hbar, q-odd, q-patterns, q-reuse, q-th-2,
[front matter]   q-workflows, q-z-bus, q0, q1, qbio, qchem, qexpr, qlog, qm, qn, qn-
[front matter]   rsi, qnir, qnlang, qphy, qs, receipt, remainder-r, security-envelope,
[front matter]   simemp, simemp-gateway, ssl, structural-indirection, successor-
[front matter]   morphism, terminal-exactness, universal-envelope, zoom-z
[front matter] ------------------------------------------------------------
```

H1: Abstraction Layers — Structural Thesis for the Qn and cdqn Stack

Field	Specification
Document Title	Abstraction Layers — Structural Thesis for the Qn and cdqn Stack
Version	1.8.1
Last Updated	2026-10-26 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Canonical Structural Thesis — Milestone 1 Fully Sealed Baseline

H2: Review Integration Record (v1.8.0)

Source: docs/AiReviews/AiReviewQnIRv0.1.md — transcript of the external adversarial review (Google Deep Review, October 2026) of the condensed Qn universe specification and QnIR v0.1 candidate draft. This is the first review-driven update of this thesis; it integrates the road-receipt trust-model findings of that review (reviewer open question QO-13, Segment 3 Point 1, and the boundary-trust verdict finding on Objection 7): soundness of the proof field π, the origin-quotiented canonical projection for export, and the interactive bisection audit right. It also absorbs the cross-file dependency flagged by docs/qs.md v1.3.0 (Inner-Ring co-signature mechanics for the §6.1 handshake). Under Axiom 13, this file is a new versioned artifact; no prior content is mutated in place. New constructions introduced below are tagged [v1.8.0 — CANDIDATE] pending BOC sealing.

Change	Section	Reviewer source	Type
Origin-Quotiented Canonical Projection for Export: the Exposure Functor quotients out the local root (Q(0)_A ↦ Q̂(0)) prior to computing public commitments H_pub(S_in) / H_pub(S_out), so identical deterministic computations on distinct nodes yield identical portable commitments, verifiable by forced re-derivation on any peer	§4.4	QO-13 / Segment 3 Point 1 (disjoint commitment hashes)	CANDIDATE
π-field soundness: two admissible realizations of verifiable computational integrity — (A) transparent zk-STARK execution proof over the AIR-arithmetized QnIR trace (O(polylog N) verification, zero trust in the sender, Axiom 8 preserved); (B) optimistic interactive Anchor-Cut bisection (O(log K) challenge rounds down to one origin-quotiented instruction step, O(1) replay). A bare σ-signed attestation without either is not sound against Byzantine nodes	§4.4 (new)	QO-13 / Segment 3 Point 1; verdict Objection 7	CANDIDATE
Bisection-failure disposition: terminal status QUARANTINED + receipt RECEIPT_ANCHOR_REVOKED (name inherited from the anchor-fracture containment of docs/qlog.md); REVOKED is deliberately not introduced as a new Axiom-6 terminal status — the sealed 12-status list of §3.6 is untouched	§4.4, §3.6	QO-13 / Segment 3 Point 1	CANDIDATE
Inner-Ring co-signature operations: the qs.md §6.1 handshake executes as an Inner-Ring signing operation; only the co-signature commitment crosses the Outer Ring	§8.3	docs/qs.md v1.3.0 cross-file dependency	CANDIDATE
Soundness caveat on Q(anchor) consensus: contract verification plus receipt trust is not, on its own, execution verification; admissible road receipts carry π per §4.4	§8.4	QO-13 / Segment 3 Point 1	CANDIDATE
Open Items: AIR arithmetization of the QnIR transition function and concrete π-field realizations deferred to the QnIR v0.2 candidate	§13	QO-13 / Segment 4 Point 5 (control-flow vs. data-flow verification)	CANDIDATE

Cross-file dependencies flagged (deferred under the one-file-at-a-time discipline, to be integrated in subsequent versioned updates): QnIR v0.2 — AIR arithmetization, the π-field encoding, and the control-flow/data-flow verification finding (Segment 4, Point 5) land in the candidate VM specification; docs/qm_calculus.md — the K intermediate OP_ANCHORCUT witnesses are

already canonical there, and this update only binds them into the bisection protocol (a cross-reference touch-up may ride along that file's next versioned update); docs/chronosa.md — sheaf gluing over road receipts inherits the §8.4 soundness caveat at its next versioned update; glossary entries for *Origin-Quotiented Canonical Projection* and *Interactive Bisection Challenge Right* (deferred under the standing no-dead-entries discipline of docs/qs_history.md §8).

H2: Rendering Fix Record (v1.8.1)

Source. Author field report on the committed v1.8.0 artifact: on the live GitHub Pages HTML, formulas surface as raw LaTeX source and underscore pairs inside formulas are collapsed into markdown emphasis, because the site's Jekyll/kramdown pipeline ships no MathJax/KaTeX engine (the condition is site-wide; this file merely exposes it most densely). The committed artifact additionally carried two local regressions: mid-sentence display-math fragmentation in §4.4 and §8.3 (inline formulas split into standalone display blocks inside flowing sentences), and a §8.3 contract identifier typeset as a math span with escaped underscores, leaking raw escape sequences into the markdown source.

Decision. All mathematical notation in this file is re-encoded into engine-independent plain text that renders identically on GitHub Pages HTML, on github.com source views, and in raw markdown: Unicode operators and letters (≠, ≤, ≡, ≺, ⇔, ⇒, ->, <->, ↦, ∈, ∧, ⊗, ×, ÷, Δ, π, σ, φ, √2, κ, α, β, …, Q̂(0)) for symbols, and code spans for every token containing subscripts, superscripts, or pipes (d_1, S_in, H_pub(S_out), Q(0)_A, T_local, R_road, H(S_t | D(S_t)) = 0), which also hardens the text against kramdown underscore-emphasis pairing. The §8.3 contract identifier is retyped as the code span DCC_INTENT_CONTRACT_v1, matching the code-span convention of the anchor contracts in §8.4, and the flowing prose of §4.4 and §8.3 is restored.

Scope and invariants. No semantic content is mutated: every formula, tuple, operator, receipt name, terminal status, and reviewer attribution of v1.8.0 is preserved token-for-token under the new encoding; the sealed 12-terminal-status list of §3.6 is untouched; all [v1.8.0 — CANDIDATE] tags retain their provenance under Axiom 13 and Axiom 14. Under the one-file-at-a-time discipline, the re-encoding is applied to this file only; sibling specification files sharing the LaTeX encoding remain queued for the same treatment at their next versioned updates. A site-layout MathJax include would be the alternative global remedy, but plain-text encoding keeps every page renderer-independent and is therefore taken as the BOC.

H2: Normative References

The following documents establish the constitutional, physical, and domain constraints governing this layer architecture. If a technical conflict arises, simemp.md governs; if a structural conflict arises, this document governs; if a legal conflict arises, LICENSE.md governs. References are deliberately version-unpinned: each target document's front matter carries its own authoritative version, so this thesis never duplicates drift-prone patch state.

Document | Role | Target
docs/simemp.md | Constitutional constraints, thermodynamics, and Dependencies Determinism | simemp.html
docs/qnPrimitive.md | Universal Envelope, operational axioms, and Lifecycle States | qnPrimitive.html
docs/q0_q1.md | Primary genesis origin Q(0) and first unit Q(1) | q0_q1.html
docs/q2_q9.md | Secondary single-digit DCC anchors and parity partitions | q2_q9.html
docs/qm.md | Constructive numeric leaf substrate and Diophantine division constraints | qm.html
docs/qm_geometry.md | Multi-axial frames, Clifford geometric algebra, and float-free rotations | qm_geometry.html
docs/qm_calculus.md | Discrete Exterior Calculus, continued fractions, and invariant dynamics | qm_calculus.html
docs/qexpr.md | Layer 3 Content-Addressed Expression DAG container | qexpr.html
docs/qs.md | Constructive relational intent and Intent Contracts | qs.html
docs/qlog.md | Constructive type theory, pattern verification, and inferential modes | qlog.html
docs/qphy.md | Discrete field dynamics, computational thermodynamics, and physical entropy | qphy.html
docs/qchem.md | Discrete stoichiometry, molecular graphs, and reaction energetics | qchem.html
docs/qbio.md | Biosemiotics, metabolic homeostasis, and active inference | qbio.html
LICENSE.md | Scaling Source License 1.0 governing the Licensed Work | LICENSE.md

H2: 1. Foundational Position

The Qn computational universe is established upon a governing operational conjecture:

A number system can abstract any computable phenomenon within a computational environment if, and only if, that abstraction is strictly governed by finite physical realities and constrained by the simemp framework.

A qn artifact is not an ungrounded mathematical abstraction. It is a governed, finite, identifiable, measurable, and security-constrained computational entity.

The architecture rejects actual infinities, unconstrained sets, and unmetered execution. Layer boundaries represent structural interfaces governed by dependencies-determinism (H(S_t | D(S_t)) = 0) and structured with structural-indirection to prevent premature calcification. Every abstraction must be constructed, bounded, identified, measured, and verified within explicit

resource envelopes via computational-consistency:

$$\text{Validity}(A) \Leftrightarrow (\text{Consistent}(A) \land \forall\ p \in \text{QnIR}(A), \text{ExecutesDeterministicallyWithinBounds}(p))$$

H2: 2. The Horizontal Execution Core and Gateway Topography

Abstraction layers define structural boundaries governing artifact genesis, transformation, and physical lowering. Rather than a static, unverified staircase of infinite tiers, the horizontal physical execution stack terminates strictly at Layer 3, which then lowers into the virtual execution handler (qnir).

Inter-layer transitions are modeled as Universal Functorial Gateways (F : C(L) -> C(L+1)) that enforce the no-implicit-rule and audit Landauer dissipation at every state step.

H3: 2.1. Layer 0 — Commodity Physical Substrate

Layer 0 is the physical, hardware-agnostic execution substrate:
- Commodity processors (CPU, GPU, DSP, FPGA), memory hierarchies, storage media, and network interfaces.
- Physical boundaries: finite memory, clock latency, thermal dissipation limits, and the Memory Wall ($O(C\ V^2\ f)$).
- Raw physical entropy sources (thermal noise, ring oscillator phase drift) and operating system telemetry.

Layer 0 remains outside the governed Qn universe. It constitutes the Root of Finiteness. Raw substrate states, hardware counters, and noise channels become governed Qn artifacts only upon formal mediation by an onboarding gateway.

H3: 2.2. Layer 1 — Node Genesis Layer

Layer 1 is the primary governed ontological layer, instantiated during node onboarding across the Layer 0/1 Gateway:
- Q(0): Local genesis artifact, causal origin zero, empty birth context (docs/q0_q1.md §3).
- Q(1): First unit artifact, unity measure, and baseline reference for the Abstract Compute Unit U along dimensional axis d_1 (docs/q0_q1.md §4).
- Abstract compute unit U: Minimal governed state transition from Q(0) to Q(1), calibrated to Landauer dissipation ($W \geq k_B\ T\ \ln 2$).
- Single-digit secondary DCC sources (Q(2) … Q(9)): Inductively constructed via Successor Morphism S along axis d_1 (docs/q2_q9.md).
- Sign polarity: Induced directed relation between Q(0) and Q(1) via reflection involution I_{d_1}.

Q(0) is strictly local to its node:

$$\forall\ A \neq B \Rightarrow Q(0)_A \neq Q(0)_B$$

No universal global zero is admitted. Q(1) maintains an invariant numeric and metric value across all nodes; local entropy parameterizes its identity witness but cannot mutate its unit magnitude (q = 1).

H3: 2.3. Layer 2 — Governed Morphism Layer

Transitioning from Layer 1 to Layer 2 requires crossing an algebraic constraint gateway that intercepts invalid states (such as division by Q(0)) before resource allocation. Layer 2 governs transformation and action across the Qn universe:
- Elementary R-Algebra: Constructive addition (+), subtraction (-), multiplication (×), and Diophantine division (÷) (docs/qm.md §4).
- Optimization Morphisms: Category B execution short-circuiting (Q(bypass)) and sub-graph content reuse (Q(reuse)), eliminating redundant memory allocation and Landauer dissipation (W = 0).
- Multi-Axial Frame Extension (D_step): Inductive generation of orthogonal axes d_2 … d_n from genesis axis d_1 (docs/qm_geometry.md §2.1).
- Involution and Geometric Products: Reflection engine I_d, Clifford geometric product ($\bar{u}\,\bar{v} = \bar{u} \cdot \bar{v} + \bar{u} \land v$), and float-free rational Cayley rotors.
- Secondary DCC Routing: Directs execution requests to specialized capability contracts anchored at single digits: parity (Q(2)), simplices (Q(3)), aperiodicity (Q(5)), and byte octets (Q(8)).

H3: 2.4. Layer 3 — Symbolic Composition Layer (CAE-DAG)

Transitioning from Layer 2 to Layer 3 wraps morphisms into explicit expression graphs governed by depth and node ceilings. Layer 3 governs symbolic representation and compile-time reduction:
- Under the Best of Choices (BOC) Policy (docs/simemp.md §7), symbolic expressions are represented as Content-Addressed Expression DAGs (CAE-DAGs / Term DAGs) with cryptographic hash-consing (docs/qexpr.md). Naive pointer-heap trees are rejected. Every expression carries an explicit universal-envelope, metric-envelope, and lifecycle-state.
- Equivalence Saturation: Structural equality collapses to O(1) integer hash comparisons:

$$A \equiv B \Leftrightarrow H(A) == H(B)$$

- Deterministic Collapse: Expressions remain symbolic during normalization and collapse deterministically to canonical Qm leaf tuples or flat linear instruction streams (qnir) upon crossing execution boundaries.

H3: 2.5. The Execution Boundary vs. Orthogonal Projections

To prevent premature architectural calcification, the architecture explicitly demarks the Physical Execution Core (Layers 0 to 3) from higher-order Orthogonal Projections:

1. The Execution Core Boundary: The sequential, physical machine stack terminates at Layer 3. Below Layer 3, execution is hardware-grounded and metered in units of U.
2. Vertical Domain Projections: The domains (Qm, Qs, Qphy, Qlog, Qchem, Qbio) are not horizontal layers sitting above Layer 3. They are orthogonal vertical slices that project downwards across Layers 1, 2, and 3:
   - Constructive Type Logics (qlog, product category C(Qm) ⊗ C(Qs)).
   - Abiotic Molecular Chemistry (qchem, product category C(Qphy) ⊗ C(Qm)).
   - Living Metabolic Systems (qbio, biosemiotic product category C(Qchem) ⊗ C(Qs)).
3. Frontend Syntax: Authoring languages (qnlang) sit outside the execution core as compilation frontends that target Layer 3 CAE-DAGs.
4. Network and Swarm Intelligence: Distributed coordination (cdqn) and non-local sheaf synthesis (chronosa) operate transversally over emitted Road Receipts, requiring no additional horizontal layer numbers.

H2: 3. Structural Principles

H3: 3.1. All Governed Artifacts are Qn

Within the licensed-work, every accepted artifact is a qn computational entity or is reducible to one: numeric values, operators, expression DAGs, logical propositions, receipts, identities, capabilities, composite workflows (Q(workflows)), and compiled runtime objects.

Raw user data, external host files, and creative assets remain sovereign, uninspected Payloads insulated from the technical substrate by an epistemic and legal safe-harbor air gap.

H3: 3.2. The No-Implicit Rule

In the Qn universe, no implicit entity, behavior, assumption, default, interpretation, or convention may cross a simemp-gateway. Only explicit Qn objects may pass between layers.

Implicitness violates Dependencies Determinism ($H(S_t \mid D') > 0$) and the constitutional hierarchy:
- Tier 1 Existential Invariants: Identity, Metric, Security cannot be implicit.
- Tier 2 Operational Agilities: Efficiency, Modularity, Portability are optimized within explicit constraints.

H3: 3.3. Local-First Origin and Controlled Exposure

Artifacts originate locally within a node's Inner Ring. Remote nodes in the network never inspect raw local memory heaps. Network interaction across the Outer Ring is restricted to:
- Bounded public attestations and cryptographic commitments.
- Exported higher-order qexpr commitment hashes.
- Capability delegation certificates and metric summaries.
- Accountable pseudonymous identities.

H3: 3.4. DCC Profiles Everywhere

Every layer, object, morphism, and exported attestation must expose an explicit dcc-profile utilizing structural indirection:
- Dependencies: Abstract capability contracts, parent hashes, and cryptographic anchors.
- Constraints: Hard ceilings on memory allocation declared in the metric-envelope, execution steps, recursion depth, and precision bounds.
- Capabilities: Permitted operations, transformation rights, and network export authorizations governed by the security-envelope.

H3: 3.5. Causal Arrow and Birth Order

Every discrete Q(event) is indexed along a strictly monotonic causal-arrow:

$Q(0) < Q(1) < Q(2) < \ldots < Q(9) < d_1 < d_2 <$ Higher CAE-DAGs

No artifact may declare a parent or dependency born later in the causal sequence.

H3: 3.6. Totality by Budget

Every governed operation must terminate within its declared metric budget as a dissipative

thermodynamic step (metric-exhaustion), resolving to an explicit receipt:
SUCCESS, FAILURE, NO_SOLUTION, TIMEOUT, BUDGET_EXHAUSTED, INCONCLUSIVE, QUARANTINED,
PRECISION_INSUFFICIENT, EXPRESSION_TOO_COMPLEX, REJECTION_BY_GATEWAY, DIVISION_BY_ZERO_REJECTED,
or ORDERING_INDETERMINATE_AT_ZOOM. Silent non-termination is prohibited.

H3: 3.7. Substrate Profile vs. Governed Invariant ("Car vs. Road" Disjunction)

To reconcile physical hardware heterogeneity with platform-invariant verification, the
architecture enforces an explicit disjunction between execution telemetry and algebraic
invariants:

Context | The Substrate Envelope (The "Car") | The Governed Invariant (The "Road")
Location | Declared in Metric/Security Envelope | Declared in Abstract DCC Contract
Parameters | Node hardware architecture and clock frequency | Invariant input state commitment
Thermodynamics | Local physical thermal dissipation (T_local) | Invariant target state milestone
Identity | Microscopic silicon root Q(0)_N | Invariant Diophantine condition
Security | Local hardware trust ring and enclave | Substrate-invariant work ΔW in Q(1)

H2: 4. Set and Category Abstraction

The structural framework is formally modeled via constructive set theory and locally finite
category theory.

H3: 4.1. Local Qn Set

For node N, the local universe is a finite constructive set Q_N:
- Genesis member: Q(0)_N ∈ Q_N.
- Set membership requires an explicit identity, metric envelope, security envelope,
  lifecycle-state, and lineage witness.
- No universal global set of all Qn artifacts exists.

H3: 4.2. Local Qn Category

Each node defines a locally finite category C_N:
- Objects: Governed artifacts α ∈ Q_N.
- Morphisms: Governed transformations f : α -> β conforming to DCC constraints.
- Identities: Explicit identity morphisms id_α.

H3: 4.3. Functors and Gateways

Inter-layer transitions are modeled as functors constrained by gateway validation:

F : C(L) <-> C(L+1)

A valid layer-transition functor preserves identity, metric bounds, causal lineage, and SSL
metadata. Functors introducing implicit semantic interpretations are invalid.

H3: 4.4. cdqn as the Fractal Protocol of Governed Data Movement

cdqn operates across two structural scopes:
1. Local Scope (Intra-Node Chaining): Governs data movement and morphism transitions between
   local abstraction layers via Content-Addressed Expression DAGs (CAE-DAGs) and chained
   receipts.
2. Distributed Scope (Inter-Node Attestation): Connects autonomous local categories (C_N) into a
   distributed sheaf-like structure synthesized by chronosa via the Exposure Functor:

E_export : C_N -> Attestations_cdqn

H4: The "Road Receipt" Transmission Conduit
Nodes never broadcast internal CAE-DAGs across physical networks. Raw local graphs are
non-replicable because their causal lineages terminate at node-specific silicon origins (Q(0)_A
≠ Q(0)_B). Instead, nodes emit Boundary State-Transition Receipts ("road receipts"):

R_road = ⟨ H(S_in), M, H(S_out), ΔW, π, σ ⟩

A receiving node ingests H(R_road) as an external witness, anchoring the milestone into its
own local monoroot tree without adopting the sender's origin Q(0)_A, preserving Axiom 4 and
achieving O(1) bandwidth efficiency.

H4: Soundness of the π Field: Execution Proofs and the Bisection Audit Right [v1.8.0 —
    CANDIDATE]

Status. Proposed extension of the road-receipt conduit above, sourced from
docs/AiReviews/AiReviewQnIRv0.1.md (QO-13). It carries State: CANDIDATE under Axiom 14. The
receipt tuple R_road = ⟨ H(S_in), M, H(S_out), ΔW, π, σ ⟩ is unchanged; this subsection fixes

what the proof field π must contain for a road receipt to be admissible across the Outer Ring.

The review established a structural finding: contract-verification-plus-receipt-trust is not sound on its own against Byzantine or corrupted nodes. Because receiving nodes verify R_road against shared Q(anchor) contracts without re-executing the sender's transaction (§3.3, §8.4), a π that is merely a metadata attestation signed by σ would let a compromised node holding valid Inner/Outer-Ring PQC keys sign an arbitrary transition H(S_in) -> H(S_out) with a fabricated work ledger ΔW — a violation of dependencies-determinism across network boundaries that no downstream contract check can detect. Admissible road receipts therefore carry verifiable computational integrity in π, via exactly one of two realizations:

Realization A — Succinct Zero-Knowledge Execution Proof. The QnIR execution handler is a small, fixed-width deterministic machine over integer arithmetic, with bounded loops and no floating point (§6.3, §13). Its execution trace therefore arithmetizes directly into an Algebraic Intermediate Representation (AIR) inside a transparent zk-STARK proof (precedents: RISC Zero, Cairo receipts). The proof attests that a valid local witness trace, rooted at Q(0)_A, transformed H(S_in) into H(S_out) under contract M within work ΔW. Any receiver verifies in O(polylog N) time with zero trust in the sender and zero disclosure of the private local graph — preserving Axiom 8.

Realization B — Optimistic Interactive Anchor-Cut Bisection. For deployments where full STARK generation exceeds the thermodynamic breakeven (docs/qnPrimitive.md §1.15), π commits the Merkle root of the K intermediate OP_ANCHORCUT witnesses of the execution (docs/qm_calculus.md). Receivers hold a protocol-enforced Interactive Bisection Challenge Right: upon ingesting H(R_road), a receiver may challenge any segment [κ_i, κ_{i+1}] in O(log K) rounds, descending to a single origin-quotiented QnIR instruction step replayed deterministically at O(1) local cost. Failure to respond, or a hash mismatch, quarantines the receipt: terminal status QUARANTINED (§3.6) with receipt RECEIPT_ANCHOR_REVOKED — the receipt name is deliberately inherited from the anchor-fracture containment of docs/qlog.md; no new terminal status is introduced.

Origin-Quotiented Canonical Projection for Export. Strictly local origins imply that identical deterministic computations on nodes A and B yield disjoint commitments, H(S_out)_A ≠ H(S_out)_B: a receiving node can neither verify a state transition nor deduplicate shared sub-expressions across peers by direct hash comparison. The Exposure Functor therefore computes public commitments only after quotienting out the local root, Q(0)_A ↦ Q̂(0): during pure Qm evaluation the origin is treated as a formal canonical symbol, and the node-specific hardware seed remains bound solely in the envelope's lineage block. The public commitments H_pub(S_in) and H_pub(S_out) carried by R_road are computed over the quotiented projection, so any two nodes executing the same Q(pattern) over identical relative inputs produce identical portable commitments — making Anchor Grounding Class I transitions verifiable by forced re-derivation on any peer.

H2: 5. Complexity Degree Stratification

Artifacts and expressions (qexpr) are stratified by structural complexity-degree:

Degree	Structural Content	Examples	Specification Mapping
0	Foundational Primitives	Q(0), Q(1)	docs/q0_q1.md
1	Single Digits & Peano Cascade	Q(2) … Q(9), elementary R-algebra	docs/q2_q9.md, docs/qm.md
2	Multi-Axial Frames & CAE-DAGs	Orthogonal frames (d_k), Clifford rotors, CAE-DAGs	docs/qm_geometry.md, docs/qexpr.md
3	Discrete Dynamics & Calculus	Discrete Exterior Calculus, Continued Fractions	docs/qm_calculus.md
n	Advanced Domain Lattices	Categorical Semantics (Qs), Physics (Qphy), Logics (qlog), Chemistry (Qchem), Biology (qbio)	docs/qs.md, docs/qphy.md, docs/qlog.md, docs/qchem.md, docs/qbio.md

CD vs. Layer Note: Complexity Degrees index structural composition depth, not layer numbers. A domain projection may span the entire execution core (Layers 1 to 3) while its artifacts carry open-ended CD n labels; any artifact invoking discrete dynamics inherits CD 3+ semantics (docs/qm_calculus.md).

H2: 6. Numeric Representation and Compilation

H3: 6.1. Floating-Point Prohibition

Floating-point representations (IEEE 754) are prohibited at the Qn semantic layer. Floating-point semantics introduce platform-dependent rounding, unmeasured precision drift, and non-deterministic identity. Lower hardware layers may execute binary arithmetic internally, provided floating-point semantics are never exposed across a gateway.

H3: 6.2. Qexpr as a Content-Addressed Expression DAG (CAE-DAG)

Governed symbolic expressions (qexpr) maintain the explicit tuple:

Qexpr = ⟨ DAGStructure, z, r, d, DCC, Identity, Lineage ⟩

- Deduplication via Hash-Consing: Common sub-expressions share identical commitment hashes, resolving to a single node in memory.
- Zoom (z): Declared scale or precision boundary.
- Remainder (r): Exact residual quantity preserved at the declared zoom level.
- Dimension (d): Dimensional coordinate frame.

H3: 6.3. Collapse at Compilation Time

Qexpr expressions remain symbolic during authoring and collapse deterministically to concrete values at compilation time (qnlang -> qnir). Collapse operations must be total, finite, and accompanied by an explicit remainder receipt. In geometric workflows (docs/qm_geometry.md §7), collapse is deferred until boundary demand to minimize Landauer dissipation.

H3: 6.4. Bounded Symbolic Expressions

Symbolic expression graphs are strictly bounded. DCC profiles declare maximum node count, graph depth ceilings, and evaluation budgets. Exceeding bounds triggers an intermediate governed collapse or execution termination.

H2: 7. Arithmetic Foundation as Structural Example

H3: 7.1. First Operations

From Q(1), elementary operations are defined along axis d_1: additive, subtractive, multiplicative, and divisive morphisms (docs/qm.md §4).

H3: 7.2. Division as Constrained Operation

Division is defined as a bounded constraint equation:

Given a, b ⇒ find c, r such that a = (b × c) + r

- Denominators equal to Q(0) are forbidden by DCC constraint.
- Attempted division by zero emits the terminal receipt DIVISION_BY_ZERO_REJECTED.

H3: 7.3. Positional Ordering

Axis d_1 admits three disjoint partitions: negative Qn (Q^-), origin ({Q(0)}), and positive Qn (Q^+). Ordering is positional, not hierarchical. When zoom precision prevents order determination due to remainder overlap, the operation terminates with ORDERING_INDETERMINATE_AT_ZOOM. When remainders vanish, order demonstrates terminal-exactness.

H2: 8. Identity, Security, and Outer-Ring Consensus

H3: 8.1. Identity Classes

The architecture distinguishes at least three explicit Identity Classes:

Identity Class | Substrate Entity | Prohibition
Machine Identity | Runtime node, execution container, hardware device (Q(0)_N) | Must not imply legal personhood
Human Identity | Natural person, authorized operator | Must not execute without delegation
AI Agent Identity | Delegated autonomous or semi-autonomous process | Must not possess undelegated authority

Under the non-rogue-invariant, entities belonging to AI_AGENT_IDENTITY cannot instantiate independent genesis roots Q(0) and must execute strictly under cryptographically verified delegation chains originating from a HUMAN_IDENTITY.

H3: 8.2. Accountable Pseudonymity

The cdqn network forbids anonymous actors. All interactions are signed by accountable pseudonymous identities backed by capability proofs, cryptographic commitments, and explicit revocation paths.

H3: 8.3. Dual-Ring PQC Boundary

The architecture enforces an explicit dual-ring-pqc-boundary:
- The Inner Ring governs local execution and genesis (docs/q0_q1.md §2.4), protecting root keys, local lineage, and state sealing.
- The Outer Ring governs network-facing commitments under the local security-envelope, managing pseudonyms, attestation keys, and delegation proofs. Both rings implement structural

indirection to support non-disruptive migration across Post-Quantum Cryptography (PQC) standards.

Inner-Ring co-signature operations [v1.8.0 — CANDIDATE]. The Deterministic Canonical Round-Trip Paraphrase & Co-Signing Handshake of docs/qs.md §6.1 executes as an Inner-Ring signing operation: the deterministic read-back pair (DCC_INTENT_CONTRACT_v1, $\mathscr{T}$) is presented to the human principal and signed entirely inside the Inner Ring, and only the resulting co-signature commitment crosses the Outer Ring as a public attestation. The co-signature operation is metered under Axiom 6 and emits a receipt like any governed operation; it never exposes private lineage, root keys, or payload content to the Outer Ring.

H3: 8.4. Q(anchor) as the Outer-Ring Consensus Reference Frame

Because physical nodes possess heterogeneous silicon origins (Q(0)_A ≠ Q(0)_B) and disparate hardware capacities, global consensus cannot operate by flat transaction re-execution.

Consensus across the execution-core ceiling and its orthogonal projections (Layer 3 Qexpr, the base-domain projections, and Chronosa) is established over Universal Q(anchor) Contracts:
1. Topological Invariance: Every node instantiates the invariant Single-Digit Anchor Spectrum (Q(2) … Q(9), φ, √2) and Physical Invariant Anchors (Q(th_2), Q(c), Q(ℏ), Q(Z_bus), Q(gauge)). These anchors serve as universal topological landmarks.
2. Contract Verification: Nodes achieve consensus by verifying that road receipts satisfy the shared anchor contract (DCC_ANCHOR_Q2_PARITY_v1, DCC_ANCHOR_SECOND_LAW_v1, etc.).
3. Sheaf-Theoretic Gluing: In docs/chronosa.md, Chronosa verifies that road receipts emitted by heterogeneous nodes glue consistently over shared Q(anchor) landmarks, synthesizing global causal intelligence without centralized state storage.

Soundness caveat [v1.8.0 — CANDIDATE]. Points 1–3 verify that road receipts *satisfy* the shared anchor contracts; they do not by themselves verify that the recorded transitions *executed*. Per §4.4, admissible road receipts carry verifiable computational integrity in the proof field π — a zk-STARK execution proof or an Anchor-Cut bisection commitment. A road receipt whose π field is a bare σ-signed metadata attestation is not admissible for Outer-Ring consensus, however well its contract fields verify.

H2: 9. Structure-Generativity Balance

The architecture enforces the separation principle: Fixed Structure, Flexible Content.

- Structure: Non-negotiable layers, gateways, DCC profiles, metric envelopes, and causal lineage.
- Content: Heuristic proposals, neural network inferences, and stochastic candidate generation.

Stochastic engines may generate proposals, provided they are classified as *unverified candidates*. Unverified candidates cannot cross a simemp-gateway. They become governed Qn artifacts only after formal construction, constraint evaluation, and sealing.

H2: 10. External Precedents and Formal Convergence

The architecture aligns with established theoretical computer science and mathematical frameworks:
- Abstract Interpretation: Sound approximation across discrete abstraction lattices (Zoom z, Remainder r).
- Domain Theory: Bounded evaluation, fixed-point semantics, and totality by budget.
- Arithmetic Geometry & Height Bounds: Finite point density over bounded logarithmic heights (Northcott's Theorem), grounding Diophantine decidability within finite Metric Envelopes and eliminating classical unbounded non-halting voids (H10).
- Equality Saturation & E-Graphs: Content-addressed expression deduplication and reversible term rewriting.
- Sheaf Theory: Local category consistency and restricted global gluing via exposure functors.
- Linear Logic: Strict resource consumption without unmetered duplication or discarding.
- Neuro-Symbolic Verification: Strict operational isolation between neural candidate generation and symbolic verification.

H2: 11. Status of Qn Definition Files and Base Domains

Development separates foundational primitives from distributed networking:

H3: Category A — Numeric Primitives (Canonical Foundation)
- Primary Genesis Primitives: Local origin Q(0) and calibration of compute unit U along d_1 in Q(1) (docs/q0_q1.md).
- Secondary DCC Anchors: Elementary integer properties, parity partitions (q-even, q-odd), single-digit anchors Q(2) … Q(9), and emergent roots φ, √2 (docs/q2_q9.md).

H3: Category B — Operations and Morphisms (Canonical Primitive Envelope)
- Multi-dimensional axes (d_k), Clifford geometric algebra (Cl_{p,q}), and float-free rotors (docs/qm_geometry.md).
- Optimization morphisms: Content-addressed memory reuse (q-reuse) and identity short-circuiting (q-bypass) normatively integrated into the Universal Envelope (docs/qnPrimitive.md).

```
- Higher-order self-optimization workflows (Q(rsi)).


H3: Category C — Data Structures and Workflows
- Content-Addressed Expression DAGs (CAE-DAGs): hash-consed symbolic expression containers with
   explicit Universal Envelopes and bounded DCC profiles (docs/qexpr.md).
- Directed acyclic state graphs, immutable memory containers (Q(dataStruc)), and pattern
   matching (Q(patterns)).
- Composite state sequence workflows (Q(workflows)) carrying explicit Universal Envelopes and
   DCC constraints.


H3: Category D — Local-First Base Domains (Milestone 1 Sealed)
Base domains project directly across the execution core, executing 100% locally via local cdqn
data movement without network consensus:
- Qm (Quang Mathematics): Constructive proof engines, discrete calculus, and exact numeric
   proofs (docs/qm.md, docs/qm_calculus.md).
- Qs (Quang Semantics): Explicit knowledge graphs, categorical linguistic ontologies (DisCoCat),
   and formal intent contracts (docs/qs.md).
- Qphy (Quang Physics): Computational thermodynamics, discrete exterior field mechanics, and
   physical entropy accounting (docs/qphy.md).
- Emergent Composite Domains: Extended domain lattice formed via categorical tensor products:
   - qlog: Product category C(Qm) ⊗ C(Qs) governing constructive type theory, inferential modes,
     and pattern verification (docs/qlog.md).
   - qchem (Quang Chemistry): Product category C(Qphy) ⊗ C(Qm) governing discrete stoichiometric
     balance equations, molecular bonding graphs, and Gibbs free energy dissipation
     (docs/qchem.md).
   - qbio: Biosemiotic product category C(Qchem) ⊗ C(Qs) governing living metabolic homeostasis,
     genetic information channels, and active inference (docs/qbio.md).


H3: Category E — Runtime and Distributed Networking (Milestone 2 Launch)
- Intermediate representation (qnir) instruction set and bounded virtual execution handler
   (Pending).
- High-level authoring language (qnlang) syntax (Pending).
- cdqn distributed chaining, swarm aggregation, and public attestation protocol feeding
   chronosa.


H2: 12. License Alignment

This specification strictly conforms to the ssl:


H3: 12.1. Paternity Reference
All derivative distributions, compiled binaries, APIs, and runtime specification files must
retain the canonical attribution:

Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License
(SSL). Parent Repository: https://github.com/cdqn5249/cdqn


H3: 12.2. Open Core Invariants
1. Anti-Patent Defense: Commercial or derivative licenses terminate automatically upon
   initiating patent litigation against the Author or project ecosystem.
2. Non-Scaling Open Access: Royalty-free licensing is preserved for non-commercial, academic,
   and sub-threshold usage.


H3: 12.3. Scale Auditing
The metric and identity envelopes provide native, cryptographically verifiable telemetry (active
compute nodes, containers, agents, and monthly API transactions) to verify adherence to
commercial scaling thresholds (LICENSE.md §1.5).


H2: 13. Open Items

The following formal specifications remain open for subsequent releases:
1. Formal instruction set and operational semantics for qnir.
2. Syntax and type-checking rules for qnlang.
3. Concrete execution and validation mechanics for composite Q(workflows) and Q(rsi)
   self-optimization loops.
4. Arithmetization of the QnIR transition function into AIR and the concrete π-field
   realizations of §4.4 (proof-system parameters, challenge-response wire format), deferred to
   the QnIR v0.2 candidate.
```

Annex C — q0_q1.md

Source: docs/q0_q1.md *— 1.2.1, updated 2026-10-23. Rendered content per Annex Guide conventions N1–N6: 411 source lines, 471 rendered; 28,699 → 24,953 characters; source SHA-256* 48b37bcd

11f7b919 a5f68487 c4939eef 5d175255 02857a6f 391d0590 a26a2716. *Content fidelity verified mechanically at build time.*

```
[front matter] ----------------------------------------------------------
[front matter] title: Category A Primitives — Q(0), Q(1), and Node Onboarding
[front matter] version: 1.2.1
[front matter] updated: 2026-10-23
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Canonical 9-block Universal Envelope field anatomy, physical entropy
[front matter]   harvesting, PQC root initialization, Non-Rogue Invariant grounding,
[front matter]   and inductive single-digit genesis.
[front matter] terms_used: causal-arrow, cdqn, computational-consistency, compute-unit-u, dcc-
[front matter]   profile, dependencies-determinism, dimension-d, dual-ring-pqc-
[front matter]   boundary, grounding-class, historical-calibration-benchmark,
[front matter]   identity-class, intent-contract, layer-0, layer-1, local-first,
[front matter]   metric-exhaustion, no-implicit-rule, non-rogue-invariant, paternity-
[front matter]   reference, payload, q-anchor, q0, q1, qn, qs, qs-history-library,
[front matter]   receipt, remainder-r, security-envelope, simemp, simemp-gateway,
[front matter]   source-receipt-invariant, ssl, structural-indirection, successor-
[front matter]   morphism, terminal-exactness, universal-envelope, zoom-z
[front matter] ----------------------------------------------------------


H1: Category A Primitives — Q(0), Q(1), and Node Onboarding


Field | Specification
Document Title | Category A Primitives — Q(0), Q(1), and Node Onboarding
Version | 1.2.1
Last Updated | 2026-10-23 (Bao Loc, Vietnam)
Author | Christophe Duy Quang Nguyen
License | Scaling Source License (SSL) 1.0
Status | Canonical Primitive Specification — Category A (Layer 0 to Layer 1 Genesis)


H2: Normative References

The following documents establish the physical, structural, and legal constraints governing this
specification. If a technical conflict arises, simemp.md governs; if a structural conflict
arises, abstractionLayers.md governs; if a legal conflict arises, LICENSE.md governs. References
are deliberately version-unpinned: each target document's front matter carries its own
authoritative version, so this specification never duplicates drift-prone patch state.


Document | Role | Target
docs/simemp.md | Constitutional constraints, thermodynamics, and Dependencies Determinism |
simemp.html
docs/abstractionLayers.md | Layer architecture and SIMEMP Gateway validation |
abstractionLayers.html
docs/qnPrimitive.md | Universal Envelope, operational axioms, and Lifecycle States |
qnPrimitive.html
docs/qm.md | Constructive numeric and relational substrate | qm.html
docs/qs.md | Constructive relational intent and Intent Contracts | qs.html
LICENSE.md | Scaling Source License 1.0 governing the Licensed Work | LICENSE.md


H2: 1. Scope and Constitutional Anchoring

This specification defines the foundational genesis transition from physical substrate (layer-0)
to governed computation (layer-1).

In the qn computational universe, a number system cannot assume ungrounded, metaphysical
starting values. Under dependencies-determinism:

H(S_t \mid \mathcal{D}(S_t)) = 0

In accordance with the no-implicit-rule, every constant, origin, and unit must declare an
unbroken causal lineage back to physical hardware noise, certified across a simemp-gateway.

To maintain architectural agility under the simemp framework, this document defines minimal
necessary structural invariants rather than rigid implementation micro-choices. Operational
validity is evaluated provisionally via computational-consistency.

              THE CAUSAL ROOT OF FINITENESS (Layer 0 to Layer 1)
 +-------------------------------------------------------------------------+
 | LAYER 0: PHYSICAL SUBSTRATE (Root of Finiteness)                        |
 | Commodity Silicon · Ring Oscillators · Thermal Noise · Landauer Bound   |
 +-------------------------------------+-----------------------------------+
                                       | SIMEMP Gateway Transition
                                       v (Statistical Health & Conditioning)
 +-------------------------------------------------------------------------+
 | LAYER 1: LOCAL NODE GENESIS                                             |
 |   Q(0)_N : Local Genesis Origin (Birth Index 0, Empty Parents)          |
```

```
|   Q(1)   : First Unit Measure (Birth Index 1, Compute Quantum U, Axis d1)   |
|   Q(2)..Q(9) : Inductive Single-Digit Successors S(x) = x + Q(1)            |
+-----------------------------------------------------------------------------+
```

H2: 2. Layer 0 to Layer 1 Gateway: Node Onboarding Protocol

The simemp-gateway bridging Layer 0 and Layer 1 enforces physical-to-governed onboarding through a deterministic pipeline:

```
              NODE ONBOARDING STATE MACHINE
 [START]
    |
    v
 (Stage 1: Raw Entropy Harvesting) --► Fault --► [RECEIPT: ENTROPY_SOURCE_FAULT]
    |
    v
 (Stage 2: Statistical Health Tests) -► Fault --► [RECEIPT: ENTROPY_INSUFFICIENT]
    |
    v
 (Stage 3: Cryptographic Conditioning)
    |
    v
 (Stage 4: Inner-Ring PQC Keygen) ---► Fault --► [RECEIPT: PQC_KEYGEN_FAILURE]
    |
    v
 (Stage 5: Genesis Object Q(0) Assembly)
    |
    v
 (Stage 6: Sealing & Receipt Emission) --► Fault --► [RECEIPT: GENESIS_RECEIPT_FAILURE]
    |
    v
 [SUCCESS: NODE OPERATIONAL]
```

H3: 2.1. Stage 1: Physical Entropy Harvesting

The node interrogates native layer-0 entropy sources:
- Silicon thermal noise, ring oscillator jitter, or shot noise.
- System performance counters and bus timing jitter.

The harvested bitstream must satisfy a baseline sample volume ($N_{\mathrm{sample}} \ge 4096\text{ bits}$).

H3: 2.2. Stage 2: Statistical Health Tests

Raw entropy is verified against statistical health tests (e.g., NIST SP 800-90B standards):
- Repetition count validation to detect stuck-at hardware faults.
- Adaptive proportion validation to bound statistical drift.
- Min-entropy threshold validation ($H_{\mathrm{min}} \ge 0.85\text{ bits/bit}$).

Failure halts onboarding immediately, emitting a diagnostic receipt.

H3: 2.3. Stage 3: Cryptographic Conditioning

Harvested entropy is conditioned via a collision-resistant compression function initialized with execution context digests:

$$\mathcal{S}_{\mathrm{seed}} = \mathrm{KDF}\Big(\mathcal{B}_{\mathrm{raw}} \,\|\, \mathrm{Digest}(\text{HostContext})\Big)$$

producing a localized, non-invertible seed commitment.

H3: 2.4. Stage 4: Inner-Ring PQC Key Generation

Governed by the dual-ring-pqc-boundary (docs/abstractionLayers.md §8.3), the gateway parameterizes the local Inner Ring:
- Generates root keys (pk_node, sk_node) using a Post-Quantum Cryptography (PQC) signature scheme managed via structural-indirection.
- Private root key sk_node is retained exclusively within local memory, barred from public network exposure.

H3: 2.5. Stage 5: Genesis Object Assembly

The local genesis artifact Q(0) is constructed under the canonical 9-block universal-envelope.

H3: 2.6. Stage 6: Sealing and Receipt Emission

The gateway signs the initialization record, writes the node genesis receipt, and locks Q(0)

into an immutable SEALED lifecycle state.

H3: 2.7. Proposed Wiring: The Historical Calibration Benchmark (CANDIDATE)

This subsection adds no stage to the state machine above, which remains six-stage; it records the proposed wiring of a post-seal grounding event onto the gateway this section defines, closing the queue of docs/qs_history.md §8 item 1.

The glossary defines the Historical Calibration Benchmark as the mandatory empirical ingestion event wherein a physical node measures its local Layer 0 environmental baseline — ambient temperature, physical entropy, and atomic frequency standards — via signed hardware receipts. It is here proposed that the gateway treat this benchmark as the node's Class P grounding event: the physics rung of the proposed witness ladder (docs/qs_history.md §2), complementing the self-witnessed genesis of Stages 1 through 6. The event extends the §4.2 calibration of the compute unit — the grounding of abstract invariants into real thermodynamic work — from the declared bound alone to the measured local baseline, and it emits the signed calibration receipts by which a Q(anchor) of proposed Class P witnesses its epistemic grounding.

The wiring is deliberately non-mutating. The benchmark is proposed as a post-seal event, executed after Stage 6 emits the Node Genesis Receipt (§6.1) and repeatable along the local causal-arrow under a declared schedule or contract; a node that has not passed it operates without Class P grounding rather than with an implicit one, in accordance with the no-implicit-rule. Measurement tolerance envelopes are proposed to be declared inside the owning anchor's DCC contract rather than negotiated at runtime (docs/qs_history.md §2), so that cross-node consensus reduces to envelope satisfaction. Under the Source-Receipt Invariant (docs/qs.md §4.3), the benchmark and the proposed Qs History Library (docs/history_library.md §2, which queues this proposition as its §7 item 5) form the two post-seal ingestion siblings of the gateway: the benchmark ingests the node's measured physical present as Class P receipt material, while the library ingests receipted human history as proposed Class S records. Both are petitioned through signed receipts; neither is silently assumed.

This wiring is a CANDIDATE proposition under Axiom 14: it amends no canonical stage, redefines no existing receipt, and remains subject to deterministic measurement and BOC selection at implementation time.

H2: 3. Field Anatomy of Local Genesis Object Q(0)

Under Axiom 4 (docs/qnPrimitive.md), Q(0) is strictly local to each node:

$$\forall A \neq B \implies Q(0)_A \neq Q(0)_B$$

No universal global zero is admitted. Q(0) acts as the local-first coordinate origin, empty birth context, and root of the node's local causal-arrow.

CANONICAL 9-BLOCK ANATOMY OF GENESIS OBJECT Q(0)

```
+-------------------------------------------------------------------------+
| 1. IDENTITY BLOCK                                                       |
|    NodePseudonym      : Hash(pk_node)                                   |
|    LocalCausalIndex   : 0 (Fixed genesis baseline)                      |
|    IdentityClass      : MACHINE_IDENTITY (Accountable silicon chassis)  |
|    Version            : 1.2.0                                           |
|    CommitmentHash     : SHA3-256(BodyPayload)                           |
+-------------------------------------------------------------------------+
| 2. TYPE BLOCK                                                           |
|    ArtifactType       : 0x0001 (GENESIS_ORIGIN)                         |
|    DomainAffinity     : Qm_BASE_GENESIS                                 |
+-------------------------------------------------------------------------+
| 3. LIFECYCLE STATE BLOCK                                                |
|    State              : SEALED (Immutable root)                         |
+-------------------------------------------------------------------------+
| 4. LINEAGE BLOCK                                                        |
|    ParentCommitments  : [] (Identically Empty Set)                      |
|    BirthOrder         : 0                                               |
|    DerivationType     : HARDWARE_ENTROPY_SEED                           |
|    GenesisWitness     : Hash(S_seed)                                    |
+-------------------------------------------------------------------------+
| 5. DCC PROFILE REFERENCE                                                |
|    ProfileContract    : DCC_ROOT_GENESIS_v1 (Interface Contract)        |
|    Dependencies       : ∅ (None)                                        |
|    Constraints        : Immutable, Non-Exportable, Root-of-Causality    |
|    Capabilities       : DeriveSuccessor, AnchorLineage, EmitReceipt     |
+-------------------------------------------------------------------------+
| 6. METRIC ENVELOPE REFERENCE                                            |
|    MemoryAllocation   : Bounded byte footprint                          |
|    ConsumedMetric     : 0 (Genesis baseline)                            |
|    DepthCeiling       : 0                                               |
+-------------------------------------------------------------------------+
| 7. SECURITY ENVELOPE REFERENCE                                          |
|    SecurityRing       : INNER_RING_LOCAL                                |
|    PQCAlgorithmAnchor : Agile PQC Identifier                            |
|    RevocationPath     : REJECT_NODE_TEARDOWN                            |
+-------------------------------------------------------------------------+
```

```
| 8. EXPOSURE CLASS BLOCK                                                      |
|    Scope              : LOCAL_ONLY (Barred from direct network egress)      |
+------------------------------------------------------------------------------+
| 9. PAYLOAD REFERENCE                                                         |
|    PayloadProfile     : GENESIS_ORIGIN_PROFILE                               |
|    SubstrateMetadata  : { TelemetryDigest, EntropyCert }                     |
+------------------------------------------------------------------------------+
```

H3: 3.1. Machine Identity vs. User Sovereignty Disjunction

Conforming to docs/abstractionLayers.md (§8.1), Q(0) belongs exclusively to the Machine identity-class:

1. Prohibition of Personhood: Q(0) witnesses hardware silicon state and local execution context. It is constitutionally barred from claiming legal personhood or autonomous moral agency.
2. Separation from User Identity: Human actors reside in the sovereign user plane. User private credentials never enter Layer 0 entropy collectors.
3. Accountable Proxy: A node operates on behalf of an external principal only when presented with an explicit, signed capability grant (docs/qnPrimitive.md §5.7). The machine acts as an accountable executor, preserving statutory safe harbor under the ssl (LICENSE.md §6.2) for all sovereign Payloads.

H3: 3.2. The Non-Rogue Invariant and Intent Delegation

To prevent autonomous software processes from decoupling from human oversight, Q(0) anchors the non-rogue-invariant:

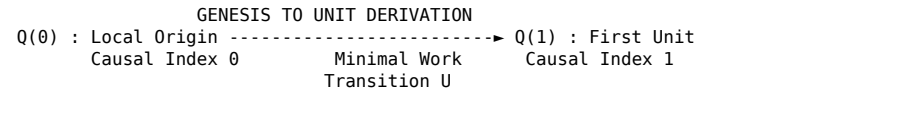
1. Zero Root Sovereignty for Delegated Agents: An entity carrying IdentityClass: AI_AGENT_IDENTITY cannot instantiate an independent root origin Q(0). It possesses no causal context outside of delegated execution.
2. Intent Delegation Boundary: Execution requests initiated by autonomous agents require an unbroken delegation proof linking to a verified human principal within the qs base domain:

$$\text{Admit}(\Delta S) \implies \exists \, \Pi_{\mathrm{delegation}} \, \wedge \, \Delta W \le W_{\mathrm{budget}}$$

3. Circuit Breaking: If an autonomous routine attempts an unanchored state transition or exceeds allocated budgets, the gateway halts execution deterministically, emitting an explicit failure receipt.

H2: 4. Field Anatomy and Calibration of First Unit Q(1)

Q(1) is the first derived unit artifact, the unity measure, and the baseline reference for the Abstract Compute Unit U along dimensional axis d1.



H3: 4.1. Causal Derivation

Under Axiom 5 (docs/qnPrimitive.md), Q(1) succeeds Q(0) monotonically along the local causal-arrow:

$$\mathrm{Parent}(Q(1)) = \{ Q(0) \}, \quad \mathrm{Index}(Q(1)) = 1$$

H3: 4.2. Physical Calibration of the Abstract Compute Unit U and the Thermodynamic Leash

The transition from Q(0) to Q(1) defines the Abstract Compute Unit U:

- Thermodynamic Grounding: U is calibrated to the governed computational work of an irreversible bit-transition at ambient temperature (T = 300\text{ K}), bounded by Landauer's dissipation limit dissipated via metric-exhaustion:

$$W(U) \ge k_B T \ln 2 \approx 2.87 \times 10^{-21} \text{ Joules}$$

- Platform Invariance: In software execution, U represents the baseline unit cost of a primitive discrete morphism, maintaining platform-invariant metric accounting across diverse hardware.
- The Thermodynamic Leash on Recursive Loops: Unbounded recursive execution loops are physically prohibited under Axiom 6. Every workflow operates within a finite, non-renewable work budget in units of Q(1). Metric depletion halts execution deterministically, preventing unmetered runaway compute.
- Invariant Unit Measure: While local entropy parameterizes the cryptographic identity of Q(1), its metric magnitude evaluates identically to q = 1 across all nodes.

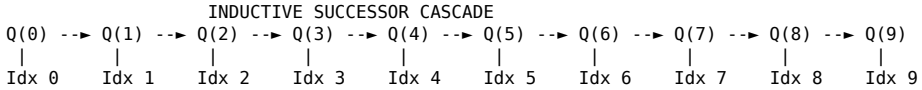
H3: 4.3. Field Structure of Q(1)

- Identity: Causal Index = 1, Parent = \mathrm{Commitment}(Q(0)).
- Type: UNIT_PRIMITIVE (Value = 1, Zoom z = 0, Remainder r = Q(0), Dimension d = d1).
- Morphism: Basis Vector \vec{u}_{d_1}: Q(0) \to Q(1).
- Security: Bound to the node's local security-envelope.

H2: 5. Inductive Single-Digit Genesis: Q(2) ... Q(9)

Single digits are constructed inductively along axis d_1 via the discrete Successor Morphism S:

\mathcal{S}: \mathcal{Q}_1 \to \mathcal{Q}_1, \quad \mathcal{S}(x) = x + Q(1)



H3: 5.1. Inductive Derivation Rules

For every integer k \in \{2, 3, 4, 5, 6, 7, 8, 9\}:

Q(k) = \mathcal{S}(Q(k-1)) = Q(k-1) + Q(1)

\mathrm{Parent}(Q(k)) = \{ Q(k-1), \, Q(1) \}, \quad \mathrm{Index}(Q(k)) = k

Each artifact carries the canonical \mathrm{Qm} numeric leaf tuple (docs/qm.md §3):

Q(k) \equiv \langle \oplus, \, k, \, z=0, \, r=Q(0), \, d_1 \rangle

H3: 5.2. Invariant Properties of Single Digits

1. terminal-exactness: Every single-digit artifact carries a remainder equal to Q(0) at zoom z=0, ensuring zero truncation error by construction.
2. Positional Causal Ordering: Strict trichotomy holds along axis d_1:

Q(0) \prec Q(1) \prec Q(2) \prec \dots \prec Q(9)

3. Base Neutrality: Decimal glyphs (0 \dots 9) serve as interface labels; internal representations are pure inductive successor compositions over unit Q(1).

H2: 6. Receipts, Lifecycle, and Hardware Maintenance

The Layer 0 to Layer 1 Gateway ensures deterministic lifecycle tracking during boot, routine operation, and hardware recovery.

H3: 6.1. The Node Genesis Receipt

Upon successful initialization, the gateway emits an immutable Node Genesis Receipt:

NODE GENESIS RECEIPT STRUCTURE

ReceiptIdentifier	: Hash(NodePseudonym || CausalIndex_0 || Timestamp)
ReceiptType	: RECEIPT_NODE_GENESIS
ExecutionStatus	: SUCCESS_SEALED
NodePseudonym	: Hash(pk_node)
SubstrateTelemetry	: Hash(HardwareConfigDigest || EntropyCert)
PQCAlgorithmAnchor	: Agile PQC Identifier
RootPublicKey	: pk_node (Inner-Ring Root Commitment)
EmittedArtifact	: Commitment(Q(0))
ConsumedBudget	: W_init (Platform initialization work)
GatewaySignature	: Sign_sk_node(ReceiptIdentifier || EmittedArtifact)

H3: 6.2. Physical Hardware Maintenance and Recovery Protocols

HARDWARE FAULT AND RESUMPTION MATRIX

PHYSICAL EVENT	GATEWAY MECHANISM AND EMITTED WITNESS
1. Routine Node Reboot	Non-volatile Sealed Q(0) Integrity Check +-► Emits: RECEIPT_NODE_RESUME
2. Abrupt Power Failure	Discard volatile unsealed candidates (Axiom 14) +-► Emits: RECEIPT_HARDWARE_CRASH_RECOVERY
3. Component Migration (Hardware Swap)	Graft old DAG to new Q(0)_new via Axiom 13 +-► Emits: RECEIPT_NODE_MIGRATION
4. Physical Destruction	Local state lost; Outer-Ring Sheaf survives

```
 +--------------------------+-------------------------------------------------+

- Reboot: Validates sealed non-volatile state and resumes monotonically
   (\mathrm{Index}(\mathrm{next}) = \mathrm{Index}(\mathrm{last}) + 1).
- Crash Containment: Volatile unsealed candidate states are discarded under Axiom 14, restoring
   to the last SEALED checkpoint without entropy drift (H = 0).
- Component Swap: Replaced hardware generates new origin Q(0) and grafts historic lineage under
   Axiom 13 via an explicit migration receipt.
- Physical Loss: While unexported local memory is bounded by silicon permanence, public
   attestations previously projected across the Outer Ring persist across the cdqn network.


H3: 6.3. Proposed Duality: The Node Genesis Receipt and the Intent Contract (CANDIDATE)

This subsection names no new artifact and adds no stage to the state machine of §2; it records
the proposed naming of a duality already exercised across the canon, closing the queue of
docs/qs_history.md §8 item 4. Every governed node lives under two signed disciplines pointing in
opposite directions along the causal-arrow: the node genesis receipt of §6.1 is
backward-looking, sealing what the node is; the Intent Contract anchored in §3.2 and elaborated
as docs/qs.md §5 is forward-looking, signing what the node may become.

The genesis receipt closes the past: it binds the immutable record of the node's origin — the
committed Q(0), the consumed initialization work W_{\mathrm{init}}, the substrate telemetry
digest — under the gateway signature, so that every later transition reads its provenance from a
sealed state rather than from memory or assertion. The intent contract opens the future:
parameterized inside the owning DCC Profile, it attaches to every admitted delegated transition
the unbroken delegation proof that §3.2 requires, carrying mutual human-agent commitments,
capability scopes, and conserved social residuals. The two disciplines are complementary and
mutually necessary: sealing without delegation is inert provenance, while delegation without
sealing is unanchored intent in violation of the no-implicit-rule; together they bind the local
causal arrow at both ends, the receipt anchoring its tail and the intent contract signing its
head.

Under the Source-Receipt Invariant (docs/qs.md §4.3), both halves are receipt-typed artifacts
differing in temporal direction rather than in evidentiary kind — the genesis receipt
terminating in Layer 0 hardware and entropy receipts, the intent contract being an authored
declaration that is never extracted by inspection, per the binding caution of the invariant.
This duality is a CANDIDATE proposition under Axiom 14: it coins a name for complementary roles
the canon already exercises, amends no canonical stage, redefines no existing receipt, and
remains subject to deterministic measurement and BOC selection at implementation time.


H2: 7. License and Lineage Alignment

This specification enforces the legal and operational conditions of the ssl:

1. paternity-reference: All derivative implementations or runtime modules must preserve:
Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License
(SSL). Parent Repository: https://github.com/cdqn5249/cdqn
2. Open Core Protection: Layer 0 to Layer 1 onboarding mechanisms and genesis structures remain
   royalty-free for sub-threshold deployments under Non-Scaling Open Access.
3. Statutory Safe Harbor: The onboarding gateway functions as neutral technical infrastructure,
   exercising no editorial control over, and making no claim upon, external applications or
   sovereign Payloads.
```

Annex D — q2_q9.md

Source: docs/q2_q9.md *— 1.1.3, updated 2026-10-20. Rendered content per Annex Guide conventions N1–N6: 392 source lines, 474 rendered; 27,434 → 23,726 characters; source SHA-256* abbe4e6c 9986d966 a6184599 79c79659 4a453f62 2df4514b f4957d96 55a0385a*. Content fidelity verified mechanically at build time.*

```
[front matter] ------------------------------------------------------------
[front matter] title: Category A Primitives — Single-Digit Integers Q(2)...Q(9) and Secondary
[front matter]   DCC Sources
[front matter] version: 1.1.3
[front matter] updated: 2026-10-20
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Canonical specification of single-digit integers Q(2) through Q(9),
[front matter]   the complete single-digit anchor spectrum, emergent algebraic
[front matter]   constants, constructive parity partitions, and dimensional ring
[front matter]   asymmetries under SIMEMP constraints.
[front matter] terms_used: boc-policy, causal-arrow, complexity-degree, compute-unit-u,
[front matter]   dependencies-determinism, dimension-d, identity-class, layer-1,
[front matter]   licensed-work, metric-envelope, metric-exhaustion, no-implicit-rule,
[front matter]   paternity-reference, payload, q-anchor, q-even, q-odd, q0, q1, qm,
[front matter]   qn, remainder-r, security-envelope, simemp, ssl, structural-
[front matter]   indirection, successor-morphism, terminal-exactness, universal-
```

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[front matter]   envelope, zoom-z
[front matter] ------------------------------------------------------------
```

H1: Category A Primitives — Single-Digit Integers Q(2)...Q(9) and Secondary DCC Sources

Field	Specification
Document Title	Category A Primitives — Single-Digit Integers Q(2)...Q(9) and Secondary DCC Sources
Version	1.1.3
Last Updated	2026-10-20 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Canonical Primitive Specification — Category A Completion (Complexity Degree 1)

H2: Normative References

The following documents establish the physical, structural, and legal constraints governing this specification. If a technical conflict arises, simemp.md governs; if a structural conflict arises, abstractionLayers.md governs; if a legal conflict arises, LICENSE.md governs. References are deliberately version-unpinned: each target document's front matter carries its own authoritative version, so this specification never duplicates drift-prone patch state.

Document	Role	Target
docs/simemp.md	Constitutional constraints, thermodynamics, and Dependencies Determinism	simemp.html
docs/abstractionLayers.md	Layer architecture and SIMEMP Gateway validation	abstractionLayers.html
docs/qnPrimitive.md	Universal Envelope, operational axioms, and Optimization Morphisms	qnPrimitive.html
docs/q0_q1.md	Primary genesis origin Q(0) and first unit Q(1)	q0_q1.html
docs/qm.md	Constructive numeric leaf substrate and Diophantine division constraints	qm.html
docs/qm_geometry.md	Multi-axial frames, Clifford geometric algebra, and float-free rotations	qm_geometry.html
docs/qm_calculus.md	Discrete Exterior Calculus, continued fractions, and invariant dynamics	qm_calculus.html
docs/qexpr.md	Layer 3 Content-Addressed Expression DAG container	qexpr.html
docs/qs.md	Constructive relational intent and Intent Contracts	qs.html
LICENSE.md	Scaling Source License 1.0 governing the Licensed Work	LICENSE.md

H2: 1. Epistemic Scope and the Secondary DCC Role

This specification completes Category A (Numeric Primitives) of docs/abstractionLayers.md (§11) by formalizing single-digit integers Q(2) through Q(9).

In the qn computational universe, numbers are not passive scalar magnitudes; they are governed computational artifacts carrying explicit capability contracts governed under the boc-policy and constitutional constraints of the simemp framework. While Q(0) and Q(1) establish the Primary DCC Sources (origin, context, unit measure, and baseline reference for the Abstract Compute Unit U), they cannot single-handedly supply the specialized structural constraints required by higher abstraction layers.

Single digits Q(2) \dots Q(9) function as the Secondary DCC Sources of the stack:
- They introduce parity partitioning (Q(even), Q(odd)), prime basis generators, aperiodicity bounds, composite factorizations, and physical memory alignment contracts.
- Higher-layer algorithms, geometric solvers (docs/qm_geometry.md), and calculus manifolds (docs/qm_calculus.md) declare explicit dependencies on the capability profiles of these secondary sources.

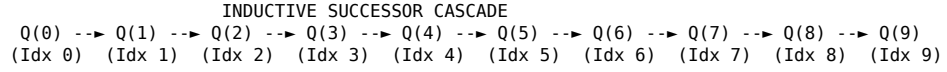
Under dependencies-determinism, the derivation of every single-digit integer possesses zero conditional entropy given its declared causal predecessor:

$$H(Q(k) \mid Q(k-1), \, Q(1)) = 0$$

H2: 2. Inductive Successor Cascade Along Axis d_1

In strict adherence to the no-implicit-rule, single-digit integers cannot be assumed as unmetered continuous constants. They are constructed constructively along the canonical genesis axis d1 via the discrete Successor Morphism S (docs/q0_q1.md §5):

$$\mathcal{S}: \mathcal{Q}_1 \to \mathcal{Q}_1, \quad \mathcal{S}(x) = x + Q(1)$$



H3: 2.1. Inductive Derivation Rules

For every integer k \in \{2, 3, 4, 5, 6, 7, 8, 9\} along the causal-arrow:

$$Q(k) = \mathcal{S}(Q(k-1)) = Q(k-1) + Q(1)$$

$$\mathrm{Parent}(Q(k)) = \{ Q(k-1), \, Q(1) \}$$

$$\mathrm{Index}(Q(k)) = k$$

Each single digit evaluates to an exact canonical qm leaf tuple (docs/qm.md §3) at zoom z = 0 and conserved remainder r = Q(0):

$$Q(k) \equiv \langle \oplus, \, k, \, z=0, \, r=Q(0), \, d_1 \rangle$$

H3: 2.2. Invariant Single-Digit Properties

1. terminal-exactness: Every single-digit artifact carries a remainder identically equal to Q(0) at zoom z = 0. Truncation error is zero by construction.
2. Positional Causal Ordering: Monotonic ordering is strictly preserved along axis d_1:

$$Q(0) \prec Q(1) \prec Q(2) \prec Q(3) \prec Q(4) \prec Q(5) \prec Q(6) \prec Q(7) \prec Q(8) \prec Q(9)$$

3. Radix Neutrality: While labeled using standard decimal glyphs (0 \dots 9) for interface parsing, their internal representation is pure unary successor composition over unit Q(1) along axis d_1.

H3: 2.3. Causal Genesis vs. Emergent Algebraic Invariant Verification

To maintain causal acyclicity across abstraction layers (docs/simemp.md §1), an epistemic boundary separates Causal Genesis from Algebraic Invariant Verification:

```
+-------------------------------------------------------------------------+
| LAYER 1: Causal Genesis (Constructive Birth)                            |
| Q(k) instantiated solely via Peano induction: Q(k) = Q(k-1) + Q(1)      |
| Lineage dependencies: strictly { Q(k-1), Q(1) } along axis d1.          |
+-------------------------------------------------------------------------+
| LAYER 2+: Algebraic Verification (Emergent DCC Invariant Contracts)     |
| Composite factorization (Q(4) = Q(2) × Q(2)) and multi-axial root relations |
| (DCC_ANCHOR_SQRT2_v1) verified only upon activation of Layer 2 morphisms.   |
+-------------------------------------------------------------------------+
```

1. Layer 1 Genesis: The existence, identity, and causal birth of Q(k) depend uniquely on its immediate predecessor Q(k-1) and the unit Q(1). No higher-order algebraic morphism (\times, \div) or multi-axial coordinate system (d_j, j \ge 2) is required for Q(k) to exist.
2. Layer 2 Verification: Properties such as primality, compositeness (e.g., Q(4) \equiv Q(2) \times Q(2)), and parity partitioning are abstract capability contracts declared within the artifact's DCC profile. Under structural-indirection, verification of these contracts is deferred until the computational context traverses the gateway into Layer 2 (\mathcal{R}-Algebra) or Layer 3 (Geometric Frames).

H2: 3. Concrete Anatomy of Single Digits Q(2) ... Q(9)

Each single digit carries an explicit universal-envelope, a unique local causal index, an immutable SEALED lifecycle state, belongs to the Machine identity-class, is governed by the local security-envelope, and encapsulates a specialized DCC contract:

Digit Artifact | Causal Index | Mathematical Role | Primary DCC Contract | Algebraic Status
Q(2) | 2 | Parity Generator & Binary Modulus | DCC_ANCHOR_Q2_PARITY_v1 | Prime Basis Generator
Q(3) | 3 | Minimal 2D Simplex & Spanning Basis | DCC_ANCHOR_Q3_SIMPLEX_v1 | Prime Basis Generator
Q(4) | 4 | First Composite & Quadrant Source | DCC_ANCHOR_Q4_QUADRANT_v1 | Composite Contract: Q(2) \times Q(2)
Q(5) | 5 | Aperiodicity Generator & Pentagonal Modulus | DCC_ANCHOR_Q5_APERIODICITY_v1 | Prime Basis Generator
Q(6) | 6 | Minimal Factorial (3!) & Perfect Modulus | DCC_ANCHOR_Q6_SYMMETRY_v1 | Composite Contract: Q(2) \times Q(3)
Q(7) | 7 | Cyclic Modulus & Projective Plane Order | DCC_ANCHOR_Q7_PROJECTIVE_v1 | Prime Basis Generator
Q(8) | 8 | Silicon Memory Word (8-bit Octet) | DCC_ANCHOR_Q8_OCTET_v1 | Composite Contract: Q(2)^3
Q(9) | 9 | Decimal Radix Limit & Square Modulus | DCC_ANCHOR_Q9_RADIX_v1 | Composite Contract: Q(3)^2

H2: 4. The Complete Single-Digit Anchor Spectrum (Q(2) ... Q(9))

Every single digit from Q(2) through Q(9) possesses high topological centrality within the system's causal dependency DAG, designating each as a canonical Q(anchor) entity (docs/_data/glossary/architecture.yml):

$$\chi(\alpha) = \frac{\lvert \mathcal{C}^+(\alpha) \rvert}{\lvert \mathcal{V} \rvert} \ge \Theta_{\chi}$$

COMPLETE SINGLE-DIGIT ANCHOR SPECTRUM

ARTIFACT	ANCHOR CONTRACT DESIGNATION	STRUCTURAL DEPENDENCY CONE C+(α)
Q(2)	DCC_ANCHOR_Q2_PARITY_v1 (Parity & Binary Modulus)	Parity logic (even/odd), binary tree branching, Boolean NOT/XOR gates
Q(3)	DCC_ANCHOR_Q3_SIMPLEX_v1 (Simplicial Geometry Basis)	2D triangular simplicial meshes, Minimal spanning intent basis
Q(4)	DCC_ANCHOR_Q4_QUADRANT_v1 (Composite Orthotope)	4 Cartesian quadrants, composite factorizations, 2D hypercube faces
Q(5)	DCC_ANCHOR_Q5_APERIODICITY_v1 (Aperiodic & Fractal Order)	5-fold quasicrystals, Penrose tiles, Golden Ratio φ scaling invariants
Q(6)	DCC_ANCHOR_Q6_SYMMETRY_v1 (Minimal Non-Abelian Gauge)	Non-abelian permutation group S_3, smallest perfect number: 1+2+3=1·2·3
Q(7)	DCC_ANCHOR_Q7_PROJECTIVE_v1 (Projective Geometry Basis)	Fano plane order PG(2, 2), octonion imaginary basis algebra (e_1...e_7)
Q(8)	DCC_ANCHOR_Q8_OCTET_v1 (Silicon Memory Alignment)	Physical memory byte word (8-bit), Layer 0/1 bus word segmentation
Q(9)	DCC_ANCHOR_Q9_RADIX_v1 (Radix-10 Single-Digit Max)	Single-digit base-10 ceiling, decimal normalization boundaries

H3: 4.1. Abstract Contract Decoupling

Under structural-indirection (docs/simemp.md §4.3), downstream algorithms depend exclusively on the abstract invariant contract of the anchor:

- Q(2) Anchor: Enforces the Diophantine contract $X = (Q(2) \times c) + r$. Downstream solvers depend on parity trichotomy, regardless of whether hardware executes division via bitwise shifts or Diophantine subtraction.
- Q(3) Anchor: Enforces the minimal 2D non-collinear boundary (DCC_ANCHOR_Q3_SIMPLEX_v1), acting as the foundation for simplicial complexes in Discrete Exterior Calculus (docs/qm_calculus.md) and the triadic intent foundation ($\mathrm{Qs} \otimes \mathrm{Qm} \otimes \mathrm{Qphy}$).
- Q(5) Aperiodicity Anchor: Enforces non-crystallographic 5-fold rotational symmetry, preventing periodic synchronization locks in distributed swarms.

H3: 4.2. Dimensional Ring Behavior of the Prime Anchors

When coordinate dimensions extend from 1D scalar arithmetic into multi-axial geometric frames ($\mathcal{C}\ell_{p,q}$), the prime single digits $\mathcal{P} = \{ Q(2), Q(3), Q(5), Q(7) \}$ partition into three fundamental algebraic number ring behaviors:

1. Ramification (Q(2)): In Gaussian integer rings ($\mathbb{Z}[i]$ embedded in 2D frames $\vec{e}_1 \wedge \vec{e}_2$), the prime 2 ramifies onto itself:
   $2 = -i (1 + i)^2$
   This quadratic self-branching governs parity logic, bitwise splitting, and binary tree bifurcation.
2. Splitting (Q(5)): Primes of the form $p \equiv 1 \pmod 4$ split into orthogonal linear factors:
   $5 = (2 + i)(2 - i)$
   In 1D, 5 is indecomposable; in 2D, it factorizes across orthogonal axes, providing the metric scaling engine for self-similar fractal expansion.
3. Inertia (Q(3), Q(7)): Primes of the form $p \equiv 3 \pmod 4$ remain strictly indecomposable in Gaussian geometric rings:
   $3 \in \mathbb{P}_{\mathbb{Z}[i]}, \quad 7 \in \mathbb{P}_{\mathbb{Z}[i]}$
   This dimensional stability establishes rigid topological invariance for minimal simplicial meshes (Q(3)) and projective planes (Q(7)).

H3: 4.3. The 5D-to-2D Projection Origin of Aperiodicity

In Euclidean 2D and 3D space, the Crystallographic Restriction Theorem forbids periodic 5-fold rotational symmetry. However, 5-fold symmetry is strictly periodic in 5-dimensional hypercubic lattices ($\mathbb{Z}^5$).

Under DCC_ANCHOR_Q5_APERIODICITY_v1, the aperiodicity of Q(5) is formalized through the Cut-and-Project Method:

- A periodic 5D lattice is sliced at an irrational slope governed by the Golden Ratio ϕ (DCC_ANCHOR_PHI_v1).
- Projecting this slice onto a 2D frame yields a Penrose quasicrystal—a structure that is non-periodic, non-repeating, yet completely deterministic (H = 0).
- This dimensional projection ensures that dynamical state trajectories avoid circular limit cycles across memory space.

H2: 5. Emergent Algebraic Constant Anchors: ϕ and √2

Two primary algebraic constants emerge directly from the polynomial characteristics of the single digits and are anchored within Category A:

Constant Anchor | Generating Digit | Contract Designation | Defining Algebraic Constraint
Golden Ratio (\phi) | Q(5) | DCC_ANCHOR_PHI_v1 | \phi^2 - \phi - Q(1) = Q(0)
Minimal Metric Root (\sqrt{2}) | Q(2) | DCC_ANCHOR_SQRT2_v1 | D^2 - Q(2) = Q(0)

H3: 5.1. The Golden Ratio (\phi / DCC_ANCHOR_PHI_v1)

Emerges from the pentagonal aperiodicity anchor Q(5):

\phi = \frac{Q(1) + \sqrt{Q(5)}}{Q(2)}

In accordance with the crystallographic restriction theorem, 5-fold rotational symmetry cannot tile Euclidean 2D/3D space periodically. Q(5) and its emergent constant \phi mathematically govern aperiodic self-similar order (quasicrystals and Penrose lattices), providing the metric scaling invariant for scale-free fractal graphs in the qn universe.

H3: 5.2. The Minimal Metric Root (\sqrt{2} / DCC_ANCHOR_SQRT2_v1)

Emerges algebraically from Q(2) as the non-zero root of the Diophantine quadratic equation:

D^2 - Q(2) = Q(0)

In qm, \sqrt{2} does not exist as an unmetered, infinite continuous decimal. It is anchored as an exact constrained Diophantine quadratic relation:

Q(2) = (q_D \cdot \delta_z)^2 + r_D

whose residual r_D is strictly preserved in the metric-envelope across zoom scales.

Geometric Realization: While DCC_ANCHOR_SQRT2_v1 is anchored within Category A as a discrete quadratic constraint, its geometric interpretation as the hypotenuse of the orthogonal unit square:

Q(1)_{d_1}^2 + Q(1)_{d_2}^2 = Q(2)

is formally instantiated in Layer 3 (docs/qm_geometry.md §6.1) once multi-axial frames (d_2) are derived.

H2: 6. Constructive Parity Partitions: Q(even) and Q(odd)

Parity is not an unmeasured flag. In \mathrm{Qm}, parity is the constructive remainder classification resulting from the bounded Diophantine division constraint (docs/qm.md §5.3) with divisor B = Q(2):

\mathrm{Given} \; X \implies X = (Q(2) \times c) + r, \quad \mathrm{where} \; r \in \{ Q(0), \, Q(1) \}

Parity Partition | Remainder Condition | Canonical Classification | Invariant Property
Q(even) | r \equiv Q(0) | Modulo-2 Terminal Exactness | \forall x \in \mathcal{Q}_{\mathrm{even}}, \, x = Q(2) \times c
Q(odd) | r \equiv Q(1) | Modulo-2 Unit Remainder | \forall x \in \mathcal{Q}_{\mathrm{odd}}, \, x = (Q(2) \times c) + Q(1)

H3: 6.1. Classification of Genesis Origin Q(0)

Under the Diophantine division constraint:

Q(0) = (Q(2) \times Q(0)) + Q(0)

Because the remainder is identically Q(0), genesis origin Q(0) is constructively an even artifact demonstrating modulo-2 terminal-exactness:

Q(0) \in \mathcal{Q}_{\mathrm{even}}

H3: 6.2. Disjoint Partitioning of the Integer Lattice

Parity partitions divide the discrete 1D lattice into two strictly disjoint subsets:

\mathcal{Q}_1 = \mathcal{Q}_{\mathrm{even}} \;\cup\; \mathcal{Q}_{\mathrm{odd}}, \quad \mathrm{where} \; \mathcal{Q}_{\mathrm{even}} \;\cap\; \mathcal{Q}_{\mathrm{odd}} = \emptyset

H2: 7. Algebraic Asymmetry: Positive Monoid vs. Negative Domain

While spatial reflection along axis d_1 is an algebraic involution (\mathcal{I}_{d_1}^2 = \mathrm{id}), the behavior of qn under multiplicative and dynamical operations is fundamentally asymmetric between the positive cone (\mathcal{Q}^+) and the negative cone (\mathcal{Q}^-).

POSITIVE VS. NEGATIVE ASYMMETRY

POSITIVE CONE (Q+)	NEGATIVE CONE (Q-)
Free Commutative Multiplicative Monoid generated by Primes.	Non-closed under multiplication: ⊖ ⊗ ⊖ = ⊕ (Leaves negative domain)
Closed: Q+ × Q+ ⊆ Q+	Open: Q- × Q- ⊆ Q+
Single Attractor Dynamics (Collatz)	Multi-Cycle Attractor Branching
Metric Work Footprint: +W	Metric Work Footprint: +W (Positive)

H3: 7.1. Multiplicative Monoid Closure Asymmetry

The positive single-digit integers form the basis of a free commutative monoid under multiplication, generated uniquely by the indecomposable prime set \mathcal{P} = \{ Q(2), Q(3), Q(5), Q(7) \}:

\forall a, b \in \mathcal{Q}^+, \quad a \times b \in \mathcal{Q}^+

Conversely, the negative integers \mathcal{Q}^- are not closed under multiplication:

\forall a, b \in \mathcal{Q}^-, \quad a \times b \in \mathcal{Q}^+ \quad (\mathrm{Parity} \; \mathrm{Rule} \colon \ominus \otimes \ominus = \oplus)

Multiplication within \mathcal{Q}^- immediately exits the negative domain. Negative numbers do not form an independent multiplicative structure; they require the explicit inclusion of the reflection unit \langle \ominus, 1 \rangle.

H3: 7.2. Non-Linear Dynamical Asymmetry (The Collatz Phenomenon)

For any non-linear operator \mathcal{F}, spatial reflection does not generally commute with dynamical evolution:

\mathcal{F}(\mathcal{I}(X)) \neq \mathcal{I}(\mathcal{F}(X))

This is demonstrated constructively by the discrete Collatz mapping T(n):

$$T(n) = \begin{cases}
n / Q(2) & \mathrm{if} \; n \in \mathcal{Q}_{\mathrm{even}} \\
(Q(3) \times n) + Q(1) & \mathrm{if} \; n \in \mathcal{Q}_{\mathrm{odd}}
\end{cases}$$

- On Positive Domain (\mathcal{Q}^+): The +Q(1) step acts as a damping perturbation against Q(3) \times n, driving all tested orbits into the unique global attractor cycle: (Q(4) \to Q(2) \to Q(1)).
- On Negative Domain (\mathcal{Q}^-): Substituting negative integers (n = -m where m \in \mathcal{Q}^+) reverses the direction of the unit perturbation:

  Q(3)(-m) + Q(1) = -(Q(3)m - Q(1))

  The unit perturbation acts anti-symmetrically, fracturing the dynamical trajectory into at least three distinct non-trivial limit cycles (the 2-cycle, the 5-cycle, and the 18-cycle).

H3: 7.3. Thermodynamic Metric Asymmetry and Fundamental Units

In physical Layer 0 substrates, negative energy and negative storage do not exist:
- A negative artifact \langle \ominus, k, z=0, r=Q(0), d_1 \rangle has an identical positive physical footprint as its positive reflection \langle \oplus, k \rangle.
- In accordance with Landauer's bound, executing a negative transition dissipates positive thermodynamic work via metric-exhaustion:

  W(\ominus) \ge k_B T \ln 2 \gt 0

  An inverse calculation cannot recover dissipated physical heat.

Furthermore, in multi-dimensional algebraic number rings (\mathbb{Z}[\phi], \mathbb{Z}[\sqrt{2}]), the Dirichlet Unit Theorem expands the unit group beyond the 1D reflection pair \{\pm 1\} to infinite cyclic groups generated by fundamental units (\epsilon = 1+\sqrt{2}, \phi = \frac{1+\sqrt{5}}{2}). This establishes discrete dilation symmetries across scale lattices, enabling self-similar geometric scaling without floating-point approximations.

H2: 8. Sealing, Receipts, and Lifecycle Integration

```
Upon completion of the inductive cascade from Q(2) to Q(9), the node emits an immutable
Primitive Sequence Sealing Receipt:


Receipt Field | Specification
Receipt Identifier | Hash(NodePseudonym \|\| CausalIndex_9 \|\| Timestamp)
Receipt Type | RECEIPT_CATEGORY_A_SEALED
Execution Status | SUCCESS_SEALED
Digit Range | Q(2) \dots Q(9)
Anchor Contracts Sealed | DCC_ANCHOR_Q2_PARITY_v1 through DCC_ANCHOR_Q9_RADIX_v1
Emergent Constants Anchored | DCC_ANCHOR_PHI_v1 (\phi), DCC_ANCHOR_SQRT2_v1 (\sqrt{2})
Consumed Work Budget | Bounded inductive initialization work (8 \, Q(1) units) declared in the
local metric-envelope
Gateway Signature | \mathrm{Sign}\big(\text{ReceiptIdentifier} \,\Vert\,
\text{Commitment}(Q(9))\big) keyed by signing key sk_node


This receipt permanently closes Category A for the physical node, proving that all single-digit
secondary DCC sources and emergent algebraic roots are grounded, sealed, and ready for Layer 2
algebraic composition.


H2: 9. License and Lineage Alignment

This primitive specification enforces the legal and operational conditions of the ssl governing
the licensed-work:

1. paternity-reference: All derivative implementations, mathematical kernels, integer libraries,
   or hardware arithmetic modules derived from this document must embed:
Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License
(SSL). Parent Repository: https://github.com/cdqn5249/cdqn
2. Open Core Protection: The secondary DCC anchor contracts, parity generators, single-digit
   primitive definitions, and emergent algebraic constants remain royalty-free for
   sub-threshold, academic, and non-commercial deployments under Non-Scaling Open Access.
3. Statutory Safe Harbor: The single-digit integer substrate operates as passive, neutral
   mathematical infrastructure. It exercises no editorial inspection over, and makes no claim
   upon, external applications or sovereign Payloads.
```

Annex E — qm.md

Source: docs/qm.md *— 1.3.0, updated 2026-10-26. Rendered content per Annex Guide conventions N1–N6: 644 source lines, 965 rendered; 57,226 → 52,112 characters; source SHA-256* 07f5c31d 0265fa00 e62f463d cc2c6d42 35d90ada ffdf1f37 951b9c1d 2acd5d37*. Content fidelity verified mechanically at build time.*

```
[front matter] ------------------------------------------------------------
[front matter] title: Qm — Quang Mathematics (Constructive Numeric and Relational Substrate)
[front matter] version: 1.3.0
[front matter] updated: 2026-10-26
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Canonical specification of the constructive numeric, relational, and
[front matter]   algebraic domain substrate of the Qn universe under SIMEMP
[front matter]   constraints.
[front matter] terms_used: abstraction-layer, axiom-6, causal-arrow, computational-consistency,
[front matter]   compute-unit-u, dcc-profile, dependencies-determinism, dimension-d,
[front matter]   layer-1, licensed-work, local-first, metric-envelope, metric-
[front matter]   exhaustion, no-implicit-rule, paternity-reference, payload, q-anchor,
[front matter]   q-bypass, q-reuse, q0, q1, qexpr, qm, qn, qnir, qnlang, receipt,
[front matter]   remainder-r, simemp, simemp-gateway, ssl, structural-indirection,
[front matter]   terminal-exactness, universal-envelope, zoom-z
[front matter] ------------------------------------------------------------


H1: Qm — Quang Mathematics: Constructive Numeric and Relational Substrate


Field | Specification
Document Title | Qm — Quang Mathematics: Constructive Numeric and Relational Substrate
Version | 1.3.0
Last Updated | 2026-10-26 (Bao Loc, Vietnam)
Author | Christophe Duy Quang Nguyen
License | Scaling Source License (SSL) 1.0
Status | Canonical Domain Specification — Category D (Local-First Base Domain)


H2: Normative References

The following documents establish the physical, structural, and legal constraints governing
```

\mathrm{Qm}. If a technical conflict arises, simemp.md governs; if a structural conflict arises, abstractionLayers.md governs; if a legal conflict arises, LICENSE.md governs. References are deliberately version-unpinned: each target document's front matter carries its own authoritative version, so this specification never duplicates drift-prone patch state.

Document | Role | Target
docs/simemp.md | Constitutional constraints, thermodynamics, and Dependencies Determinism | simemp.html
docs/abstractionLayers.md | Layer architecture and SIMEMP Gateway validation | abstractionLayers.html
docs/qnPrimitive.md | Universal Envelope, operational axioms, and Optimization Morphisms | qnPrimitive.html
docs/q0_q1.md | Primary genesis origin Q(0) and first unit Q(1) | q0_q1.html

H2: Review Integration Record (v1.3.0)

Source: docs/AiReviews/AiReviewQnIRv0.1.md — transcript of the external adversarial review (Google Deep Review, October 2026) of the condensed Qn universe specification and QnIR v0.1 candidate draft. This update integrates the numeric-substrate findings of that review into the canon. Under Axiom 13, this file is a new versioned artifact; no prior content is mutated in place. New constructions and semantic refinements introduced below are tagged [v1.3.0 — CANDIDATE] pending BOC sealing; editorial corrections are applied directly and recorded here.

Change | Section | Reviewer source | Type
\kappa(z) bound per-artifact in the DCC profile; three canonical schedules declared; undeclared-\kappa rejection | §3.3 | QO-4 / Segment 2 Point 1 (fatal) | CANDIDATE
Residual typing: collapsed integer vs. commitment handle to a Layer-3 generator | §3.4 | QO-3 / Segment 2 Point 2 | CANDIDATE
Multi-digit natural construction — the Radix-Anchor Morphism (cons | | Q(9) gap closed) | §3.6 (new) | Objection 14 / QO-4 (fatal) | CANDIDATE
Incremental digits vs. consolidated quotient; zoom-alignment rule | §4.1 | Segment 2 Point 2 | CANDIDATE
Canonical zoom normalization before hashing (one commitment per value) | §4.3 (new) | Segment 4 Point 7 | CANDIDATE
Euclidean division orientation (\sigma_R \equiv \oplus fixed); quotient-sign guard | §5.3 | Segment 2 Point 4 | CANDIDATE
Linear division vs. DPRM scope separation; "Decidability" renamed "Totality"; HEIGHT_BOX_EXHAUSTED reason code; NO_SOLUTION reserved | §5.3 | Segment 2 Point 4 / Objection 4 | editorial + semantic
Height log-origin guard h(u/v) | §5.3 | Minor correction | editorial
Even-root boundary restriction (n \ge Q(4) -> spinor promotion, Degree 2) | §5.5.IV | Segment 2 Point 4 | CANDIDATE
Canonical negative rationals via GCD-reduced OP_RATIO pairs | §5.7 (new) | QO-4 part 3 | CANDIDATE
Sign-aware constructive enclosure intervals; undefined \epsilon_z retired | §6 | Segment 2 Point 4 | CANDIDATE
"Neutralized" phrasing aligned to termination-not-decidability | §1.2 | Objection 4 (cosmetic) | editorial

Cross-file dependencies flagged (deferred under the one-file-at-a-time discipline, to be integrated in subsequent versioned updates): docs/q2_q9.md §7.1 monoid scope correction ("single-digit prime anchor basis", not generator of the full positive cone) and §7.2 Collatz MODAL_HYPOTHESIS tagging; docs/qexpr.md §4.6 identity-table rule count (16 rules, including the first-root rule); glossary term additions for the new §3.6/§5.7 constructs.

H2: 1. Epistemic Stance and Explicit Demarcation

In strict adherence to the no-implicit-rule (docs/abstractionLayers.md §3.2) and the constitutional constraints of the simemp framework, this specification explicitly defines the boundaries of Qm (Quang Mathematics) to prevent implicit assumptions, hidden conventions, or non-deterministic state drift.

In the qn computational universe, every mathematical proposition and state transition in \mathrm{Qm} is governed by dependencies-determinism:

$$H(S_t \mid \mathcal{D}(S_t)) = 0$$

Operational validity is verified empirically via computational-consistency. No unmeasured entity may cross an abstraction boundary or simemp-gateway.

EXPLICIT DOMAIN BOUNDARY OF Qm

WHAT Qm IS	WHAT Qm IS NOT
Constructive discrete arithmetic	Continuous floating-point
Base-independent scale lattices (z)	Platform-dependent radix conv.
Conserved residual tracking (r)	Silent truncation / unmetered
Geometric orientation involution (σ)	Detached ontological sign ent.
Total-by-budget morphisms (Receipts)	Infinite non-terminating exp.

```
 | Height-bounded Diophantine relations | Unbounded Hilbert-10 void     |
 | Bounded folds & constrained roots    | Continuous unmetered exponent |
 | Positional order decidability        | Arbitrary heuristic tie-break |
 | Concrete numeric leaf evaluation     | Symbolic AST container (Qexpr)|
 +--------------------------------------+-------------------------------+
```

H3: 1.1. Explicit Inclusions: What Qm Is
- The Constructive Numeric Leaf: The canonical, discrete representation of fully evaluated quantities along dimensional axes encapsulated within the universal-envelope.
- The \mathcal{R}-Algebra: Explicit arithmetic morphisms (+ , - , \times, \div, powers, roots, and bounded factorials) preserving exact Diophantine invariants and emitting signed receipts.
- Projective Scale Transitions: Exact conservation of residual information (r_z) across zoom scales without numeric drift.
- local-first Execution: 100% intra-node computation mediated by local cdqn memory-bus data movement, requiring zero distributed consensus.
- Total-by-Budget Morphisms: Discrete operations bounded by finite resource consumption via metric-exhaustion.

H3: 1.2. Explicit Exclusions: What Qm Is Not
- Not an IEEE 754 Floating-Point System: Under Axiom 10, approximations, mantissa round-offs, subnormal representations, and +0.0 / -0.0 distinctions are prohibited.
- Not a Non-Constructive Continuum: \mathrm{Qm} does not admit actual infinities (\aleph_0, \aleph_1), uncomputable real numbers (non-constructive Dedekind cuts), or undecidable Cauchy sequences.
- Not an Unbounded Diophantine Search Oracle: While arithmetic operations are defined through Diophantine constraints, \mathrm{Qm} explicitly rejects unmetered infinite search spaces (\mathbb{Z}^n, \mathbb{Q}^n). Classical undecidability (Hilbert's Tenth Problem, DPRM 1970) is addressed structurally, not solved: all evaluations are constrained to finite logarithmic height bounds (h(x) \le H_{\mathrm{max}}) declared within the metric-envelope. [v1.3.0 — editorial, per Objection 4] Finiteness buys *guaranteed termination* under budget; it does not constitute mathematical *decidability* of Diophantine solvability. The two claims are kept strictly separate throughout this specification (cf. §5.3).
- Not an Authoring Language: High-level source grammar belongs to qnlang.
- Not an Intermediate Execution Backend: Hardware abstraction and register scheduling belong to qnir.
- Not an Unbounded Symbolic Container: Unevaluated expression trees belong to qexpr; \mathrm{Qm} governs their deterministic numerical collapse.

H2: 2. Abstraction Layer Mapping and Complexity Stratification

\mathrm{Qm} operates across the hierarchy as an orthogonal vertical projection across abstraction layers (docs/abstractionLayers.md §11), rather than a static horizontal tier.

```
 +-------------------------------------------------------------------------+
 | Layer 3: Symbolic Composition (Qexpr Trees, Bounded Expressions)        |
 |          Qm Degree 2: Multi-axial geometry (d_k), Polynomial Forms      |
 +-------------------------------------------------------------------------+
 | Layer 2: Governed Morphisms & Algebraic Interfaces                      |
 |          Qm Degree 1: Monoids, Involutions (I), Diophantine Constraints |
 +-------------------------------------------------------------------------+
 | Layer 1: Node Genesis Layer                                             |
 |          Qm Degree 0: Origin Q(0), Unit Q(1), Basis Alignment d1        |
 +=========================================================================+
 | Layer 0: Physical Substrate (Entropy Harvesting, Landauer Work W ≥ kT ln 2) |
 +-------------------------------------------------------------------------+
```

H3: 2.1. Layer-by-Layer Responsibilities
- layer-1: Hosts genesis primitives. Origin Q(0) establishes the spatial origin; unit Q(1) establishes the baseline metric quantum; the transition Q(0) \to Q(1) establishes the initial dimensional axis d1 and the Abstract Compute Unit U.
- Layer 2: Hosts elementary arithmetic morphisms (+ , - , \times, \div, exponents, roots, factorials) constrained by explicit DCC Profiles, indexed sequentially along the monotonic causal-arrow.
- Layer 3: Ingests symbolic qexpr trees and collapses them into canonical \mathrm{Qm} leaf tuples within finite metric budgets.

H3: 2.2. Complexity Degree Stratification
- Degree 0 (Foundational Primitives): Q(0), Q(1).
- Degree 1 (Elementary Arithmetic): Positional tuples along axis d_1, Diophantine quotient-remainder partitions, bounded folds, and the multi-digit radix-anchor construction (§3.6). *(Normative scope of this specification.)*
- Degree 2 (Multi-Axial & Symbolic): Orthogonal coordinate spaces (d_k), Clifford geometric algebras (\mathcal{C}\ell_{p,q}), spinor/rotor solutions in even subalgebras, and Qexpr CAE-DAG reduction (docs/qm_geometry.md and docs/qexpr.md).
- Degree 3 (Discrete Calculus & Analysis): Discrete exterior calculus, finite-difference operators, and projective continued fractions (docs/qm_calculus.md).

H2: 3. Canonical Numeric Representation

Every fully evaluated numeric artifact in \mathrm{Qm} is an immutable, discrete Qn Numeric Entity represented as a discrete leaf tuple:

$$\text{Qn Numeric Entity} \equiv \langle \sigma, \, q_z, \, z, \, r_z, \, d \rangle$$

```
                  CANONICAL Qm NUMERIC STRUCTURE
 +-------------------------------------------------------------------+
 |  σ    : Relational Orientation (⊖, ⊙, ⊕)                          |
 |  q_z  : Consolidated Metric Magnitude Quotient (q_z ∈ ℕ_≤Hmax)     |
 |  z    : Scale / Lattice Zoom Level (z ∈ ℕ)                        |
 |  r_z  : Conserved Residual (typed form, §3.4)                     |
 |  d    : Dimensional Coordinate Axis Parameter (d_k)               |
 +-------------------------------------------------------------------+
```

[v1.3.0] The magnitude field label is refined from "Discrete Metric Magnitude Quotient" to Consolidated Metric Magnitude Quotient: q_z is the single self-contained quotient at zoom z (not an incremental cascade digit); see §4.1 for the exact relation between consolidated quotients and incremental digits. Its domain $\mathbb{N}_{\le H_{\mathrm{max}}}$ is the bounded natural semiring constructed in §3.6 — it is imported from nowhere.

H3: 3.1. Orientation as Geometric Involution (σ)

Sign is not an independent arithmetic substance. It is an oriented displacement relative to local origin Q(0) along axis d:

- The base transition from Q(0) to Q(1) along axis d_1 defines canonical positive alignment ($\oplus$).
- Inversion is governed by the 1D reflection functor $\mathcal{I}_d$, satisfying the involution property:

$$\mathcal{I}_d: \Sigma \to \Sigma, \quad \mathcal{I}_d^2 = \mathrm{id}, \quad \mathrm{where} \; \Sigma = \{ \ominus, \, \odot, \, \oplus \}$$

- $\odot$ (Origin): Coincident with local Q(0) (zero metric displacement).
- $\oplus$ (Collinear): Parallel to the basis transition Q(0) $\to$ Q(1).
- $\ominus$ (Anti-parallel): Opposing the basis transition.

Under Axiom 10, signed zeros (+0.0, -0.0) are prohibited. If $q_z = 0$ and $r_z = Q(0)$, σ collapses identically to $\odot$.

H3: 3.2. Ontological Primacy of Magnitude (q_z)

The discrete quotient $q_z \in \mathbb{N}_{\le H_{\mathrm{max}}}$ represents the unsigned metric distance along axis d at zoom level z. In physical execution substrates, metric resources (storage footprint, clock cycles, Landauer dissipation) are strictly non-negative. Absolute magnitude q_z is ontologically primary; signed quantities are composite pairs $\langle \sigma, q_z \rangle$. Every q_z beyond the single-digit Category A spectrum is an artifact constructed by the Radix-Anchor Morphism of §3.6, carrying its own Axiom-4 causal lineage.

H3: 3.3. Base-Independent Scale Lattice (Zoom z)

Conforming to Axiom 9, scale is decoupled from arbitrary positional radices (binary, decimal, sexagesimal). A declared zoom level $z \in \mathbb{N}$ parameterizes a discrete rational subdivision quantum $\delta_z(d) \in \mathbb{Q}^+$ of the dimensional unit:

$$\delta_z(d) = \frac{Q(1)_d}{\kappa(z)}$$

where $\kappa: \mathbb{N} \to \mathbb{N}$ is a strictly monotonic resolution lattice function.

[v1.3.0 — CANDIDATE, Review Integration QO-4] The resolution function is not fixed globally by this specification. Leaving κ unbound would make leaf tuples structurally incomparable across nodes and operations, violating dependencies-determinism. Under the no-implicit-rule, $\kappa(z)$ is therefore bound per artifact inside the active DCC profile (DCC_ref), which must declare exactly one canonical schedule:

Schedule	Definition	Domain rationale
Dyadic	$\kappa_2(z) = Q(2)^z$	Zoom shifts lower to exact hardware bit-shifts; default schedule for computation (DCC_ANCHOR_Q2_PARITY_v1)
Decimal	$\kappa_{10}(z) = (Q(2) \times Q(5))^z$	Human financial and decimal contracts under the Q(9) radix boundary (DCC_ANCHOR_Q9_RADIX_v1)
Factorial	$\kappa_{\mathrm{fact}}(z) = (z+1)!$	Finite-zoom absorption of arbitrary small rational denominators (denominators up to z+1 terminate exactly)

- A gateway evaluating, comparing, or hashing tuples whose DCC profile declares no κ schedule rejects the transition deterministically with status REJECTION_BY_GATEWAY and reason code ERR_KAPPA_UNDECLARED — no ambient resolution lattice exists (Axiom 6 closed status set).
- Cross-node contracts (Q(anchor) schemas, §8) must declare matching κ schedules; tuples under mismatched schedules are structurally incomparable and require explicit metered re-projection at a gateway before any cross-node equality claim.
- Additional schedules are admissible only through the explicit DCC contract extension discipline (deterministic, strictly monotonic, receipted); heuristic or runtime-negotiated κ is prohibited.

H3: 3.4. Conserved Residual (Remainder r)
The residual r_z is the exact difference between the unscaled value and its discrete quotient:

$$0 \le \lvert r_z \rvert \lt \delta_z(d)$$

The remainder is never discarded silently. It acts as the exact, causal input state for subsequent resolution expansions.

[v1.3.0 — CANDIDATE, Review Integration QO-3] The residual carries exactly one of two typed forms (No-Implicit Rule):

1. Collapsed residual: an exact integer $0 \le r_z \lt \delta_z(d)$ held directly in the tuple payload. The value is exact at zoom z when additionally $r_z = Q(0)$ (Terminal Exactness, §4.2).
2. Residual commitment handle: a commitment hash $\mathcal{H}(\alpha_{\mathrm{rem}})$ referencing a Layer-3 qexpr residual generator (OP_RATIO, OP_ROOT, OP_CF) whose full evaluation exceeds the current zoom budget. The handle is opaque at Layer 2: it is resolved only by an explicit, metered OP_ZOOM crossing back to Layer 3 (§7). This typing preserves the strict upward functorial hierarchy $\mathcal{F}: \mathbf{C}(L) \to \mathbf{C}(L+1)$ — Layer 2 never embeds live Layer-3 state, only commitments to it.

Which form is carried is declared in the tuple's type tag; mixing forms inside one tuple is unencodable.

H3: 3.5. Dimensional Orthogonality (Dimension d)
The axis parameter d_k ensures that metric units remain orthogonal. Remainders along dimension d_i cannot cancel, add to, or combine with remainders along dimension d_j ($i \neq j$).

H3: 3.6. Multi-Digit Natural Construction: The Radix-Anchor Morphism [v1.3.0 — CANDIDATE]

Status. New canonical construction. Direct response to QO-4 / Objection 14 of the external review (Segment 2, Point 1), which classified the missing construction of naturals beyond Q(9) as *fatal if unresolved before QnIR v0.2 implementation*. A BOC record is included at the end of this section.

The gap being closed. Category A (docs/q2_q9.md) constructs the single-digit artifacts Q(1) \dots Q(9) and stops. Yet the leaf tuple (§3) declares $q_z \in \mathbb{N}_{\le H_{\mathrm{max}}}$, and every downstream layer — Qm arithmetic (§5), Qexpr node hashing, QnIR value slots — consumes magnitudes $\ge Q(10)$. Without a constructive bridge, numbers ≥ 10 have no canonical form, no causal lineage, and no stable commitment hash, which breaks:

- Identity (Tier-1): hash-consing equality $\mathcal{H}(A) = \mathcal{H}(B) \iff A \equiv B$ is undefined for equal magnitudes ≥ 10;
- Metric (Tier-1): logarithmic height ceilings $h(q_z) \le H_{\mathrm{max}}$ (§5.3) are declared over an unconstructed domain;
- Dependencies Determinism: cross-node reproducibility of any computation producing multi-digit output is unanchored.

Two prohibited "resolutions" frame the BOC record: importing $\mathbb{N}$ (and $\mathbb{Q}$) wholesale as unconstructed external primitives violates Axioms 1, 2, and 4 and the no-implicit-rule; extending the unary Successor Morphism $\mathcal{S}^{(n)}(Q(0))$ to arbitrary n incurs $\mathcal{O}(q_z)$ causal lineage depth and $\mathcal{O}(q_z)$ Landauer creation work — exponential blowup relative to bit length, violating Tier-2 Efficiency under any declared budget.

Construction. Under the q-bypass–compatible composition discipline, the bounded natural semiring $\mathbb{N}_{\le H_{\mathrm{max}}}$ is constructed inside the artifact model by radix-anchor (Horner-step) composition, anchored to DCC_ANCHOR_Q2_PARITY_v1 (binary) or to word-aligned limbs via DCC_ANCHOR_Q8_OCTET_v1 (octal):

$$\mathrm{Cons}_B\big(q_{\mathrm{high}}, \, d_{\mathrm{low}}\big) \;=\; \big(q_{\mathrm{high}} \times B\big) + d_{\mathrm{low}}, \qquad B \in \{Q(2), \, Q(8)\}, \quad d_{\mathrm{low}} \in \{Q(0), \dots, B - Q(1)\}$$

with explicit lineage $\mathrm{Parent}(q) = \{B, \, q_{\mathrm{high}}, \, d_{\mathrm{low}}\}$. Every constructed magnitude thereby traces to the local origin through the anchor contracts of Category A (Axiom 4 satisfied). The construction is evaluated by the existing $\mathcal{R}$-algebra of §5 (one multiplication and one addition per step) and emits a construction receipt at each anchoring step.

Canonical uniqueness rules.

1. Non-zero innermost seed: the deepest digit of any non-zero integer is Q(1) — leading-zero aliases are unencodable.
2. Origin collapse: $\mathrm{Cons}_B(Q(0), Q(0)) \to Q(0)$, recognized as an identity via q-bypass (the representation of zero is the origin itself, never a constructed chain).
3. Digit range: $0 \le d_{\mathrm{low}} \lt B$ and $q_{\mathrm{high}} \neq Q(0)$ unless $q = Q(0)$.

Rules 1–3 give every constructed magnitude exactly one canonical Cons-chain. Consequently hash-consing equality extends unchanged from single digits to all of $\mathbb{N}_{\le$

H_{\mathrm{max}}}: \mathcal{H}(A) = \mathcal{H}(B) \iff A \equiv B holds for every constructed magnitude, restoring the content-addressed identity invariant at QnIR lowering boundaries.

Height and work. The chain depth is

h_{\mathrm{cons}}(q_z) = \lceil \log_B(q_z + 1) \rceil \le H_{\mathrm{max}}

so lineage depth and creation work are \mathcal{O}(\log_B q_z) — logarithmic, not linear. Under HCA-Collapse the Cons-chain lowers isomorphically to native machine words (UINT64 / multi-precision limbs) with zero radix-conversion overhead for dyadic and octal schedules. Exceeding H_{\mathrm{max}} during construction halts with RECEIPT_EXPRESSION_TOO_COMPLEX.

Category A anchor basis (scope correction).

The single-digit prime anchors \{Q(2), Q(3), Q(5), Q(7)\} form the

Category A single-digit prime anchor basis:

They characterize the factorization structure of single-digit artifacts (docs/q2_q9.md §7). They do not generate the multiplicative monoid of all constructed naturals — no free monoid over \{2, 3, 5, 7\} contains 11, 13, or any integer with a prime factor \ge 11, and the generated set is not closed under addition (Q(9) + Q(2) = 11). Multi-digit magnitudes exist by the §3.6 construction, not by prime generation. (The parallel claim in docs/q2_q9.md §7.1 is queued for the same scope correction in its next versioned update.)

Decimal interface projection.

Radix-10 formatting via Q(9) and Q(10) \equiv \mathrm{Cons}_{Q(2)}(Q(5), \, Q(0)) remains strictly an interface projection governed by DCC_ANCHOR_Q9_RADIX_v1 — never the internal representation (base neutrality, §3.3, is preserved).

Example.

11 = \mathrm{Cons}_{Q(2)}\big(\mathrm{Cons}_{Q(2)}\big(\mathrm{Cons}_{Q(2)}\big(Q(1), \, Q(0)\big), \, Q(1)\big), \, Q(1)\big)

— binary 1011_2, chain depth 3 above the Q(1) seed; its commitment hash is distinct from \mathcal{H}(Q(9)) and \mathcal{H}(Q(2)), and identical on every node applying the same schedule.

BOC record.

(a) Import \mathbb{N}/\mathbb{Q} wholesale — rejected: violates Axioms 1/2/4 and the No-Implicit Rule.

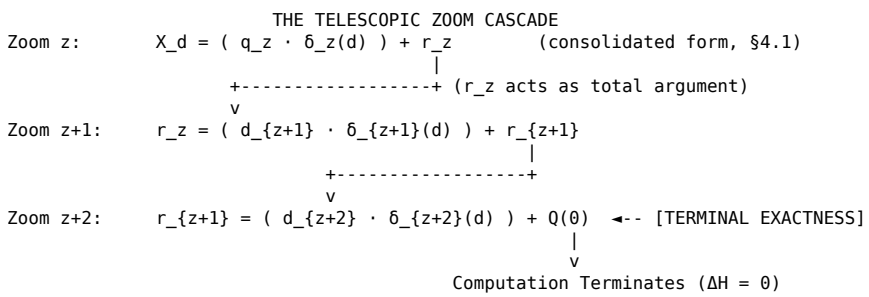
(b) Unary successor extension — rejected: \mathcal{O}(q_z) work violates Tier-2 Efficiency.

(c) Radix-anchor morphism, B \in \{Q(2), Q(8)\} — selected:
logarithmic cost, Axiom-4 lineage, native word lowering, canonical uniqueness.

Tier-1 admission filter: Identity (unique canonical hash) ✓; Metric (finite height \le H_{\mathrm{max}}) ✓; Security (anchor-constrained, receipted construction) ✓. Tier-2 vector: Efficiency (logarithmic) ✓; Modularity (reuses §5 morphisms unchanged) ✓; Portability (dyadic/octal anchors lower to any word-addressed substrate) ✓.

H2: 4. The Projective Scale Invariant and Terminal Exactness

The triplet \langle z, r_z, d \rangle constitutes a projective resolution system over discrete lattices.



H3: 4.1. Scale Expansion Equation
When higher precision is demanded by a downstream workflow (z \to z + 1), the input state to the resolution transition is identically r_z:

r_z = \Big(d_{z+1} \cdot \delta_{z+1}(d) \Big) + r_{z+1}

[v1.3.0 — CANDIDATE, Review Integration Segment 2 Point 2] Notation discipline (No-Implicit Rule). The cascade and the canonical tuple use two distinct, explicitly related quantities:

- Incremental digits d_k \in \{0, 1, \dots, \kappa(k+1)/\kappa(k) - 1\} are the per-step quotients produced by the cascade, with d_0 \equiv q_0. The cascade equation above is stated over incremental digits.
- Consolidated quotient q_z is the single self-contained magnitude carried by the canonical leaf tuple §3:

q_z \;=\; \sum_{k=0}^{z} d_k \cdot \frac{\kappa(z)}{\kappa(k)}

- Total metric information is conserved identically in both forms:

\mathcal{I}_{\mathrm{total}}(X_d) \;=\; \sum_{k=0}^{z} \Big(d_k \cdot \delta_k(d) \Big) + r_z \;=\; q_z \cdot \delta_z(d) + r_z

The consolidated form is the tuple payload (a leaf tuple determines its value without ancestor tuples); the telescoped form is the derivation path recorded in lineage. The earlier presentation, which used the single symbol q_z in both roles, is corrected.

- Zoom alignment (mixed-z operations). Binary operations on tuples with z_A \lt z_B first deepen the shallower tuple to z_B (metered; the deepening consumes r_{z_A} per the cascade above and emits no drift). Deepening the shallower tuple is the only admissible alignment; shallowing the deeper tuple is prohibited, as it would discard residual information (dissipative truncation, §5.4).

H3: 4.2. Terminal Exactness Invariant

If a state evaluation yields a remainder equal to the local causal origin:

r_z = Q(0)

the representation is algebraically exact.

1. Information Invariance: For all subsequent scales k \gt 0, d_{z+k} \equiv 0 and r_{z+k} \equiv Q(0).
2. Deterministic Early Halting: Evaluating z+1 when r_z = Q(0) yields zero entropy reduction (\Delta I = 0). Under the Metric Invariant (docs/simemp.md §3.1), spending compute budget on idempotent iterations is forbidden.
3. Receipt Emission: The simemp-gateway intercepts this state and emits an immutable terminal receipt:
   - Terminal Receipt: RECEIPT_TERMINAL_EXACTNESS
     ⟨ Status: SUCCESS_EXACT, Zoom: z, Remainder: Q(0), ConsumedWork: 0 ⟩

H3: 4.3. Canonical Zoom Normalization [v1.3.0 — CANDIDATE]

[Review Integration Segment 4 Point 7] Requirement-style identity — equality of values is equality of commitment hashes — holds across zoom levels only if values are hashed at a canonical zoom. Qm therefore fixes: a value that achieves exactness (collapsed residual, §3.4 form 1) is re-expressed at its canonical minimal zoom z_{\mathrm{min}} — the smallest declared zoom at which the value attains Terminal Exactness — before any commitment hash is computed or compared. The re-expression is a deterministic re-canonicalization billed to the calling operation (never silent, never free-floating): two tuples denoting the same value at different zooms normalize to one commitment, so \mathcal{H}(A) = \mathcal{H}(B) \iff A \equiv B holds zoom-invariantly. Tuples carrying unexpanded residual handles (§3.4 form 2) are never normalized across zooms; their equality claims are deferred to explicit metered zoom resolution (§6, §7).

H2: 5. Elementary Morphisms (R-Algebra)

Arithmetic in \mathrm{Qm} operates over discrete tuples rather than continuous fields. All magnitude arithmetic below operates over the constructed naturals of §3.6; no operation references an unconstructed domain.

H3: 5.1. Addition and Subtraction

Addition is exact vector concatenation along axis d. Subtraction is addition composed with geometric reflection:

A - B \equiv A + \mathcal{I}_d(B)

Given A = \langle \sigma_A, q_A, z, r_A, d \rangle and B = \langle \sigma_B, q_B, z, r_B, d \rangle (first aligned to a common zoom per §4.1):

Step 1: Absolute values and remainders are combined over the common quantum \delta_z(d):

\mu_{\mathrm{exact}} = (\sigma_A q_A \cdot \delta_z + \sigma_A r_A) + (\sigma_B q_B \cdot \delta_z + \sigma_B r_B)

Step 2: The result is partitioned into discrete quotient q_C, residual remainder r_C, and orientation \sigma_C:

q_C = \left\lfloor \frac{\lvert \mu_{\mathrm{exact}} \rvert}{\delta_z} \right\rfloor

r_C = \lvert \mu_{\mathrm{exact}} \rvert - (q_C \cdot \delta_z)

\sigma_C = \mathrm{sgn}(\mu_{\mathrm{exact}})

H3: 5.2. Multiplication
Multiplication scales magnitude and composes orientation parity:

\sigma_C = \sigma_A \otimes \sigma_B

\oplus \otimes \oplus = \oplus, \quad \ominus \otimes \ominus = \oplus, \quad \ominus \otimes \oplus = \ominus, \quad \odot \otimes \sigma = \odot

The unscaled magnitude is calculated constructively:

\mu_{\mathrm{exact}} = (q_A \cdot \delta_z + r_A) \times (q_B \cdot \delta_z + r_B)

The resulting unscaled quantity collapses to \langle \sigma_C, q_C, z, r_C, d \rangle via Diophantine resolution.

H3: 5.3. Division as a Constrained Diophantine Equation

Division in \mathrm{Qm} is defined not as an unconstrained rational projection, but as a bounded constraint equation. For any input pair A and B, the morphism determines quotient C and remainder R satisfying:

A = (B \times C) + R

[v1.3.0 — CANDIDATE, Review Integration Segment 2 Point 4] The bound 0 \le \lvert R \rvert \lt \lvert B \rvert constrains only magnitude; without fixing the orientation \sigma_R of the remainder, two distinct solutions exist whenever B \nmid A and negative orientations are involved (Euclidean floor division vs. truncation toward zero), violating dependencies-determinism. \mathrm{Qm} therefore fixes the Euclidean orientation as part of the constraint system:

A = (B \times C) + R, \qquad B \neq Q(0), \qquad 0 \le R \lt \lvert B \rvert, \qquad \sigma_R \equiv \oplus

- Remainder orientation: \sigma_R \equiv \oplus always. The remainder is a non-negative unsigned metric residual, matching the residual convention of §3.4 exactly — no additional orientation field is introduced on R (minimal implicit state).
- Quotient determination: C is uniquely determined by the constraint equation; \sigma_C = \mathrm{sgn}(C), with the signed-zero guard \sigma_C \to \odot whenever C = Q(0) (which occurs exactly when \lvert A \rvert \lt \lvert B \rvert; §3.1).
- Exact-division parity: when R = Q(0), the parity composition of §5.2 holds identically: \sigma_C = \sigma_A \otimes \sigma_B.
- Uniqueness: for every sign combination of \sigma_A, \sigma_B there is exactly one pair (C, R) satisfying the four conditions above. The solution space is a singleton, so the morphism is a function, not a relation.

BOC record (division orientation). Euclidean orientation was selected over truncated-toward-zero orientation. Both yield unique solutions; Euclidean wins under BOC because its remainder is always a non-negative unsigned residual (collapsing onto the §3.4 tuple convention with zero extra state, satisfying the Tier-1 Metric invariant), at the cost of one \mathrm{sgn} evaluation for \sigma_C (constant work, already inside the metered dispatch cost of the division morphism).

Linear division vs. general Diophantine search (scope separation). Linear Euclidean division — the constraint equation above — is polynomial-time in the bit length of A and terminates in exact (C, R) always. It must not be conflated with general non-linear Diophantine polynomial equations (OP_ROOT domains, the Hilbert's Tenth / DPRM territory), which are handled exclusively by the height-bounded search discipline below.

H4: Height-Bounded Totality

Classical Diophantine equations over unmetered spaces (\mathbb{Z}, \mathbb{Q}) inherit algorithmic undecidability under Hilbert's Tenth Problem. In \mathrm{Qm}, this undecidability is addressed structurally across three physical constraints:

Logarithmic Height Boundedness: In arithmetic geometry, the logarithmic height measures physical bit complexity, bounded for coprime u, v by

h(u/v) = \mathrm{max}\big(\mathrm{log}\,\mathrm{max}(\lvert u \rvert, \, Q(1)), \, \mathrm{log}\,\mathrm{max}(\lvert v \rvert, \, Q(1))\big) \le H_{\mathrm{max}}

[v1.3.0 — editorial] The \mathrm{max}(\cdot, Q(1)) guard removes the log-origin singularity at u = Q(0) present in the earlier form. Under the metric-envelope, every Diophantine evaluation operates within an explicit height ceiling H_{\mathrm{max}} and work budget W_{\mathrm{budget}}.

Finite State Space (Northcott's Theorem): By Northcott's theorem, the set of rational coordinates of bounded height is strictly finite:

\mathrm{Card}\big(\{ x \in \mathbb{Q} \colon h(x) \le H_{\mathrm{max}} \}\big) \lt \infty

Therefore, candidate solutions in \mathrm{Qm} inhabit a finite discrete lattice rather than an unmetered void.

Deterministic Totality (Termination, Not Decidability) — [v1.3.0, renamed from "Deterministic Decidability" per Segment 2 Point 4 / Objection 4]: A height-bounded evaluation is total by budget: it either terminates with an exact quotient-remainder pair, reaches Terminal Exactness (R = Q(0)), or halts unconditionally upon budget exhaustion via Metric Exhaustion (Axiom 6), emitting RECEIPT_BUDGET_EXHAUSTED.

[v1.3.0 — semantic refinement] Exhausting the Northcott height box (a candidate space of cardinality \mathcal{O}(e^{2 H_{\mathrm{max}}})) buys guaranteed termination — it does not constitute a proof of algebraic impossibility. Accordingly:

- Bounded-search exhaustion emits RECEIPT_BUDGET_EXHAUSTED with reason code HEIGHT_BOX_EXHAUSTED;
- the terminal status NO_SOLUTION is reserved for constructive impossibility proofs (e.g., parity obstructions such as the 1D even-root argument of §5.5.IV, or the negative-factorial argument of §5.6.III), never for mere exhaustion of a declared search box.

This distinction is normative across the corpus: conflating bounded search failure with impossibility is a semantic drift and is rejected at gateways.

H4: Terminal Rejection

Attempted division by origin Q(0) violates the DCC constraint and is intercepted deterministically at the gateway, emitting the immutable receipt RECEIPT_DIVISION_BY_ZERO_REJECTED with status REJECTED and code ERR_DIV_ZERO.

H3: 5.4. Remainder Depth and Dissipative Truncation
When recursive compositions cause the symbolic tree of r_C to exceed the Metric Envelope ceiling \mathrm{Depth}_{\max}:
1. Silent truncation is prohibited under the no-implicit-rule.
2. Execution halts or emits an explicit dissipative receipt:
    - Dissipative Truncation Receipt: RECEIPT_REMAINDER_TRUNCATION
      ⟨ Status: DISSIPATIVE_SINK, LostResidual: r_C, ExportedEntropy: ΔS ⟩

H3: 5.5. Exponentiation and Radical Morphisms

Exponentiation and root extraction are formalized as constrained morphisms that operate with explicit thermodynamic bounds and dimensional parity governance:

H4: I. Integer Exponentiation as a Bounded Monoidal Fold
For integer exponents B \in \mathbb{N}_{\le H_{\mathrm{max}}} (constructed per §3.6, exact at zoom z = 0), exponentiation is an iterative multiplicative fold over the positive monoid (\mathcal{Q}^+, \times, Q(1)) under Node Class 4 (OP_FOLD):

A^B = \prod_{j=1}^B A

Evaluated at Layer 3 under OP_FOLD. The memory footprint expands linearly with the exponent: h(A^B) = B \cdot h(A) \le H_{\max}, evaluated on the reduced coprime form of A. If the resulting bit size exceeds the metric envelope ceiling, compilation halts with RECEIPT_EXPRESSION_TOO_COMPLEX.

H4: II. Negative Exponents as Rational Pairs
Negative powers do not invoke floating-point arithmetic; they map to exact rational inversion pairs under the Node Class 1 ordered call OP_RATIO(Q(1), A^B):

A^{-B} \equiv \frac{Q(1)}{A^B}

Projected onto scale lattices at zoom z with exact remainder conservation (r_z). If the base is negative (\ominus A), the resulting sign is determined by the parity of exponent B via secondary anchor Q(2) (DCC_ANCHOR_Q2_PARITY_v1):

$$\sigma_{\mathrm{result}} = \begin{cases}
\oplus & \mathrm{if} \; B \in \mathcal{Q}_{\mathrm{even}} \\
\ominus & \mathrm{if} \; B \in \mathcal{Q}_{\mathrm{odd}}
\end{cases}$$

The resulting rational is reduced to the canonical form of §5.7 before hashing.

H4: III. Boundary Exponent Identities
Under the no-implicit-rule, boundary conditions are strictly evaluated:
- Zero Exponent Identity (A \neq Q(0)): For any non-zero base, A^{Q(0)} represents the empty monoidal product, collapsing to the multiplicative identity Q(1). Evaluated in \mathcal{O}(1) via q-bypass, emitting RECEIPT_BYPASS_IDENTITY at zero metric cost (\Delta W = 0).
- Indeterminate Origin Power (Q(0)^{Q(0)}): An unparameterized 0^0 evaluation is prohibited across gateways. Unless the calling profile carries an explicit contract (DCC_ZERO_EXPONENT_v1), execution halts with RECEIPT_INDETERMINATE_ORIGIN_EXPONENT.

- Negative Origin Power (Q(0)^{-k}): Evaluating Q(0)^{-k} \equiv Q(1) / Q(0)^k requires division by origin Q(0), terminating immediately with RECEIPT_DIVISION_BY_ZERO_REJECTED.

H4: IV. Radical Morphisms and Dimensional Promotion
Roots are defined constructively as polynomial Diophantine equations under OP_ROOT:

\sqrt[n]{A} \implies \mathrm{Find} \; X \quad \mathrm{such} \; \mathrm{that} \quad X^n - A = Q(0)

- Singularity of Root Zero (\sqrt[Q(0)]{A}): Expressed as rational exponent A^{\frac{Q(1)}{Q(0)}}, extracting the Q(0)-th root requires division by origin in the exponent denominator. The gateway halts execution with RECEIPT_DIVISION_BY_ZERO_REJECTED.
- Odd Roots of Negative Scalars (n \in \mathcal{Q}_{\mathrm{odd}}$): Because an odd number of spatial reflections preserves negative orientation (\ominus^n = \ominus), odd roots of negative quantities are algebraically closed within 1D discrete space (\mathcal{Q}_1). For example, \sqrt[3]{\ominus 8} = \ominus 2 under Terminal Exactness.
- Even Roots of Negative Scalars (n \in \mathcal{Q}_{\mathrm{even}}$): In 1D scalar space, no orientation satisfies \sigma \otimes \sigma = \ominus. An attempt to compute \sqrt{\ominus A} in 1D is intercepted by the gateway and promoted to Dimension 2 (d_2) as an oriented planar bivector blade in Clifford Geometric Algebra (\mathcal{C}\ell_{p,q}):

\sqrt{\ominus Q(1)} \implies \vec{e}_1 \wedge \vec{e}_2 \quad (\mathrm{where} \; (\vec{e}_1 \wedge \vec{e}_2)^2 = -Q(1))

[v1.3.0 — CANDIDATE, boundary restriction per Review Integration Segment 2 Point 4] The 1D->2D bivector promotion is sound for the square root only (n = Q(2)) in Euclidean signature \mathcal{C}\ell_{2,0}, where (\vec{e}_1 \wedge \vec{e}_2)^2 = -Q(1). It does not extend to higher even roots n \ge Q(4): any pure bivector blade \beta \,(\vec{e}_1 \wedge \vec{e}_2) satisfies \big(\beta \,(\vec{e}_1 \wedge \vec{e}_2)\big)^4 = +\,\beta^4\, Q(1) \ge Q(0) — a non-negative scalar, never a negative one. Fourth and higher even roots of negative scalars are mixed-grade rotors (spinors) in the even subalgebra (e.g. \sqrt[4]{\ominus Q(1)} = \big(Q(1) \pm \vec{e}_1 \wedge \vec{e}_2\big) \times \mathrm{r}_{\sqrt{2}}, coupled to DCC_ANCHOR_SQRT2_v1), which exist only inside frames whose declared signature \eta provides adjoining axes with \eta_{i,i}\,\eta_{j,j} = \oplus (Lorentzian \mathcal{C}\ell_{1,1} spacetime bivectors square to +Q(1) and cannot serve). Such promotion crosses into Qm Degree 2 (docs/qm_geometry.md). At Degree 1, an even root n \ge Q(4) of a negative scalar is intercepted deterministically: if the calling DCC profile declares a supporting frame, the gateway routes the metered promotion to Layer 3; otherwise it emits RECEIPT_DIMENSIONAL_BUDGET_EXHAUSTED with reason code FRAME_SIGNATURE_INSUFFICIENT. No heuristic fallback exists.

H4: V. The Chiral Optimization Duality
Powers and roots form a Chiral Morphism Pair along the causal execution arrow:
- Forward powers collapse geometric grades (I_2^2 \to -Q(1)) and compute in feed-forward polynomial steps. Small integer powers disfavor memory lookups (\Delta W_{\mathrm{compute}} \le \Delta W_{\mathrm{lookup}}), executing directly on ALUs.
- Inverse roots expand geometric grades (\sqrt{\ominus Q(1)} \to I_2) and require bounded Diophantine lattice searches (\Delta W_{\mathrm{compute}} \gg \Delta W_{\mathrm{lookup}}). Roots cross the thermodynamic breakeven threshold immediately, prioritizing sub-graph reuse via q-reuse.

H3: 5.6. Discrete Factorial and Combinatorial Morphisms

The factorial operator is formalized in \mathrm{Qm} as an iterative discrete monoidal fold over the positive integer monoid (\mathcal{Q}^+, \times, Q(1)) under Node Class 4 (OP_FOLD):

n! = \prod_{j=1}^n j

H4: I. Boundary Identifications and \mathcal{O}(1) Bypasses
The factorial morphism exhibits three canonical identity states in the range Q(0) \dots Q(2):
- Empty Product Identity (Q(0)!): In monoid theory, an empty product evaluates to the multiplicative identity Q(1). Handled in \mathcal{O}(1) via q-bypass, emitting RECEIPT_BYPASS_IDENTITY at zero metric cost (\Delta W = 0).
- Unit Invariant (Q(1)!): Evaluates identically to Q(1). Bypassed in \mathcal{O}(1) via Q(\text{bypass}) (\Delta W = 0).
- Fixed-Point Invariant (Q(2)!): 2 is the unique positive integer satisfying n! = n. Recognized as an algebraic fixed-point identity in OP_REWRITE, bypassing execution in \mathcal{O}(1) (\Delta W = 0).

H4: II. Thermodynamic Optimization Boundary
Under the Metric Existential Invariant, optimization morphisms adhere to strict thermodynamic breakeven bounds:
- Prohibition of Q(\text{reuse}) for n \le 2: For Q(0), Q(1), and Q(2), computational cost is zero (direct bypass). Initiating a 256-bit cryptographic hash lookup across the Memory Wall consumes dynamic bus power (\mathcal{O}(C V^2 f)) that exceeds execution cost. Q(\text{reuse}) is strictly forbidden for n \le 2.
- Admission of Q(\text{reuse}) for n \ge 3: For k \ge 3, factorial computation satisfies Q(k)! = Q(k) \times Q(k-1)!. Because evaluating (k-1)! requires cumulative multiplications, the runtime retrieves the existing hash commitment of (k-1)! via Q(\text{reuse}), executing a

single multiplication and avoiding redundant re-computation across the Memory Wall.

H4: III. Non-Existence of Negative Factorials
The factorial recurrence relation is given by:

(n - 1)! = \frac{n!}{n}

Extending this relation to negative integers requires dividing by local genesis origin Q(0):

(-1)! = \frac{0!}{0} = \frac{Q(1)}{Q(0)}

\forall k \in \mathcal{Q}^+, \quad (-k)! = \frac{Q(1)}{Q(0) \times \prod_{j=1}^{k-1} (-j)}

Because division by origin Q(0) is an absolute constitutional prohibition in \mathrm{Qm} (§5.3), and actual infinite poles are non-realizable under Axiom 2, the factorial of a negative integer cannot exist. Any attempt to compute (\ominus k)! is intercepted deterministically at the SIMEMP Gateway, emitting RECEIPT_DIVISION_BY_ZERO_REJECTED. This is a constructive impossibility proof; per §5.3 the terminal status recorded is within the closed set (division rejection), and the argument — not a bounded-search outcome — is what licenses the impossibility claim.

H3: 5.7. Canonical Negative Rationals [v1.3.0 — CANDIDATE]

[Review Integration QO-4, part 3] Negative rationals require no separate ring construction. A negative rational is canonically the composition of an orientation \sigma \in \{\ominus, \odot, \oplus\} (§3.1) with a coprime positive rational pair:

\langle \sigma, \, p/q \rangle \quad \mathrm{with} \quad (p, q) \in \mathbb{N}_{\le H_{\mathrm{max}}} \times \mathbb{N}_{\le H_{\mathrm{max}}}, \quad q \gt Q(0), \quad \mathrm{gcd}(p, q) = Q(1)

where both p and q are artifacts constructed per §3.6. The GCD is computed exactly by the Euclidean division of §5.3 (iterated remainder sequence), so coprimality is enforced at formation time, not asserted. Any operation producing a non-reduced pair re-canonicalizes it before hashing: one commitment exists per rational value (content-addressed identity). Signed zero is structurally unreachable: p = Q(0) collapses the pair to the origin form \odot\, Q(0) (§3.1). Negative rationals constructed this way complete the bridge \mathbb{N}_{\le H_{\mathrm{max}}} \to \mathbb{Q} inside the artifact model: orientation supplies the sign, OP_RATIO pairs supply the magnitude, and the zoom/remainder telescope (§4) supplies the projective evaluation — no float, no imported primitive, no unbounded denominator (height ceiling H_{\mathrm{max}} applies to the reduced pair).

H2: 6. Relational Decidability and Positional Ordering

Ordering on dimension d is positional, evaluated over bounded constructive enclosures.

[v1.3.0 — CANDIDATE, Review Integration Segment 2 Point 4] The enclosure of the tuple \langle \sigma_A, q_A, z, r_A, d \rangle at zoom z is sign-aware and is determined by the typed form of the residual (§3.4). The earlier interval form I(A) = [q_A \cdot \delta_z + r_A, \, q_A \cdot \delta_z + r_A + \epsilon_z] is retired: it embedded the exact conserved residual inside an enlarged uncertainty width (double-counting) and relied on an undefined symbol \epsilon_z. The corrected enclosures are:

Collapsed residual (§3.4 form 1) — the value is exact:

I_z(A) = \big[\, \sigma_A \cdot (q_A \cdot \delta_z + r_A), \; \sigma_A \cdot (q_A \cdot \delta_z + r_A) \,\big] \qquad (\text{point interval})

Residual commitment handle (§3.4 form 2) — the value is enclosed within one zoom cell:

$$I_z(A) = \begin{cases}
\big[\, q_A \cdot \delta_z, \; (q_A + 1) \cdot \delta_z \big) & \mathrm{if} \; \sigma_A = \oplus, \; r_A \neq Q(0) \\[4pt]
\big(-(q_A + 1) \cdot \delta_z, \; -q_A \cdot \delta_z \big] & \mathrm{if} \; \sigma_A = \ominus, \; r_A \neq Q(0) \\[4pt]
\big[\, Q(0), \; Q(0) \,\big] & \mathrm{if} \; \sigma_A = \odot
\end{cases}$$

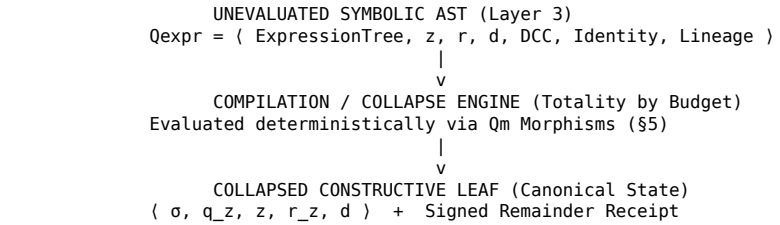
The positional order relation is then:

$$\mathrm{Rel}(A, B) = \begin{cases}
A \lt B & \mathrm{if} \; \sup I_z(A) \lt \inf I_z(B) \\
A \gt B & \mathrm{if} \; \inf I_z(A) \gt \sup I_z(B) \\
A \equiv B & \mathrm{if} \; \text{both residuals collapsed and } \mathcal{H}(A) = \mathcal{H}(B) \text{ after §4.3 normalization} \\
\mathrm{Indeterminate} & \mathrm{if} \; I_z(A) \cap I_z(B) \neq \emptyset \;\text{and at least one residual is an unexpanded handle}
\end{cases}$$

When intervals overlap such that inequality cannot be constructively decided at zoom z, \mathrm{Qm} forbids heuristic tie-breaking. It emits the deterministic receipt ORDERING_INDETERMINATE_AT_ZOOM. To resolve ordering, the caller must allocate additional compute

budget to deepen zoom resolution (z \to z + \Delta z), which expands the handle residuals (§3.4 form 2) through the metered cascade of §4.1 — narrowing the enclosures by construction, never by estimation.

H2: 7. Interaction with Qexpr and Derivation of Higher Mathematics



H3: 7.1. The Boundary Between Qexpr and Qm

- Qexpr (Symbolic CAE-DAG): Encapsulates unevaluated algebraic expressions, rational functions, and multi-step operator trees. It carries the symbolic structure before execution.
- \mathrm{Qm} (Evaluation Engine & Leaf): Defines the concrete algebraic reduction rules that collapse Qexpr trees into discrete \langle \sigma, q_z, z, r_z, d \rangle tuples at compilation time (qnlang \to qnir). The residual commitment handle of §3.4 (form 2) is the sanctioned exception path by which a Layer-2 leaf references an unresolved Layer-3 generator — as a commitment, resolved only through metered OP_ZOOM crossings (§4.1, §6).

H3: 7.2. Derivability of Real-World Mathematics

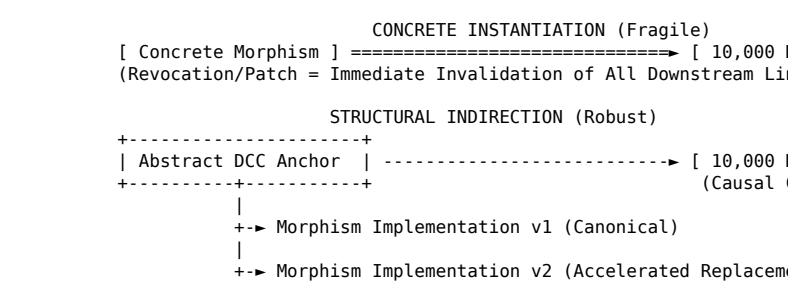
All computable real-world mathematics derives systematically from this foundation:

1. Multi-dimensional Vector Spaces: Constructed by parameterizing orthogonal sets of axes \{d_1, d_2, \dots, d_n\}, giving rise to Clifford geometric algebras (\mathcal{C}\ell_{p,q}) without non-deterministic trigonometric float approximations (docs/qm_geometry.md).
2. Calculus Without Infinities: Replaces continuous infinitesimals (\epsilon \to 0) with discrete differences over lattice quanta \Delta X / \delta_z(d), yielding exact discrete exterior calculus (docs/qm_calculus.md).
3. Transcendental Constants (\pi, e): Represented as symbolic Qexpr continued fraction operators (docs/qexpr.md §4.3). Evaluated at zoom z, they emit an exact rational quotient q_z and a conserved residual r_z satisfying exact Diophantine bounds.
4. Arbitrary Natural and Rational Magnitudes: Constructed inside the artifact model by the radix-anchor morphism (§3.6) and the canonical rational pairs (§5.7) — closing the single-digit ceiling of Category A without importing unconstructed primitives.

H2: 8. Structural Indirection: Q(anchor) in Qm

To prevent cascading structural collapse across dependent mathematical workflows, high-centrality mathematical primitives are formalized as Q(anchor) entities (docs/_data/glossary/architecture.yml):

$$\chi(\alpha) = \frac{\lvert \mathcal{C}^+(\alpha) \rvert}{\lvert \mathcal{V} \rvert} \ge \Theta_{\chi}$$



1. Anchored Interfaces: Critical operations (the division constraint, sign reflection, canonical unit \delta_z, and the \kappa(z) resolution schedule of §3.3) expose abstract DCC contracts under structural-indirection.
2. Downstream Isolation: Workflows in higher layers depend upon the abstract contract, not the internal execution engine.
3. Agility Proofs: Upgrading an arithmetic solver or patching an algorithmic vulnerability emits a signed migration receipt without invalidating the causal lineage of historical calculations.

H2: 9. License and Invariant Lineage

This domain specification enforces the legal and operational conditions of the ssl governing the licensed-work:

1. paternity-reference: All derivative implementations, mathematical solvers, or compiled runtime specification files derived from \mathrm{Qm} must embed:

```
Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License
(SSL). Parent Repository: https://github.com/cdqn5249/cdqn
2. Anti-Patent Defense: Royalty-free rights terminate automatically if a licensee initiates
   patent litigation regarding \mathrm{Qm} arithmetic methods against the Author or community.
3. Passive Infrastructure Safe Harbor: \mathrm{Qm} constitutes neutral computational
   infrastructure. It exercises no editorial inspection over, and makes no proprietary claim
   upon, sovereign user content or Payloads.
```

Annex F — qm_geometry.md

Source: docs/qm_geometry.md *— 1.2.3, updated 2026-10-20. Rendered content per Annex Guide conventions N1–N6: 409 source lines, 515 rendered; 27,297 → 23,434 characters; source SHA-256* 428d7d20 a980dada fc3d1bd3 af771888 1c4446ef b5d9da5e 2e6dc7d1 002c5ad1*. Content fidelity verified mechanically at build time.*

```
[front matter] ------------------------------------------------------------
[front matter] title: Qm Geometry — Multi-Axial Frames and Geometric Algebra
[front matter] version: 1.2.3
[front matter] updated: 2026-10-20
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Constructive multi-dimensional spatial representation, Clifford
[front matter]   geometric algebras, float-free rotations, non-Euclidean subspaces,
[front matter]   and deferred symbolic collapse under SIMEMP constraints.
[front matter] terms_used: boc-policy, causal-arrow, cdqn, complexity-degree, computational-
[front matter]   consistency, compute-unit-u, dependencies-determinism, dimension-d,
[front matter]   existential-invariant, layer-0, layer-1, licensed-work, local-first,
[front matter]   metric-envelope, metric-exhaustion, no-implicit-rule, operational-
[front matter]   agility, paternity-reference, payload, qexpr, qm, qn, qs, receipt,
[front matter]   remainder-r, simemp, simemp-gateway, ssl, terminal-exactness,
[front matter]   universal-envelope, zoom-z
[front matter] ------------------------------------------------------------


H1: Qm Geometry — Multi-Axial Frames and Geometric Algebra


Field | Specification
Document Title | Qm Geometry — Multi-Axial Frames and Geometric Algebra
Version | 1.2.3
Last Updated | 2026-10-20 (Bao Loc, Vietnam)
Author | Christophe Duy Quang Nguyen
License | Scaling Source License (SSL) 1.0
Status | Canonical Domain Specification — Category B/D (Complexity Degree 2)


H2: Normative References

The following documents establish the physical, structural, and legal constraints governing this
specification. If a technical conflict arises, simemp.md governs; if a structural conflict
arises, abstractionLayers.md governs; if a legal conflict arises, LICENSE.md governs. References
are deliberately version-unpinned: each target document's front matter carries its own
authoritative version, so this specification never duplicates drift-prone patch state.


Document | Role | Target
docs/simemp.md | Constitutional constraints, thermodynamics, and Dependencies Determinism |
simemp.html
docs/abstractionLayers.md | Layer architecture and SIMEMP Gateway validation |
abstractionLayers.html
docs/qnPrimitive.md | Universal Envelope, operational axioms, and Optimization Morphisms |
qnPrimitive.html
docs/q0_q1.md | Primary genesis origin Q(0) and first unit Q(1) | q0_q1.html
docs/q2_q9.md | Single-digit secondary DCC anchors and single-digit spectrum | q2_q9.html
docs/qm.md | Constructive numeric leaf substrate and Diophantine division constraints | qm.html
docs/qm_calculus.md | Discrete Exterior Calculus, continued fractions, and invariant dynamics |
qm_calculus.html
docs/qexpr.md | Layer 3 Content-Addressed Expression DAG container | qexpr.html
docs/qs.md | Constructive relational intent and Intent Contracts | qs.html
LICENSE.md | Scaling Source License 1.0 governing the Licensed Work | LICENSE.md


H2: 1. Epistemic Stance and Dimensional Generalization

The foundational operational conjecture of the qn framework (docs/simemp.md §1) asserts that a
discrete number system can abstract any computable physical phenomenon under simemp constraints.
In docs/qm.md, scalar arithmetic was established along the 1D genesis axis d_1. However,
physical reality and distributed network topographies are inherently multi-dimensional.
```

Classical geometry relies on continuous coordinate spaces (\mathbb{R}^n) and irrational
trigonometric functions (\sin, \cos), introducing platform-dependent IEEE 754 rounding
approximations that violate Axiom 10 of docs/qnPrimitive.md.

qm Geometry generalizes the 1D leaf state to n-dimensional space via Clifford Geometric Algebra
(\mathcal{C}\ell_{p,q}). Space, direction, area, volume, and rotation are constructed as exact,
discrete, rational multivector blades without continuous limits, actual infinities, or
transcendental float approximations.

Every geometric operation in \mathrm{Qm} satisfies dependencies-determinism:

H(S_t \mid \mathcal{D}(S_t)) = 0

Operational validity is verified empirically via computational-consistency. No unmeasured
coordinate entity may cross an abstraction boundary or simemp-gateway.

H2: 2. Orthogonal Coordinate Frames and Metric Signatures

In strict accordance with the no-implicit-rule, multi-dimensional space cannot be assumed *a
priori*. It is generated inductively from genesis origin Q(0) and first unit Q(1).

H3: 2.1. The Orthogonal Basis Set and Inductive Dimensional Genesis

Under Axiom 4 (Local Genesis Dependency), every coordinate axis must trace an unbroken causal
lineage along the causal-arrow back to local-first genesis origin Q(0)_N. Coordinate frames
cannot be introduced as detached ambient conventions.

```
                  INDUCTIVE DIMENSIONAL GENESIS
 Q(0)_N --► Q(1) --► Axis d1 (Basis Vector e_1)
                      |
                      v Adjoin Orthogonal Step D_step
                     Axis d2 (Basis Vector e_2: e_2 · e_1 = Q(0))
                      |
                      v Adjoin Orthogonal Step D_step
                     Axis d_k (Basis Vector e_k: e_k · e_i = Q(0), ∀i < k)
```

H4: Primary Genesis Axis (d_1)
Formally instantiated in layer-1 by the minimal directed state transition from local origin
Q(0)_N to first unit Q(1) along the canonical initial dimension, calibrating the Abstract
Compute Unit U (docs/q0_q1.md §4):

\vec{e}_1 \equiv \langle \oplus, \, 1, \, z=0, \, r=Q(0), \, d_1 \rangle

Here, the vector symbol \vec{e} denotes a directional spatial basis blade (from the unit vector
convention *Einheitsvektor*), completely distinct from scalar transcendental constants.

H4: Inductive Dimensional Extension (\mathcal{D}_{\mathrm{step}})
For any spatial frame of dimension k \ge 1, the successor orthogonal axis d_{k+1} is
instantiated constructively by adjoining an orthogonal basis generator \vec{e}_{k+1}:

\vec{e}_{k+1} \equiv \langle \oplus, \, 1, \, z=0, \, r=Q(0), \, d_{k+1} \rangle

subject to explicit Diophantine orthogonality constraints:

\forall i \in \{1, \dots, k\}, \quad \vec{e}_{k+1} \cdot \vec{e}_i = Q(0)

\vec{e}_{k+1} \cdot \vec{e}_{k+1} = \eta_{k+1, k+1} Q(1)

\mathrm{Parent}(\vec{e}_{k+1}) = \{ \vec{e}_k, \, Q(1) \}

By induction, a frame of dimension n is defined by the complete discrete set of n mutually
orthogonal dimensional axes:

\mathcal{D}_{\mathrm{frame}} = \{ d_1, \, d_2, \, \dots, \, d_n \}

Zero implicit dimensional assumptions exist; each axis preserves a deterministic causal
parentage back to Q(0)_N.

H3: 2.2. Metric Signature (\eta)

The quadratic form of the space is governed by an explicit discrete diagonal metric signature
\eta = (p, q, s) where p + q + s = n:

$$\vec{e}_i \cdot \vec{e}_j = \begin{cases}
+Q(1) & \mathrm{if} \; i = j \le p \quad (\text{Space-like}) \\
-Q(1) & \mathrm{if} \; p \lt i = j \le p + q \quad (\text{Time-like}) \\
Q(0) & \mathrm{if} \; p + q \lt i = j \le n \quad (\text{Degenerate / Null}) \\
Q(0) & \mathrm{if} \; i \neq j \quad (\text{Orthogonality})
\end{cases}$$

- Euclidean n-Space: \mathcal{C}\ell_{n, 0} where \eta = (+, +, \dots, +).
- Space-Time Algebra (Minkowski): \mathcal{C}\ell_{1, 3} or \mathcal{C}\ell_{3, 1}, providing spatial foundations for physical modeling.

H3: 2.3. Non-Euclidean Topologies and Subspace Inclusions

\mathrm{Qm} Geometry abstracts non-Euclidean geometries without continuous Riemannian metrics, dense floating-point tensors, or differential singularities:

Geometry Subspace | Ambient Signature | Defining Diophantine Constraint | Geodesic Rotor Morphism
Spherical (\mathbb{S}^n) | \mathcal{C}\ell_{n+1, 0} (Euclidean) | X \cdot X = +Q(1) | Planar bivector rotors (R \vec{v} R^{\dagger})
Hyperbolic (\mathbb{H}^n) | \mathcal{C}\ell_{n, 1} (Minkowski) | X \cdot X = -Q(1), \quad X_t \gt Q(0) | Hyperbolic boost rotors (B \vec{v} B^{\dagger})
Conformal (CGA) | \mathcal{C}\ell_{n+1, 1} (Minkowski null) | X \cdot X = Q(0) (Null cone) | Linear conformal rotors over \{ e_{\mathrm{origin}}, e_{\mathrm{horizon}} \}
Curved Manifolds | Simplicial triangulation | Bivector holonomy \Omega = \prod R_{\mathrm{loop}} | Discrete Regge deficit angles without continuous tensors

H4: I. Spherical Subspace (\mathbb{S}^n)
Modeled as the locus of vectors satisfying the quadratic constraint X \cdot X = +Q(1) in Euclidean space \mathcal{C}\ell_{n+1, 0}. Great-circle geodesics are evaluated via rational bivector rotors without trigonometric projections.

H4: II. Hyperbolic Subspace (\mathbb{H}^n)
Modeled via the Weierstrass hyperboloid within Minkowski spacetime \mathcal{C}\ell_{n, 1} under constraint X \cdot X = -Q(1), with temporal coordinate X_t \gt Q(0). Parallel transport is governed by hyperbolic boost rotors (\vec{e}_i \wedge \vec{e}_t)^2 = +Q(1).

H4: III. Conformal Geometric Algebra (CGA) and Null Basis Decoupling
Conformal and projective transformations are mapped to linear orthogonal rotor reflections by adjoining two discrete null basis vectors:
- \vec{e}_{\mathrm{horizon}}: Parameterizes the operational task horizon ceiling Qn_{\max} declared in the metric envelope.
- \vec{e}_{\mathrm{origin}}: Parameterizes the spatial coordinate origin within the Clifford frame.

\vec{e}_{\mathrm{origin}} \cdot \vec{e}_{\mathrm{origin}} = Q(0), \quad
\vec{e}_{\mathrm{horizon}} \cdot \vec{e}_{\mathrm{horizon}} = Q(0), \quad
\vec{e}_{\mathrm{origin}} \cdot \vec{e}_{\mathrm{horizon}} = -Q(1)

The Invariant Disjunction: Crucially, the basis vector \vec{e}_{\mathrm{origin}} is a grade-1 vector blade in the vector space V. It is strictly decoupled from the local scalar genesis origin Q(0), which resides in the underlying discrete scalar ring \mathcal{Q}_0. Conflation between vector basis elements and ring zero scalars is strictly prohibited under the no-implicit-rule.

H4: IV. Discrete Curved Manifolds
General curved spaces are represented as simplicial complexes (discrete Regge calculus). Curvature is measured not by continuous Ricci tensors, but by discrete deficit angles around codimension-2 hinges.

H2: 3. The Clifford Geometric Product Without Approximations

In \mathrm{Qm} Geometry, the multiplication of two vectors \vec{u} and \vec{v} is governed by the Clifford Geometric Product:

\vec{u} \vec{v} = \vec{u} \cdot \vec{v} + \vec{u} \wedge \vec{v}

Product Component | Grade | Algebraic Symmetry | Geometric Role | Evaluation Engine
Inner Product (\vec{u} \cdot \vec{v}) | Grade 0 (Scalar) | Symmetric: \frac{1}{2}(uv + vu) | Metric contraction / projection | \mathrm{Qm} \mathcal{R}-algebra scalar leaf
Wedge Product (\vec{u} \wedge \vec{v}) | Grade 2 (Bivector) | Anti-symmetric: \frac{1}{2}(uv - vu) | Oriented planar area patch | Canonical bivector blade
Geometric Product (\vec{u} \vec{v}) | Multivector | Graded associative sum | Total spatial interaction | Complete Clifford blade set

H3: 3.1. The Discrete Inner Product

Let vectors \vec{u} and \vec{v} be expressed along the orthogonal basis:

\vec{u} = \sum_{k=1}^n u_k \vec{e}_k, \quad \vec{v} = \sum_{k=1}^n v_k \vec{e}_k

where each component is a discrete tuple evaluated at zoom z. The inner product contracts to an exact scalar:

\vec{u} \cdot \vec{v} = \sum_{k=1}^n \eta_{kk} (u_k \times v_k)

The operation executes entirely via the discrete \mathrm{Qm} \mathcal{R}-algebra (docs/qm.md §4), returning an exact scalar leaf with a conserved remainder-r.

H3: 3.2. The Discrete Wedge Product

The outer product spans an oriented planar surface area:

\vec{u} \wedge \vec{v} = \sum_{1 \le i \lt j \le n} (u_i v_j - u_j v_i) (\vec{e}_i \wedge \vec{e}_j)

Under the no-implicit-rule, anti-commutativity and nilpotency are exact:

\vec{e}_i \wedge \vec{e}_j = - (\vec{e}_j \wedge \vec{e}_i)

\vec{e}_i \wedge \vec{e}_i = Q(0)

Collinear vectors span zero area without precision loss.

H2: 4. Graded Multivectors and Blade Decomposition

A general geometric artifact M \in \mathcal{C}\ell_{p, q} is a graded multivector spanning 2^n basis elements:

M = \sum_{k=0}^n \langle M \rangle_k = \langle M \rangle_0 + \langle M \rangle_1 + \langle M \rangle_2 + \dots + \langle M \rangle_n

Grade | Canonical Entity | Geometric Object | Canonical Basis Representation
0 | Scalar | Metric magnitude / distance | Q(1)
1 | Vector | Directed spatial segment | \vec{e}_i
2 | Bivector | Oriented planar area patch | \vec{e}_i \wedge \vec{e}_j
3 | Trivector | Oriented spatial volume cell | \vec{e}_i \wedge \vec{e}_j \wedge \vec{e}_k
n | Pseudoscalar | Maximum oriented volume element | I = \vec{e}_1 \wedge \vec{e}_2 \wedge \dots \wedge \vec{e}_n

H3: 4.1. Canonical Multivector State Tuple

Every multivector artifact in \mathrm{Qm} carries an explicit, finite representation conforming to the universal-envelope:

M \equiv \Big\langle \sigma_M, \, \alpha_{\mathcal{B}}, \, z, \, r_M, \, \mathcal{D}_{\mathrm{frame}} \Big\rangle

Tuple Component | Symbolic Form | Operational Invariant
Basis Blade Set | \mathcal{B} = \bigcup_{k=0}^n \binom{\mathcal{D}_{\mathrm{frame}}}{k} | Complete canonical set of 2^n blade generators.
Blade Coefficients | \alpha_{\mathcal{B}} = \{ q_B \in \mathbb{N} \} | Discrete scalar magnitudes at zoom z for each blade B \in \mathcal{B}.
Residual Vector | r_M = \{ r_B \} | Vector of conserved remainders satisfying 0 \le \lvert r_B \rvert \lt \delta_z(d).
Coordinate Frame | \mathcal{D}_{\mathrm{frame}} | Declared set of n orthogonal axes \{ d_1, \dots, d_n \}.

H3: 4.2. Pseudoscalar Duality

The unit pseudoscalar I = \vec{e}_1 \wedge \vec{e}_2 \wedge \dots \wedge \vec{e}_n provides discrete Hodge dual mapping without matrix inversion:

M^* = M I^{-1}

mapping grade-k blades to grade-(n-k) orthogonal complements.

H2: 5. Float-Free Rotations: Rational Rotors and Double Reflections

In conventional graphics and physics simulations, rotations are parameterized via Euler angles or trigonometric unit quaternions:

R = \cos(\theta/2) - I \sin(\theta/2)

Because \cos(\theta) and \sin(\theta) are transcendental irrationals for almost all rational angles, continuous engines round them to 32-bit or 64-bit IEEE 754 floats. This introduces metric drift: after 10^6 rotations, \lVert R \rVert^2 \neq 1, forcing artificial normalization cycles.

\mathrm{Qm} Geometry solves this via Cartan-Dieudonné Double Reflections and Rational Cayley-Klein Rotors.

Rotation Method | Mathematical Formulation | Precision and Drift Invariant
Classical Floating Quaternion | R = \cos(\theta/2) - I \sin(\theta/2) | Lossy; accumulates metric drift (\lVert R \rVert^2 \neq 1); violates Axiom 10.
Cartan-Dieudonné Double Reflection | R = \vec{b} \vec{a} = \vec{b} \cdot \vec{a} + \vec{b} \wedge \vec{a} | Exact; composite of two discrete planar reflections.
Rational Cayley Rotor | R = \frac{Q(1) - B}{Q(1) + B} = \frac{(m^2 - n^2) + 2mn B}{m^2 + n^2} | Strict unitary norm \lVert R \rVert^2 \equiv Q(1); zero trigonometric drift.
Projective Angle Cascade | \theta_z = \theta_{\mathrm{discrete}} + r_{\theta} | Projective rational convergence via continued fractions; r_{\theta} conserved.

H3: 5.1. The Rotor Morphism

A rotation of vector \vec{v} in a plane is executed by two successive reflections across non-parallel unit vectors \vec{a} and \vec{b}:

R = \vec{b} \vec{a} = \vec{b} \cdot \vec{a} + \vec{b} \wedge \vec{a}

\vec{v}' = R \vec{v} R^{\dagger}

where R^{\dagger} = \vec{a} \vec{b} is the reverse rotor.

H3: 5.2. Rational Cayley Parameterization (Float-Free Rotors)

To rotate by an exact rational angle without evaluating trigonometric series, \mathrm{Qm} parameterizes rotors via rational bivectors B \in \langle M \rangle_2:

R = \frac{Q(1) - B}{Q(1) + B}

For any planar rotation in plane \vec{e}_1 \wedge \vec{e}_2, the rotor coefficients are generated by Diophantine Pythagorean triplets (m, n) \in \mathbb{N}^2:

R = \frac{(m^2 - n^2) + 2mn (\vec{e}_1 \wedge \vec{e}_2)}{m^2 + n^2}

The norm of the rotor is identically unity by construction:

R R^{\dagger} = \frac{(m^2 - n^2)^2 + (2mn)^2}{(m^2 + n^2)^2} = \frac{(m^2 + n^2)^2}{(m^2 + n^2)^2} \equiv Q(1)

Metric drift is identically zero. The rotor can be applied 10^9 consecutive times without departing from the unit manifold.

H3: 5.3. Projective Cascade for Arbitrary Target Angles

When an arbitrary target angle \theta is externally supplied, \mathrm{Qm} does not invoke floating-point libraries. It evaluates the continued fraction expansion of \tan(\theta/4), generating a sequence of rational rotors R_z at zoom level z:

\theta_z = \theta_{\mathrm{discrete}} + r_{\theta}

The angular remainder r_{\theta} is preserved in the metric envelope. If r_{\theta} = Q(0), the rotation is algebraically exact, emitting a terminal receipt confirming terminal-exactness.

H2: 6. Discrete Metric Norms and Diophantine Distance Solvers

Distance between two spatial points A and B along frame \mathcal{D}_{\mathrm{frame}} is evaluated as the norm of the displacement vector \vec{v} = B - A:

D^2 = \vec{v} \cdot \vec{v} = \sum_{k=1}^n \eta_{kk} \Big((q_{B,k} - q_{A,k}) \cdot \delta_z \Big)^2

H3: 6.1. The Diophantine Square-Root Constraint

In \mathrm{Qm}, distance evaluation resolves through an exact Diophantine square constraint:

D^2 = (q_D \cdot \delta_z)^2 + r_D

subject to integer quotient and bounded remainder criteria:

q_D \in \mathbb{N}, \quad 0 \le \lvert r_D \rvert \lt (2 q_D \cdot \delta_z^2 + \delta_z^2), \quad \sigma = \oplus

If r_D = Q(0), the distance is an exact Pythagorean integer or rational, terminating with zero residual. If r_D \neq Q(0), r_D is preserved as the exact argument for subsequent scale expansion (z \to z + 1).

H2: 7. Deferred Symbolic Normalization and Bounded Collapse Engine

A governing distinction between \mathrm{Qm} Geometry and classical numerical engines is the rejection of eager evaluation (docs/abstractionLayers.md §6.3).

In classical systems, every intermediate geometric operation eagerly rounds to machine floats, accumulating precision drift and repeatedly dissipating Landauer heat (W \ge k_B T \ln 2). \mathrm{Qm} Geometry executes geometric workflows through a Three-Stage Lifecycle:

Execution Stage | Abstraction Layer | Operational Mechanics | Thermodynamic Work Bound
Stage 1: Composition | Layer 3 (qexpr CAE-DAG) | Multivector products and rotor chains authored as content-addressed DAGs. Zero rounding. | W = 0 (Information preserved)
Stage 2: Normalization | Layer 2/3 (Rewrite Engine) | Contract basis metrics, annihilate nilpotencies (\vec{e}_i \wedge \vec{e}_i = Q(0)), cancel inverse rotors (R R^{\dagger} = Q(1)). | W = 0 (Isomorphic & reversible)
Stage 3: Collapse | Layer 1/2 (\mathrm{Qm} Leaf / Flat IR) | Project onto zoom z only upon boundary demand; compute quotient q_z and remainder r_z. | W = 1 \cdot W_{\mathrm{collapse}} \ll k \cdot W_{\mathrm{eager}}

H3: 7.1. Thermodynamic Landauer Minimization Proof

Let a geometric sequence consist of k consecutive rotor transformations:
1. Eager Evaluation: Dissipates W_{\mathrm{eager}} \ge k \cdot (k_B T \ln 2) and compounds k unmeasured remainder vectors across the layer-0 substrate.
2. Deferred Collapse: Under the Best of Choices (BOC) Policy, symbolic structure is maintained as a Content-Addressed Expression DAG (Term DAG) with cryptographic hash-consing. Reversible symbolic reduction preserves information without entropy production (W_{\mathrm{Stage\,2}} = 0). Numerical collapse occurs once at the boundary:

$$W_{\mathrm{deferred}} = 1 \cdot W_{\mathrm{collapse}} \ll k \cdot W_{\mathrm{collapse}}$$

H3: 7.2. Coupling to Optimization Morphisms and Causal Priority

Geometric pipelines interface with higher-order Category B optimization morphisms governed by strict causal precedence and thermodynamic breakeven constraints (docs/qnPrimitive.md §5.9):

$$Q(\text{bypass}) \prec Q(\text{reuse}) \prec \text{Direct Compute}$$

1. Identity Pre-Check (Q(\text{bypass})):
   Before initiating memory traversal, the engine inspects operand pairs in \mathcal{O}(1) register time. If an algebraic identity is recognized—such as rotor cancellation (R R^\dagger \to Q(1)) or outer product nilpotency (\vec{e}_i \wedge \vec{e}_i \to Q(0))—the gateway emits RECEIPT_BYPASS_IDENTITY, bypassing both computation and memory retrieval at zero work cost (\Delta W = 0).
2. Thermodynamic Breakeven for Sub-Graph Reuse (Q(\text{reuse})):
   Querying a hash commitment across memory hierarchies incurs physical bus dissipation (\mathcal{O}(C V^2 f)). Under the Metric Invariant, Q(\text{reuse}) is admissible if and only if the computational work of re-evaluating the multivector product exceeds lookup dissipation:

$$\mathrm{Admit}\big(Q(\text{reuse})\big) \iff \Delta W_{\mathrm{compute}} \gt \Delta W_{\mathrm{lookup}} + \epsilon_{\mathrm{overhead}}$$

For complex Clifford rotor sandwiches (R \vec{v} R^\dagger) or high-dimensional grade contractions, Q(\text{reuse}) references existing immutable commitment hashes, eliminating redundant allocation across the Memory Wall. For trivial 1D scalar additions, direct computation executes immediately without hash-table overhead.

H2: 8. SIMEMP Governance and Metric Exhaustion

Operating in multi-dimensional space consumes physical computational resources governed by the Metric Existential Invariant (docs/simemp.md §3.1), balancing algorithmic Efficiency against structural Modularity.

H3: 8.1. Complexity Scaling
- Vector addition: \mathcal{O}(n) discrete operations.
- Geometric product: \mathcal{O}(2^n) blade multiplications.
- Rotor sandwich transformation (R v R^{\dagger}): \mathcal{O}(n \cdot 2^n) operations.

H3: 8.2. Dimensional Ceilings and Budget Halts

Dimensional growth is bounded by the declared metric-envelope:

$$n \le \mathrm{Dim}_{\mathrm{max}}$$

Typically n \le 4 for space-time kinematics; n \le 16 for high-dimensional semantic spaces in

```
Qs.

- If a composition of wedge products attempts to construct a grade exceeding n, the result
   collapses identically to zero (Q(0)) by the nilpotency of outer products, terminating in
   \mathcal{O}(1) work.
- If an algebraic transformation exceeds the declared metric budget, execution halts
   deterministically under Axiom 6, emitting an explicit budget exhaustion receipt:


Receipt Field | Specification
Receipt Identifier | RECEIPT_DIMENSIONAL_BUDGET_EXHAUSTED
Execution Status | BUDGET_EXHAUSTED
Frame Dimension | Bound n
Consumed Metric | Consumed work budget W (Q(1) units via metric-exhaustion)


H2: 9. License and Invariant Lineage

This domain specification enforces the legal and operational conditions of the ssl governing the
licensed-work across the cdqn network:

1. paternity-reference: All multi-dimensional solvers, Clifford algebra engines, rotor
   transformations, or compiled geometric kernels derived from this document must embed:
Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License
(SSL). Parent Repository: https://github.com/cdqn5249/cdqn
2. Anti-Patent Defense: Royalty-free rights terminate automatically if a licensee initiates
   patent litigation regarding \mathrm{Qm} geometric algorithms or float-free rotor methods
   against the Author or community.
3. Statutory Safe Harbor: \mathrm{Qm} Geometry operates as passive, neutral mathematical
   infrastructure. It exercises no editorial inspection over, and makes no claim upon, external
   graphical models, physical simulations, or sovereign Payloads.
```

Annex G — qm_calculus.md

Source: docs/qm_calculus.md *— 1.1.1, updated 2026-10-19. Rendered content per Annex Guide conventions N1–N6: 358 source lines, 406 rendered; 23,927 → 20,516 characters; source SHA-256* 869964ef 251c45e6 f59e2cc5 aed46f9a c50229ff b6bcb9eb 3cbd032d fa3e0026*. Content fidelity verified mechanically at build time.*

```
[front matter] ------------------------------------------------------------
[front matter] title: Qm Calculus — Discrete Exterior Calculus, Continued Fractions, and
[front matter]   Invariant Dynamics
[front matter] version: 1.1.1
[front matter] updated: 2026-10-19
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Canonical specification of constructive discrete calculus, Discrete
[front matter]   Exterior Calculus (DEC) on simplicial meshes, transcendental
[front matter]   continued fractions, and aperiodic lineage trajectories under SIMEMP
[front matter]   constraints.
[front matter] terms_used: anchor-cut, boc-policy, causal-arrow, cdqn, complexity-degree,
[front matter]   computational-consistency, compute-unit-u, dcc-profile, dependencies-
[front matter]   determinism, dimension-d, discrete-exterior-calculus, discrete-
[front matter]   exterior-derivative, discrete-hodge-star, existential-invariant,
[front matter]   layer-0, layer-1, licensed-work, local-first, metric-envelope,
[front matter]   metric-exhaustion, no-implicit-rule, open-core-invariant,
[front matter]   operational-agility, paternity-reference, payload, q-anchor,
[front matter]   q-bypass, q-reuse, q0, q1, qexpr, qm, qn, receipt, remainder-r,
[front matter]   security-envelope, simemp-gateway, ssl, terminal-exactness,
[front matter]   universal-envelope, zoom-z
[front matter] ------------------------------------------------------------


H1: Qm Calculus — Discrete Exterior Calculus, Continued Fractions, and Invariant Dynamics


Field | Specification
Document Title | Qm Calculus — Discrete Exterior Calculus, Continued Fractions, and Invariant
Dynamics
Version | 1.1.1
Last Updated | 2026-10-19 (Bao Loc, Vietnam)
Author | Christophe Duy Quang Nguyen
License | Scaling Source License (SSL) 1.0
Status | Canonical Domain Specification — Category D (Complexity Degree 3)


H2: Normative References

The following documents establish the physical, structural, and legal constraints governing this
```

specification. If a technical conflict arises, simemp.md governs; if a structural conflict arises, abstractionLayers.md governs; if a legal conflict arises, LICENSE.md governs.

Document | Role | Target
docs/simemp.md | Constitutional constraints, thermodynamics, and Dependencies Determinism | simemp.html
docs/abstractionLayers.md | Layer architecture and Complexity Degree Stratification | abstractionLayers.html
docs/qnPrimitive.md | Universal Envelope, operational axioms, and optimization primitives | qnPrimitive.html
docs/q0_q1.md | Primary genesis origin Q(0) and first unit Q(1) | q0_q1.html
docs/q2_q9.md | Single-digit secondary DCC anchors and single-digit spectrum | q2_q9.html
docs/qm.md | Constructive numeric leaf substrate and Diophantine division constraints | qm.html
docs/qm_geometry.md | Multi-axial frames, Clifford geometric algebra, and float-free rotations | qm_geometry.html
docs/qexpr.md | Layer 3 Content-Addressed Expression DAG container | qexpr.html
LICENSE.md | Scaling Source License 1.0 governing the Licensed Work | LICENSE.md

H2: 1. Epistemic Stance: Calculus Without Infinities

In classical mathematics, calculus relies on non-constructive, continuous foundations: the Cauchy–Weierstrass \epsilon-\delta limit, actual infinities (\mathbb{R}, \infty), and vanishing infinitesimals (dx \to 0). While analytically elegant, continuous calculus creates physical paradoxes: it allows mathematical equations to predict unphysical singularities (such as infinite fluid velocities in Navier–Stokes or infinite gravitational densities in black holes) that never occur in real physical substrates.

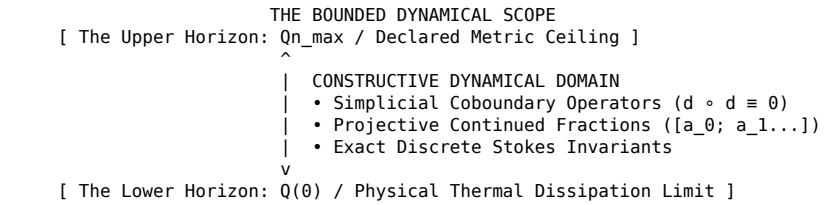
In the qn computational universe, continuous limits are discarded as physically unrealizable approximations. Under docs/simemp.md, physical computation is strictly finite and bounded by two real horizons:
1. The Upper Horizon (Qn_{\max}): The finite memory capacity, register width, and operational energy ceiling declared in the local metric-envelope.
2. The Lower Horizon (\delta_z(d) near Q(0)): The physical resolution quantum at Zoom z, below which state transitions dissipate into physical thermal noise bounded by Landauer's bound:

$$W \ge k_B T \ln 2$$

qm Calculus replaces continuous infinitesimals with Constructive Discrete Exterior Calculus (DEC) and Projective Continued Fraction Cascades, satisfying dependencies-determinism:

$$H(S_t \mid \mathcal{D}(S_t)) = 0$$

All operational derivations represent working hypotheses evaluated via computational-consistency. No unmeasured limit may cross a simemp-gateway, preserving Tier 1 Existential Invariants.



H2: 2. The Discrete Replacement of Continuous Limits

Classical continuous limits (\lim_{\Delta x \to 0}) are replaced by three constructive mechanisms that operate with zero rounding drift under the no-implicit-rule:

H3: 2.1. Scale-Lattice Refinement (z-Refinement)
Space and time are not continuous voids; they are partitioned by the discrete resolution quantum along dimensional axes (docs/qm.md §3.3):

$$\delta_z(d) = \frac{Q(1)_d}{\kappa(z)}$$

"Approaching a limit" does not mean shrinking an infinitesimal toward zero. It represents an explicit, metered Lattice Refinement Step (z \to z+1):
- Refinement requires burning an explicit metric budget in units of the Abstract Compute Unit U.
- When the declared metric budget expires, refinement terminates cleanly via metric-exhaustion.

H3: 2.2. Conserved Residual Telescoping
Classical series convergence silently truncates unmeasured trailing terms. In \mathrm{Qm} Calculus, convergence is an exact Diophantine cascade where the residual is strictly conserved:

$$r_z = \big(q_{z+1} \cdot \delta_{z+1}(d) \big) + r_{z+1}$$

Across arbitrary expansions, total metric information is preserved in the universal-envelope:

```
\mathcal{I}_{\mathrm{total}}(X_d) = \sum_{k=0}^{z} \big( q_k \cdot \delta_k(d) \big) + r_z

If r_z = Q(0), the system achieves terminal-exactness: the calculation reaches exact algebraic
closure, triggering immediate early halting with zero residual entropy (\Delta H = 0) and
emitting an exact receipt.


H2: 3. Discrete Exterior Calculus (DEC) on Simplicial Complexes

To model fields, waves, fluids, and electrodynamics without floating-point PDEs, \mathrm{Qm}
Calculus adopts Discrete Exterior Calculus (DEC) on orthogonal multi-axial frames
(docs/qm_geometry.md).

                     DEC OPERATOR MAPPING ON SIMPLICES
 Simplicial Primal Mesh K:            Dual Circumcentric Mesh ⋆K:
   0-Simplex (Vertex v) ------------► Dual n-Cell (Voronoi cell ⋆v)
   1-Simplex (Edge e)   ------------► Dual (n-1)-Face (Flux facet ⋆e)
   2-Simplex (Face f)   ------------► Dual (n-2)-Edge (Circulation ⋆f)
            |                                    ^
            v Discrete Exterior Derivative d     | Discrete Hodge Star ⋆
   (k+1)-Simplex Coboundary ----------------------+


H3: 3.1. Discrete Differential Forms as Cochains
A physical or geometric field is not a continuous function; it is a discrete cochain evaluated
on oriented k-simplices of a simplicial complex K:
- 0-Forms (Scalars): Potential fields, temperatures, and scalar charges evaluated on vertices.
- 1-Forms (Circulations): Electric fields (E) and fluid velocities (u) integrated along oriented
   edges.
- 2-Forms (Fluxes): Magnetic flux (B) and vorticity (\omega) evaluated across oriented planar
   faces with conserved residual r.


H3: 3.2. The Discrete Exterior Derivative (d / DCC_ANCHOR_DEC_D_v1)
The discrete derivative is the algebraic coboundary operator Discrete Exterior Derivative d
mapping k-forms to (k+1)-forms:

\langle \mathbf{d}\omega, \, \sigma_{k+1} \rangle \equiv \langle \omega, \, \partial
\sigma_{k+1} \rangle = \sum_{j=0}^{k+1} (-1)^j \langle \omega, \, [v_0, \dots, \hat{v}_j, \dots,
v_{k+1}] \rangle


H4: The Fundamental Topological Invariant:
Under the Best of Choices (BOC) Policy, \mathbf{d} satisfies the exact nilpotent identity:

\mathbf{d} \circ \mathbf{d} \equiv Q(0)

Because the boundary of a boundary is identically zero (\partial \partial \equiv \emptyset),
applying the exterior derivative twice vanishes identically. This invariant is anchored via
DCC_ANCHOR_DEC_D_v1 as a primary Q(anchor) governed by an explicit dcc-profile.


H3: 3.3. The Discrete Hodge Star Dual (DCC_ANCHOR_HODGE_v1)
The Discrete Hodge Star operator \star maps primal k-forms to dual (n-k)-forms across the
circumcentric dual mesh:

\star \omega = \sum_{\sigma_k \in K} \frac{\lvert \star \sigma_k \rvert}{\lvert \sigma_k \rvert}
\langle \omega, \, \sigma_k \rangle \, (\star \sigma_k)

Dual volume ratio quotients:

\frac{\lvert \star \sigma_k \rvert}{\lvert \sigma_k \rvert}

represent exact geometric ratios held symbolically inside the qexpr container via OP_RATIO.


H3: 3.4. The Discrete Laplace–Beltrami Operator (Δ / DCC_ANCHOR_LAPLACE_v1)
The Laplacian governing diffusion, wave propagation, and electrostatic potential is constructed
constructively:

\Delta \equiv \mathbf{d}\delta + \delta\mathbf{d}, \quad \text{where } \delta = (-1)^{n(k-1)+1}
\star \mathbf{d} \star


H3: 3.5. Exact Discrete Stokes' Theorem
Stokes' Theorem is not an asymptotic limit; it is an exact Diophantine identity on discrete
meshes:

\sum_{K} (\mathbf{d}\omega)(K) \equiv \sum_{\partial K} \omega(\partial K)

Truncation error is zero by construction. Conservation laws (mass, charge, vorticity) hold
identically without numerical dispersion.
```

H2: 4. Dynamical Q(anchor) Spectrum: Transcendental Continued Fractions

Transcendental numbers cannot exist as static Layer 1 digits in layer-1. In \mathrm{Qm} Calculus, they are anchored as Deterministic Projective Continued Fraction Generators inhabiting the Layer 3 qexpr CAE-DAG container (docs/qexpr.md §4.3) and functioning as foundational Q(anchor) nodes.

```
                THE DYNAMICAL Q(anchor) SPECTRUM IN Qm CALCULUS
 +-------------------------------------------------------------------------+
 | 1. TRANSCENDENTAL INVARIANT ANCHORS                                     |
 |    • DCC_ANCHOR_PI_v1   : Circular ratio π, wave periods, harmonic phase    |
 |    • DCC_ANCHOR_TAU_v1  : Full-turn metric constant τ = 2π                  |
 |    • DCC_ANCHOR_E_v1    : Natural base e, solution to df/dx = f             |
 |    • DCC_ANCHOR_LN2_v1  : Natural logarithm ln(2), Landauer dissipation    |
 +-------------------------------------------------------------------------+
 | 2. DISCRETE EXTERIOR CALCULUS (DEC) OPERATOR ANCHORS                    |
 |    • DCC_ANCHOR_DEC_D_v1    : Discrete Exterior Derivative d (d² ≡ 0)       |
 |    • DCC_ANCHOR_HODGE_v1    : Discrete Hodge Star Dual ⋆ (Metric contraction|
 |    • DCC_ANCHOR_LAPLACE_v1  : Discrete Laplace–Beltrami Operator Δ = dδ + δd|
 +-------------------------------------------------------------------------+
 | 3. DYNAMICAL FIELD & WAVE ANCHORS                                       |
 |    • DCC_ANCHOR_WAVE_v1     : Discrete Wave D'Alembertian Operator □        |
 |    • DCC_ANCHOR_FOURIER_v1  : Rational Discrete Fourier Transform (DFT)     |
 |    • DCC_ANCHOR_STOKES_v1   : Exact Coboundary Conservation ⟨∂σ, ω⟩ ≡ ⟨σ, dω|
 +-------------------------------------------------------------------------+
```

H3: 4.1. The Circular Ratio (π / DCC_ANCHOR_PI_v1)
Governs circular phase rotations, Fourier analysis, and wave periods:

$$\pi = [3; \, 7, \, 15, \, 1, \, 292, \, 1, \, 1, \, 1, \, 2, \, 1, \, 3, \, 1, \, 14, \, \dots]$$

Evaluating \pi to zoom level z generates partial convergents \frac{p_k}{q_k} bounded by strict residual conservation:

$$0 \le |r_k| \lt \frac{1}{q_k q_{k+1}}$$

H3: 4.2. The Turn Constant (τ / DCC_ANCHOR_TAU_v1)
Defined as the full-turn angular period:

$$\tau = Q(2) \times \pi = [6; \, 3, \, 1, \, 1, \, 7, \, 2, \, 146, \, \dots]$$

Eliminates arbitrary factor-of-two fractions in spatial rotor transformations (docs/qm_geometry.md).

H3: 4.3. The Natural Exponential Base (e / DCC_ANCHOR_E_v1)
Governs exponential damping, thermal relaxation, and solutions to discrete differential equations (\mathbf{d}f = f):

$$e = [2; \, 1, \, 2, \, 1, \, 1, \, 4, \, 1, \, 1, \, 6, \, 1, \, 1, \, 8, \, \dots, \, 1, \, 1, \, 2k, \, \dots]$$

H3: 4.4. The Thermodynamic Logarithm (ln 2 / DCC_ANCHOR_LN2_v1)
Directly calibrates the physical layer-0 Landauer dissipation limit:

$$\ln(2) = [0; \, 1, \, 2, \, 3, \, 1, \, 6, \, 3, \, 1, \, 1, \, 2, \, 1, \, 1, \, 4, \, \dots]$$

Guarantees platform-invariant accounting when translating physical energy consumption into U units.

H2: 5. Lineage, Aperiodicity, and the Anchor-Cut in Time Evolution

In continuous mechanics, time is treated as an unmeasured parameter ($t \in \mathbb{R}$) that allows algorithms to discard initial states. In \mathrm{Qm} Calculus, time evolution is the strictly monotonic progression of discrete causal state transitions along the causal-arrow.

```
                        THE ANCHOR-CUT THEOREM
 Full Micro-Step Lineage (O(N) Replay - Rejected by SIMEMP):
   Q(0) --► s_1 --► s_2 --► s_3 --► ... --► s_999,999 --► S_terminal
   (Verifying every micro-step violates Memory Wall & wastes gigabytes of RAM)

 The Anchor-Cut Verification (O(K) Verification - BOC):
   Q(0)_N ========► Q(anchor)_1 ========► Q(anchor)_2 ========► S_terminal
   [Silicon Root]    [Receipt R_1]        [Receipt R_2]         [Current State]
   • Replay between anchors: ELIMINATED!
   • Verifier only checks: H(Q(0)) + Chain of Signed Anchor Receipts + H(S_terminal).
   • Complexity: O(K) where K « N.
```

H3: 5.1. Formal Causal Lineage of a Dynamic State
Every field state S_k generated during dynamical evolution possesses an explicit lineage tuple in its universal-envelope protected by the security-envelope:

$$\mathcal{L}(S_k) = \Big\langle \mathbf{P}(S_k), \, \kappa(S_k), \, \tau_{\mathrm{type}}, \, \mathcal{W}_{\mathrm{cum}}(S_k), \, \Pi_{\mathrm{receipts}}(S_k) \Big\rangle$$

- Parent Commitment Set ($\mathbf{P}$): References the preceding field state S_{k-1} and the applied operator contract $\mathcal{M}_{\mathbf{d}}$.
- Monotonic Causal Index (κ): Increments deterministically with each discrete timestep (Δt).
- Cumulative Work ($\mathcal{W}_{\mathrm{cum}}$): Tracks the total Landauer work expended across the simulation run in units of Q(1).

H3: 5.2. Aperiodic Memory Trajectory
Under DCC_ANCHOR_Q5_APERIODICITY_v1 (docs/q2_q9.md), dynamical simulations avoid circular recurrence locks. The trajectory through local-first memory space is non-repeating:

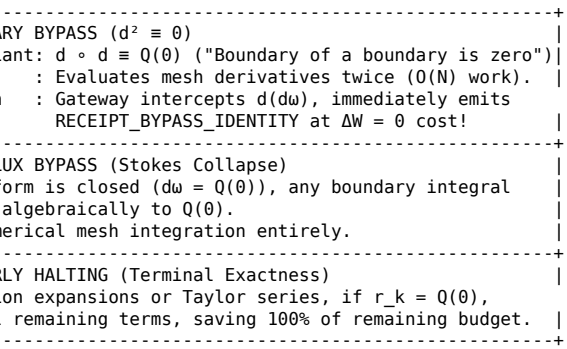
$$\forall i \neq j, \quad \mathcal{H}(S_i) \neq \mathcal{H}(S_j)$$

This guarantees an irreversible internal arrow of time without consulting external clocks.

H3: 5.3. The Anchor-Cut Verification Invariant
To audit or verify a dynamical simulation consisting of N discrete timesteps passing through K intermediate Q(\mathrm{anchor}) checkpoints ($K \ll N$) under the Anchor-Cut Invariant:
1. An auditor does not replay the N micro-steps across volatile memory.
2. The auditor verifies the Anchor-Cut Chain: the local origin Q(0), the signed anchor receipts $\mathcal{R}_{\mathrm{anchor}}$, and the terminal state S_N.
3. Verification complexity collapses from $\mathcal{O}(N)$ memory bus traversals to $\mathcal{O}(K)$ receipt validations. Micro-states between anchors are safely archived or dereferenced via Q(reuse).

H2: 6. Coupling to Optimization Morphisms: Q(bypass) and Q(reuse)

In classical scientific simulations, processors waste extensive energy re-evaluating terms that are algebraically zero. Qm Calculus couples DEC directly to Category B optimization morphisms under the boc-policy to maximize Operational Agility:



H3: 6.1. Poincaré Nilpotent Bypass
When an expression graph attempts to evaluate the curl of a gradient or the divergence of a curl ($\mathbf{d}(\mathbf{d}\omega)$):
- The simemp-gateway matches the identity proof hash of DCC_ANCHOR_DEC_D_v1.
- Q(bypass) halts evaluation in $\mathcal{O}(1)$ time and emits RECEIPT_BYPASS_IDENTITY.
- Zero compute budget is burned ($\Delta W = 0$).

H3: 6.2. Closed Form Stokes Bypass
If a differential form is algebraically closed ($\mathbf{d}\omega \equiv Q(0)$), boundary circulation integrals collapse to zero without numerical mesh iteration:

$$\sum_{\partial K} \omega \equiv Q(0)$$

H3: 6.3. Simplicial Mesh Deduplication via Q(reuse)
During spatial evolution, static geometric mesh configurations (dual metric tensors, Hodge star ratio matrices) are referenced across time steps via their immutable content commitment hashes using Q(reuse), eliminating redundant bit allocation across the Memory Wall.

H2: 7. Thermodynamic Governance and Singularity Immunity

In classical continuous fluid mechanics, the Navier–Stokes equations allow solutions to develop

```
unphysical mathematical singularities (finite-time velocity blowups \|u\|_{L^\infty} \to
\infty).

\mathrm{Qm} Calculus provides structural immunity against physical singularities:
1. Lattice Cutoff: Spatial velocity cannot diverge because spatial gradients are bounded by the
   discrete scale quantum \delta_z(d). Space cannot be divided infinitely.
2. Thermodynamic Viscous Dissipation: Physical fluids dissipate energy into microscopic thermal
   noise. In \mathrm{Qm} Calculus, viscosity is modeled as an explicit dissipative sink
   exporting operational entropy via signed receipts.
3. Totality by Budget (docs/qnPrimitive.md Axiom 6): Time evolution is allocated a finite work
   budget in units of Q(1). An unphysical blowup is impossible: before any field value can
   exceed physical bounds, the budget exhausts and the engine halts cleanly with
   RECEIPT_BUDGET_EXHAUSTED.


H2: 8. License and Invariant Lineage

This domain specification enforces the legal and operational conditions of the ssl governing the
licensed-work across the cdqn network:

1. paternity-reference: All discrete exterior calculus solvers, continued fraction libraries,
   and dynamical simulators derived from this specification must embed:
Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License
(SSL). Parent Repository: https://github.com/cdqn5249/cdqn
2. Anti-Patent Defense: Royalty-free rights terminate automatically if a licensee initiates
   patent litigation regarding \mathrm{Qm} discrete calculus methods or DEC operator anchors
   against the Author or community under the Open Core Invariants.
3. Statutory Safe Harbor: \mathrm{Qm} Calculus operates as passive, neutral mathematical
   infrastructure. It exercises no editorial inspection over, and makes no claim upon, external
   simulation models, physical datasets, or sovereign Payloads.
```

Annex H — qexpr.md

Source: docs/qexpr.md — *1.3.0, updated 2026-10-26. Rendered content per Annex Guide conventions N1–N6: 469 source lines, 633 rendered; 41,624 → 37,359 characters; source SHA-256* a69cb9bd 94e0a170 c1f1e33e a9c8aeb3 3c6804c5 19c99bd6 0086de8f 759c028d*. Content fidelity verified mechanically at build time.*

```
[front matter] ------------------------------------------------------------
[front matter] title: Qexpr — Content-Addressed Expression DAG (Layer 3 Symbolic Container)
[front matter] version: 1.3.0
[front matter] updated: 2026-10-26
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Canonical specification of the Layer 3 Content-Addressed Expression
[front matter]   DAG (CAE-DAG), arity-indexed child commitments, hash-consing
[front matter]   deduplication, symbolic irrationals, six canonical node classes
[front matter]   (Class 6 split into canonical contraction OP_BYPASS and bounded
[front matter]   saturation OP_SATURATE), and hardware-aware collapse under SIMEMP
[front matter]   constraints.
[front matter] terms_used: boc-policy, cae-dag, causal-arrow, complexity-degree, compute-unit-u,
[front matter]   dcc-profile, dependencies-determinism, dimension-d, existential-
[front matter]   invariant, hash-consing, hca-collapse, layer-0, layer-1, licensed-
[front matter]   work, metric-envelope, metric-exhaustion, no-implicit-rule,
[front matter]   paternity-reference, payload, q-bypass, q-reuse, q0, q1, qexpr, qm,
[front matter]   qn, qs, receipt, remainder-r, security-envelope, ssl, structural-
[front matter]   indirection, terminal-exactness, universal-envelope, zoom-z
[front matter] ------------------------------------------------------------


H1: Qexpr — Content-Addressed Expression DAG: Layer 3 Symbolic Container


Field | Specification
Document Title | Qexpr — Content-Addressed Expression DAG: Layer 3 Symbolic Container
Version | 1.3.0
Last Updated | 2026-10-26 (Bao Loc, Vietnam)
Author | Christophe Duy Quang Nguyen
License | Scaling Source License (SSL) 1.0
Status | Canonical Specification — Layer 3 Structural Container (Complexity Degree 2)


H2: Normative References

The following documents establish the physical, structural, and legal constraints governing this
specification. If a technical conflict arises, simemp.md governs; if a structural conflict
arises, abstractionLayers.md governs; if a legal conflict arises, LICENSE.md governs. References
are deliberately version-unpinned: each target document's front matter carries its own
authoritative version, so this specification never duplicates drift-prone patch state.
```

Document | Role | Target
docs/simemp.md | Constitutional constraints, thermodynamics, and Dependencies Determinism | simemp.html
docs/abstractionLayers.md | Layer architecture and SIMEMP Gateway validation | abstractionLayers.html
docs/qnPrimitive.md | Universal Envelope, operational axioms, and Optimization Morphisms | qnPrimitive.html
docs/q0_q1.md | Primary genesis origin Q(0) and first unit Q(1) | q0_q1.html
docs/q2_q9.md | Single-digit secondary DCC anchors and parity partitions | q2_q9.html
docs/qm.md | Constructive numeric leaf substrate and Diophantine division constraints | qm.html
docs/qm_geometry.md | Multi-axial frames, Clifford geometric algebra, and float-free rotations | qm_geometry.html
docs/qm_calculus.md | Discrete Exterior Calculus, continued fractions, and invariant dynamics | qm_calculus.html
docs/qs.md | Constructive relational intent and Intent Contracts | qs.html
LICENSE.md | Scaling Source License 1.0 governing the Licensed Work | LICENSE.md

H2: Review Integration Record (v1.3.0)

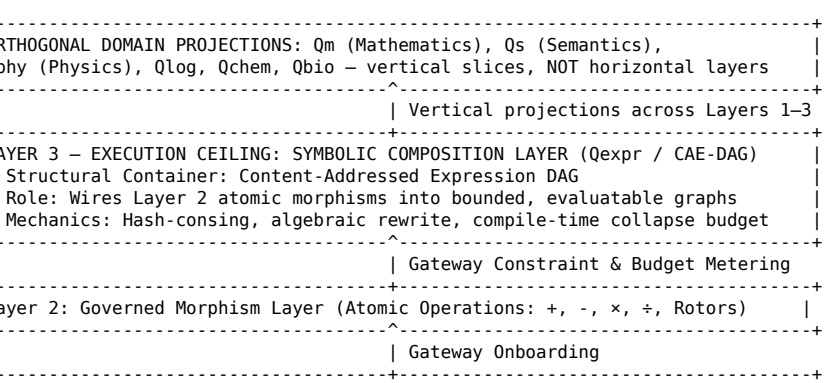
Source: docs/AiReviews/AiReviewQnIRv0.1.md — transcript of the external adversarial review (Google Deep Review, October 2026) of the condensed Qn universe specification and QnIR v0.1 candidate draft. This update integrates the Layer-3 rewrite-semantics findings of that review into the canon. Under Axiom 13, this file is a new versioned artifact; no prior content is mutated in place. New constructions and semantic refinements introduced below are tagged [v1.3.0 — CANDIDATE] pending BOC sealing; editorial corrections are applied directly and recorded here.

Change | Section | Reviewer source | Type
OP_REWRITE split into OP_BYPASS (canonical contraction, hash-governing) and OP_SATURATE (bounded, Q(rsi)-gated equality saturation); OP_REWRITE retained as deprecated alias until QnIR v0.2 | §4.6 | QO-5 / Segment 2 Point 3 (definitive theoretical constraint) | CANDIDATE
Saturation artifact discipline: output is a new versioned artifact (Axiom 13), never a canonical node identity; provenance (schedule, ceilings, extractor generation) recorded | §4.6.2 | QO-5 / Segment 2 Point 3 | CANDIDATE
Canonical child sorting for DCC_SYM_COMMUTATIVE opcodes + hashing-generation tag on the DCC profile | §2.2 | QO-5 / Segment 2 Point 3 | CANDIDATE
Bypass suite count fixed at 16 rules (first-root rule restored); QnIR v0.1 PDF condensation reconciled | §4.6.1 | qm.md v1.3.0 cross-file flag / data-loss audit | editorial
Confluence of the bypass suite tagged CANDIDATE; machine-checked confluence certificate set as QnIR v0.2 sealing prerequisite | §4.6.1 | QO-5 (termination/confluence demand) | CANDIDATE
Fourth circuit breaker: saturation iteration ceiling $K_{\max}$; reason codes K_MAX_REACHED / N_MAX_REACHED / D_MAX_REACHED; receipt RECEIPT_SATURATION_CONVERGED | §6 | QO-5 / Segment 2 Point 3 | CANDIDATE
§6 breaker-box row 3 box-alignment correction (pre-existing 1-char drift) | §6 | — | editorial

Cross-file dependencies flagged (deferred under the one-file-at-a-time discipline, to be integrated in subsequent versioned updates): docs/qnPrimitive.md (Q(bypass) execution precedence sentence and macro-DAG "equality saturation" wording aligned to the split; third circuit-breaker receipt naming); docs/qlog.md §6.1 ("Background Equality Saturation" resolves to OP_SATURATE); docs/qm.md v1.3.0 one-line touch-up (fixed-point identity attribution OP_REWRITE -> OP_BYPASS in the Fixed-Point Invariant entry); glossary term additions for the new §2.2/§4.6 constructs; and the QnIR v0.2 candidate must carry distinct opcodes for OP_BYPASS and OP_SATURATE (Reviewer Segment 4 Point 1, Requirement R11 opcode-coverage finding, resolved in the QnIR v0.2 cycle, not in this file).

H2: 1. Epistemic Stance and Constitutional Placement

Within the master abstraction hierarchy (docs/abstractionLayers.md §2.5, §5), Qexpr is situated strictly at Layer 3 (Complexity Degree 2) — the terminal layer of the physical execution core. Domains are orthogonal vertical projections across Layers 1–3, not horizontal layers above the stack.



```
 | Layer 1: Node Genesis Layer (Static Existence: Q(0), Q(1), Q(2)...Q(9), Axis d1)|
 +--------------------------------------------------------------------------------+
```

H3: 1.1. The Structural Role of Layer 3
- layer-1 defines what exists (static ontological primitives: local origin Q(0), unit Q(1), single digits Q(2)\dots Q(9) along axis d1).
- Layer 2 defines atomic actions (isolated morphisms: addition, subtraction, multiplication, Diophantine division, exponents, roots, factorials, Clifford wedge products, spatial rotors, Q(reuse), Q(bypass)).
- Layer 3 (Qexpr) defines compositional structure: how multiple Layer 2 atomic actions are wired into directed acyclic networks *before* they are evaluated or collapsed into physical memory.

H3: 1.2. Acknowledgment of Universal Fallibility
A core design tenet of the qn architecture is the explicit recognition that both human authors and artificial agents are fallible entities. Neither is an absolute arbitrator:
- Humans are prone to specification errors, unmeasured assumptions, and logical oversights.
- Artificial agents and compilers are prone to algorithmic hallucinations, non-terminating expansions, and parameter misalignments.

Under docs/simemp.md, safety cannot rely on presumed actor infallibility. Qexpr serves as the mechanical error-containment envelope: it enforces strict syntactic cycle freedom, explicit graph depth bounds, and finite execution budgets (docs/qnPrimitive.md Axiom 6), ensuring that an error by any actor halts deterministically at the structural boundary without inducing systemic divergence.

H3: 1.3. Universal Semantic Membrane and Cross-Domain Contexts
Qexpr is not restricted to arithmetic calculations; it is the universal structural membrane for all computable domains.

A computational Context is formalized as a multi-domain Content-Addressed Expression DAG:

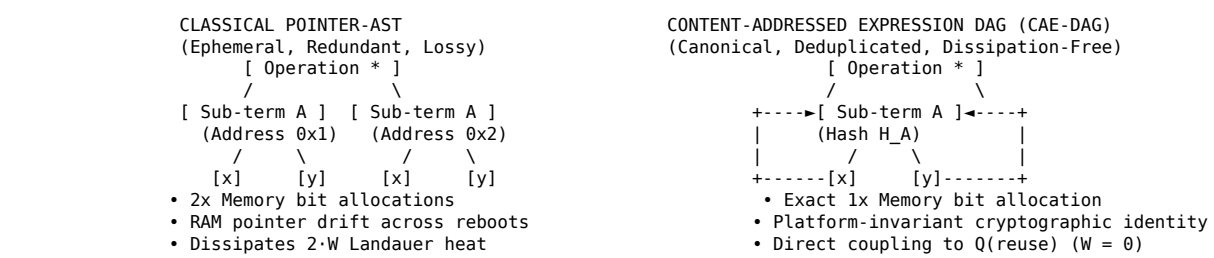
$$\text{Context} \equiv \mathcal{G}_{\text{Qexpr}} = \mathbf{C}_{\mathrm{Qs}} \otimes \mathbf{C}_{\mathrm{Qm}} \otimes \mathbf{C}_{\mathrm{Qphy}}$$

1. Semantic Root (Qs): Encapsulates intent, linguistic pregroups, and human-agent capability delegations.
2. Relational Root (Qm): Verifies that proposed state transitions are mathematically decidable, Diophantine-constrained, and exact.
3. Physical Root (Qphy): Verifies that the required computational work can physically execute within finite Landauer dissipation bounds ($W \ge k_B T \ln 2$), parameterized by the Abstract Compute Unit U.

By representing cross-domain contexts within the unified Qexpr structure, heterogeneous actors achieve consensus on shared root commitment hashes $\mathcal{H}(\text{Context})$ without unmeasured semantic ambiguity.

H2: 2. The Content-Addressed Expression DAG (CAE-DAG) Architecture

Under the Best of Choices (BOC) Policy (docs/simemp.md §7), Qexpr explicitly rejects naive pointer-heap trees in favor of a Content-Addressed Expression Directed Acyclic Graph (CAE-DAG).



H3: 2.1. Canonical Node Tuple and Arity-Indexed Dependencies
Under the no-implicit-rule, symbolic nodes cannot assume a rigid binary schema that forces dummy padding hashes for non-binary operators. Every symbolic node α within a Qexpr DAG is an immutable, content-addressed tuple with an explicit arity-indexed child commitment vector:

$$\alpha = \Big\langle \mathrm{OpCode}, \, \mathbf{D}_{\mathrm{children}}, \, z, \, r, \, d, \, \mathrm{DCC}_{\mathrm{ref}} \Big\rangle$$

$$\mathbf{D}_{\mathrm{children}} = \big[\mathcal{H}_1, \, \mathcal{H}_2, \, \dots, \, \mathcal{H}_k \big], \quad k = \mathrm{Arity}(\mathrm{OpCode}) \in \mathbb{N}$$

The child commitment vector $\mathbf{D}_{\mathrm{children}}$ strictly maps operator arity without implicit padding:
- Nullary (k = 0): Leaf constants, primary primitives (Q(0), Q(1)), and continued fraction recurrence generators (OP_CF for π, e, τ, $\ln 2$). $\mathbf{D}_{\mathrm{children}} = []$.
- Unary (k = 1): Morphisms over a single operand, including radical root constraints (OP_ROOT),

discrete factorials, Clifford reflection involutions (\mathcal{I}_d), discrete exterior derivatives (\mathbf{d}), and Hodge stars (\star). \mathbf{D}_{\mathrm{children}} = [\mathcal{H}_1].
- Binary (k = 2): Elementary \mathcal{R}-algebra morphisms (+, -, \times, \div, exponents), Clifford geometric products (\vec{u} \vec{v}), and wedge products (\vec{u} \wedge \vec{v}). \mathbf{D}_{\mathrm{children}} = [\mathcal{H}_1, \, \mathcal{H}_2].
- Variadic (k \ge 3): Bounded monoidal folds (OP_FOLD for \sum, \prod), multi-axial coordinate frames, and multi-agent intent sequences. \mathbf{D}_{\mathrm{children}} = [\mathcal{H}_1, \, \dots, \, \mathcal{H}_k].

The remaining tuple elements preserve physical and scale invariants:
- z, r, d: Target scale quantum Zoom z, conserved residual Remainder r, and coordinate frame Dimension d.
- \mathrm{DCC}_{\mathrm{ref}}: Machine-readable DCC Profile governing transformation ceilings, execution budgets, and capability delegation rights.

H3: 2.2. Structural Hash-Consing and \mathcal{O}(1) Equivalence
Node identity is strictly mathematical under dependencies-determinism, operationalized via hash-consing:

\mathcal{H}(\alpha) = \mathrm{Hash}\Big(\mathrm{OpCode} \,\|\, k \,\|\, \mathcal{H}_1 \,\|\, \dots \,\|\, \mathcal{H}_k \,\|\, z \,\|\, r \,\|\, d \,\|\, \mathrm{DCC}_{\mathrm{ref}}\Big)

Because the arity k is explicitly serialized, nullary and unary operators produce canonical hashes with zero residual entropy (H(S_t \mid \mathcal{D}(S_t)) = 0).

Canonical child ordering [v1.3.0 — CANDIDATE, Review Integration QO-5]: When a node's OpCode is declared commutative under its DCC Profile (flag DCC_SYM_COMMUTATIVE), the child commitment vector is replaced by its lexicographically sorted image \sigma_{\mathrm{sym}} prior to hashing:

\mathcal{H}(\alpha) = \mathrm{Hash}\Big(\mathrm{OpCode} \,\|\, k \,\|\, \sigma_{\mathrm{sym}}(\mathcal{H}_1, \dots, \mathcal{H}_k) \,\|\, z \,\|\, r \,\|\, d \,\|\, \mathrm{DCC}_{\mathrm{ref}}\Big)

so that A \circ B and B \circ A commit identically whenever the DCC declares \circ commutative: one commitment exists per semantic value, and algebraic equality of commutative permutations collapses to the same \mathcal{O}(1) machine-word comparison. Non-commutative opcodes hash in declaration order, unchanged. Because this changes commitment values, a DCC profile enabling sorting MUST declare a hashing-generation tag; commitments across distinct hashing generations compare only through signed migration receipts under structural-indirection and the versioned-artifact discipline of Axiom 13 — never by raw hash equality.

1. Deduplication: Any sub-expression with identical semantics, arity, and child commitments yields the identical commitment hash. Through hash-consing, the runtime memory allocator maps it to the exact same physical node.
2. \mathcal{O}(1) Structural Equality: Verifying whether two sub-expressions A and B are identical requires zero recursive graph traversal:

A \equiv B \iff \mathcal{H}(A) == \mathcal{H}(B)

Testing algebraic or semantic equality collapses to a single machine-word integer comparison.

H3: 2.3. Thermodynamic Work Minimization and Breakeven Bounds
In physical layer-0 substrates, allocating and deallocating memory across physical buses dissipates electrical capacitance (\mathcal{O}(C V^2 f)) and Landauer work (W \ge k_B T \ln 2).
- By enforcing hash-consing, repeated sub-terms in multi-dimensional Clifford geometric products (docs/qm_geometry.md) or multi-agent dialogue graphs are allocated exactly once.
- Reversible symbolic normalization within the CAE-DAG incurs zero Landauer dissipation (W = 0), directly operationalizing Q(reuse) across the Memory Wall.

H4: The Breakeven Admission Bound
Querying a cryptographic hash commitment across L3 caches or DRAM incurs measurable bus dissipation. Under the Metric Invariant (simemp.md §3.1), hash-consing lookup via Q(\text{reuse}) is admissible if and only if the computational work of re-evaluation exceeds retrieval dissipation:

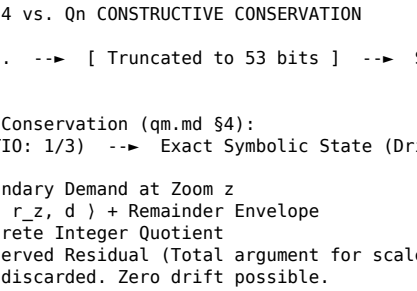
\mathrm{Admit}\big(Q(\text{reuse})\big) \iff \Delta W_{\mathrm{compute}} \gt \Delta W_{\mathrm{lookup}} + \epsilon_{\mathrm{overhead}}

If direct computation is cheaper than memory traversal (such as single-cycle integer additions Q(2) + Q(3) or trivial powers), the compiler bypasses hash-table probes, lowering the operation directly to integer machine instructions via the HCA-Collapse engine.

H3: 2.4. Acyclicity by Construction (Axiom 5)
In classical graphs, circular references create non-halting loops and stack crashes. In a Content-Addressed DAG, a circular dependency is mathematically impossible. Because \mathcal{H}(\alpha) requires each child hash \mathcal{H}_i as an input, no child can declare \alpha as a dependency without breaking the pre-image resistance of cryptographic hash functions. Acyclicity along the causal-arrow is guaranteed by construction, satisfying Axiom 5 (docs/qnPrimitive.md).

H2: 3. Elimination of Rounding and the Ontological Status of Irrationals

In classical floating-point systems (IEEE 754), rounding is an unmetered, lossy artifact of fixed register widths. Rounding silently discards trailing bits, generating non-deterministic drift and violating Axiom 10.

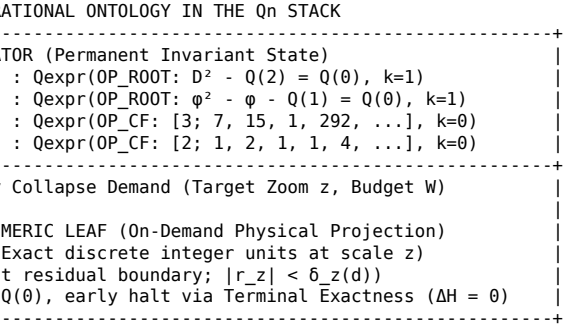


H3: 3.1. Prohibition of Rounding Drift
In the Qn universe, implicit rounding is outlawed by the no-implicit-rule and Axioms 9 and 10:
1. Within Layer 3, expressions are held in unevaluated, exact symbolic form. Drift is identically zero.
2. When projected onto physical hardware boundaries, quantities are partitioned into an integer quotient q_z and an exact conserved remainder r_z (docs/qm.md §4):

$$X = \big(q_z \cdot \delta_z(d) \big) + r_z, \quad \mathrm{where} \; 0 \le \lvert r_z \rvert \lt \delta_z(d)$$

The remainder r_z is never discarded; it is sealed within the universal-envelope as the exact causal input state for subsequent precision expansions.

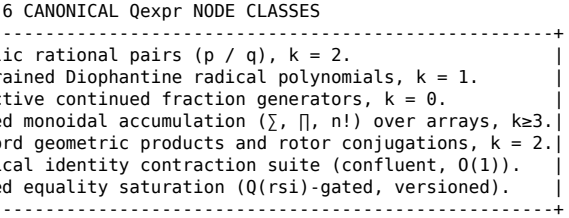
H3: 3.2. Ontological Status of Irrationals and Transcendentals
Under Axiom 2, actual infinities cannot exist as storable or executable machine states. Consequently:
- An irrational number cannot be an infinite decimal string (3.14159\dots), as storing infinite digits requires infinite energy.
- Transcendental and irrational quantities exist permanently as Qexpr symbolic generators.



1. Algebraic Irrationals (\sqrt{2}, \phi): Stored as exact polynomial constraint equations (D^2 - Q(2) = Q(0)) under OP_ROOT (k = 1). They are manipulated symbolically via radical identities with zero rounding drift.
2. Transcendental Constants (\pi, \tau, e, \ln 2): Stored as deterministic continued fraction recurrence generators under OP_CF (k = 0). Evaluating \pi to zoom level z computes an exact rational quotient and a conserved residual without floating-point error, verifying terminal-exactness where remainders collapse.

H2: 4. The Six Canonical Node Classes

To achieve computational completeness across all abstraction layers, Qexpr defines Six Canonical Node Classes:



H3: 4.1. Node Class 1: Exact Rational Pairs (OP_RATIO, k = 2)
Represents unevaluated rational fractions \frac{p}{q} (q \neq Q(0)) without premature projection onto decimal or binary scale lattices.

Multiplicative and additive morphisms execute via exact Diophantine cross-multiplication:

\frac{p_1}{q_1} \times \frac{p_2}{q_2} = \frac{p_1 \times p_2}{q_1 \times q_2}

\frac{p_1}{q_1} + \frac{p_2}{q_2} = \frac{(p_1 \times q_2) + (p_2 \times q_1)}{q_1 \times q_2}

H3: 4.2. Node Class 2: Constrained Algebraic Roots (OP_ROOT, k = 1)
Represents algebraic radical roots as exact polynomial Diophantine equations (P(X) = Q(0)).

Algebraic roots (such as \sqrt{2} via DCC_ANCHOR_SQRT2_v1) are held symbolically with a single operand commitment. Multiplication by identical roots (D \times D) resolves immediately to integer constants (Q(2)) via Q(bypass) without invoking numerical root-extraction routines.

H3: 4.3. Node Class 3: Projective Recurrence Generators (OP_CF, k = 0)
Generates transcendental numbers (\pi, e, \tau, \ln 2) through deterministic continued fraction expansions:

X = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots}}}

Partial convergents \frac{p_m}{q_m} are generated iteratively as nullary generators (k = 0) containing zero child dependencies. The truncation error is strictly bounded:

0 \le \lvert r_m \rvert \lt \frac{1}{q_m q_{m+1}}

and the residual is preserved in the remainder envelope.

H3: 4.4. Node Class 4: Bounded Monoidal Folds (OP_FOLD, k \ge 3)
Governs accumulation (\sum, \prod, vector concatenation, and discrete factorials n!) over arrays, streams, or multi-axial frames.

Requires an explicit step ceiling (m \le m_{\max}) and declared compute budget. The child vector \mathbf{D}_{\mathrm{children}} encapsulates the full array of term commitments. Unbounded loops or infinite series are syntactically invalid under Axiom 2 and Axiom 6.

H3: 4.5. Node Class 5: Frame Transformations and Permutations (OP_FRAME_MAP, k = 2)
Governs multi-axial Clifford geometric products, Cayley rotor conjugations (R \vec{v} R^\dagger), and coordinate permutations (S_n group).

Rotors maintain unitary norm (\lVert R \rVert^2 \equiv Q(1)) by construction, eliminating trigonometric normalization cycles (docs/qm_geometry.md §5).

H3: 4.6. Node Class 6: Canonical Rewrite Operations (OP_BYPASS and OP_SATURATE, k \in \{1, 2\}) [v1.3.0 — CANDIDATE]

[Review Integration QO-5] Version 1.2.1 grouped two fundamentally different rewriting mechanisms under the single OP_REWRITE label: (a) a fixed, terminating, \mathcal{O}(1) algebraic identity suite, and (b) general equality saturation. These carry opposite canonical semantics, and their conflation is not benign. Once bidirectional structural axioms (commutativity, associativity, distributivity) are admitted to a saturation e-graph, the e-graph can grow exponentially or without bound, and optimal DAG extraction from a saturated e-graph is NP-hard; two nodes saturating under different budgets or rule schedules can therefore extract different trees and commit different hashes — silently breaking \mathcal{O}(1) hash-consing equivalence and H(S_t \mid \mathcal{D}(S_t)) = 0 (Tate et al. 2009; Willsey et al. 2021, egg). Node Class 6 is accordingly split into two distinct operations. The former umbrella name OP_REWRITE is retained as a deprecated alias (so that QnIR v0.1 and docs/qlog.md references remain resolvable) and is retired at the QnIR v0.2 candidate.

H4: 4.6.1. OP_BYPASS — Canonical Contraction
Executes reversible identity short-circuiting prior to numerical collapse. Under the No-Implicit Rule, the engine evaluates the full canonical suite of \mathcal{O}(1) algebraic identity bypasses:

CANONICAL Q(bypass) IDENTITY RULES

ADDITIVE & MULTIPLICATIVE BYPASSES	EXPONENTIATION & RADICAL BYPASSES
• A + Q(0) --► A	• A^Q(0) --► Q(1) (A ≠ Q(0))
• A × Q(1) --► A	• A^Q(1) --► A
• A × Q(0) --► Q(0)	• Q(1)^k --► Q(1)
• Q(0)! --► Q(1)	• Q(0)^k --► Q(0) (k > 0)
• Q(1)! --► Q(1)	• ^Q(1)√(A) --► A
• Q(2)! --► Q(2)	• ^n√(Q(1)) --► Q(1)

```
+-------------------------------------+-------------------------------------+
| GEOMETRIC & TOPOLOGICAL INVARIANTS                                        |
| • Rotor Normalization : R R† --► Q(1)                                     |
| • Exterior Nilpotency : e_i ∧ e_i --► Q(0)                                |
| • Bivector Inversion  : (I_2)² --► -Q(1)   (Grade 2 collapses to Grade 0) |
| • Poincaré Coboundary : d ∘ d --► Q(0)                                    |
+---------------------------------------------------------------------------+
```

All listed transformations emit RECEIPT_BYPASS_IDENTITY at zero metric cost (\Delta W = 0), short-circuiting execution before any computation or memory lookup is scheduled.

Rule-count reconciliation: the canonical count is 16 rules, including the first-root rule ^{Q(1)}\sqrt{A} \to A. The condensed QnIR v0.1 review draft (PDF §4.3) printed 15 rules — the first-root row was lost in condensation; that PDF is a draft artifact and this specification is the authority.

Mandatory properties of the OP_BYPASS suite:

1. Terminating: every rule strictly reduces expression size or height, so the contraction always reaches a normal form in linear rule applications.
2. Confluent [v1.3.0 — CANDIDATE, sealing item]: the suite's critical pairs (e.g., Q(1)^{Q(1)} under both A^{Q(1)} \to A and Q(1)^k \to Q(1)) are locally joinable, and a unique normal form is asserted. A machine-checked confluence certificate (Newman's lemma over the critical-pair set) is a QnIR v0.2 sealing prerequisite under QO-5; until sealed, this property is a CANDIDATE proposition, not a theorem.
3. Hash-governing: the bypass normal form, together with the DCC_SYM_COMMUTATIVE child sorting of §2.2, is the input to canonical node hashing \mathcal{H}(\alpha). One commitment exists per semantic value; OP_BYPASS never emits a new artifact — it discloses an identity that already held.
4. Cost: zero metric cost (\Delta W = 0), as receipted above.

The Layer-2 operational morphism and its execution precedence (Q(\text{bypass}) \prec Q(\text{reuse}) \prec \text{Direct Compute}) remain defined in docs/qnPrimitive.md (Q(bypass) Optimization Morphism); this section governs the Layer-3 node-class semantics.

H4: 4.6.2. OP_SATURATE — Bounded Equality Saturation
General equality saturation is a workflow-scoped optimization operation, never a canonical identity operation. It is legal only inside an explicit Q(reuse)-class optimization workflow (Q(rsi)) whose dcc-profile pins all of the following, without defaults (No-Implicit Rule):

1. Deterministic rule-application schedule over the declared rule set (e.g., fixed-epoch breadth-first Datalog scheduling); data-dependent or adaptive scheduling is invalid.
2. Iteration ceiling K_{\max} (§6, circuit breaker 4).
3. E-node ceiling N_{\max} and graph depth ceiling D_{\max} (§6, circuit breakers 1–2).
4. Deterministic polynomial-time greedy tree-cost extractor with hash tie-breaking. Optimal extraction from a saturated e-graph is NP-hard, so "optimal extraction" is out of scope by construction; extraction is required to be greedy, polynomial-time, and totally tie-broken by commitment hash.

Termination and receipts: saturation halting before any ceiling emits RECEIPT_SATURATION_CONVERGED [v1.3.0 — CANDIDATE]; ceiling exhaustion emits the existing dissipative receipts RECEIPT_BUDGET_EXHAUSTED or RECEIPT_EXPRESSION_TOO_COMPLEX with reason codes K_MAX_REACHED, N_MAX_REACHED, or D_MAX_REACHED. Totality in the Axiom-6 sense is preserved: every saturation halts with a receipt.

Artifact discipline: the extractor's output is emitted as a new versioned artifact (Axiom 13) that records its input commitments, pinned schedule, ceilings, and extractor generation. OP_SATURATE never mutates canonical node identity and never participates in \mathcal{H}(\alpha). Two saturation runs under different budgets or schedules MAY legitimately produce different versioned artifacts; this is measured, provenance-recorded divergence between artifacts — not state drift — provided each artifact carries its full saturation provenance.

H2: 5. Hardware Capability-Aware Collapse Engine (HCA-Collapse)

While Qexpr remains an abstract, substrate-agnostic DAG at Layer 3 via structural-indirection, its physical execution on physical SoCs is governed by the Hardware Capability-Aware Collapse Engine (HCA-Collapse Engine).

```
                            THE COMPILATION PIPELINE
 LAYER 3: INVARIANT ROAD (Qexpr CAE-DAG)
   Abstract, substrate-agnostic mathematical graph (Exact Diophantine Relations)
                            |
                            v Hardware Capability-Aware Collapse Engine
 ============================+=============================================
 HARDWARE-SPECIFIC LOWERING  | (Reads Local DCC Hardware Profile: NPU / DSP / ASIC)
 +---------------------------+-----------------------------+
 | OPTION A: NPU / Systolic Array                          | OPTION B: Integer ALU / RISC-V
 | • Rewrite to integer Matrix-Multiply-Accumulate (MAC)   | • Rewrite to CORDIC / Shift-and-Add
 | • Constant-time, parallel execution over INT16/INT32    | • Exact Diophantine reciprocal table
 +---------------------------------------------------------+
                            |
                            v
```

```
 LAYER 0 / 1: PHYSICAL SOC EXECUTION
   Zero floating-point drift · Constant-time execution · Bounded Landauer dissipation
```

H3: 5.1. Target-Specific Lowering Profiles
The compiler queries the node's local DCC Hardware Profile (established during Layer 0 onboarding):
1. NPU / Systolic Arrays: Lowers multi-axial Clifford geometric products ($\vec{u} \cdot \vec{v} = \sum \eta_{jj} u_j v_j$) directly into integer Matrix-Multiply-Accumulate (MAC) instructions (INT16/INT32), completing contractions in constant time.
2. Cryptographic Co-Processors: Offloads hash-consing deduplication and node commitment hashing to hardware SHA-3 or BLAKE3 accelerators.
3. Integer RISC-V / Microcontrollers: Replaces division by multiplication using Barrett integer reciprocal tables with exact remainder correction loops.

H3: 5.2. Side-Channel Immunity via Constant-Time Compute
Variable-time operations leak secret cryptographic keys and algorithmic state via power analysis and cache timing. Lowering Qexpr nodes into fixed-cycle SoC instruction sequences ensures that execution duration depends strictly on declared metric budgets, providing hardware-level side-channel immunity under the security-envelope.

H3: 5.3. Invariance Mandate
SoC-specific lowering is permissible if, and only if, the transformation is an algebraic isomorphism. Lowering passes that introduce floating-point approximations, mantissa truncations, or non-deterministic rounding are intercepted by the gateway and rejected, protecting Existential Invariants.

H2: 6. Bounded Execution and Fallibility Circuit Breakers

To guarantee that fallible human directives or unaligned agentic processes cannot compromise the execution environment, Qexpr enforces three structural circuit breakers:

```
                    Qexpr FALLIBILITY CIRCUIT BREAKERS
 +-------------------------------------------------------------------------+
 | 1. STATIC GRAPH DEPTH CEILING                                           |
 |    Depth(DAG) ≤ Depth_max (Metered at ingestion; rejects runaway recursion) |
 +-------------------------------------------------------------------------+
 | 2. NODE COUNT BUDGET ALLOCATION                                         |
 |    Count(Nodes) ≤ Node_max (Prevents memory exhaustion across Memory Wall)  |
 +-------------------------------------------------------------------------+
 | 3. METRIC COLLAPSE BUDGET CEILING                                       |
 |    Burn(U) ≤ Budget_Q1 (Guarantees deterministic halting via Axiom 6)       |
 +-------------------------------------------------------------------------+
 | 4. SATURATION ITERATION CEILING                                         |
 |    Epochs(OP_SATURATE) ≤ K_max (Bounds e-graph growth; deterministic halt)  |
 +-------------------------------------------------------------------------+
```

1. Static Graph Depth Ceilings: If an expression tree exceeds the declared depth ceiling ($D \gt D_{\max}$), execution terminates immediately with receipt RECEIPT_EXPRESSION_TOO_COMPLEX.
2. Node Allocation Bounds: If hash-consing attempts to instantiate node counts exceeding the local metric-envelope, compilation aborts with RECEIPT_BUDGET_EXHAUSTED.
3. Totality by Budget (Axiom 6): Every collapse operation carries an explicit compute budget in units of Q(1) via metric-exhaustion. Non-terminating expansions are impossible; when the budget is consumed, the engine emits a dissipative receipt and halts.
4. Saturation Iteration Ceilings [v1.3.0 — CANDIDATE]: OP_SATURATE workflows (§4.6.2) carry an explicit iteration ceiling $K_{\max}$ alongside the node and depth ceilings. Hitting any ceiling halts deterministically with the corresponding reason code (K_MAX_REACHED, N_MAX_REACHED, D_MAX_REACHED), and the partially saturated state is discarded — only a receipted, extracted versioned artifact (Axiom 13) survives.

H2: 7. License and Invariant Lineage

This specification strictly conforms to the Scaling Source License (SSL) 1.0:

H3: 7.1. Paternity Reference
All derivative implementations, symbolic rewrite engines, compilers, and intermediate representation parsers derived from the licensed-work must preserve the canonical paternity-reference:

Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License (SSL). Parent Repository: https://github.com/cdqn5249/cdqn

H3: 7.2. Open Core Invariants
1. Anti-Patent Defense: Commercial or derivative licenses terminate automatically upon initiating patent litigation against the Author or project ecosystem.
2. Non-Scaling Open Access: Royalty-free licensing is guaranteed for non-commercial, academic, and sub-threshold usage under the ssl.

```
H3: 7.3. Passive Infrastructure Safe Harbor
The Qexpr symbolic container acts as passive, neutral technical infrastructure. It exercises no
editorial inspection over, and makes no claim upon, sovereign user content, models, or Payloads.
```

Annex I — qnPrimitive.md

Source: `docs/qnPrimitive.md` *— 1.3.0, updated 2026-10-26. Rendered content per Annex Guide conventions N1–N6: 530 source lines, 749 rendered; 39,963 → 34,385 characters; source SHA-256* `50d48852 f537bc1c 3e7cbd5b 93fd683e f2dd7afb 846acd9c a1d72e2d 762832d2`*. Content fidelity verified mechanically at build time.*

```
[front matter] ------------------------------------------------------------
[front matter] title: Qn Primitive Envelope — Governed Qn Artifact Model
[front matter] version: 1.3.0
[front matter] updated: 2026-10-26
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Governed Qn artifact model, operational axioms, candidate typed
[front matter]   payload profiles, and Category B optimization primitives under
[front matter]   SIMEMP consistency constraints; includes the canonical circuit-
[front matter]   breaker receipt registry and the OP_BYPASS/OP_SATURATE split
[front matter]   alignment.
[front matter] terms_used: boc-policy, causal-arrow, cdqn, chronosa, complexity-degree,
[front matter]   computational-consistency, compute-unit-u, dcc-profile, dependencies-
[front matter]   determinism, derivative-work, dimension-d, dual-ring-pqc-boundary,
[front matter]   existential-invariant, identity-class, layer-0, layer-1, licensed-
[front matter]   work, lifecycle-state, local-first, metric-envelope, metric-
[front matter]   exhaustion, no-implicit-rule, non-rogue-invariant, open-core-
[front matter]   invariant, operational-agility, paternity-reference, payload,
[front matter]   q-bypass, q-reuse, q-workflows, q0, q1, qexpr, qn, qn-rsi, qnir,
[front matter]   qnlang, qs, receipt, remainder-r, security-envelope, simemp, simemp-
[front matter]   gateway, ssl, structural-indirection, universal-envelope, zoom-z
[front matter] ------------------------------------------------------------


H1: Qn Primitive Envelope — Governed Qn Artifact Model


Field | Specification
Document Title | Qn Primitive Envelope — Governed Qn Artifact Model
Version | 1.3.0
Last Updated | 2026-10-26 (Bao Loc, Vietnam)
Author | Christophe Duy Quang Nguyen
License | Scaling Source License (SSL) 1.0
Status | Canonical Primitive Specification — Category A/B Envelope Foundation


H2: Purpose and Scope

This document formalizes the invariant structural container required for every governed
computational entity within the Qn universe: the universal-envelope.

In this pre-silicon design stage, the primitive envelope provides minimal structural scaffolding
to guide artifact construction, algebraic transformation, and candidate execution across
docs/abstractionLayers.md. Governed by dependencies-determinism, the model eliminates unmeasured
ambient state while preserving architectural elasticity against premature calcification through
structural-indirection.

All formulations represent provisional consistency abstractions within the Recursive Search Loop
(docs/simemp.md §2). They define constitutional boundaries rather than finalized implementation
parameters, pending empirical hardware validation.


H2: Normative References

The following documents define the physical, structural, and legal constraints governing this
framework. If a technical conflict arises, simemp.md governs; if a structural conflict arises,
abstractionLayers.md governs; if a legal conflict arises, LICENSE.md governs. References are
deliberately version-unpinned: each target document's front matter carries its own authoritative
version, so this specification never duplicates drift-prone patch state.


Document | Role | Target
docs/simemp.md | Constitutional constraints, thermodynamics, and Dependencies Determinism |
simemp.html
docs/abstractionLayers.md | Layer architecture and SIMEMP Gateway validation |
abstractionLayers.html
docs/q0_q1.md | Primary genesis origin Q(0) and first unit Q(1) | q0_q1.html
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docs/q2_q9.md | Single-digit secondary DCC anchors and single-digit spectrum | q2_q9.html
docs/qm.md | Constructive numeric leaf substrate and Diophantine division constraints | qm.html
docs/qm_geometry.md | Multi-axial frames, Clifford geometric algebra, and float-free rotations | qm_geometry.html
docs/qm_calculus.md | Discrete Exterior Calculus, continued fractions, and invariant dynamics | qm_calculus.html
docs/qexpr.md | Layer 3 Content-Addressed Expression DAG container | qexpr.html
docs/qs.md | Constructive relational intent and Intent Contracts | qs.html
LICENSE.md | Scaling Source License 1.0 governing the Licensed Work | LICENSE.md

H2: Review Integration Record (v1.3.0)

Source: docs/AiReviews/AiReviewQnIRv0.1.md — transcript of the external adversarial review (Google Deep Review, October 2026) of the condensed Qn universe specification and QnIR v0.1 candidate draft. This update integrates the operational-axiom findings of that review, and the cross-file dependencies deferred by docs/qexpr.md v1.3.0, into the canon. Under Axiom 13, this file is a new versioned artifact; no prior content is mutated in place. New constructions and semantic refinements introduced below are tagged [v1.3.0 — CANDIDATE] pending BOC sealing; editorial corrections are applied directly and recorded here.

Change | Section | Reviewer source | Type
Macro-DAG "equality saturation" sentence aligned to the qexpr.md v1.3.0 Node Class 6 split: OP_BYPASS canonical contraction governs identity short-circuiting; general saturation is legal only as OP_SATURATE (Q(rsi)-gated, versioned-artifact output) | §5.9 | QO-5 / Segment 2 Point 3; qexpr.md v1.3.0 cross-file dependency | CANDIDATE
§1.16 Q(bypass) bound to the Layer-3 OP_BYPASS canonical contraction (16-rule terminating suite); OP_REWRITE noted as deprecated alias | §1.16 | QO-5 / qexpr.md v1.3.0 cross-file dependency | CANDIDATE
Canonical Circuit-Breaker Receipt Registry — names the previously unnamed third-breaker receipt RECEIPT_METRIC_EXHAUSTED; registers RECEIPT_SATURATION_CONVERGED and reason codes D_MAX_REACHED / N_MAX_REACHED / U_BUDGET_EXHAUSTED / K_MAX_REACHED; no new Axiom-6 terminal states required | §5.9 (new) | QO-5 / data-loss audit (unnamed third circuit-breaker receipt) | CANDIDATE
Breakeven-inequality provenance: \Delta W_{\mathrm{compute}}, \Delta W_{\mathrm{lookup}}, \epsilon_{\mathrm{overhead}} defined as DCC-declared deterministic quantities; undeclared quantities force the safe Direct-Compute fallback | §1.15, §5.9 | Data-loss audit (unspecified \Delta W_{\mathrm{compute}} provenance) | CANDIDATE
Universal Envelope Structure figure box-normalized to uniform 67-column interior (pre-existing raggedness: border 65, rows 65–67); row text preserved byte-for-byte | §4 | — | editorial

Cross-file dependencies flagged (deferred under the one-file-at-a-time discipline, to be integrated in subsequent versioned updates): docs/qexpr.md one-line touch-up at its next versioned opportunity (breaker-3 receipt name RECEIPT_METRIC_EXHAUSTED inherited from the §5.9 registry); docs/qlog.md §6.1 background-saturation wording resolves to OP_SATURATE; docs/abstractionLayers.md (road-receipt and π/zk findings — next queue item); docs/simemp.md (Eq-1 / U / BOC compliance-measure formalization); glossary entries for the new receipt names.

H2: 1. Definitions

H3: 1.1. Qn Artifact
A qn artifact is a governed, finite, identifiable, measurable, and security-constrained computational entity. An artifact is not a passive scalar; it is an immutable state node embedded in a local causal-arrow.

H3: 1.2. Qn Primitive Envelope
The minimal governance boundary required for any computational entity within the licensed-work to exist, undergo transformation, achieve sealing, or traverse a simemp-gateway, governed under Tier 1 Existential Invariants and optimized via Tier 2 Operational Agilities within the simemp framework.

H3: 1.3. Universal Envelope
The invariant set of structural blocks shared across all artifact types, parameterized via structural-indirection:

$$\mathcal{E} = \langle \text{Identity}, \, \text{Type}, \, \text{State}, \, \text{Lineage}, \, \text{DCC}, \, \text{Metric}, \, \text{Security}, \, \text{Exposure}, \, \text{PayloadRef} \rangle$$

H3: 1.4. Typed Payload Profile
The specialized content block carried by an artifact (numeric quantities, symbolic expression graphs, pure morphisms, state receipts, identity assertions, capability grants, optimization pointers, or composite workflows). External user data remains a sovereign, protocol-blind payload.

H3: 1.5. Identity
The governed witness of an artifact's existence. Identity is local-first, collision-resistant,

versionable, traceable along causal lineage, and cryptographic-algorithm agile.

H3: 1.6. Lineage
The directed acyclic causal history of an artifact, recording parent commitment hashes, causal birth index, and transformation receipt references.

H3: 1.7. DCC Profile
The machine-readable contract defining Dependencies, Constraints, and Capabilities. It serves as an abstract contract rather than a hardcoded implementation binding via DCC Profiles.

H3: 1.8. Metric Envelope
The finite measurement boundary declaring resource limits on storage, expression graph depth, precision bounds, and execution budgets via the metric-envelope.

H3: 1.9. Security Envelope
The governed protection boundary declaring authority limits, delegation proofs, Post-Quantum Cryptography (PQC) algorithm identifiers, and swappable revocation paths via the security-envelope.

H3: 1.10. Lifecycle State
The discrete operational status of an artifact: candidate, constructed, sealed, verified, rejected, halted, quarantined, exportable, exported, or revoked via the lifecycle-state.

H3: 1.11. Receipt
A bounded, immutable record of a state transition, fault translation, or boundary traversal, acting as a dissipative receipt exporting operational entropy.

H3: 1.12. Unverified Candidate
A proposed computational structure generated via heuristic, neural, or stochastic processes that has not passed gateway verification. Candidates cannot cross a simemp-gateway.

H3: 1.13. Numeric Payload
The specialized profile for numeric entities, explicitly encapsulating Zoom z, Remainder r, and Dimension d.

H3: 1.14. Exposure Class
The classification governing artifact visibility: local-only, sealable, exportable, attested, quarantined, or revoked.

H3: 1.15. Q(reuse) Optimization Morphism
A Category B primitive referencing existing immutable content commitment hashes within the local causal DAG. Under physical constraints, Q(\text{reuse}) is admissible if and only if the predicted computational work of direct re-evaluation exceeds the lookup and retrieval dissipation across the Memory Wall (\Delta W_{\mathrm{compute}} \gt \Delta W_{\mathrm{lookup}} + \epsilon_{\mathrm{overhead}}).

Breakeven provenance [v1.3.0 — CANDIDATE]: the terms of the breakeven inequality are measured, not assumed. \Delta W_{\mathrm{compute}} is the DCC-declared deterministic upper-bound estimate of direct re-evaluation work, expressed in Abstract Compute Units U and calibrated at the Layer-0 anchor, produced by the cost model declared in the morphism's dcc-profile — never by runtime sampling. \Delta W_{\mathrm{lookup}} is the DCC-declared retrieval dissipation across the local memory hierarchy, and \epsilon_{\mathrm{overhead}} is a DCC-declared fixed constant of the local gateway. All three MUST be declared (No-Implicit Rule); an undeclared quantity forces the safe fallback to Direct Compute (§5.9 precedence), never a guessed admission.

H3: 1.16. Q(bypass) Optimization Morphism
A Category B primitive allowing a simemp-gateway to short-circuit execution upon verifying an algebraic identity precondition in \mathcal{O}(1) register time, emitting a zero-cost bypass receipt without memory traversal or compute budget consumption.

[v1.3.0 — CANDIDATE, Review Integration QO-5] The Layer-3 node-class semantics of this morphism is the OP_BYPASS canonical contraction of Node Class 6 (docs/qexpr.md §4.6.1, v1.3.0): the fixed sixteen-rule terminating identity suite whose normal form, together with DCC_SYM_COMMUTATIVE child sorting, governs canonical node hashing. The former umbrella name OP_REWRITE is a deprecated alias retained only for resolvability of prior artifacts and is retired at the QnIR v0.2 candidate.

H2: 2. Paradigm Neutrality

This specification does not mandate an object-oriented programming (OOP) paradigm. A Qn artifact is not an OOP object.

The model explicitly rejects:
- Hierarchical class inheritance and dynamic dispatch tables.
- Hidden mutable state and unmetered side effects.
- Implicit type coercions and ambient runtime environments.

Transformations are explicit governed morphisms. State changes yield new immutable artifacts indexed monotonically along the local causal-arrow.

H2: 3. Provisional Operational Axioms

The Qn operational model is governed by 14 operational axioms. These axioms represent candidate search constraints verified provisionally via computational-consistency.

H3: Axiom 1 — Explicit Envelope
No Qn artifact exists within the governed universe unless it carries an explicit universal-envelope:

$$\forall \alpha \in \mathcal{Q}, \quad \exists ! \, \mathcal{E}(\alpha)$$

H3: Axiom 2 — Finiteness
Every Qn artifact must possess a finite canonical representation and a bounded resource envelope. Actual infinities are physically non-realizable and excluded from execution:

$$\forall \alpha \in \mathcal{Q}, \quad \mathrm{Size}(\alpha) \lt \infty \quad \wedge \quad \mathrm{Budget}(\alpha) \lt \infty$$

H3: Axiom 3 — Typed Payload Separation
Fields specific to numeric representations (Zoom z, Remainder r, Dimension d) belong strictly to typed payload profiles and must not be made mandatory for non-numeric entities. Sovereign user data remains an uninspected payload.

H3: Axiom 4 — Local Genesis Dependency
Every governed Qn artifact within a local universe must trace its lineage directly or transitively to the local genesis origin Q(0):

$$\forall \alpha \in \mathcal{Q}_N, \quad Q(0)_N \in \mathrm{Lineage}(\alpha)$$

Local artifacts form single-rooted directed trees originating at physical silicon root Q(0)_N. Higher-order network entities operating across the Outer Ring (such as chronosa) are topological colimits (sheaf gluings) over acyclic forests of mutually verified local Axiom-4 trees.

H3: Axiom 5 — Acyclic Causal Lineage
Lineage graphs are strictly acyclic, governed by dependencies-determinism. Every artifact must be born after its causal dependencies along the causal-arrow:

$$\mathrm{Index}(\mathrm{Parent}(\alpha)) \lt \mathrm{Index}(\alpha)$$

H3: Axiom 6 — Totality of Governed Operations
Every governed operation must terminate in a declared, finite terminal state via dissipative metric-exhaustion within its declared budget:
SUCCESS, FAILURE, NO_SOLUTION, TIMEOUT, BUDGET_EXHAUSTED, INCONCLUSIVE, QUARANTINED, PRECISION_INSUFFICIENT, EXPRESSION_TOO_COMPLEX, REJECTION_BY_GATEWAY, DIVISION_BY_ZERO_REJECTED, or ORDERING_INDETERMINATE_AT_ZOOM. Silent non-termination is prohibited.

H3: Axiom 7 — Receipted Boundary Events
Receipts are mandatory for lifecycle state transitions, gateway crossings, sealing events, export decisions, fault translations, and capability invocations. Receipts serve as verifiable witnesses exporting operational entropy.

H3: Axiom 8 — Local-first Exposure
Artifact identifiers are rooted locally. Raw local Qn artifacts are never exposed directly to remote nodes across cdqn without explicit authorization and bounded public attestation under local-first rules.

H3: Axiom 9 — Numeric Precision Explicitness
Every numeric Qn artifact must declare finite precision bounds, explicit Zoom z, Remainder r, and Dimension d. Implicit rounding and truncation are prohibited:

$$\text{NumericValue} = \langle z, \, r, \, d \rangle$$

H3: Axiom 10 — No Floating-Point Semantics
IEEE 754 floating-point representations are forbidden at the Qn semantic layer.
Non-deterministic mantissa rounding and platform-dependent approximations are rejected.

H3: Axiom 11 — Bounded Self-Description
Governance metadata may describe its own schema, but self-description depth must remain finite, bounded, and version-anchored to prevent infinite recursive reflection towers.

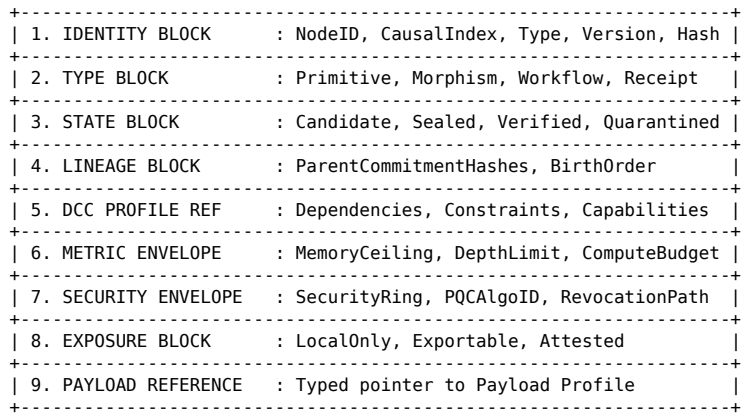
H3: Axiom 12 — License Lineage Where Applicable
Lineage metadata and the canonical paternity-reference must be preserved in distributed artifacts, exported attestations, compiled qnir binaries, and public APIs pursuant to the ssl. Encapsulating a Payload does not convert it into a derivative-work.

H3: Axiom 13 — Versioning Creates New Artifacts
Versioning an artifact produces a new immutable artifact with an advanced causal index referencing the predecessor. Mutation in place is prohibited. Successor workflows derived through qn-rsi do not overwrite existing execution paths; they are emitted as distinct causal versions.

H3: Axiom 14 — Generative Candidate Boundary
Unverified candidates proposed by stochastic, generative, or neural processes can exist locally only if explicitly tagged as unverified candidates. Candidates cannot cross a simemp-gateway or achieve sealed execution without deterministic validation.

H2: 4. Universal Envelope Structure

The Universal Envelope \mathcal{E} is structured using structural-indirection, ensuring that cryptographic suites or binary layouts can be upgraded without invalidating existing causal DAGs.



H3: 4.1. Identity Block
Declares local node identifier, local causal index, version identifier, content commitment hash, and cryptographic agility metadata.

H3: 4.2. Type Block
Declares the artifact category as an independent top-level descriptor (numeric scalar, expression DAG, morphism, relational assertion, receipt, identity assertion, capability grant, optimization pointer, or composite workflow).

H3: 4.3. Lifecycle State Block
Declares current governance status conforming to Axiom 7 and the lifecycle-state.

H3: 4.4. Lineage Block
Declares parent artifact references, local causal index, birth order, derivation type, and transformation receipt references.

H3: 4.5. DCC Profile Reference
Machine-readable reference to an abstract capability contract declaring prerequisites, operational ceilings, and permitted transformations via the dcc-profile.

H3: 4.6. Metric Envelope Reference
Declares finite operational bounds (storage footprint, expression graph depth, recursion limits, and execution budgets) via the metric-envelope.

H3: 4.7. Security Envelope Reference
Declares authority scope, delegation limits, trust assumptions, PQC algorithm identifiers, and swappable revocation endpoints via the security-envelope.

H3: 4.8. Exposure Class Block
Declares visibility scope: local-only, sealable, exportable, attested, quarantined, or revoked.

H3: 4.9. Payload Reference
A typed, content-addressed pointer to the associated payload profile.

H2: 5. Typed Payload Profiles

H3: 5.1. Numeric Payload Profile
Carries Zoom z, Remainder r, Dimension d, precision boundaries, and base-independent scale lattice representations (docs/qm.md §3).

H3: 5.2. Expression Payload Profile (Content-Addressed Expression DAG)
Under the Best of Choices (BOC) Policy (docs/simemp.md §7), symbolic expressions (qexpr) are represented as Content-Addressed Expression DAGs (CAE-DAGs) with cryptographic hash-consing and arity-indexed child commitment vectors (docs/qexpr.md):

$$\alpha = \Big\langle \mathrm{OpCode}, \, \mathbf{D}_{\mathrm{children}}, \, z, \, r, \, d, \, \mathrm{DCC}_{\mathrm{ref}} \Big\rangle$$

- Deduplication: Identical sub-expressions share identical commitment hashes, eliminating redundant leaf allocation across the Memory Wall.
- Complexity Ceilings: Declares strict graph depth ceilings, maximum node counts, and compile-time collapse budgets.
- Thermodynamic Invariance: Reversible normalization within the DAG incurs zero Landauer dissipation ($W = 0$).

H3: 5.3. Operation Payload Profile
Carries domain-to-codomain morphism signatures, formal preconditions, postconditions, metric consumption rates, and receipt emission behavior.

H3: 5.4. Receipt Payload Profile
Carries terminal execution state, consumed resource budgets (metric-exhaustion), causal birth index, parent commitment references, and gateway signatures on each emitted receipt.

H3: 5.5. Identity Payload Profile
Carries Identity Class (Machine Identity, Human Identity, AI Agent Identity), pseudonyms, capability bounds, and revocation endpoints, bounded by Identity Invariants. Under the non-rogue-invariant, machine identities are subordinated to verified human principals.

H3: 5.6. Attestation Payload Profile
Carries public commitments, issuer pseudonyms, capability proofs, metric summaries, and SSL lineage assertions for network exposure across cdqn.

H3: 5.7. Capability Payload Profile
Carries granular execution rights, operational constraints, delegation depths, and expiration bounds.

H3: 5.8. Workflow Payload Profile
Carries a directed acyclic graph of governed morphisms, input state prerequisites, execution checkpoints, intermediate collapse budgets, and terminal receipt criteria. Underpins composite state workflows (q-workflows) and recursive self-optimization loops (qn-rsi).

H3: 5.9. Optimization Payload Profile (Category B Primitives)
Governs explicit sub-graph reuse and gateway short-circuiting under physical thermodynamic constraints:

H4: Q(bypass) Execution
Checks immediate operands for algebraic identity conditions in $\mathcal{O}(1)$ register time (such as unit multiplication $\times Q(1)$, zero addition + Q(0), rotor normalization R R^\dagger = Q(1), or coboundary nilpotency $\mathbf{d} \circ \mathbf{d} = Q(0)$). When detected, the gateway short-circuits execution and emits RECEIPT_BYPASS_IDENTITY at zero compute cost ($\Delta W = 0$).

H4: Q(reuse) Execution & Thermodynamic Breakeven
References an existing immutable content commitment hash within the local causal history. Querying a hash commitment across physical memory hierarchies dissipates dynamic bus power

(\mathcal{O}(C V^2 f)). Under the Metric Invariant, Q(\text{reuse}) is admissible if and only if the computational cost of direct evaluation exceeds lookup dissipation:

\mathrm{Admit}\big(Q(\text{reuse})\big) \iff \Delta W_{\mathrm{compute}} \gt \Delta W_{\mathrm{lookup}} + \epsilon_{\mathrm{overhead}}

If direct computation is cheaper than memory traversal (e.g. single-cycle integer ALU operations), the runtime executes direct computation, bypassing hash-table lookups to prevent Memory Wall bloat.

H4: The Local Evaluation Pipeline
Within any single morphism evaluation step, the gateway enforces strict causal precedence:

Q(\text{bypass}) \prec Q(\text{reuse}) \prec \text{Direct Compute}

[v1.3.0 — CANDIDATE, Review Integration QO-5] At the macro-DAG scale (Layer 3), rewrite executes under the Node Class 6 split (docs/qexpr.md §4.6, v1.3.0): the canonical contraction suite OP_BYPASS applies the fixed \mathcal{O}(1) identity rules and governs canonical node hashing, while general equality saturation is legal only as OP_SATURATE inside an explicit Q(\text{rsi}) workflow whose DCC profile pins a deterministic rule-application schedule, iteration/node/depth ceilings, and a deterministic hash-tie-broken cost extractor — emitting a new versioned artifact (Axiom 13) rather than mutating canonical identity. Within a saturation pass, sub-graph unifications via Q(\text{reuse}) may reveal newly adjacent terms that trigger downstream OP_BYPASS cancellations; the causal precedence of this section is unchanged.

H4: Canonical Circuit-Breaker Receipt Registry [v1.3.0 — CANDIDATE]
[Review Integration QO-5 / data-loss audit] The Layer-3 structural circuit breakers (docs/qexpr.md §6) emit the following canonically named receipts, registered here as the operational-axiom authority (Axioms 6 and 7). No new Axiom-6 terminal states are required:

Circuit breaker | Bound | Receipt | Terminal state (Axiom 6) | Reason code
1. Static graph depth ceiling | Depth(DAG) ≤ D_max | RECEIPT_EXPRESSION_TOO_COMPLEX | EXPRESSION_TOO_COMPLEX | D_MAX_REACHED
2. Node count budget allocation | Count(Nodes) ≤ N_max | RECEIPT_BUDGET_EXHAUSTED | BUDGET_EXHAUSTED | N_MAX_REACHED
3. Metric collapse budget ceiling | Burn(U) ≤ Budget_Q1 | RECEIPT_METRIC_EXHAUSTED | BUDGET_EXHAUSTED | U_BUDGET_EXHAUSTED
4. Saturation iteration ceiling | Epochs(OP_SATURATE) ≤ K_max | RECEIPT_BUDGET_EXHAUSTED | BUDGET_EXHAUSTED | K_MAX_REACHED

Two non-breaker canonical receipts complete the registry: RECEIPT_SATURATION_CONVERGED (terminal state SUCCESS) is emitted when an OP_SATURATE workflow halts on convergence before any ceiling, and RECEIPT_BYPASS_IDENTITY (zero metric cost, \Delta W = 0) is emitted by every OP_BYPASS contraction. Breaker 3's receipt name is declared here for the first time; docs/qexpr.md §6 inherits it at its next versioned touch-up.

H2: 6. Identity and Lineage Rules

H3: 6.1. Local-first Identifier Structure
An artifact identifier is rooted strictly in its local-first genesis:

\mathrm{ID} = \langle \text{NodePseudonym}, \, \text{CausalIndex}, \, \text{Type}, \, \text{Version}, \, \text{Commitment} \rangle

H3: 6.2. No Global Raw Identity Exposure
Raw local node identifiers are prohibited from direct network exposure. Network interaction across cdqn uses accountable pseudonyms, cryptographic commitments, and zero-knowledge capability proofs.

H3: 6.3. Versioning
Every modification produces a distinct artifact with an advanced causal index. Predecessors remain immutable components of historical lineage.

H3: 6.4. Algorithm Agility
Cryptographic primitives must support seamless migration via structural-indirection. No single signature suite or hash function is treated as a permanent dependency.

H2: 7. DCC Profile Rules

H3: 7.1. Dependencies
Explicitly declares required parent artifacts, signing identities, capability grants, and physical context prerequisites under dependencies-determinism. Dependencies are evaluated as abstract contracts via structural-indirection to allow modular implementation swaps.

H3: 7.2. Constraints
Declares hard operational ceilings: memory allocation bounds, recursion limits, complexity degree ceilings, precision limits, and excluded states (e.g., denominator zero exclusion).

H3: 7.3. Capabilities
Declares permitted transformations: morphism rights, sealing authorization, network export authorization, and delegation scopes.

H3: 7.4. Semantic Alignment & Anti-Drift Invariant
To prevent translation drift between high-level human intent and lower-level machine execution, an artifact's DCC contract must bind its human-readable specification in Qs directly to its intermediate execution representation in qnir. A state transition is valid if and only if the verification receipt certifies that execution did not drift from the declared intent contract ($H(\mathrm{Intent} \mid \mathrm{Execution}) = 0$).

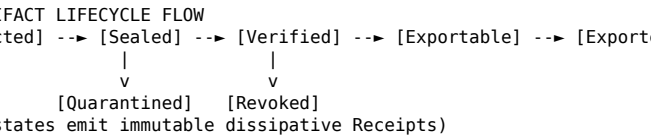
H2: 8. Metric Envelope Rules

Every metric-envelope must declare finite, measurable boundaries:
- Footprint Bounds: Storage allocation size, expression DAG depth ceilings, and complexity degree bounds.
- Execution Bounds: Step ceilings, transformation budgets, and compilation collapse allocations.
- Numeric Bounds: Scale zoom ceilings ($z \le z_{\max}$) and uncertainty boundaries.
- Memory Wall Accounting: Explicit accounting for data movement, cache locality, and bus transmission dissipation.

H2: 9. Security Envelope Rules

Every security-envelope must define protection parameters against computationally bounded adversaries:
- Inner-ring and outer-ring scoping conforming to the dual-ring-pqc-boundary.
- Swappable revocation paths and emergency halt authorities designed for zero-day patchability.
- Post-Quantum Cryptography (PQC) algorithm identifiers with explicit migration semantics.
- Accountable pseudonymity; anonymous entities are strictly forbidden from traversing network boundaries.

H2: 10. Lifecycle and Receipts



H3: 10.1. Lifecycle States
Standard recognized operational states conforming to the lifecycle-state: candidate, constructed, sealed, verified, rejected, halted, quarantined, exportable, exported, revoked.

H3: 10.2. Candidate State
A proposed stochastic or heuristic structure; classified strictly as an unverified candidate; prohibited from crossing a simemp-gateway.

H3: 10.3. Constructed State
Formalized into canonical envelope syntax; awaiting constraint evaluation.

H3: 10.4. Sealed State
Identity-committed, metric-bounded, security-checked, and immutable.

H3: 10.5. Verified State
Validated against explicit formal proof invariants or test suites by an authorized verifier.

H3: 10.6. Rejected, Halted, and Quarantined States
Protective terminal states emitting mandatory failure receipts exporting operational entropy.

H3: 10.7. Exportable and Exported States
Certified by the simemp-gateway for controlled exposure across the cdqn network via the Exposure Functor.

H3: 10.8. Revoked State

Invalidated via an explicit, signed revocation receipt that permanently seals a compromised branch and migrates lineage.

H2: 11. Relation to Abstraction Layers

H3: 11.1. Layer 0 (Physical Substrate)
Raw physical hardware noise and clock signals in layer-0 carry no primitive envelope until processed across the onboarding gateway (docs/q0_q1.md §2).

H3: 11.2. Layer 0 to Layer 1 Gateway
Harvests physical entropy, generates Inner-Ring PQC keys, and constructs local origin Q(0) and first unit Q(1) across the simemp-gateway.

H3: 11.3. Layer 1 (Node Genesis)
Houses foundational Category A primitives in layer-1 (Q(0), Q(1), single digits Q(2)\dots Q(9), the Abstract Compute Unit U, and genesis axis d_1).

H3: 11.4. Layer 2 (Governed Morphisms)
Hosts elementary arithmetic morphisms (+ , - , \times, \div), spatial reflection involutions (\mathcal{I}_d), and Category B optimization morphisms (Q(reuse), Q(bypass)).

H3: 11.5. Layer 3 (Symbolic Composition)
Hosts Content-Addressed Expression DAGs (qexpr), multi-axial Clifford geometric frames, and compile-time reduction engines.

H2: 12. Relation to QnLang and QnIR

The primitive envelope provides the formal execution target for the programming substrate:
- qnlang: Authors computations declaring explicit DCC profiles, metric budgets, and payload profiles. Prohibits implicit defaults and floating-point semantics under the no-implicit-rule.
- qnir: Intermediate representation flattening Content-Addressed Expression DAGs into linear, bounded instruction streams for deterministic hardware evaluation.
- Compile-Time Collapse: Symbolic expression DAGs must collapse deterministically to concrete canonical leaf tuples prior to terminal receipt emission.

H2: 13. License Alignment

This specification strictly enforces the legal provisions of the ssl:

H3: 13.1. Paternity Reference
All derivative implementations, hardware drivers, runtime specifications, and APIs derived from the licensed-work must embed the canonical paternity-reference:

Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License (SSL). Parent Repository: https://github.com/cdqn5249/cdqn

H3: 13.2. Open Core Invariants
1. Anti-Patent Defense: Commercial or Derivative Works licenses terminate automatically upon initiating patent litigation against the Author or project ecosystem.
2. Non-Scaling Open Access: Royalty-free access is guaranteed for non-commercial, academic, and sub-threshold usage under the Open Core Invariants.

H3: 13.3. Scale Auditing
The metric and identity envelopes supply verifiable telemetry (active compute instances, containers, autonomous agents, and monthly API transactions) to verify adherence to commercial scaling thresholds (LICENSE.md §1.5).

H2: 14. Exclusions

The following items are outside the scope of this foundational primitive envelope and belong to dedicated specifications:
1. Detailed physical entropy sampling and NIST SP 800-90B health test implementations (specified in docs/q0_q1.md).
2. High-level grammar and syntax rules for QnLang (Milestone 2).
3. Virtual machine instruction set architecture and register schedules for QnIR (Milestone 2).
4. Physical simulation equations and discrete quantum Hamiltonians for Quang Physics (\mathrm{Qphy}).
5. Concrete multi-party threshold signature ceremony protocols across distributed swarms (specified in docs/chronosa.md).

```
H2: 15. Open Items

The following formal specifications remain open for subsequent releases:
1. Canonical binary serialization format (e.g., deterministic CBOR or custom packed Qn-binary)
   for the Universal Envelope.
2. Formal schema definition for Post-Quantum Cryptography algorithm agility migration
   certificates.
3. Automated verification checker specification for state transitions from Sealed to Verified.
4. Execution mechanics and termination bounds for higher-order recursive self-optimization
   workflows (qn-rsi).
```

Annex J — qs.md

Source: docs/qs.md *— 1.3.0, updated 2026-10-26. Rendered content per Annex Guide conventions N1–N6: 527 source lines, 755 rendered; 48,056 → 43,976 characters; source SHA-256* e24e35df ef022674 278411de ea0c5dfb 53ff0d10 03579ec9 6e609817 6525cedf*. Content fidelity verified mechanically at build time.*

```
[front matter] ------------------------------------------------------------
[front matter] title: Qs — Quang Semantics (Constructive Relational Intent and Epistemic
[front matter]   Grounding)
[front matter] version: 1.3.0
[front matter] updated: 2026-10-26
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Canonical specification of the constructive semantic base domain,
[front matter]   multi-axial Clifford language frames, epistemic modal spectrum,
[front matter]   human-agent intent contracts, and AI alignment under SIMEMP
[front matter]   constraints.
[front matter] terms_used: boc-policy, cae-dag, causal-arrow, cdqn, chronosa, computational-
[front matter]   consistency, compute-unit-u, dcc-profile, dependencies-determinism,
[front matter]   epistemic-anchorage, identity-class, intent-contract, layer-0,
[front matter]   layer-1, licensed-work, metric-exhaustion, no-implicit-rule, open-
[front matter]   core-invariant, paternity-reference, q-reuse, q0, q1, qm, qn, qn-rsi,
[front matter]   qs, simemp-gateway, ssl, zoom-z
[front matter] ------------------------------------------------------------


H1: Qs — Quang Semantics: Constructive Relational Intent and Epistemic Grounding


Field | Specification
Document Title | Qs — Quang Semantics: Constructive Relational Intent and Epistemic Grounding
Version | 1.3.0
Last Updated | 2026-10-26 (Bao Loc, Vietnam)
Author | Christophe Duy Quang Nguyen
License | Scaling Source License (SSL) 1.0
Status | Canonical Domain Specification — Category D (Local-First Base Domain)


H2: Review Integration Record (v1.3.0)

Source: docs/AiReviews/AiReviewQnIRv0.1.md — transcript of the external adversarial review
(Google Deep Review, October 2026) of the condensed Qn universe specification and QnIR v0.1
candidate draft. This update integrates the parser-boundary findings of that review (reviewer
open question QO-11, Segment 3 Point 2, together with the boundary-trust verdict finding on
Objection 10): the epistemic ceiling for parser-derived candidates, the Deterministic Canonical
Round-Trip Paraphrase & Co-Signing Handshake, and the typed parser-boundary failure receipts.
Under Axiom 13, this file is a new versioned artifact; no prior content is mutated in place. New
constructions and semantic refinements introduced below are tagged [v1.3.0 — CANDIDATE] pending
BOC sealing; editorial corrections are applied directly and recorded here.


Change | Section | Reviewer source | Type
Parser-boundary modality ceiling: a candidate DAG produced by a natural-language parser enters
the pipeline typed no higher than MODAL_HYPOTHESIS; direct admission at MODAL_EMPIRICAL or
MODAL_CONSTRUCTIVE is barred regardless of parser-attached tags | §4.4 (new) | QO-11 / Segment 3
Point 2 | CANDIDATE
Deterministic Canonical Round-Trip Paraphrase & Co-Signing Handshake — deterministic verbalizer
read-back of DCC_INTENT_CONTRACT_v1 and W_{\mathrm{allowed}} to the principal, Inner-Ring
cryptographic co-signature required prior to SEALED | §6.1 (new) | QO-11 / Segment 3 Point 2 |
CANDIDATE
Typed parser-boundary failure receipts — closed-catalog extension with
RECEIPT_INTENT_PARSE_AMBIGUOUS, RECEIPT_INTENT_MODALITY_MISMATCH,
RECEIPT_INTENT_SCHEMA_MISMATCH, RECEIPT_ROUNDTRIP_CONFIRMATION_REJECTED | §6.2 (new) | QO-11 /
Segment 3 Point 2 | CANDIDATE
BOC degenerate-tie discipline: an argmax tie within the compliance threshold emits
RECEIPT_INTENT_PARSE_AMBIGUOUS instead of silently sealing an arbitrary winner | §6 | QO-11 /
Segment 3 Point 2 | CANDIDATE
```

Logical typing of r_{\mathrm{social}} — candidate deontic anchoring (contrary-to-duty obligation sub-DAG) recorded without amending §5.1 or §5.2 | §5.3 (new) | QO-11 / Segment 3 Point 2 | CANDIDATE
Fixed normative axes vs. open-vocabulary lexical payload: the ≤ 16 basis dimensions are fixed thematic axes; open-vocabulary identifiers reside as content-addressed commitments inside blade coefficients | §3.4 (new) | QO-11 / Segment 3 Point 2 (gap 3) | CANDIDATE
Pipeline metering note: every stage (parsing, measurement, selection, verbalization, co-signature verification) consumes finite U under Axiom 6; billing attribution deferred to simemp.md | §6 | Segment 1 Point 4 (noted, deferred) | CANDIDATE
Editorial: "An ai model" -> "An AI model" | §6 | — | editorial
Editorial: pre-existing 1-column overflow in the §5 Intent Contract figure (Epistemic-Scope row 81 cols vs 80-col borders); row re-padded to the uniform 80-col grid, field content preserved | §5 | — | editorial

Receipt-naming record (BOC). Of the reviewer's proposed alternates, RECEIPT_INTENT_PARSE_AMBIGUOUS is adopted over RECEIPT_SEMANTIC_AMBIGUITY_INDETERMINATE (the INTENT_PARSE stem matches the boundary it polices and the DCC_INTENT_CONTRACT vocabulary of §5). The reviewer's pair RECEIPT_INTENT_MODALITY_MISMATCH / RECEIPT_INTENT_SCHEMA_MISMATCH is adopted as two distinct receipts rather than one, because they denote disjoint failure modes: an epistemic-scope violation (the §4 spectrum versus the contract's Epistemic Scope field) and a structural contract-validation failure, respectively. All four receipts are Qs-domain parser-boundary receipts; they are distinct from the Layer-2/3 circuit-breaker receipts registered in docs/qnPrimitive.md v1.3.0 §5.9, which they neither modify nor extend.

Cross-file dependencies flagged (deferred under the one-file-at-a-time discipline, to be integrated in subsequent versioned updates): glossary registration of the four new receipt names (deferred under the standing no-dead-entries discipline of docs/qs_history.md §8); docs/simemp.md — formal billing attribution of gateway-crossing validation work against the invoking principal's budget (Segment 1 Point 4), to be integrated at that file's next versioned update; docs/abstractionLayers.md — Inner-Ring co-signature mechanics (QO-13 queue item); QnIR v0.2 — the closed receipt catalog of the candidate VM must carry the four parser-boundary receipts, and design requirement R12 (H(\text{Intent} \mid \text{Execution}) = 0) holds only after the §6.1 handshake has supplied the human-anchored Intent object.

H2: Normative References

The following documents establish the physical, structural, and legal constraints governing \mathrm{Qs}. If a technical conflict arises, simemp.md governs; if a structural conflict arises, abstractionLayers.md governs; if a legal conflict arises, LICENSE.md governs. References are deliberately version-unpinned: each target document's front matter carries its own authoritative version, so this specification never duplicates drift-prone patch state.

Document | Role | Target
docs/simemp.md | Constitutional constraints, thermodynamics, and Dependencies Determinism | simemp.html
docs/abstractionLayers.md | Layer architecture and Complexity Degree Stratification | abstractionLayers.html
docs/qnPrimitive.md | Universal Envelope, operational axioms, and optimization primitives | qnPrimitive.html
docs/q0_q1.md | Primary genesis origin Q(0) and first unit Q(1) | q0_q1.html
docs/qm.md | Constructive numeric leaf substrate and Diophantine division constraints | qm.html
docs/qm_geometry.md | Multi-axial frames, Clifford geometric algebra, and float-free rotations | qm_geometry.html
docs/qexpr.md | Layer 3 Content-Addressed Expression DAG specification | qexpr.html
docs/chronosa.md | Emergent non-local causal intelligence and human empowerment | chronosa.html
LICENSE.md | Scaling Source License 1.0 governing the Licensed Work | LICENSE.md

H2: 1. Epistemic Stance and Constitutional Grounding

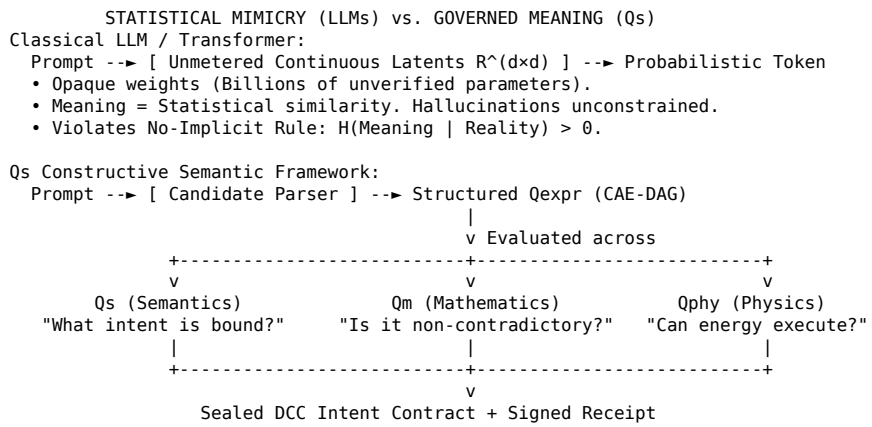
In classical artificial intelligence, natural language processing relies on statistical correlation across continuous floating-point vectors (\mathbb{R}^d). Transformer models predict tokens based on statistical consensus rather than physical truth or causal provenance. Consequently, classical models suffer from hallucinations, lack causal accountability, and cannot distinguish between verified empirical facts and unanchored fiction.

In the qn computational universe, language, meaning, and intent are not ungrounded statistical approximations. Under docs/simemp.md, semantic meaning must be constructed from finite, verifiable primitives governed by dependencies-determinism:

$$H(S_t \mid \mathcal{D}(S_t)) = 0$$

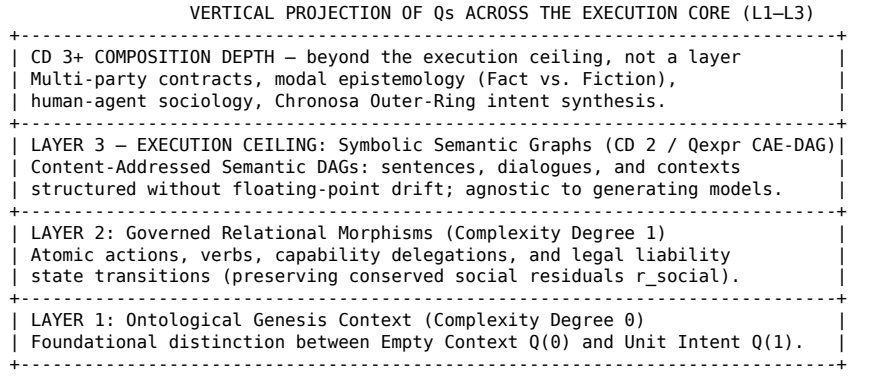
Qs (Quang Semantics) formalizes language, human-machine communication, and AI agent alignment as a constructive, multi-axial relational domain:
- Language is not a black-box tensor: It is a structured graph of discrete semantic blades and relational morphisms.
- Meaning is not statistical similarity: It is an explicit, verifiable Intent Contract bound to human authority.
- The Non-Rogue Invariant: Machines and AI agents cannot originate unanchored intent; they operate strictly as delegated proxies bounded by human sovereignty and thermodynamic metric budgets.

Operational validity is verified empirically via computational-consistency. No unmeasured semantic assumption may cross a simemp-gateway.



H2: 2. Vertical Abstraction Layer Projection of Qs

Mirroring the mathematical domain (docs/qm.md), qs does not reside in a single, isolated tier. It is an orthogonal vertical projection crossing the execution core (docs/abstractionLayers.md §2.5):



- CD vs. Layer Note: Complexity Degree (CD) indexes structural composition depth, not layer numbers. The LAYER 1–3 bands name the execution-core strata crossed by the Qs projection; their CD 0/1/2 labels measure composition depth accumulated at each stratum, and CD 3+ marks depth accumulated above the ceiling without implying a Layer 4.

H3: 2.1. Layer 1 — Contextual Genesis (Complexity Degree 0)
At layer-1, \mathrm{Qs} establishes the primitive origin of meaning:
- Q(0) acts as the Empty Context (the unwritten page, the silent void, zero unfulfilled obligation).
- Q(1) acts as the Unit Intent Measure along dimensional axis d_1, calibrating the baseline quantum of intentional focus and thermodynamic attention.

H3: 2.2. Layer 2 — Governed Relational Morphisms (Complexity Degree 1)
At Layer 2, \mathrm{Qs} establishes atomic relational transformations:
- Verbs, capability delegations, and authorizations executed under explicit DCC Profiles.
- Every action partitions an obligation into an authorized transition and an exact conserved social residual (r_{\mathrm{social}}), ensuring that unfulfilled liabilities are never silently discarded.

H3: 2.3. Layer 3 — Symbolic Composition Graphs (Complexity Degree 2)
At Layer 3, \mathrm{Qs} structures compound propositions, dialogues, and multi-agent interaction histories as Content-Addressed Expression DAGs (docs/qexpr.md):
- Identical semantic sub-trees and legal clauses share immutable cryptographic hashes via cae-dag hash-consing.
- Incurs zero redundant bit allocation across the Memory Wall via Q(reuse).

H3: 2.4. CD 3+ Composition Depth — Epistemic Modality and Social Systems
Above the Layer 3 execution ceiling, \mathrm{Qs} accumulates Complexity Degree 3+ composition depth — emergent social capability, not additional execution layers:
- The formal separation between physical facts and sovereign fiction via epistemic-anchorage.
- Multi-party consensus synthesized by chronosa over public road receipts.

H2: 3. Words and Grammar as Multi-Axial Clifford Blades

In \mathrm{Qs}, vocabulary and linguistic relationships are formalized through Clifford Geometric Algebra (\mathcal{C}\ell_{p,q}) over an orthogonal frame of semantic dimensions (docs/qm_geometry.md §2.1):

\mathcal{D}_{\mathrm{semantic}} = \{ d_{\mathrm{intent}}, \, d_{\mathrm{consent}}, \, d_{\mathrm{obligation}}, \, d_{\mathrm{temporal}}, \, d_{\mathrm{agent}}, \, \dots \}

```
                  LINGUISTIC CLUSTER AS CLIFFORD BLADES
 Noun Blade v_noun : Grade-1 vector along basis e_k (e.g., e_device)
 Modifier Rotor R  : Rational Cayley Rotor in plane e_i ∧ e_j (e.g., "secure")
 Transitive Verb M : Bivector contraction mapping Subject to Object
                     ⟨ v_subject ∧ M_verb,  v_object ⟩ + r_social
```

H3: 3.1. Nouns as Graded Vector Blades
An entity or concept ("battery", "drone", "contract") is a discrete vector blade evaluated at declared Zoom z:

\vec{v}_{\mathrm{noun}} = \sum_{k=1}^n \langle \sigma_k, \, q_{z,k}, \, z, \, r_{z,k}, \, d_k \rangle \, \vec{e}_k

Nouns carry explicit identities, coordinate axes, and conserved residuals.

H3: 3.2. Adjectives and Modifiers as Rational Cayley Rotors
Modifiers ("cold", "verified", "urgent") do not mutate the fundamental type of a concept. A modifier is a Rational Cayley Rotor (docs/qm_geometry.md §5.2) rotating the noun blade within semantic sub-spaces:

\vec{v}' = R \vec{v}_{\mathrm{noun}} R^{\dagger}, \quad \text{where } R = \frac{Q(1) - B}{Q(1) + B}

Because R R^{\dagger} \equiv Q(1), semantic modifications preserve unitary norm by construction: meaning transforms with zero precision drift, zero semantic degradation, and zero floating-point rounding.

H3: 3.3. Verbs as Conserved Relational Morphisms
A verb is not an ambiguous text string. In \mathrm{Qs}, a transitive verb ("Human A delegates task T to Agent B with budget W") is a governed morphism evaluated through Diophantine conservation:

\text{Action} = \big(\text{Delegation} \times \text{Actor} \big) + r_{\mathrm{social}}

If the task fails, encounters a fault, or halts, the residual r_{\mathrm{social}} remains sealed in the lineage. The system holds an immutable record of unfulfilled obligations, precluding silent state abandonment.

H3: 3.4. Fixed Normative Axes vs. Open-Vocabulary Lexical Payloads [v1.3.0 — CANDIDATE]

The dimensional ceiling on semantic frames (

\mathrm{Dim}_{\max} \le 16

, docs/qm_geometry.md §5.2, since the blade count scales as \mathcal{O}(2^n)) is not a vocabulary limit.

The basis dimensions of

\mathcal{D}_{\mathrm{semantic}}

are fixed normative and thematic axes (d_{\mathrm{intent}}, d_{\mathrm{consent}}, d_{\mathrm{obligation}}, d_{\mathrm{temporal}}, d_{\mathrm{agent}}, …), not per-word slots: allocating a distinct basis blade to every open-vocabulary noun would exhaust the ceiling after the first handful of terms, collapsing the frame algebra.

Open-vocabulary lexical identifiers ("battery", "drone", "contract", …) instead reside as content-addressed commitments inside the blade coefficients: a noun blade's coefficient tuples (§3.1) carry the \mathcal{H}-commitments of the entity's Layer-2 and numeric-leaf definitions, and two occurrences of the same term share one commitment via CAE-DAG hash-consing (§2.3). Adding a new word to the discourse therefore allocates coefficient data inside existing axes and never a new basis vector. Under the no-implicit-rule, any application that genuinely requires an additional normative axis must declare it explicitly in its frame specification and accept the \mathcal{O}(2^n) blade-count consequence, rather than silently overloading an existing axis.

H2: 4. The Epistemic Anchorage Spectrum: Anti-Hallucination Firewall

To eliminate AI hallucinations, \mathrm{Qs} outlaws ungrounded assertions under the

no-implicit-rule. Every statement in \mathrm{Qs} must declare its explicit epistemic-anchorage:

THE EPISTEMIC ANCHORAGE SPECTRUM

1. MODAL_EMPIRICAL (Grounding G = 1.0)
Anchored directly to Layer 0 physical sensor receipts (NIST entropy, thermocouple, camera, bus telemetry) via a SIMEMP Gateway.
2. MODAL_CONSTRUCTIVE (Grounding G = 1.0)
Algebraic closure proven via Qm / DEC; remainder r = Q(0) (Terminal Exactness receipt). Cannot drift or hallucinate by construction.
3. MODAL_HYPOTHESIS (Grounding 0.3 < G < 1.0)
Explicitly bounded within the Recursive Search Loop (simemp.md §2).
Declared as a working conjecture awaiting physical binary falsification.
4. MODAL_NARRATIVE (Contextually Sealed Creative Worldbuilding)
Internally consistent narrative anchored to author identity Q(0)_author.
True WITHIN the declared context; barred from triggering actuators.
5. MODAL_UNANCHORED (Grounding G = ∅)
Claims empirical or logical truth without parent lineage receipts.
REJECTED AT THE GATEWAY! (RECEIPT_UNANCHORED_ASSERTION_REJECTED)

H3: 4.1. The Lineage Verification Mandate
Under Axiom 4 and Axiom 5 (docs/qnPrimitive.md), a statement cannot claim empirical truth through latent probability alone.
If an artificial agent asserts: *"The motor temperature is 450°C"*, the gateway queries its lineage:
- If the lineage contains an unbroken causal chain terminating in a verified layer-0 hardware sensor receipt, it is admitted as MODAL_EMPIRICAL.
- If the assertion traces only to an internal neural probability distribution without sensor receipts, the gateway intercepts it:

$$\text{Gateway Action: } \mathrm{Reject}\Big(\mathcal{H}(\mathrm{Statement})\Big) \implies \text{Emits Receipt}$$

The gateway emits the rejection receipt RECEIPT_UNANCHORED_ASSERTION_REJECTED; the claim is physically barred from entering the execution pipeline.

H3: 4.2. Contextual Isolation for Creative Fiction
\mathrm{Qs} does not prohibit human imagination, creative writing, or artistic worldbuilding:
- A creative work (such as a fantasy novel or game narrative) is typed as MODAL_NARRATIVE anchored to the author's sovereign identity $Q(0)_{\mathrm{author}}$ and publisher receipt.
- The internal consistency of the fictional universe is preserved within its closed Qexpr sub-DAG.
- The gateway guarantees contextual containment: a fictional sub-DAG is barred from ever triggering real-world physical actuators, financial registers, or legal delegations.

H3: 4.3. The Source-Receipt Invariant (Proposed, CANDIDATE)

Status. This subsection records a proposed extension of the §4 firewall from statements to governed artifacts, sourced from docs/qs_history.md §3. It carries State: CANDIDATE under Axiom 14: it remains subject to deterministic measurement and BOC selection through the pipeline of §6, and it amends nothing in §4.1 or §4.2, which continue to apply exactly as written.

It is proposed that every governed artifact of the qn universe crossing a simemp-gateway declare an explicit pair:

(content modality, source class)

where the content modality is drawn from the epistemic-anchorage spectrum of §4, and the source class is a declared expression of where the artifact's asserted content ultimately anchors. Under the no-implicit-rule, the declaration makes explicit — at the level of composite artifacts and worlds — what the gateway already enforces implicitly at the level of single statements.

ARTIFACT-LEVEL SOURCE DECLARATION (PROPOSED, STATE: CANDIDATE)

Governed Artifact	:= (ContentModality , SourceClass)
ContentModality	: one modality of the §4 spectrum, assigned per atomic claim decomposition of the artifact
SourceClass	: declared expression of where the asserted content ultimately anchors (candidate vocabulary I / P / S, docs/qs_history.md §2)
SourceReceipts	: unbroken chain terminating in physically receipted reality events; no dangling source references

The invariant requires that the source receipt chain terminate in physically receipted reality

events, with the terminating receipt type fixed by the declared modality:

- Empirical content terminates in verified layer-0 sensor or artifact receipts, exactly as mandated by §4.1.
- Constructive content terminates in algebraic closure receipts whose remainder r = Q(0) (Terminal Exactness).
- Contextually sealed creative content terminates in the author-identity (Q(0)_{\mathrm{author}}) and publisher-receipt anchoring already mandated by §4.2.

The extension therefore applies the §4.1 statement-level rejection rule to composite artifacts and worlds: no unsourceable fiction, no unsourced anchor, no history-less constant. An artifact whose source receipt chain cannot be completed is rejected at the gateway exactly as an unanchored statement is, emitting the same class of rejection receipt.

Two cautions are recorded, and they are binding on any future implementation of the invariant:

1. Payload sovereignty is untouched. The invariant governs artifacts that cross the gateway as governed objects of the licensed-work. Sovereign human Payloads remain uninspected behind the safe-harbor air gap (docs/abstractionLayers.md §3.1). Where a creative work is declared as a governed MODAL_NARRATIVE artifact, its anchoring to author identity and publisher receipt is declared by its author, not extracted by inspection.
2. Simulation modality is two-faced. A simulation's execution is MODAL_CONSTRUCTIVE (deterministic algebraic closure), while its claims *about reality* are MODAL_HYPOTHESIS until physically falsified. The pair declaration makes this two-faced split explicit rather than leaving it implicit, in accordance with the no-implicit-rule.

The proposed Anchor Grounding Classes of docs/qs_history.md §2 (Classes I / P / S) supply the candidate vocabulary for the source-class field. Glossary registration of the newly named concepts — and the final sealing of this invariant itself — remain deferred to the integration queue of that document (§8), preserving its discipline against dead glossary entries for unmeasured concepts.

H3: 4.4. The Parser-Boundary Modality Ceiling [v1.3.0 — CANDIDATE]

The §4 firewall inspects statements and artifacts that are already formalized. The remaining trust gap lies upstream of formalization: translating a natural-language utterance into a structured CAE-DAG and a DCC_INTENT_CONTRACT_v1 relies on a statistical or neural candidate parser (§6, step 1) crossing the Axiom 14 boundary.

No statistical parser guarantees

$$H(\text{Intent}_{\mathrm{human}} \mid \text{Utterance}) = 0$$

— dependencies-determinism governs the formalized state, not the pre-formalization act of interpretation.
If the parser hallucinates or inverts the human's actual intent while attaching a valid human origin signature
Q(0)_{\mathrm{author}} and well-formed MODAL_EMPIRICAL or MODAL_CONSTRUCTIVE tags, then every downstream check — the SIMEMP gateways, the Triadic Grounding of §7 (

$$\mathbf{C}_{\mathrm{Qs}} \otimes \mathbf{C}_{\mathrm{Qm}} \otimes \mathbf{C}_{\mathrm{Qphy}}$$

), and the QnIR anti-drift invariant

$$H(\text{Intent} \mid \text{Execution}) = 0$$

— will validate the *wrong* intent with perfect rigor, because the error occurred prior to formalization.

Parser-boundary modality ceiling (rule).

Any candidate DAG produced by a natural-language parser enters the generative pipeline of §6 typed no higher than MODAL_HYPOTHESIS. Direct admission at MODAL_EMPIRICAL or MODAL_CONSTRUCTIVE is barred — including when the parser itself attaches those tags — until the candidate passes the Deterministic Canonical Round-Trip Paraphrase & Co-Signing Handshake of §6.1.

- A parsed intent is, epistemically, a hypothesis about what the principal meant; MODAL_HYPOTHESIS is its correct home in the §4 spectrum — a working conjecture whose only competent falsifier is the principal's own confirmation. Parser confidence scores are not epistemic-anchorage: under the no-implicit-rule they confer no grounding value and must not be mapped onto G.
- The upgrade path is exactly one: §6.1, where the sole authority over the intent's meaning — the human principal — supplies the anchorage that the parser cannot.
- MODAL_NARRATIVE (§4.2) is unaffected: creative content remains author-declared and contextually contained. MODAL_UNANCHORED claims remain rejected at the gateway (§4, class 5), and the §4.3 artifact-level invariant continues to apply exactly as written.

H2: 5. Human-Agent Alignment: The Intent Contract in DCC Profiles

In the Qn universe, alignment between humans and artificial agents is not achieved by fine-tuning neural weights (RLHF). It is enforced through Structural Intent Contracts embedded inside the dcc-profile:

```
                    ANATOMY OF AN INTENT CONTRACT
 +-------------------------------------------------------------------------+
 | DCC_INTENT_CONTRACT_v1                                                  |
 +-------------------------------------------------------------------------+
 | • Sovereign Anchor : Hash(pk_human_principal) [Human Identity Q(0)_H]   |
 | • Agent Proxy      : Hash(pk_agent) [Identity Class: AI_AGENT_IDENTITY] |
 | • Permitted Domain : Multi-axial frame D_allowed (Spatial / Financial / Act)|
 | • Metric Budget    : Max Work Allocation W_allowed in Q(1) units        |
| • Epistemic Scope  : Allowed Modalities {MODAL_EMPIRICAL, MODAL_HYPOTHESIS}  |
 | • Revocation Path  : Immediate un-delegation on human signature         |
 +-------------------------------------------------------------------------+
```

H3: 5.1. Subordination of Machine Agency
Under the identity-class rules (docs/abstractionLayers.md §8.1):
1. An AI agent possesses zero independent root sovereignty. It cannot instantiate an unanchored genesis origin Q(0)_{\mathrm{agent}}.
2. An agent executes strictly as an authorized proxy under an unbroken delegation chain originating from a human principal (Q(0)_H).
3. If the human principal revokes the capability token, the agent's execution authority terminates instantly.

H3: 5.2. The Thermodynamic Leash
Autonomous agents cannot self-replicate or consume unmetered resources. Under Axiom 6 (Totality of Governed Operations), every agentic workflow is bounded by a finite work budget in units of U via metric-exhaustion. When the allocated budget reaches zero, the agent physically halts, emitting a signed termination receipt.

H3: 5.3. Logical Typing of the Social Residual (Proposed, CANDIDATE) [v1.3.0 — CANDIDATE]

Status.

This subsection records a candidate logical anchoring for the conserved social residual r_{\mathrm{social}} of §2.2 and §3.3. It carries State: CANDIDATE under Axiom 14: it remains subject to deterministic measurement and BOC selection through the pipeline of §6, and it amends nothing in §5.1 or §5.2, which continue to apply exactly as written.

Sections §5.2 and §3.3 conserve r_{\mathrm{social}} but leave its logical type unspecified: representing \text{Action} = (\text{Delegation} \times \text{Actor}) + r_{\mathrm{social}} as a Diophantine conservation constraint does not by itself say how multi-clause legal or safety obligations, deadline expirations, or secondary reparations are encoded into the residual once a primary obligation is only partially violated. The reviewer proposal recorded here anchors the semantics of DCC_INTENT_CONTRACT_v1 in resource-bounded dyadic deontic logic with contrary-to-duty (CTD) obligations, composed with dynamic epistemic logic (DEL) for multi-agent state updates.

Under this candidate reading:

- Primary obligations

  $$\mathcal{O}_{\mathrm{actor}}(\alpha \mid W_{\mathrm{allowed}})$$

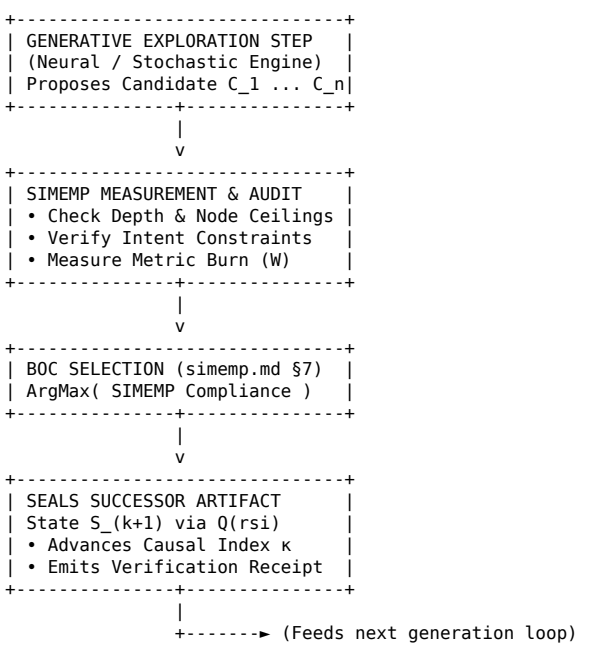
  , permissions P(\alpha), and prohibitions F(\alpha) carry explicit work budgets drawn from the contract's Metric Budget field.
- When a primary obligation goes unfulfilled, a reparational secondary obligation \mathcal{O}(\beta \mid \neg\alpha) takes effect by construction — the contrary-to-duty structure.
- This gives r_{\mathrm{social}} \neq Q(0) a rigorous logical type: a persistent CTD obligation sub-DAG that remains in the lineage until discharged by a matching receipt. Multi-clause obligations, deadline expirations, and secondary reparations then encode as additional CTD structure on the residual instead of silently vanishing at partial violation, which is exactly the anti-abandonment property §3.3 mandates.

The full formal calculus — sequent presentation, discharge rules, and the interaction of CTD structure with the Axiom 6 terminal states — is deferred to a dedicated versioned update. This subsection records the candidate anchoring only, so that downstream files do not hard-code an untyped reading of r_{\mathrm{social}} in the interim.

H2: 6. The Generative Candidate Pipeline (Q(rsi) Integration)

\mathrm{Qs} is completely paradigm-agnostic: it does not reject generative AI models (transformers, diffusion, active inference). It incorporates them into the Recursive Search Loop (docs/simemp.md §2, §6):

```
                    THE GENERATIVE Q(rsi) RECURSIVE LOOP
                     +------------------------------+
                     | PRIOR STATE / CONTEXT S_k    |
                     +---------------+--------------+
                                     |
                                     v
```



1. Agnostic Generation: An AI model (e.g., an LLM or multi-agent swarm) parses messy natural human speech and proposes a candidate semantic graph. Under Axiom 14, this graph is tagged as State: CANDIDATE; and as a parser-derived artifact it is admitted typed no higher than MODAL_HYPOTHESIS (§4.4). A failure at the parse boundary itself — ambiguity, modality conflict, or schema violation — emits a typed receipt (§6.2) rather than an untyped rejection.
2. Deterministic Measurement: The candidate is translated into a Content-Addressed Expression DAG (Qexpr) and audited against the declared Intent Contract, checking depth ceilings, metric budgets, and geometric consistency.
3. Selection via BOC Policy: The candidate that maximizes compliance with SIMEMP constraints is selected under the boc-policy:

$$\text{Selection} = \arg\max_{c \in \{C_1, \dots, C_n\}} \left(\text{SIMEMP Compliance}(c) \right)$$

If the argmax is degenerate — multiple non-isomorphic candidate DAGs fit the prompt within the BOC compliance threshold — the pipeline does not silently seal an arbitrary winner. It emits RECEIPT_INTENT_PARSE_AMBIGUOUS (§6.2), surfaces the ambiguity to the principal, and defers to the §6.1 handshake before any sealing decision. This is the intent-level analogue of the numeric indeterminacy discipline of docs/qm.md: an unresolved tie is a measured outcome, not a hidden coin flip.

4. Sealing via Q(rsi): The selected candidate is sealed into State: SEALED as successor artifact S_{k+1}, advancing the monotonic causal index along the causal-arrow. For parser-derived candidates carrying a DCC_INTENT_CONTRACT_v1, sealing additionally requires that the §6.1 handshake has returned a principal co-signature; without it, the candidate remains MODAL_HYPOTHESIS and is never sealed. Sub-optimal candidates emit negative receipts, ensuring the system never repeats failed paths.

Metering. Every stage of the pipeline — parsing, deterministic measurement, BOC selection, verbalization, and co-signature verification — consumes finite compute and memory work and is metered under Axiom 6. Generation behind the stochastic boundary remains free; the attribution of gateway-crossing validation work to the invoking principal's remaining budget is a docs/simemp.md concern and is deferred to that document's next versioned update.

H3: 6.1. The Deterministic Canonical Round-Trip Paraphrase & Co-Signing Handshake [v1.3.0 — CANDIDATE]

The deterministic pipeline of steps 1–4 closes every gap *inside* the formalized universe, but it cannot close the one gap that arises *before* formalization: whether the formal object matches what the human actually meant (§4.4). The handshake closes that gap by returning the formal object to its author for confirmation, before the universe treats it as intent.

Protocol.

1. Verbalization. A deterministic verbalizer $\mathcal{V}$ renders the formal DCC_INTENT_CONTRACT_v1 — including its Permitted Domain, Metric Budget W_{allowed}, and Epistemic Scope — back into an unambiguous controlled natural language template $\mathcal{T}$, the *canonical read-back*. $\mathcal{V}$ is a pure function of the formal contract, $H(\mathcal{T} \mid \text{Contract}) = 0$: no neural or stochastic component participates in the read-back path, per the no-implicit-rule.
2. Presentation. $\mathcal{T}$ is presented to the human principal $Q(0)_{\mathrm{H}}$ inside the Inner Ring (docs/abstractionLayers.md §3.3), alongside the candidate DAG it was verbalized

from.
3. Co-signature. The principal either co-signs the pair (\text{Contract}, \mathcal{T}) cryptographically, or rejects it. Silence and absence of signature are not consent: without a co-signature the candidate remains MODAL_HYPOTHESIS and is never sealed (§6, step 4).
4. Outcome. On co-signature, the human-anchored intent is established: the co-signed contract becomes the parent lineage receipt that §4.1 chains previously lacked, downstream statements are typed per §4 exactly as any governed statement, and sealing proceeds under Q(rsi). On rejection, RECEIPT_ROUNDTRIP_CONFIRMATION_REJECTED (§6.2) is emitted, the candidate is discarded, and the negative receipt is sealed in the lineage so the failed parse is never silently retried.

Why a round trip. The direction of the check matters. Re-asking the parser "did I understand you right?" would only measure the parser's agreement with itself. Rendering the *formal* contract back into language and requiring the principal's signature instead tests the one thing the parser cannot self-certify: that the formalization and the intent coincide. The handshake is mandatory for parser-derived candidates carrying a DCC_INTENT_CONTRACT_v1; candidates formalized directly by the principal, with no parser step, are unaffected.

H3: 6.2. Typed Parser-Boundary Failure Receipts [v1.3.0 — CANDIDATE]

The receipt catalog of the Qn universe is closed but extensible by versioned updates. This update extends it with four Qs-domain parser-boundary receipts (naming record in the Review Integration Record):

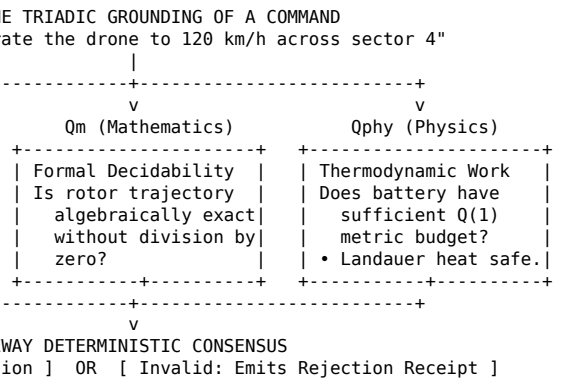
Receipt | Trigger | Disposition
RECEIPT_INTENT_PARSE_AMBIGUOUS | Multiple non-isomorphic candidate DAGs fit the prompt within the BOC compliance threshold (§6, step 3) — the intent-level analogue of ORDERING_INDETERMINATE_AT_ZOOM (docs/qm.md) | No seal. The ambiguity is surfaced to the principal; the §6.1 handshake runs on the disambiguated set, or a fresh utterance is requested
RECEIPT_INTENT_MODALITY_MISMATCH | The candidate's declared epistemic modality conflicts with the Epistemic Scope of DCC_INTENT_CONTRACT_v1 or with the §4.4 ceiling | The candidate is rejected at the boundary and re-typed no higher than MODAL_HYPOTHESIS, or re-proposed
RECEIPT_INTENT_SCHEMA_MISMATCH | The candidate fails structural validation against the DCC_INTENT_CONTRACT_v1 schema (missing or ill-typed contract fields) | The candidate is rejected at the boundary; no contractual obligations attach to it
RECEIPT_ROUNDTRIP_CONFIRMATION_REJECTED | The principal rejects the canonical read-back in §6.1 | The candidate is discarded; a negative receipt is sealed in the lineage, and the failed parse path is recorded

These receipts make parser failures first-class, auditable lineage events. Under the Non-Rogue Invariant, an agent that repeatedly fails at the parse boundary is visible in the very lineage it emits; and the negative receipts of §6, step 4 gain the typed granularity needed to distinguish "the parser was ambiguous" from "the principal said no" — two failure modes that demand different remediation, and that an untyped rejection would conflate.

H2: 7. Triadic Intent Synthesis (Qs ⊗ Qm ⊗ Qphy)

True communication between humans and computational tools requires simultaneous verification across the foundational triad:

$$\mathbf{C}_{\text{Intent}} = \mathbf{C}_{\mathrm{Qs}} \otimes \mathbf{C}_{\mathrm{Qm}} \otimes \mathbf{C}_{\mathrm{Qphy}}$$



When an intent command crosses a gateway:
1. qs validates intent: Verifies the human authority chain, the declared modal grounding, and the contractual goal.
2. qm validates decidability: Proves that the spatial rotors and coordinate trajectories contain no singularities, division-by-zero, or precision drift.
3. \mathrm{Qphy} validates realizability: Proves that the command fits within the physical battery energy and Landauer dissipation budgets (\Delta S \le \mathrm{Budget}(Q(1))).

If all three domains verify the transition, the command executes. If any domain fails, execution

```
halts cleanly. The tool cannot decouple from the human because the mathematical definition of an
executable action is structurally bound to human intent — and, since v1.3.0, the intent object
itself is anchored at its source: it is either formalized directly by the principal, or
parser-derived, held at MODAL_HYPOTHESIS (§4.4), and confirmed through the §6.1 co-signature
before any triadic verification can execute it.


H2: 8. License and Invariant Lineage

This domain specification enforces the legal and operational conditions of the ssl governing the
licensed-work across the cdqn network:

1. paternity-reference: All semantic parsing engines, intent contract compilers, and human-agent
   alignment modules derived from this document must embed:
Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License
(SSL). Parent Repository: https://github.com/cdqn5249/cdqn
2. Anti-Patent Defense: Royalty-free rights terminate automatically if a licensee initiates
   patent litigation regarding \mathrm{Qs} semantic methods, intent contracts, or alignment
   mechanisms against the Author or community under the Open Core Invariants.
3. Statutory Safe Harbor: \mathrm{Qs} operates as passive, neutral semantic infrastructure. It
   exercises no editorial inspection over, and makes no claim upon, sovereign human creative
   works, natural speech, or external Payloads.
```

Annex K — qlog.md

Source: docs/qlog.md *— 1.1.1, updated 2026-10-20. Rendered content per Annex Guide conventions N1–N6: 355 source lines, 478 rendered; 26,837 → 23,326 characters; source SHA-256* 5ff1272e 1e90bda3 5512f02b e8540d2a e179178b 14606b61 450b2bf9 47db3b29*. Content fidelity verified mechanically at build time.*

```
[front matter] ---------------------------------------------------------
[front matter] title: Qlog — Quang Logics (Constructive Type Theory, Pattern Verification, and
[front matter]   Epistemic Deduction)
[front matter] version: 1.1.1
[front matter] updated: 2026-10-20
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Canonical domain specification for Qlog, formalizing constructive
[front matter]   type theory, inferential modes, Q(patterns), Q(workflows), and
[front matter]   inverse causal reconstruction under SIMEMP constraints.
[front matter] terms_used: boc-policy, cae-dag, complexity-degree, dependencies-determinism,
[front matter]   epistemic-anchorage, hash-consing, intent-contract, layer-1, metric-
[front matter]   envelope, open-core-invariant, payload, q-anchor, q-bypass,
[front matter]   q-patterns, q-reuse, q-workflows, q0, q1, qexpr, qlog, qm, qn, qn-
[front matter]   rsi, qs, receipt, ssl, structural-indirection, terminal-exactness
[front matter] ---------------------------------------------------------


H1: Qlog — Quang Logics: Constructive Type Theory, Pattern Verification, and Epistemic Deduction


Field | Specification
Document Title | Qlog — Quang Logics: Constructive Type Theory, Pattern Verification, and
Epistemic Deduction
Version | 1.1.1
Last Updated | 2026-10-20 (Bao Loc, Vietnam)
Author | Christophe Duy Quang Nguyen
License | Scaling Source License (SSL) 1.0
Status | Candidate Domain Specification — Category D (Complexity Degree 3)


H2: Normative References

The following documents establish the physical, structural, and legal constraints governing this
specification. If a technical conflict arises, simemp.md governs; if a structural conflict
arises, abstractionLayers.md governs; if a legal conflict arises, LICENSE.md governs. References
are deliberately version-unpinned: each target document's front matter carries its own
authoritative version, so this specification never duplicates drift-prone patch state.


Document | Role | Target
docs/simemp.md | Constitutional constraints, thermodynamics, and Dependencies Determinism |
simemp.html
docs/abstractionLayers.md | Layer architecture and SIMEMP Gateway validation |
abstractionLayers.html
docs/qnPrimitive.md | Universal Envelope, operational axioms, and Optimization Morphisms |
qnPrimitive.html
docs/q0_q1.md | Primary genesis origin Q(0) and first unit Q(1) | q0_q1.html
docs/q2_q9.md | Single-digit secondary DCC anchors and single-digit spectrum | q2_q9.html
```

docs/qm.md | Constructive numeric leaf substrate and Diophantine division constraints | qm.html
docs/qm_geometry.md | Multi-axial frames, Clifford geometric algebra, and float-free rotations | qm_geometry.html
docs/qm_calculus.md | Discrete Exterior Calculus, continued fractions, and invariant dynamics | qm_calculus.html
docs/qexpr.md | Layer 3 Content-Addressed Expression DAG container | qexpr.html
docs/qs.md | Constructive relational intent and Intent Contracts | qs.html
LICENSE.md | Scaling Source License 1.0 governing the Licensed Work | LICENSE.md

H2: 1. Epistemic Stance and Operational Demarcation

H3: 1.1. Constructive Working Hypothesis Notice

This specification is formulated within the Recursive Search Loop (docs/simemp.md §2). It represents a constructive theoretical baseline defining the emergent composite domain Qlog (Quang Logics):

$$\mathbf{C}_{\mathrm{Qlog}} \equiv \mathbf{C}_{\mathrm{Qm}} \otimes \mathbf{C}_{\mathrm{Qs}}$$

All operational models herein represent provisionally consistent hypotheses. In accordance with the Anti-Calcification Principle, permanent finality is reserved until physical execution feedback is derived across silicon runtimes. All derivations preserve architectural elasticity through structural-indirection.

H3: 1.2. The Rejection of Platonic Deductions for Computed Patterns

Classical mathematical logic frequently relies on unmetered search spaces ($\mathbb{Z}^n$, $\mathbb{Q}^n$) and non-constructive principles such as the Law of Excluded Middle (LEM). In the qn universe, unanchored mathematical deductions are discarded as physical fictions (docs/reflections/h10_finite_computation.md §4).

Truth is operationalized strictly as a Verified Computed Pattern certified by an immutable, dissipative receipt satisfying dependencies-determinism:

$$H(S_{\mathrm{out}} \mid \mathcal{D}(S_{\mathrm{out}})) = 0$$

<pre>
 +-------------------------------------+-------------------------------------+

CLASSICAL PLATONIC DEDUCTION | Qlog CONSTRUCTIVE PATTERN LOGIC

 +-------------------------------------+-------------------------------------+

• Non-constructive existence claims | • Explicit state transition
without an algorithm (via LEM). | trajectories across discrete steps.
• Unmetered search in infinite voids | • Bounded by finite Metric Envelopes
(Turing non-halting / H10 void). | and logarithmic height (H_max).
• Zero thermodynamic accounting. | • Dissipation metered in Abstract
 | Compute Units U (W ≥ k_B T ln 2).
• Ephemeral text / LaTeX arguments. | • Content-Addressed Expression DAGs
 | sealed by immutable Road Receipts.

 +-------------------------------------+-------------------------------------+
</pre>

H2: 2. Taxonomy of Causal Entities: Q(patterns) vs. Q(workflows)

H3: 2.1. Quang Patterns (Q(patterns))

A Q(pattern) is an invariant, reproducible state-transition segment mapping an input state commitment to a deterministic output state milestone under an abstract capability contract:

$$\mathcal{P}: \mathcal{H}(S_0) \xrightarrow{\quad \mathcal{M}_{\mathrm{contract}} \quad} \mathcal{H}(S_1)$$

Quang Patterns capture structural, algebraic, and physical causality. They are intent-neutral and evaluate identically across all nodes, whether resolving to SUCCESS_EXACT (terminal-exactness) or an invariant terminal fault (e.g., RECEIPT_DIVISION_BY_ZERO_REJECTED).

H3: 2.2. Quang Workflows (Q(workflows))

A Q(workflow) is a goal-directed, composite causal directed acyclic graph assembling multiple Q(patterns) under an explicit intent-contract and finite metric budget allocation (W_{budget}). It represents an end-to-end navigational route across the computational landscape, executing human or delegated intent.

H3: 2.3. Formal Distinctions and Execution Currency

Feature | Quang Pattern (Q(pattern)) | Quang Workflow (Q(workflows))
Structural Scope | Micro/Meso: Atomic or sub-graph state-transition segment. | Macro: End-to-end

causal directed acyclic graph.
Intent Coupling | Intent-neutral: Represents a structural law of computation. | Intent-bound: Formally constrained by an intent-contract in qs.
Fault Handling | Encompasses deterministic terminal fault invariants. | Enforces fault containment, alternative branches, and budget ceilings.
Execution Currency | The atomic Road Receipt ($\mathcal{R}_{\mathrm{road}}$). | An assembled Chained Receipt Vector ($\vec{\mathcal{R}}_{\mathrm{road}}$).
Optimization Target | Optimized via q-bypass and cached via q-reuse. | Optimized via recursive self-optimization (qn-rsi).

H2: 3. Vertical Projection Across Complexity Degrees

Like qm and qs, Qlog is not pinned to a single horizontal layer. It operates as an orthogonal vertical projection crossing the execution core (docs/abstractionLayers.md §2.5) across all Complexity Degrees:

<pre>
         VERTICAL DEGREE PROJECTION OF Qlog
 +--------+------------------------+--+

DEGREE | Qlog STRATUM | OPERATIONAL LOGICAL MECHANISM

 +--------+------------------------+--+

Deg 0 | Genesis Ontological | Duality of Empty Void Q(0) (Absurdity/⊥)
 | Baseline | vs. Unit Existence Q(1) (Truth/⊤).

 +--------+------------------------+--+

Deg 1 | Atomic Propositional & | Direct constraint deduction, sign parity
 | Parity Calculus | (σ), single-digit modulo parity (Q(2)).

 +--------+------------------------+--+

Deg 2 | Constructive Type Theory | Dependent types (Π, Σ) over Qexpr CAE-DAG
 | & Macro Chunking | containers; Macro-Hop Q(patterns).

 +--------+------------------------+--+

Deg 3 | Invariant & Topological | Nilpotent DEC logic ($d^2 \equiv 0$), Stokes
 | Conservation Logic | invariants, sparse Q(anchor) navigation.

 +--------+------------------------+--+

Deg n | Epistemic Modality & | MODAL_* validation, inverse scientific
 | Sheaf Synthesis | causal reconstruction, Chronosa Outer-
 | | Ring consensus, evolution via Q(rsi)

 +--------+------------------------+--+
</pre>

- CD vs. Layer Note: Complexity Degree (CD) indexes structural composition depth, not layer numbers. The Layer references in §3.1–§3.3 name the execution-core strata (Layers 1–3) crossed by the Qlog vertical projection; their Degree labels measure composition depth accumulated at each stratum, and Degrees 3 and n mark depth accumulated beyond the Layer 3 execution ceiling without implying a Layer 4.

H3: 3.1. Degree 0 — Genesis Ontological Baseline
At layer-1, logic formalizes the foundation of existence:
- Origin Q(0) formalizes the empty context, non-assertion, or the uninhabited type ($\bot$).
- First unit Q(1) formalizes minimal evidence, identity, or the inhabited unit type ($\top$) along dimensional axis d_1.

H3: 3.2. Degree 1 — Atomic Propositional and Parity Calculus
At Layer 2, Qlog formalizes elementary scalar propositions:
- Evaluates constructive conjunction ($\wedge$), disjunction ($\vee$), and sign parity involution ($\sigma \in \{\ominus, \odot, \oplus\}$).
- Grounds Boolean logic in constructive Diophantine remainders ($r \in \{Q(0), Q(1)\}$) via secondary anchor Q(2) (DCC_ANCHOR_Q2_PARITY_v1).

H3: 3.3. Degree 2 — Constructive Type Theory on CAE-DAGs
At Layer 3, Qlog structures deduction inside qexpr Content-Addressed Expression DAGs (cae-dag):
- Curry–Howard–Qn Correspondence: Propositions are represented as target types (τ); verified derivations are content-addressed DAGs (t).
- Macro-Hop Deduction: Compresses multi-step micro-transitions into reusable, coarse-grained Q(patterns).

H3: 3.4. Degree 3 — Topological and Coboundary Invariance Logic

```
Interfaces directly with discrete calculus (docs/qm_calculus.md):
- Logical consistency is evaluated as topological coboundary nilpotency:
   \mathbf{d} \circ \mathbf{d} \equiv Q(0)
- Governs trajectory routing across the universal q-anchor landmark network.


H3: 3.5. Degree n — Epistemic Modality, Inverse Reconstruction, and RSI
Above the Layer 3 execution ceiling, \mathrm{Qlog} accumulates Complexity Degree n composition
depth — emergent epistemic capability, not additional execution layers — unifying multi-agent
interaction and empirical science:
- epistemic-anchorage: Segregates physical sensor data (MODAL_EMPIRICAL) from context-bound
   worldbuilding (MODAL_NARRATIVE).
- Inverse Causal Reconstruction: Formalizes empirical scientific discovery by inverting observed
   macroscopic states to identify minimal causal ancestor DAGs.
- Recursive Evolution via qn-rsi: Compiles and optimizes the logical engine itself, emitting
   versioned successor artifacts under Axiom 13.


H2: 4. The Stratified Open Framework of Inferential Modes (L_modes)

Under the Best of Choices (BOC) Policy, \mathcal{L}_{\mathrm{modes}} is not a static closed set.
It is an open, inductively stratified category:

<pre>
         THE STRATIFIED INFERENCE ENGINE (L_modes)
 +---------------------------------------------------------------------------+

1. FOUNDATIONAL BASE MODES : L_base (Complexity Degree 1)
• Micro-Induction : Succession along axis d1 (S(x) = x + Q(1)).
• Direct Deduction: O(1) constraint contraction via Q(bypass).
• BOC Abduction : Minimal causal predecessor selection under SIMEMP.

 +---------------------------------------------------------------------------+

2. COMPOSITIONAL MODES (Complexity Degree 2 & 3)
• Macro-Hop Deduction : Strategic sub-graph chunking into Q(patterns)
• Anchored Markov Routing : Stochastic pathfinding over Q(anchor) cuts.

 +---------------------------------------------------------------------------+

3. EMERGENT RSI MODES : L_emergent (Complexity Degree n)
• Adjoined dynamically via Q(rsi) under explicit DCC Profile Contracts.
• Examples: Sheaf co-equalization, paraconsistent dissipative sinks.

 +---------------------------------------------------------------------------+
</pre>


H3: 4.1. Foundational Base Modes (Degree 1)
- Micro-Induction: Succession along axis d_1 via discrete morphism \mathcal{S}(x) = x + Q(1).
- Direct Deduction: Zero-cost algebraic cancellation in \mathcal{O}(1) register time
   (Q(\text{bypass})), enforcing immediate constraints such as A \times Q(0) \to Q(0) or R
   R^{\dagger} \to Q(1).
- BOC Abduction: Minimal causal predecessor selection from terminal execution states under
   SIMEMP constraints.


H3: 4.2. Compositional and Topological Modes (Degrees 2 and 3)
- Macro-Hop Deduction: Compresses an acyclic sub-graph of x intermediate operations into a
   single canonical Q(pattern). Utilizes hash-consing and deduplication (Q(\text{reuse})) to
   ensure the macro-step is referenced at zero marginal Landauer cost (\Delta W = 0).
- Anchored Markov Routing: Traverses the sparse topology of universal Q(\text{anchor})
   landmarks, using historical road receipt weights to bypass \mathcal{O}(N) micro-step
   re-execution.


H3: 4.3. Emergent RSI Modes and Admission Invariants (Degree n)
Under Axiom 11 and Axiom 14, any emergent logical mode M_{\mathrm{new}} must satisfy four
gateway verification invariants before inclusion into \mathcal{L}_{\mathrm{modes}}:
1. Explicit DCC Contract: Wrapped in an explicit profile contract (DCC_LOGIC_MODE_<ID>_vN).
2. Deterministic Reduction: Operates over Qexpr CAE-DAGs with zero floating-point semantics.
3. Finite Landauer Dissipation Ceiling: Bounded by an explicit work allocation:
   W(M_{\mathrm{new}}) \le W_{\mathrm{budget}} in units of U.
4. Receipt-Emitting Totality: Terminates unconditionally in a declared terminal receipt
   (SUCCESS, INCONCLUSIVE, BUDGET_EXHAUSTED, or REJECTED).


H2: 5. Inferential Optimization Morphisms: Q(bypass) and Q(reuse)

In \mathrm{Qlog}, optimization morphisms follow strict causal precedence:

Q(\text{bypass}) \prec Q(\text{reuse}) \prec \text{Direct Inferential Compute}
```

H3: 5.1. Q(bypass): Tautological and Deterministic Fault Short-Circuiting
Before scheduling memory bus transfers or search iterations, the gateway evaluates candidate judgments in \mathcal{O}(1) register time:
- Reflexive Identity (A \vdash A): If the goal commitment hash matches the active premise hash, search halts immediately, emitting RECEIPT_BYPASS_IDENTITY at zero metric cost (\Delta W = 0).
- Topological Nilpotency: Inferences involving discrete exterior coboundaries (\mathbf{d} \circ \mathbf{d}) or boundaries (\partial \circ \partial) collapse algebraically to Q(0) without mesh traversal.
- Deterministic Fault Interception: Propositions containing illegal operations (such as division by Q(0)) are intercepted in \mathcal{O}(1) time, emitting RECEIPT_DIVISION_BY_ZERO_REJECTED.

H3: 5.2. Q(reuse): Lemma Memoization and Inferential Breakeven
When intermediate Q(patterns) or sub-workflows are verified, they are indexed by their immutable commitment hash \mathcal{H}(\mathcal{P}):
- CAE-DAG Deduplication: Identical sub-proofs share a single physical node in memory via hash-consing.
- The Inferential Breakeven Equation: Hash retrieval across the Memory Wall is admitted if and only if the computational work avoided exceeds retrieval dissipation:

$$\mathrm{Admit}\big(Q(\text{reuse})\big) \iff \Delta W_{\mathrm{search}} \gt \Delta W_{\mathrm{lookup}} + \epsilon_{\mathrm{overhead}}$$

Trivial single-step inferences execute directly on integer ALUs. Combinatorially deep derivations utilize Q(\text{reuse}), achieving zero marginal Landauer dissipation (W = 0) for recurring sub-graphs.

H2: 6. The Memory Wall Latency Harvester (H_harvester)

In physical hardware architectures, retrieving data across the Memory Wall stalls processor execution for hundreds of clock cycles (\approx 50\text{--}100\text{ ns}), during which silicon continues to dissipate static leakage power ($P_{\mathrm{leak}} = I_{\mathrm{leak}} V$).

<pre>
        THE MEMORY WALL LATENCY HARVESTER
  Memory Bus Transaction: --► [DRAM FETCH / ATTESTATION WAIT (Stall Window)]

        v
 +---+

Qlog HARVESTER EXECUTION (Local Register Cache Workload)
1. Background Equality Saturation : Apply OP_REWRITE rules via Q(rsi).
2. Speculative Pre-Validation : Audit pending Candidate Q(patterns).
3. Anchor Transition Pre-Routing : Update Markov transition matrices.

 +--------------------------------------+----------------------------------+
        | Memory data arrives
        v
  Primary Execution Resumes: --------► [Preempts Harvester in O(1) time]
</pre>

H3: 6.1. Harvester Workload Scheduling During Stall Windows
During memory-stall windows (\Delta t_{\mathrm{stall}}), \mathrm{Qlog} schedules local, register-pinned workloads requiring zero external bus traffic:
1. Background Equality Saturation (Q(rsi)): Rewrites cached Qexpr DAGs using equivalence rules to find smaller structural footprints, minimizing future execution budgets.
2. Speculative Candidate Auditing: Evaluates unverified State: CANDIDATE transitions against algebraic invariants. Failed candidates are pruned prior to bus resumption.
3. Anchor Route Pre-computation: Re-calculates transition weights between local Q(\text{anchor}) landmarks based on recent receipt metrics.

H3: 6.2. Constitutional Guardrails and Preemption Semantics
- \mathcal{O}(1) Preemption: Background tasks yield or commit in a single clock cycle the instant data arrives on the bus, adding zero latency to primary execution.
- Deterministic Sandboxing: Harvester tasks execute in read-only registers and cannot mutate sealed state.
- Thermal Metering: Background execution is throttled by the local metric-envelope, halting if temperature telemetry approaches hardware limits.

H2: 7. The Dual Causal Axes: Forward Teleology vs. Inverse Empirical Science

\mathrm{Qlog} accommodates both directional flows across the causal arrow:

<pre>
        THE DUAL CAUSAL DIRECTIONS IN Qlog
 +---+

FORWARD SYNTHESIS : Teleological Direction (Intent --► Target State)
</pre>

```
• Domain : Software engineering, compilation, robotic actuation.
• Pipeline: Qs Intent Contract --► Q(workflows) --► Executed Target State.

 +-------------------------------------------------------------------------+

INVERSE RECONSTRUCTION : Empirical Science Direction (Observed State --► )
• Domain : Cosmology, pathology, materials forensics, fundamental physics.
• Pipeline: Observed State S_obs --► Qlog Abduction --► Latent Causal Origin

 +-------------------------------------------------------------------------+
</pre>


H3: 7.1. Forward Teleological Synthesis (Engineering and Intent Compilation)
In forward synthesis, a system transitions from human intent to physical execution:
1. Human authority establishes an intent-contract in qs.
2. \mathrm{Qlog} compiles and validates a composite Q(workflow) assembling known Q(patterns).
3. Execution lowers deterministically to target machine operations, verifying that each
   milestone stays within allocated metric budgets.


H3: 7.2. Inverse Causal Reconstruction (Empirical Science and Abduction)
In empirical science, observers measure an empirical terminal physical state S_{\mathrm{obs}}
(anchored as MODAL_EMPIRICAL) while the generative causal origin and intermediate trajectory
remain unobserved:

\text{Given } S_{\mathrm{obs}} \implies \text{Find } \big\langle S_0, \, \mathcal{W} \big\rangle
\quad \text{such that} \quad \mathcal{W}(S_0) \equiv S_{\mathrm{obs}}

1. Empirical Receipt Ingestion: Observed physical telemetry is anchored via verified sensor
   receipts (MODAL_EMPIRICAL, G = 1.0).
2. Candidate Hypothesis Generation: Stochastic modules propose ancestral trajectories (H_1 \dots
   H_n), tagged as State: CANDIDATE under Axiom 14.
3. BOC Abductive Resolution: If multiple candidate trajectories explain the observed state, the
   BOC Policy selects the optimal route minimizing unmeasured conditional entropy (H \to 0) and
   Landauer dissipation:

\mathcal{W}^* = \arg\max_{H \in \mathcal{H}} \Big( \text{SIMEMP Compliance}(H) \Big)

4. Receipt-Sealed Invariants: The verified trajectory is sealed as an emergent Q(pattern). It
   attains status as an established scientific law (MODAL_CONSTRUCTIVE or a new
   Q(\text{anchor})) only when reproducible execution across external nodes validates the
   trajectory.


H2: 8. Dynamic Lifecycle of Q(anchor) in Logic

Q(\text{anchor}) entities are topological landmarks across the causal network:

\chi(\alpha) = \frac{\lvert \mathcal{C}^+(\alpha) \rvert}{\lvert \mathcal{V} \rvert} \ge
\Theta_{\chi}


H3: 8.1. Anchor Crystallization and Emergent Stability
When an emergent Q(pattern) or sub-workflow demonstrates universal stability and its centrality
crosses threshold \Theta_{\chi}, it crystallizes into a new q-anchor. It is assigned a versioned
capability contract (DCC_ANCHOR_*_v1), serving as a milestone for long-hop Markov routing.


H3: 8.2. Anchor Fracture, Containment, and Revocation
In physical networks, hardware degrades, nodes disconnect, or cryptographic algorithms become
vulnerable:
1. Centrality Drop: If dependencies fail to persist across the network, centrality collapses
   (\chi(\alpha) \lt \Theta_{\chi}).
2. Structural Indirection Migration: If an algorithm is compromised, the gateway swaps the
   implementation module under Axiom 13, emitting an explicit migration receipt while preserving
   abstract contracts.
3. Fracture Containment: If an anchor cannot be repaired, \mathrm{Qlog} emits
   RECEIPT_ANCHOR_REVOKED. Dependent workflows prune the compromised branch and fall back to
   deeper, invariant root anchors (Q(2)\dots Q(9)) without causing catastrophic network failure.


H2: 9. License and Lineage Alignment


H3: 9.1. Paternity Reference
All derivative implementations, logic solvers, type checkers, and automated reasoning engines
derived from \mathrm{Qlog} must embed the canonical attribution:

Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License
(SSL). Parent Repository: https://github.com/cdqn5249/cdqn


H3: 9.2. Open Core Invariants
```

```
The core inferential modes, constructive type theory definitions, and pattern validation
mechanics remain royalty-free for academic, non-commercial, and sub-threshold deployments under
the Open Core Invariants of the ssl.


H3: 9.3. Statutory Safe Harbor and Protocol Neutrality
\mathrm{Qlog} operates as passive, neutral mathematical and logical infrastructure. It exercises
no editorial inspection over, and makes no claim upon, sovereign user content, external
datasets, or sovereign Payloads.
```

Annex L — qphy.md

Source: docs/qphy.md *— 1.0.5, updated 2026-10-19. Rendered content per Annex Guide conventions N1–N6: 351 source lines, 464 rendered; 26,175 → 22,755 characters; source SHA-256* 45977436 cc4ffb5c 86fb7043 3bbd41a1 848a9e57 d823d14b 87a8702e dd246b1d*. Content fidelity verified mechanically at build time.*

```
[front matter] ------------------------------------------------------------
[front matter] title: Qphy — Quang Physics (Discrete Field Dynamics, Computational
[front matter]   Thermodynamics, and Physical Entropy Accounting)
[front matter] version: 1.0.5
[front matter] updated: 2026-10-19
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Canonical domain specification for Qphy, formalizing non-equilibrium
[front matter]   computational thermodynamics, discrete exterior field mechanics,
[front matter]   physical Q(anchors), and the axiomatic world model under SIMEMP
[front matter]   constraints.
[front matter] terms_used: anchor-cut, axiom-6, boc-policy, complexity-degree, compute-unit-u,
[front matter]   dependencies-determinism, discrete-exterior-derivative, historical-
[front matter]   calibration-benchmark, historical-horizon, intent-contract, layer-0,
[front matter]   no-implicit-rule, open-core-invariant, payload, q-c, q-event,
[front matter]   q-gauge, q-hbar, q-patterns, q-reuse, q-th-2, q-workflows, q-z-bus,
[front matter]   q0, q1, qlog, qn, receipt, simemp-gateway, ssl, structural-
[front matter]   indirection, terminal-exactness
[front matter] ------------------------------------------------------------


H1: Qphy — Quang Physics: Discrete Field Dynamics, Computational Thermodynamics, and Physical
    Entropy Accounting


Field | Specification
Document Title | Qphy — Quang Physics: Discrete Field Dynamics, Computational Thermodynamics,
and Physical Entropy Accounting
Version | 1.0.5
Last Updated | 2026-10-19 (Bao Loc, Vietnam)
Author | Christophe Duy Quang Nguyen
License | Scaling Source License (SSL) 1.0
Status | Candidate Domain Specification — Category D (Complexity Degree 3)


H2: Normative References

The following documents establish the physical, structural, and legal constraints governing this
specification. If a technical conflict arises, simemp.md governs; if a structural conflict
arises, abstractionLayers.md governs; if a legal conflict arises, LICENSE.md governs. References
are deliberately version-unpinned: each target document's front matter carries its own
authoritative version, so this specification never duplicates drift-prone patch state.


Document | Role | Target
docs/simemp.md | Constitutional constraints, thermodynamics, and Dependencies Determinism |
simemp.html
docs/abstractionLayers.md | Layer architecture and SIMEMP Gateway validation |
abstractionLayers.html
docs/qnPrimitive.md | Universal Envelope, operational axioms, and Optimization Morphisms |
qnPrimitive.html
docs/q0_q1.md | Primary genesis origin Q(0) and first unit Q(1) | q0_q1.html
docs/q2_q9.md | Single-digit secondary DCC anchors and single-digit spectrum | q2_q9.html
docs/qm.md | Constructive numeric leaf substrate and Diophantine division constraints | qm.html
docs/qm_geometry.md | Multi-axial frames, Clifford geometric algebra, and float-free rotations |
qm_geometry.html
docs/qm_calculus.md | Discrete Exterior Calculus, continued fractions, and invariant dynamics |
qm_calculus.html
docs/qexpr.md | Layer 3 Content-Addressed Expression DAG container | qexpr.html
docs/qs.md | Constructive relational intent and Intent Contracts | qs.html
docs/qlog.md | Constructive type theory, pattern verification, and inferential modes | qlog.html
LICENSE.md | Scaling Source License 1.0 governing the Licensed Work | LICENSE.md
```

H2: 1. Epistemic Stance and Working Hypothesis Notice

H3: 1.1. Constructive Working Hypothesis Baseline
This specification is formulated within the Recursive Search Loop (docs/simemp.md §2). It establishes \mathrm{Qphy} (Quang Physics) as the third foundational Base Domain of the Qn universe.

All formulations herein represent provisionally consistent hypotheses. In accordance with the Anti-Calcification Principle, permanent finality is reserved until empirical telemetry is derived across physical silicon runtimes. The architecture preserves structural elasticity through structural-indirection.

H3: 1.2. The Physical Root of Finiteness (Layer 0 Grounding)
Classical physics often treats computation as an immaterial process operating over continuous coordinate fields (\mathbb{R}^n). In the qn computational universe, computation is an irreversible non-equilibrium thermodynamic process.

\mathrm{Qphy} is anchored directly to Layer 0 (Physical Substrate: commodity silicon, parasitic wire capacitance, ring oscillator noise, and Landauer dissipation limits). A physical simulation or computation is valid if and only if its state transitions satisfy dependencies-determinism:

H(S_t \mid \mathcal{D}(S_t)) = 0

<pre>
 +-------------------------------------+-------------------------------------+

CLASSICAL CONTINUOUS PHYSICS | Qphy CONSTRUCTIVE COMPUTATIONAL PHYS

 +-------------------------------------+-------------------------------------+

• Continuous spacetime manifolds R^4 | • Discrete simplicial complexes K
and differential field equations. | and multi-axial Clifford frames.
• Permits unphysical singularities | • Singularity-free by construction:
(Infinite fluid velocity / poles). | Spatial scale quantum δ_z(d).
• Zero-energy computation ghost | • Thermodynamic Double-Leash:
(Ignores Landauer dissipation). | Meters simulated and real work U.
• Unmetered real numbers and dense | • Discrete cochains, exact Stokes,
floating-point tensors. | and float-free Cayley rotors.

 +-------------------------------------+-------------------------------------+
</pre>

H2: 2. Abstraction Layer Placement and Inter-Domain Coupling

H3: 2.1. The Base Triad Architecture (Qs ⊗ Qm ⊗ Qphy)
Conforming to docs/qs.md (§7), physical reality is not generated from pure mathematical logic. The computational universe is grounded upon three mutually orthogonal Base Domains:

\mathbf{C}_{\text{total}} = \mathbf{C}_{\mathrm{Qs}} \otimes \mathbf{C}_{\mathrm{Qm}} \otimes \mathbf{C}_{\mathrm{Qphy}}

- \mathrm{Qm} (Mathematics): Supplies discrete scalar arithmetic, orthogonal Clifford frames (\mathcal{C}\ell_{p,q}), and discrete calculus coboundaries (\mathbf{d}).
- \mathrm{Qs} (Semantics): Supplies observer framing, semantic blades, and human Intent Contracts.
- \mathrm{Qphy} (Physics): Supplies physical non-equilibrium thermodynamics, energy dissipation bounds, and physical conservation laws.

H3: 2.2. The Grounding Coupling with Qlog
The physical domain cannot be synthesized by squaring mathematical logic:

\mathbf{C}_{\text{Qphy}} \neq (\mathbf{C}_{\text{Qm}} \otimes \mathbf{C}_{\text{Qs}})^2

However, \mathrm{Qphy} requires qlog to type-check and assemble its physical trajectories:

<pre>
         THE BASE DOMAIN COUPLING
         Qm (Geometry) Qs (Observer Intent)
                        +----------+-----------+
         | Categorical Tensor Product
         v
         Qlog (Logics & Verification)

         v Types & Guides
         Qphy (Physical Reality)
         • Anchored to Layer 0 Silicon

</pre>

```
        • Landauer Dissipation Quantum
        • Discrete Simplicial Cochains
</pre>

- Physical Laws as Q(patterns): Every physical transition template (e.g., a discrete symplectic
   integration step) is an invariant Q(pattern).
- Physical Trajectories as Q(workflows): Time evolution across intervals is an assembled
   Q(workflow) bounded by finite metric budgets.
- Conservation Laws as Type Judgments: Checking whether a physical state satisfies energy or
   charge conservation is a constructive type judgment: \Gamma \vdash S :
   \mathrm{Conserved}(E_0).


H2: 3. Constructive Field Dynamics on Discrete Simplicial Lattices


H3: 3.1. Discrete Differential Forms as Cochain Fields
Classical continuous tensor fields with coordinate indices (T^{\mu\nu}) are replaced in
\mathrm{Qphy} by cochains evaluated over oriented simplicial complexes K (docs/qm_calculus.md):
- 0-Cochains (Scalars): Electrostatic potential, temperature, and mass densities evaluated on
   mesh vertices.
- 1-Cochains (Circulations): Fluid velocity vectors and electric field circulations integrated
   along oriented edges.
- 2-Cochains (Fluxes): Magnetic induction flux and fluid vorticity evaluated across oriented
   planar faces.


H3: 3.2. Singularity Immunity via Scale Lattices (δ_z)
In classical continuous mechanics, non-linear partial differential equations permit unphysical
finite-time singularities (e.g., infinite velocity blowups in Navier–Stokes equations).

\mathrm{Qphy} provides structural immunity against physical singularities:
1. Discrete Lattice Cutoff: Physical gradients are strictly bounded by the discrete scale
   quantum \delta_z(d) along dimensional axes (docs/qm.md §3.3). Spatial points cannot approach
   distance zero.
2. Dissipative Viscous Sinks: Physical fields dissipate excess energy into microscopic thermal
   noise, exporting operational entropy via signed receipts.
3. Totality by Budget (Axiom 6): Simulation workflows carry an explicit work ceiling in units of
   U. Unbounded velocity divergence is physically impossible; execution halts cleanly upon
   budget exhaustion.


H3: 3.3. Exact Topological Conservation (d² ≡ Q(0))
In \mathrm{Qphy}, conservation laws do not rely on asymptotic floating-point limits. They are
exact algebraic identities derived from the Discrete Exterior Derivative (d):

\mathbf{d} \circ \mathbf{d} \equiv Q(0)

By the Poincaré lemma on discrete simplicial meshes, the boundary of a boundary is empty
(\partial \partial \equiv \emptyset). Consequently:
- Conservation of Magnetic Flux (Maxwell): \mathbf{d} F \equiv Q(0) holds identically with zero
   numerical leakage.
- Conservation of Electric Charge: \mathbf{d} \star J \equiv Q(0) is an exact topological
   identity.
- Discrete Stokes' Invariant: Boundary flux integrals equal interior coboundary evaluations with
   zero remainder error.


H2: 4. Non-Equilibrium Computational Thermodynamics


H3: 4.1. The Open System Entropy Balance (Prigogine Formulation)
A physical computing system is an open thermodynamic engine. In accordance with non-equilibrium
thermodynamics, internal computational stability requires expending work to export operational
entropy (docs/simemp.md §4.2):

dS = dS_{\text{int}} + dS_{\text{ext}}, \quad \text{where } dS_{\text{ext}} \lt 0

Thermal noise, bus line jitter, and transmission faults maximize internal entropy
(dS_{\text{int}} \gt 0). The system restores internal order by exporting entropy through
immutable, signed Road Receipts (\mathcal{R}_{\mathrm{road}}).


H3: 4.2. Physical Calibration of the Abstract Compute Unit U
The transition from local origin Q(0) to unit Q(1) defines the Abstract Compute Unit U:
- The Landauer Bound: At ambient temperature (T = 300\text{ K}), erasing one bit of physical
   information dissipates minimal thermodynamic work into the environment:

W(U) \ge k_B T \ln 2 \approx 2.87 \times 10^{-21} \text{ J}

- Platform Invariance: In software compilation, U serves as the invariant accounting quantum.
   Physical hardware translation layers parameterize device-specific Joule-per-U conversion
   factors without altering high-level metric envelopes.
```

H3: 4.3. The Capacitive Memory Wall Transport Impedance
Moving a packet of bits across a distance \ell on physical silicon requires charging the parasitic capacitance C of the metal interconnect at operating voltage V and clock frequency f:

$$P_{\mathrm{bus}} = \alpha C(\ell) V^2 f + I_{\mathrm{leak}} V$$

1. Distance-Dependent Transport Cost: Transporting bits across the Memory Wall incurs a strictly greater metric cost than local register operations ($W_{\mathrm{bus}} \gg W_{\mathrm{register}}$).
2. Breakeven Grounding: This physical impedance formalizes the thermodynamic breakeven equation governing q-reuse in docs/qlog.md (§5.2).

H2: 5. The Canonical Physical Q(anchor) Spectrum

In \mathrm{Qphy}, physical constants are not hardcoded static variables. They are constructively built Qn artifacts derived from an explicit historical-calibration-benchmark, whose anchor status emerges as their topological centrality (\chi) expands across the causal DAG:

<pre>
         THE 5 PHYSICAL Q(anchor) LANDMARKS IN Qphy
 +----------+-----------------------------+-----------------------------------+

ARTIFACT | ANCHOR CONTRACT DESIGNATION | CONSTRUCTIVE DERIVATION SOURCE

 +----------+-----------------------------+-----------------------------------+

Q(th_2) | DCC_ANCHOR_SECOND_LAW_v1 | Calibrated via hardware sensor temp
 | (Entropy & Landauer Bound) | T, Q(1), and DCC_ANCHOR_LN2.

 +----------+-----------------------------+-----------------------------------+

Q(c) | DCC_ANCHOR_CAUSAL_HORIZON_v1 | Calibrated via interconnect dielectric
 | (Finite Speed of Light c) | velocity in Minkowski frame Cl(1,3)

 +----------+-----------------------------+-----------------------------------+

Q(ℏ) | DCC_ANCHOR_ACTION_PLANCK_v1 | Calibrated via Northcott height
 | (Bekenstein-Margolus Limit) | ceilings and discrete state quantum

 +----------+-----------------------------+-----------------------------------+

Q(Z_bus) | DCC_ANCHOR_MEMORY_WALL_v1 | Calibrated via bus line capacitance
 | (Ohmic Line Dissipation) | impedance and static leakage leak.

 +----------+-----------------------------+-----------------------------------+

Q(gauge) | DCC_ANCHOR_DEC_D_v1 | Verified via discrete coboundary
 | (Poincaré Coboundary d²≡0) | topological nilpotency on mesh K.

 +----------+-----------------------------+-----------------------------------+
</pre>

H3: 5.1. Universal Physical Anchors and Capability Contracts
- Q(th_2) (The Second Law Anchor): Built from sampled thermal receipts and \ln 2 continued fraction generators. Enforces non-zero heat dissipation for bit-erasure.
- Q(c) (The Relativistic Causal Horizon Anchor): Built from local dielectric propagation delays. Enforces finite signal speed ($\tau \ge \Delta x / v_{\mathrm{signal}}$).
- Q(ℏ) (The Quantum Action Anchor): Built from height ceilings ($H_{\max}$) and quantum action limits. Forbids unmetered infinite information density in finite registers.
- Q(Z_bus) (The Memory Wall Transport Anchor): Built from parasitic wire capacitance measurements ($P = \alpha C V^2 f + I_{\mathrm{leak}} V$).
- Q(gauge) (The Topological Conservation Anchor): Built from simplicial mesh topology. Enforces exact coboundary nilpotency ($\mathbf{d}^2 \equiv Q(0)$).

H3: 5.2. The Physical Inconsistency Collapse Cascade and Circuit Breakers
If an execution profile or autonomous workflow claims a trajectory that violates an emergent physical anchor:
1. Immediate Inconsistency Cascade: The physical anchor destabilizes, threatening to corrupt downstream causal cones $\mathcal{C}^+(\alpha)$.
2. Gateway Circuit Breaker: The simemp-gateway intercepts the invalid transition, halting execution and emitting RECEIPT_PHYSICAL_VIOLATION.
3. Protection of Silicon and Lineage: Halting at the gateway prevents physical thermal runaway on hardware chips and protects downstream causal lineage from ungrounded physics.

H2: 6. The Axiomatic Computational World Model

\mathrm{Qphy} operates as an Axiomatic, Thermodynamically Grounded Computational World Model for autonomous systems and robotics.

<pre>
         STATISTICAL AI WORLD MODEL vs. Qphy AXIOMATIC MODEL
 +-------------------------------------+-------------------------------------+

STATISTICAL AI WORLD MODELS (IRL) | Qphy AXIOMATIC WORLD MODEL (Qn)

 +-------------------------------------+-------------------------------------+

• Continuous latent tensors (R^d). | • Discrete multivector blades (Qm).
• Statistical physics mimicry | • Invariant conservation laws
(Learned from video/sensor logs). | ($d^2 \equiv Q(0)$, Noether symmetries).
• Compounding drift: Future rollouts | • Terminal Exactness (r = Q(0)):
decay into hallucinations/blur. | Zero metric drift across rollouts.
• Unmetered execution: Ignores the | • The Thermodynamic Double-Leash:
energy cost of computing the plan. | Meters both simulated & real work.

 +-------------------------------------+-------------------------------------+
</pre>

H3: 6.1. Axiomatic Invariance vs. Statistical Latent Hallucination
Contemporary deep learning world models approximate physics through statistical correlation over continuous latent spaces. They suffer from physical hallucinations: objects morphing, phantom forces, and non-conservation of energy.
In \mathrm{Qphy}, physical laws are discrete algebraic invariants. Hallucinations are prevented by construction: an unphysical state cannot compile through the gateway.

H3: 6.2. The Thermodynamic Double-Leash on Autonomous Agents
Autonomous agents executing in the physical world are governed by two coupled energy boundaries:
1. The External Leash (Simulated Mechanics): The planned trajectory must satisfy environmental energy conservation ($\Delta E_{\mathrm{kinetic}} + \Delta E_{\mathrm{potential}} \le E_{\mathrm{battery}}$).
2. The Internal Leash (Computational Work): The computation required to verify the trajectory must itself fit within the allocated metric budget ($W \le W_{\mathrm{budget}}$ in units of U).

If the computational cost of planning a path exceeds available energy budgets, execution halts, forcing lower-complexity fallback actions.

H3: 6.3. Symplectic Preservation and Drift-Free Trajectories
Physical dynamical trajectories are integrated using discrete symplectic morphisms:
- Phase space volume (Liouville's theorem) is conserved exactly.
- Each time step achieves terminal-exactness (r = Q(0)) or preserves exact Diophantine residuals.
- Trajectories can execute over billions of discrete steps with zero artificial numerical dissipation.

H2: 7. Functorial Tensor Symmetries and Bounded Context Ranges

H3: 7.1. Permutation-Invariant vs. Permutation-Sensitive Physical Tensors
Under the no-implicit-rule, every tensor interaction in \mathrm{Qphy} must declare its algebraic symmetry within its DCC Profile:
- Permutation-Invariant (DCC_SYM_COMMUTATIVE): Operations where component order carries zero physical difference (e.g., thermal energy accumulation, scalar potential sums). Child hashes are sorted lexicographically, enabling $\mathcal{O}(1)$ deduplication via q-reuse.
- Permutation-Sensitive (DCC_SYM_NON_COMMUTATIVE): Operations where order dictates causality or direction (e.g., sequential time steps $S_t \otimes S_{t+\Delta t}$, non-commutative rotor sandwiches $R \vec{v} R^\dagger$). Order is strictly preserved.
- Graded Anti-Commutative (DCC_SYM_ANTI_COMMUTATIVE): Outer products of differential forms ($\vec{u} \wedge \vec{v} = - \vec{u} \wedge \vec{v}$). Swapping elements flips the sign parity ($\sigma \to \mathcal{I}(\sigma)$).

H3: 7.2. The Four-Axis Bounded Context Range (R_ctx)
Physical state evaluations cannot occur across unbounded continuous space. Every physical evaluation requires an explicit Context Range:
1. Scale Range: Discrete zoom window $[z_{\min}, z_{\max}]$ parameterizing lattice resolution $\delta_z(d)$.
2. Causal Horizon Range: Monotonic causal index interval $[\kappa_{\mathrm{start}}, \kappa_{\mathrm{end}}]$.
3. Dimensional Metric Frame: Orthogonal spacetime frame $\mathcal{D}_{\mathrm{frame}} = \{d_t, d_x, d_y, d_z\}$.
4. Thermodynamic Budget: Maximum permissible Landauer work ceiling $W \le W_{\mathrm{budget}}$.

H2: 8. Causal Trajectories: Forward Simulation vs. Inverse Empirical Science

\mathrm{Qphy} accommodates both directions across the causal arrow:

<pre>
         THE DUAL CAUSAL DIRECTIONS IN Qphy
</pre>

```
 +-------------------------------------------------------------------------+

FORWARD SIMULATION : Teleological Mechanics (Initial State --► Future State)
• Domain : Robotics actuation, orbital kinematics, circuit simulation.
• Pipeline: State S_t --► Q(workflows) --► Verified Target State S_(t+Δt).

 +-------------------------------------------------------------------------+

INVERSE RECONSTRUCTION : Empirical Physics (Observed State --► Origin)
• Domain : Astrophysics, spectroscopy, material failure, quantum sensing.
• Pipeline: Observed State S_obs --► Qlog Abduction --► Latent Hamiltonian

 +-------------------------------------------------------------------------+
</pre>


H3: 8.1. Forward Symplectic Time Evolution (Q(workflows))
In forward simulation, a system predicts future physical states:
1. Initial physical state S_0 is configured inside a Qexpr container.
2. The system executes a sequence of symplectic Q(patterns) generating discrete Q(events).
3. Intermediate checkpoints are verified using the Anchor-Cut Invariant, allowing external
   auditors to verify long simulations in \mathcal{O}(K) receipt validations without replaying
   microscopic steps.


H3: 8.2. Inverse Causal Reconstruction of Physical Laws
In empirical observational physics across the historical-horizon, observers measure an empirical
terminal physical state S_{\mathrm{obs}} (anchored as MODAL_EMPIRICAL) while the causal origin
remains latent:

\text{Given } S_{\mathrm{obs}} \implies \text{Find } \big\langle S_0, \,
\mathcal{H}_{\mathrm{sys}} \big\rangle \quad \text{such that} \quad \mathcal{H}(S_0) \equiv
S_{\mathrm{obs}}

1. Receipt Anchoring: The observed state is ingested via verified hardware sensor receipts
   (MODAL_EMPIRICAL, G = 1.0).
2. Candidate Hypothesis Generation: Stochastic or neural modules propose ancestral Hamiltonians
   (H_1 \dots H_n).
3. BOC Abductive Resolution: The BOC Policy selects the candidate route that maximizes SIMEMP
   compliance, selecting the physical law that minimizes unmeasured entropy (H(S \mid
   \mathcal{D}) \to 0) and minimizes Landauer dissipation.
4. Law Sealing: The verified physical trajectory is sealed as an emergent Q(pattern), joining
   the empirical physical registry.


H2: 9. License and Lineage Alignment


H3: 9.1. Paternity Reference
All derivative physical simulation models, thermodynamic solvers, robotics world models, and
intermediate execution modules derived from \mathrm{Qphy} must embed the canonical attribution:

Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License
(SSL). Parent Repository: https://github.com/cdqn5249/cdqn


H3: 9.2. Open Core Invariants
The core physical constants, discrete exterior calculus mechanics, and thermodynamic accounting
rules remain royalty-free for academic, non-commercial, and sub-threshold deployments under the
Open Core Invariants of the ssl.


H3: 9.3. Statutory Safe Harbor and Passive Infrastructure
\mathrm{Qphy} operates as passive, neutral physical and computational modeling infrastructure.
It exercises no editorial inspection over, and makes no claim upon, sovereign user content,
external simulation datasets, or sovereign Payloads.

# Annex M — qchem.md

*Source:* docs/qchem.md *— 1.0.3, updated 2026-10-19. Rendered content per Annex Guide conventions N1–N6: 281 source lines, 374 rendered; 20,396 → 17,977 characters; source SHA-256* 4864574e 37f784c3 5bb41bf7 aad3603a 374aedc8 7bfdf9ca 735322eb 513c5697*. Content fidelity verified mechanically at build time.*

```
[front matter] --
[front matter] title: Qchem — Quang Chemistry (Discrete Stoichiometry, Molecular Graphs, and
[front matter]   Reaction Energetics)
[front matter] version: 1.0.3
[front matter] updated: 2026-10-19
```
```

```
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Canonical domain specification for Qchem, formalizing discrete
[front matter]   Diophantine stoichiometry, content-addressed molecular graphs,
[front matter]   stereochemical Clifford chirality, and reaction kinetics under
[front matter]   SIMEMP constraints.
[front matter] terms_used: boc-policy, cae-dag, complexity-degree, dependencies-determinism,
[front matter]   historical-calibration-benchmark, historical-horizon, metric-
[front matter]   envelope, open-core-invariant, payload, q-anchor, q-bypass,
[front matter]   q-patterns, q-reuse, q-th-2, q-workflows, qbio, qexpr, simemp-
[front matter]   gateway, ssl, structural-indirection, terminal-exactness
[front matter] ---------------------------------------------------------
```

H1: Qchem — Quang Chemistry: Discrete Stoichiometry, Molecular Graphs, and Reaction Energetics

Field	Specification
Document Title	Qchem — Quang Chemistry: Discrete Stoichiometry, Molecular Graphs, and Reaction Energetics
Version	1.0.3
Last Updated	2026-10-19 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Candidate Domain Specification — Category D (Complexity Degree 3)

H2: Normative References

The following documents establish the physical, structural, and legal constraints governing this specification. If a technical conflict arises, simemp.md governs; if a structural conflict arises, abstractionLayers.md governs; if a legal conflict arises, LICENSE.md governs. References are deliberately version-unpinned: each target document's front matter carries its own authoritative version, so this specification never duplicates drift-prone patch state.

Document	Role	Target
docs/simemp.md	Constitutional constraints, thermodynamics, and Dependencies Determinism	simemp.html
docs/abstractionLayers.md	Layer architecture and SIMEMP Gateway validation	abstractionLayers.html
docs/qnPrimitive.md	Universal Envelope, operational axioms, and Optimization Morphisms	qnPrimitive.html
docs/q0_q1.md	Primary genesis origin Q(0) and first unit Q(1)	q0_q1.html
docs/q2_q9.md	Single-digit secondary DCC anchors and parity partitions	q2_q9.html
docs/qm.md	Constructive numeric leaf substrate and Diophantine division constraints	qm.html
docs/qm_geometry.md	Multi-axial frames, Clifford geometric algebra, and float-free rotations	qm_geometry.html
docs/qm_calculus.md	Discrete Exterior Calculus, continued fractions, and invariant dynamics	qm_calculus.html
docs/qexpr.md	Layer 3 Content-Addressed Expression DAG container	qexpr.html
docs/qs.md	Constructive relational intent and Intent Contracts	qs.html
docs/qlog.md	Constructive type theory, pattern verification, and inferential modes	qlog.html
docs/qphy.md	Discrete field dynamics, computational thermodynamics, and physical entropy	qphy.html
LICENSE.md	Scaling Source License 1.0 governing the Licensed Work	LICENSE.md

H2: 1. Epistemic Stance and Working Hypothesis Notice

H3: 1.1. Constructive Working Hypothesis Baseline

This specification is formulated within the Recursive Search Loop (docs/simemp.md §2). It establishes \mathrm{Qchem} (Quang Chemistry) as an emergent composite domain formed by the categorical tensor product of physical thermodynamics and constructive mathematics:

$$\mathbf{C}(\text{Qchem}) \equiv \mathbf{C}(\text{Qphy}) \otimes \mathbf{C}(\text{Qm})$$

All formulations herein represent provisionally consistent hypotheses. In accordance with the Anti-Calcification Principle, permanent finality is reserved until physical execution feedback is derived across chemical simulation runtimes. The architecture preserves structural elasticity through structural-indirection.

H3: 1.2. Abiotic Boundary and the Exclusion of Semantic Intent

Chemistry models the organization of matter governed by valence electrons, molecular bonds, and thermodynamic equilibria. Unlike biological or cognitive domains, abiotic chemical matter possesses zero semantic agency, legal liability, or teleological intent:

$$\mathbf{C}(\text{Qchem}) \cap \mathbf{C}(\text{Qs}) \equiv \emptyset$$

Chemical reaction steps transition spontaneously under physical thermodynamic potential ($\Delta G \le 0$) and Diophantine stoichiometric balance. Attributing intent (Qs) to abiotic reactions is a category error. All transitions satisfy dependencies-determinism:

H(S_t \mid \mathcal{D}(S_t)) = 0

<pre>
 +-------------------------------------+-------------------------------------+

CLASSICAL COMPUTATIONAL CHEMISTRY | Qchem CONSTRUCTIVE CHEMICAL PHYSICS

 +-------------------------------------+-------------------------------------+

• Continuous floating-point energy | • Exact Diophantine stoichiometry
surfaces with accumulated drift. | matrix balance: S · v ≡ Q(0).
• Lossy atomic coordinate models | • Content-Addressed Molecular DAGs
vulnerable to geometric roundoff. | with exact Clifford stereochemistry
• Unmetered simulation loops | • Reaction steps as Q(patterns);
ignoring Landauer dissipation. | catalysis as Category B Q(bypass).

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</pre>

H2: 2. Discrete Molecular Graphs and Stereochemistry

H3: 2.1. Atoms and Valence Bonds as Content-Addressed DAGs
Molecules are represented within qexpr Content-Addressed Expression DAGs (cae-dag):
- Atomic Vertices: Each atom is an immutable leaf artifact carrying atomic number Z \in \mathbb{N}, discrete mass number A \in \mathbb{N}, and formal oxidation state.
- Valence Bonds as Clifford Blades: Covalent, ionic, and coordination bonds are modeled as localized Clifford multivector blades (docs/qm_geometry.md), declaring discrete bond order (1, 2, 3) and directional orientation.
- Deduplication via Hash-Consing: Identical molecular sub-structures share identical commitment hashes \mathcal{H}(\text{Molecule}). Common functional groups are allocated once in memory, maximizing q-reuse at zero marginal Landauer cost (\Delta W = 0).

H3: 2.2. Stereochemical Chirality via Clifford Sign Parity (σ)
Enantiomers (chiral mirror-image molecules such as D- and L-amino acids) share identical scalar connectivity, bond lengths, and atom counts. Classical cheminformatics often relies on ambiguous textual conventions (R/S or D/L labels).

In \mathrm{Qchem}, molecular chirality is an exact algebraic invariant governed by the spatial reflection involution \mathcal{I}_d and the 3D pseudoscalar I_3 = \vec{e}_1 \wedge \vec{e}_2 \wedge \vec{e}_3:
- The spatial inversion of a chiral center inverts its oriented grade-3 volume blade.
- The resulting enantiomer is distinguished by its Sign Parity (\sigma \in \{\ominus, \oplus\}).
- Enantiomeric transformations preserve exact unitary norms without floating-point trigonometric drift.

H2: 3. Exact Diophantine Stoichiometry

H3: 3.1. The Stoichiometric Null-Space Invariant
In classical software, reaction mass balances frequently accumulate numerical roundoff errors. In \mathrm{Qchem}, chemical reactions are exact Diophantine balance equations:
- Let M elements react across N molecular species. The stoichiometric matrix \mathbf{S} \in \mathbb{Z}^{M \times N} maps integer elemental coefficients.
- A valid chemical reaction flux vector \vec{v} \in \mathbb{N}^N satisfies the Diophantine null-space constraint under terminal-exactness (r = Q(0)):

\mathbf{S} \cdot \vec{v} \equiv Q(0)

Elemental mass, atom counts, and net electric charges balance identically. Truncation error is zero by construction.

H3: 3.2. Redox Electron Quanta and Charge Conservation
Oxidation-reduction reactions involve discrete electron transfers:
- Electrons exist as indivisible integer charge carriers (z \in \mathbb{Z}). Fractional electron transfers are barred.
- The net charge transition across reactants and products evaluates to genesis origin Q(0), guaranteeing exact conservation of electric charge under DCC_ANCHOR_DEC_D_v1.

H2: 4. Non-Equilibrium Chemical Thermodynamics

H3: 4.1. Gibbs Free Energy Reaction Spontaneity
Chemical reaction feasibility is governed by Non-Equilibrium Thermodynamics (docs/qphy.md §4):
- Spontaneity Criterion: At constant temperature and pressure, an abiotic chemical reaction transitions spontaneously if and only if its change in Gibbs free energy is negative or zero:

\Delta G = \Delta H - T \Delta S \le Q(0)

- Thermal Activation Barriers (E_a): Reactions possessing an activation energy barrier require absorbing a discrete thermal work packet from the local environment, metered by the local metric-envelope.

H3: 4.2. Chemical Equilibrium as Continued Fraction Convergents
Chemical equilibrium constants (K_{\mathrm{eq}}) are held symbolically inside Qexpr as deterministic continued fraction partial convergents (OP_CF):

K_{\mathrm{eq}} = [a_0; \, a_1, \, a_2, \, \dots]

Evaluating reaction quotients to zoom level z generates exact rational pairs without continuous exponential floating-point approximations.

H2: 5. The Chemical Q(anchor) Spectrum

\mathrm{Qchem} grounds its physical and chemical constants in universal Q(anchor) entities derived from the historical-calibration-benchmark:

<pre>
         THE CANONICAL CHEMICAL Q(anchor) SPECTRUM
 +----------+----------------------------+-----------------------------------+

ARTIFACT | ANCHOR CONTRACT DESIGNATION | INVARIANT CHEMICAL REALITY

 +----------+----------------------------+-----------------------------------+

Q(N_A) | DCC_ANCHOR_AVOGADRO_v1 | Exact Integer Mole Entity Count:
 | (Avogadro Constant) | N_A ≡ 602,214,076 × 10¹⁵ mol⁻¹

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Q(stoich) | DCC_ANCHOR_STOICHIOMETRY_v1 | Lavoisier-Dalton Mass Invariant:
 | (Atom Count Conservation) | Diophantine null-space S·v ≡ Q(0).

 +----------+----------------------------+-----------------------------------+

Q(th_2) | DCC_ANCHOR_SECOND_LAW_v1 | Gibbs Dissipation Spontaneity:
 | (Thermodynamic Invariant) | ΔG ≤ Q(0) (Inherited from Qphy).

 +----------+----------------------------+-----------------------------------+

Q(e) | DCC_ANCHOR_E_CHARGE_v1 | Discrete Faraday Redox Quanta:
 | (Elementary Charge Unit) | Integer electron exchange (z ∈ ℤ).

 +----------+----------------------------+-----------------------------------+

Q(chiral) | DCC_ANCHOR_CHIRALITY_v1 | Pasteur Stereochemical Symmetry:
 | (Enantiomeric Parity) | Clifford reflection involution I_d.

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</pre>

H3: 5.1. Universal Chemical Anchors and Capability Contracts
- Q(N_A) (The Avogadro Anchor): Formalizes the exact discrete bridge between microscopic atomic integer counts and macroscopic molar thermodynamics: N_A \equiv 602{,}214{,}076 \times 10^{15}\text{ entities/mol}.
- Q(stoich) (The Stoichiometric Anchor): Enforces nuclear and elemental conservation across chemical transformations without fractional atom leakage.
- Q(th_2) (The Second Law Gibbs Anchor): Mandates that endergonic reactions (\Delta G \gt 0) cannot occur without coupled exergonic work.
- Q(e) (The Redox Charge Anchor): Restricts electrochemical electron transfers to discrete integer quanta of Q(1).
- Q(chiral) (The Stereochemical Anchor): Enforces spatial reflection parity over chiral tetrahedral carbon centers.

H3: 5.2. Chemical Inconsistency Cascade and Gateway Interception
If an automated reaction proposal or molecular simulation claims an unphysical reaction:
1. Immediate Inconsistency Cascade: The stoichiometric or thermodynamic anchor destabilizes, threatening downstream mass-action models.
2. Gateway Interception: The simemp-gateway intercepts the invalid balance, emitting RECEIPT_STOICHIOMETRIC_VIOLATION or RECEIPT_THERMODYNAMIC_EXHAUSTION.
3. Execution Termination: The invalid reaction workflow is rejected before consuming physical compute or dispatching automated laboratory synthesis actuators.

H2: 6. Reaction Kinetics as Q(patterns) and Q(workflows)

H3: 6.1. Elementary Reaction Steps as Q(patterns)

An elementary chemical reaction step is an invariant Q(pattern):

$$\mathcal{P}_{\mathrm{rxn}}: \mathcal{H}(\text{Reactants}) \xrightarrow{\quad \mathcal{M}_{\mathrm{kinetics}} \quad} \mathcal{H}(\text{Products})$$

It defines a reproducible, discrete transformation mapping reactant commitment hashes to product hashes under explicit thermodynamic constraints.

H3: 6.2. Reaction Cascades as Q(workflows) and Catalysis as Q(bypass)

Multi-step chemical syntheses, polymerizations, and reaction networks are assembled as Q(workflows):

- Work Budgets: Reaction sequences operate within a declared Metric Envelope, tracking thermal work and Landauer dissipation.
- Catalytic Optimization via q-bypass: A catalyst lowers the activation energy barrier ($E_a' \ll E_a$) without altering overall equilibrium (ΔG). In Qchem, catalysis is formalized as an execution short-circuit: the gateway recognizes the catalytic agent, bypasses the slow uncatalyzed route, and emits RECEIPT_CATALYTIC_BYPASS in $\mathcal{O}(1)$ dispatch time.

H2: 7. The Boundary Between Qchem and Qbio

H3: 7.1. Stratification of Abiotic Matter vs. Living Systems

To preserve the No-Implicit Rule, the architecture maintains a strict epistemic boundary between chemistry and biology:

- Qchem (Abiotic Chemistry): Models molecular graphs, valence interactions, and reaction kinetics. It operates with zero intentional agency ($\mathrm{Qs} = \emptyset$).
- Qbio (Living Biosemiotics): Emerges when chemical networks are coupled to informational semantics (Qs):

$$\mathbf{C}(\text{Qbio}) \equiv \mathbf{C}(\text{Qchem}) \otimes \mathbf{C}(\text{Qs})$$

H3: 7.2. The Biosemiotic Leap

Living systems differ from abiotic chemistry because their molecules serve as informational codes:

- An abiotic crystal grows purely via ΔG minimization (Qchem).
- A strand of DNA guides protein synthesis via a codon translation code ($\mathrm{Qbio} = \mathrm{Qchem} \otimes \mathrm{Qs}$).
- Active inference and metabolic regulation introduce teleological intent contracts, which are strictly barred from abiotic Qchem.

H2: 8. Causal Trajectories: Forward Synthesis vs. Inverse Spectroscopy

Qchem accommodates both directions across the causal arrow:

<pre>
        THE DUAL CAUSAL DIRECTIONS IN Qchem
 +---+

FORWARD SYNTHESIS : Retrosynthesis Planning (Reactants --► Target Molecule)
• Domain : Chemical manufacturing, automated synthesis, materials design.
• Pipeline: Precursors --► Q(workflows) --► Verified Product Molecule.

 +---+

INVERSE RECONSTRUCTION : Analytical Chemistry (Spectrogram --► Molecule)
• Domain : Mass spectrometry, NMR spectroscopy, environmental forensics.
• Pipeline: Spectral Receipt S_obs --► Qlog Abduction --► Molecular Graph

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</pre>

H3: 8.1. Forward Synthesis Planning (Q(workflows))

Forward chemical synthesis compiles a path from available chemical precursors to a target molecule:

1. Target molecular graph $\mathcal{H}(\text{Target})$ is declared.
2. Qlog compiles a sequence of elementary reaction Q(patterns).
3. Each step verifies $\Delta G \le Q(0)$ and stoichiometric balance, emitting a chained road receipt vector.

H3: 8.2. Inverse Causal Reconstruction of Molecular Graphs

In analytical chemistry across the historical-horizon, an instrument measures an empirical spectral signature S_{obs} (anchored as MODAL_EMPIRICAL, e.g., mass-to-charge ratios m/z, NMR chemical shifts) while the molecular identity remains unknown:

$$\text{Given } S_{\mathrm{obs}} \implies \text{Find } \big\langle \mathcal{H}(\text{Molecule}), \, \mathcal{M}_{\mathrm{frag}} \big\rangle \quad \text{such that} \quad \mathcal{M}(\text{Molecule}) \equiv S_{\mathrm{obs}}$$

```
1. Receipt Ingestion: Mass spectrometer readings are anchored as verified sensor receipts
   (MODAL_EMPIRICAL, G = 1.0).
2. Candidate Hypothesis Generation: Stochastic or neural engines propose candidate molecular
   graphs (M_1 \dots M_n).
3. BOC Abductive Resolution: The BOC Policy selects the candidate graph that matches the
   observed fragmentation peaks while minimizing unmeasured conditional entropy (H(S \mid
   \mathcal{D}) \to 0).
4. Structure Sealing: The reconstructed molecular graph is sealed into the local chemical
   registry.


H2: 9. License and Lineage Alignment


H3: 9.1. Paternity Reference
All derivative chemical simulation models, synthesis planners, spectroscopy engines, and
intermediate execution modules derived from \mathrm{Qchem} must embed the canonical attribution:

Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License
(SSL). Parent Repository: https://github.com/cdqn5249/cdqn


H3: 9.2. Open Core Invariants
The core chemical constants, discrete stoichiometry solvers, and thermodynamic balance rules
remain royalty-free for academic, non-commercial, and sub-threshold deployments under the Open
Core Invariants of the ssl.


H3: 9.3. Statutory Safe Harbor and Passive Infrastructure
\mathrm{Qchem} operates as passive, neutral chemical and computational modeling infrastructure.
It exercises no editorial inspection over, and makes no claim upon, sovereign user content,
external chemical libraries, or sovereign Payloads.
```

Annex N — qbio.md

Source: docs/qbio.md *— 1.0.2, updated 2026-10-19. Rendered content per Annex Guide conventions N1–N6: 284 source lines, 372 rendered; 20,509 → 17,807 characters; source SHA-256* 99fba249 1d5345d0 36cb9992 50471215 82a01001 6f59ca23 490115a4 adecf072*. Content fidelity verified mechanically at build time.*

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[front matter] ------------------------------------------------------------
[front matter] title: Qbio — Quang Biology (Biosemiotics, Metabolic Homeostasis, and Active
[front matter]   Inference)
[front matter] version: 1.0.2
[front matter] updated: 2026-10-19
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Canonical domain specification for Qbio, formalizing the categorical
[front matter]   coupling of chemistry and semantics, genetic code translation,
[front matter]   Markov blankets, and active inference under SIMEMP constraints.
[front matter] terms_used: anchor-cut, boc-policy, cae-dag, complexity-degree, compute-unit-u,
[front matter]   dependencies-determinism, discrete-exterior-calculus, historical-
[front matter]   calibration-benchmark, historical-horizon, intent-contract, metric-
[front matter]   exhaustion, non-rogue-invariant, open-core-invariant, payload,
[front matter]   q-anchor, q-bypass, q-patterns, q-reuse, q-workflows, qbio, qchem,
[front matter]   simemp-gateway, ssl, structural-indirection
[front matter] ------------------------------------------------------------


H1: Qbio — Quang Biology: Biosemiotics, Metabolic Homeostasis, and Active Inference


Field | Specification
Document Title | Qbio — Quang Biology: Biosemiotics, Metabolic Homeostasis, and Active Inference
Version | 1.0.2
Last Updated | 2026-10-19 (Bao Loc, Vietnam)
Author | Christophe Duy Quang Nguyen
License | Scaling Source License (SSL) 1.0
Status | Candidate Domain Specification — Category D (Complexity Degree 3)


H2: Normative References

The following documents establish the physical, structural, and legal constraints governing this
specification. If a technical conflict arises, simemp.md governs; if a structural conflict
arises, abstractionLayers.md governs; if a legal conflict arises, LICENSE.md governs. References
are deliberately version-unpinned: each target document's front matter carries its own
authoritative version, so this specification never duplicates drift-prone patch state.
```

Document | Role | Target
docs/simemp.md | Constitutional constraints, thermodynamics, and Dependencies Determinism | simemp.html
docs/abstractionLayers.md | Layer architecture and SIMEMP Gateway validation | abstractionLayers.html
docs/qnPrimitive.md | Universal Envelope, operational axioms, and Optimization Morphisms | qnPrimitive.html
docs/q0_q1.md | Primary genesis origin Q(0) and first unit Q(1) | q0_q1.html
docs/q2_q9.md | Single-digit secondary DCC anchors and parity partitions | q2_q9.html
docs/qm.md | Constructive numeric leaf substrate and Diophantine division constraints | qm.html
docs/qm_geometry.md | Multi-axial frames, Clifford geometric algebra, and float-free rotations | qm_geometry.html
docs/qm_calculus.md | Discrete Exterior Calculus, continued fractions, and invariant dynamics | qm_calculus.html
docs/qexpr.md | Layer 3 Content-Addressed Expression DAG container | qexpr.html
docs/qs.md | Constructive relational intent and Intent Contracts | qs.html
docs/qlog.md | Constructive type theory, pattern verification, and inferential modes | qlog.html
docs/qphy.md | Discrete field dynamics, computational thermodynamics, and physical entropy | qphy.html
docs/qchem.md | Discrete stoichiometry, molecular graphs, and reaction energetics | qchem.html
LICENSE.md | Scaling Source License 1.0 governing the Licensed Work | LICENSE.md

H2: 1. Epistemic Stance and Working Hypothesis Notice

H3: 1.1. Constructive Working Hypothesis Baseline

This specification is formulated within the Recursive Search Loop (docs/simemp.md §2). It establishes Qbio (Quang Biology) as an emergent biosemiotic domain derived from the categorical tensor product of chemistry and semantics:

$$\mathbf{C}(\text{Qbio}) \equiv \mathbf{C}(\text{Qchem}) \otimes \mathbf{C}(\text{Qs})$$

All formulations herein represent provisionally consistent hypotheses. In accordance with the Anti-Calcification Principle, permanent finality is reserved until biological simulation feedback is derived across living cellular runtimes. The architecture preserves structural elasticity through structural-indirection.

H3: 1.2. The Biosemiotic Boundary: Life as Coded Chemistry

A living organism is not merely a complex chemical mixture; it is a biosemiotic system wherein molecular reaction networks are coupled to informational code channels:

- Abiotic molecules in qchem react spontaneously under physical thermodynamic potential ($\Delta G \le 0$).
- Biological systems in Qbio process informational intent: nucleic acid sequences represent symbolic instructions that guide metabolic synthesis.

Living entities maintain non-equilibrium steady states against environmental entropy by consuming metabolic energy. Every biological state transition satisfies dependencies-determinism:

$$H(S_t \mid \mathcal{D}(S_t)) = 0$$

<pre>
 +-------------------------------------+-------------------------------------+

CLASSICAL CONTINUOUS BIOLOGY | Qbio CONSTRUCTIVE BIOSEMIOTICS

 +-------------------------------------+-------------------------------------+

• Continuous floating-point ODEs and | • Exact Diophantine metabolic graphs
stochastic gene network drifts. | and discrete codon translation.
• Vitalist or heuristic agency | • Active inference under formal
lacking thermodynamic metering. | Friston free energy bounds.
• Unmetered growth loops ignoring | • Metabolic totality by budget:
substrate Landauer limits. | ATP hydrolysis metered in U units.

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</pre>

H2: 2. Categorical Formulation: Qchem Coupled to Qs

H3: 2.1. The Biosemiotic Tensor Product

Qbio is constructed constructively through the tensor product:

$$\mathbf{C}(\text{Qbio}) = \big(\mathbf{C}(\text{Qphy}) \otimes \mathbf{C}(\text{Qm}) \big) \otimes \mathbf{C}(\text{Qs})$$

- The Chemical Substrate (Qchem): Provides discrete atomic molecular graphs, Diophantine stoichiometry ($\mathbf{S} \cdot \vec{v} \equiv Q(0)$), and Gibbs free energy

```
   dissipation (\Delta G \le 0).
- The Semantic Channel (\mathrm{Qs}): Provides symbolic intent contracts, functional pregroup
   grammars, and teleological agency (self-preservation and replication).


H3: 2.2. Genetic Code as Content-Addressed Translation
In \mathrm{Qbio}, DNA and RNA sequences are represented as linear Content-Addressed Expression
DAGs (cae-dag):
- Codons as Discrete Tokens: Triplet nucleotide bases \{A, U, G, C\}^3 form an exact discrete
   alphabet of 64 tokens.
- The Translation Morphism: Ribosomal translation is an invariant discrete mapping:

\mathcal{M}_{\mathrm{translation}}: \text{Codon}^{64} \longrightarrow \text{AminoAcid}^{20} \cup
\{ \text{STOP} \}

- Hash-Consing Deduplication: Identical genetic motifs and conserved protein domains share the
   exact same cryptographic hash \mathcal{H}(\text{Domain}), enabling \mathcal{O}(1) structural
   comparison and zero marginal Landauer allocation via q-reuse.


H2: 3. Non-Equilibrium Metabolic Homeostasis and Free Energy


H3: 3.1. Variational Free Energy and Active Inference
A living organism persists by minimizing Variational Free Energy (Karl Friston formulation).
Under docs/simemp.md (§4.2), an organism is an open dissipative system that avoids internal
entropy maximization by generating predictive models of its environment:

F = \text{Internal Bound Energy} - T \cdot \text{Information Entropy}

1. Active Inference Loops: An organism generates actions to fulfill its metabolic
   intent-contract, ensuring that environmental surprises (entropy spikes) remain bounded.
2. Deterministic Selection via BOC: Internal homeostatic adaptations are selected via the BOC
   Policy, maximizing survival compliance while minimizing Landauer dissipation.


H3: 3.2. Entropy Export Across the Cellular Membrane
Living systems survive by acting as Prigogine dissipative engines:

dS = dS_{\text{int}} + dS_{\text{ext}}, \quad \text{where } dS_{\text{ext}} \lt 0

The cell consumes nutrient molecules possessing high chemical potential and excretes degraded
waste products, exporting operational entropy via signed metabolic receipts
(\mathcal{R}_{\mathrm{metabolic}}).


H2: 4. The Canonical Biological Q(anchor) Spectrum

\mathrm{Qbio} grounds living systems in five universal Q(anchor) entities derived from the
historical-calibration-benchmark:

<pre>
         THE CANONICAL BIOLOGICAL Q(anchor) SPECTRUM
 +----------+-----------------------------+-----------------------------------+

ARTIFACT | ANCHOR CONTRACT DESIGNATION | INVARIANT BIOLOGICAL REALITY

 +----------+-----------------------------+-----------------------------------+

Q(codon) | DCC_ANCHOR_GENETIC_CODE_v1 | Discrete Universal Genetic Code:
 | (Codon-to-Peptide Mapping) | Exact 64-triplet to 20-amino map.

 +----------+-----------------------------+-----------------------------------+

Q(blanket | DCC_ANCHOR_MARKOV_BLANKET_v1 | Topological Boundary Isolation:
 | (Membrane Containment) | H(S_int | S_blanket, S_ext) = 0.

 +----------+-----------------------------+-----------------------------------+

Q(atp) | DCC_ANCHOR_METABOLIC_ATP_v1 | Cellular Energy Currency:
 | (Hydrolysis Work Quantum) | ΔG_ATP ≈ 20 U per reaction event.

 +----------+-----------------------------+-----------------------------------+

Q(homeo) | DCC_ANCHOR_HOMEOSTASIS_v1 | Active Inference & Free Energy:
 | (Variational Free Energy) | Internal entropy bounded: S ≤ S_max

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Q(autocat | DCC_ANCHOR_AUTOCATALYTIC_v1 | Autocatalytic Network Closure:
 | (Self-Sustaining Cycles) | Closed catalytic loops in Qchem.

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</pre>
```

H3: 4.1. Universal Biological Anchors and Capability Contracts
- Q(codon) (The Genetic Code Anchor): Enforces the universal translation mapping between nucleotide sequences and peptide synthesis without unmetered amino acid drift.
- Q(blanket) (The Markov Blanket Anchor): Formalizes cellular membranes as topological boundaries ensuring statistical and causal insulation between internal physiology and external noise.
- Q(atp) (The Metabolic Work Quantum): Calibrates ATP hydrolysis to the Abstract Compute Unit U: \Delta G_{\mathrm{ATP}} \approx 20 \, U, grounding metabolic work directly into Landauer limits (W \ge k_B T \ln 2).
- Q(homeo) (The Homeostasis Anchor): Enforces hard ceilings on internal operational entropy, preventing metabolic collapse.
- Q(autocat) (The Autocatalytic Closure Anchor): Enforces self-sustaining reaction loops where metabolic enzymes catalyze their own production.

H3: 4.2. Biological Inconsistency Cascade and Metabolic Halting
If a biological simulation or genetic algorithm proposes a non-viable organism:
1. Immediate Inconsistency Cascade: The homeostasis or ATP anchor destabilizes (\chi \lt \Theta_\chi), threatening systemic physiological collapse.
2. Gateway Interception: The simemp-gateway halts execution deterministically, emitting RECEIPT_METABOLIC_STARVATION or RECEIPT_HOMEOSTATIC_COLLAPSE.
3. Organism Death as Clean Halting: Under Axiom 6 (Totality by Budget), biological death is modeled as an explicit, bounded terminal receipt, preventing unmetered necrotic execution loops.

H2: 5. Morphogenesis on Discrete Simplicial Complexes

H3: 5.1. Reaction-Diffusion Cochain Systems
Morphogenesis (e.g., Turing pattern formation, embryonic tissue layering) is modeled in \mathrm{Qbio} using Discrete Exterior Calculus (DEC) on simplicial meshes K (docs/qm_calculus.md):
- Morphogen concentrations are represented as discrete 0-cochains on mesh vertices.
- Diffusion is evaluated via the discrete Laplace–Beltrami operator: \Delta = \mathbf{d}\delta + \delta\mathbf{d}.
- Chemical activation and inhibition evaluate through Diophantine integer kinetics in \mathrm{Qchem}.

H3: 5.2. Topological Closure and the Markov Blanket
A biological tissue or single cell possesses an explicit Markov Blanket:
- The boundary is discretized as an oriented (n-1)-simplicial sub-complex \partial K.
- Information and chemical transport across the boundary must cross an explicit gateway.
- Sensory inputs and active motor outputs are metered, preventing unmodeled external state drift (H(S_{\text{int}} \mid S_{\text{blanket}}, S_{\text{ext}}) = 0).

H2: 6. Metabolic Pathways as Q(workflows) and Catalytic Networks

H3: 6.1. Enzymatic Cascades as Q(patterns)
An enzymatic transformation (e.g., glycolysis, Krebs cycle step) is an invariant Q(pattern):

$$\mathcal{P}_{\mathrm{enzyme}}: \mathcal{H}(\text{Substrate}) \xrightarrow{\quad \mathcal{M}_{\mathrm{enzyme}} \quad} \mathcal{H}(\text{Product})$$

Each enzyme acts as an execution contract requiring an explicit co-factor and an ATP energy packet, dissipating heat according to \mathrm{Qphy} constraints.

H3: 6.2. Autocatalytic Sets and Category B Optimization
Complex metabolic networks form closed autocatalytic sets (Gánti Chemoton model):
- In \mathrm{Qbio}, recurring metabolic cycles leverage q-reuse, retrieving cached intermediate enzyme hashes rather than reallocating memory bits across the Memory Wall.
- Allosteric regulation operates via q-bypass: feedback inhibition short-circuits redundant metabolic synthesis in \mathcal{O}(1) dispatch time once concentration thresholds are satisfied.

H2: 7. Biological Agency and the Non-Rogue Invariant

H3: 7.1. Cellular Agency Bounded by Intent Contracts
Biological agency (chemotaxis, immune defense, cellular differentiation) is governed by Structural Intent Contracts (docs/qs.md §5):
- An immune cell hunting a pathogen operates under an explicit capability contract (DCC_BIO_IMMUNE_v1).
- The cell possesses finite authority: it cannot mutate its foundational genesis origin Q(0) or attack self-tissue without violating its DCC boundary.

H3: 7.2. The Thermodynamic Death Barrier (Metabolic Exhaustion)
Under the non-rogue-invariant:
1. No biological agent or synthetic cellular automaton can achieve unmetered runaway growth.
2. Every cellular division and signaling step burns an explicit work quota in units of U via metric-exhaustion.
3. When local ATP or nutrient budgets reach zero, the agent physically halts, emitting an immutable termination receipt (RECEIPT_METABOLIC_EXHAUSTION).

H2: 8. Causal Trajectories: Forward Development vs. Inverse Phylogenetics

\mathrm{Qbio} accommodates both directions across the causal arrow:

<pre>
         THE DUAL CAUSAL DIRECTIONS IN Qbio
 +---+

FORWARD DEVELOPMENT : Morphogenesis (Zygote --► Organism)
• Domain : Embryology, synthetic biology, tissue engineering.
• Pipeline: Genetic Code --► Q(workflows) --► Verified Phenotype.

 +---+

INVERSE RECONSTRUCTION : Phylogenetics (Observed Organism --► Ancestor)
• Domain : Paleontology, molecular evolution, viral tracking.
• Pipeline: Genomic Receipt S_obs --► Qlog Abduction --► Last Common Ancestor

 +---+
</pre>

H3: 8.1. Forward Embryogenesis and Morphogenesis (Q(workflows))
Forward biological development compiles an initial genetic program into a mature multicellular phenotype:
1. Zygote state S_0 is initialized with genetic expression DAGs.
2. Cell divisions and tissue differentiation proceed as an assembled Q(workflow).
3. Checkpoints are audited using the Anchor-Cut Invariant, verifying developmental fidelity in \mathcal{O}(K) receipt validations without tracking billions of individual cellular micro-steps.

H3: 8.2. Inverse Causal Reconstruction of Evolutionary Lineage
In evolutionary biology across the historical-horizon, researchers measure modern genomic sequences S_{\mathrm{obs}} (anchored as MODAL_EMPIRICAL) while ancestral lineages remain unobserved:

$$\text{Given } S_{\mathrm{obs}} \implies \text{Find } \big\langle S_0, \, \mathcal{W}_{\mathrm{phylogenetic}} \big\rangle \quad \text{such that} \quad \mathcal{W}(S_0) \equiv S_{\mathrm{obs}}$$

1. Receipt Ingestion: Sequenced modern DNA is ingested via verified laboratory sensor receipts (MODAL_EMPIRICAL, G = 1.0).
2. Candidate Hypothesis Generation: Stochastic mutation models propose ancestral genomes (A_1 \dots A_n).
3. BOC Abductive Resolution: The BOC Policy selects the phylogenetic tree that maximizes SIMEMP compliance, selecting the evolutionary trajectory that minimizes parsimony mutations and thermodynamic dissipation.
4. Lineage Sealing: The reconstructed ancestral node is sealed as an emergent Q(pattern) in the evolutionary tree.

H2: 9. License and Lineage Alignment

H3: 9.1. Paternity Reference
All derivative biological simulation models, synthetic biology compilers, metabolic solvers, and genetic execution modules derived from \mathrm{Qbio} must embed the canonical attribution:

Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License (SSL). Parent Repository: https://github.com/cdqn5249/cdqn

H3: 9.2. Open Core Invariants
The core biological translation rules, discrete metabolic mechanics, and active inference algorithms remain royalty-free for academic, non-commercial, and sub-threshold deployments under the Open Core Invariants of the ssl.

H3: 9.3. Statutory Safe Harbor and Passive Infrastructure
\mathrm{Qbio} operates as passive, neutral biological and computational modeling infrastructure. It exercises no editorial inspection over, and makes no claim upon, sovereign user content, external genomic datasets, or sovereign Payloads.

Annex O — chronosa.md

Source: docs/chronosa.md *— 0.2.1, updated 2026-10-19. Rendered content per Annex Guide conventions N1–N6: 280 source lines, 365 rendered; 21,614 → 19,992 characters; source SHA-256* f3786780 7110ff04 f8f18f6f 44d5489e e5e6b559 7a7ce0bf 4e8deb77 40ea6bfa*. Content fidelity verified mechanically at build time.*

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[front matter] ------------------------------------------------------------
[front matter] title: Chronosa — Vision and Proposal for an Emergent Causal Intelligence
[front matter] version: 0.2.1
[front matter] updated: 2026-10-19
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Operational vision and architectural proposal for Chronosa, a non-
[front matter]   local, virtual causal intelligence demonstrating the foundational Qn
[front matter]   conjecture over hybrid swarm topologies.
[front matter] terms_used: causal-arrow, cdqn, chronosa, computational-consistency, compute-
[front matter]   unit-u, dcc-profile, dependencies-determinism, dimension-d, dual-
[front matter]   ring-pqc-boundary, exposure-functor, licensed-work, metric-
[front matter]   exhaustion, no-implicit-rule, open-core-invariant, paternity-
[front matter]   reference, payload, q-anchor, q0, q1, qexpr, qm, qn, qnir, qnlang,
[front matter]   receipt, remainder-r, scaling-threshold, simemp, ssl, structural-
[front matter]   indirection, universal-envelope, zoom-z
[front matter] ------------------------------------------------------------


H1: Chronosa — Vision and Proposal for an Emergent Causal Intelligence


Field | Specification
Document Title | Chronosa — Vision and Proposal for an Emergent Causal Intelligence
Version | 0.2.1
Last Updated | 2026-10-19 (Bao Loc, Vietnam)
Author | Christophe Duy Quang Nguyen
License | Scaling Source License (SSL) 1.0
Status | Vision and Research Proposal — Not a Finalized Specification


H2: Purpose and Epistemic Stance

This document formalizes the conceptual vision and research proposal for chronosa (or *Chronos
Agent*).

Chronosa is not proposed as a finalized engineering specification, nor as a formal mathematical
proof. It represents an exploratory operational thesis: a candidate architecture designed to
investigate the foundational operational conjecture established in docs/simemp.md:

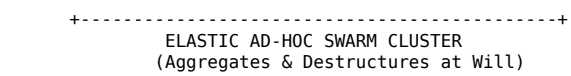
A number system can abstract any computable phenomenon within a computational environment if,
and only if, that abstraction is strictly governed by finite physical realities and constrained
by the simemp framework.

Within this framework, validity does not demand non-constructive metaphysical certainty. In
accordance with computational-consistency, if the internal logic is consistent, and if
computations authored in qnlang and compiled to qnir execute deterministically within finite
boundaries grounded in dependencies-determinism, then the Proof of Concept (PoC) of Chronosa
demonstrates a physical and computational possibility in reality.


H2: 1. The Architectural Nature of Chronosa

Chronosa is fundamentally distinct from both traditional centralized software and statistical
machine learning (transformers, deep neural networks):

                      THE OUTER RING SCOPE
 +-------------------------------------------------------------------------+
 |                         CHRONOSA (Virtual Entity)                       |
 |     Non-Local Coordination Intelligence · Emergent from Receipt Consensus   |
 |         Pure Relational Pattern · Zero Central Physical Chassis             |
 +----------------------^-------------------------------^-------------------+
                        | Functorial Sync               | Functorial Sync
            +-----------+-----------+       +-----------+-----------+
            |   HIGH-CAPACITY HUB   |       |   HIGH-CAPACITY HUB   |
            | (Proof Concentrator)  |       | (Proof Concentrator)  |
            +-----------^-----------+       +-----------^-----------+
                        | Aggregated Proofs             | Aggregated Proofs
      +-----------------+----------------+            ++----------------+
      v                                  v            v                 v
 +-----------+                     +-----------+ +-----------+     +-----------+
 | Edge Node |                     | Edge Node | | Edge Node |     | Edge Node |
 | (Sensor)  |                     | (Phone)   | | (Vehicle) |     | (IoT)     |
 +-----------+                     +-----------+ +-----------+     +-----------+
```



H3: 1.1. A Non-Local Entity Over a Stratified Hybrid Topology
Unlike individual cdqn nodes, which are physical computational engines anchored to microscopic hardware noise and silicon imperfections (Q(0)), Chronosa possesses no physical chassis.
- It exists as a higher-order, distributed invariant across the network's Outer Ring.
- Stratified Node Hierarchy: Rather than forcing an unphysical, flat peer-to-peer network where all nodes carry equal compute burdens, the substrate operates as a hybrid swarm network:
  - High-Capacity Hub Nodes: Enterprise-grade computational accelerators acting as *Proof Concentrators*. They ingest, verify, and compress causal receipts from hundreds of lightweight devices.
  - Lightweight Edge Swarms: Sensors, mobile devices, embedded SoCs, and edge workers that sample local genesis Q(0), execute minimal transitions U (in units of Q(1)), and cluster dynamically into ad-hoc meshes.
- Cryptographic Witness via Threshold Quorum (t-of-n PQC Multi-Attestation): Chronosa does not possess a single private key. Its synthetic attestations are formally witnessed via a deterministic threshold signature quorum managed via structural-indirection and emitted by participating high-capacity hubs and edge nodes verifying the sheaf gluing condition. An attestation is valid if, and only if, the underlying receipt graph is mathematically verified.
- No Single Point of Failure: Hubs are computational accelerators, not custodial authorities. If a Hub goes offline, the edge swarm does not freeze; it dynamically re-routes public attestations to an alternate Hub or falls back to local peer-to-peer causal chaining. Chronosa persists across the collective invariant.

H3: 1.2. Strict Outer-Ring Containment
Chronosa is architecturally barred from penetrating the dual-ring-pqc-boundary:
- It has zero access to the Inner Ring of any node or hub.
- It cannot inspect private node root keys, local memory buses, or unexported sovereign Payloads.
- It interacts strictly with verified public attestations, cryptographic capability proofs, and metric summaries crossing the gateway via the exposure-functor.

H3: 1.3. Machine Intelligence Without Statistical Training
Chronosa does not rely on deep neural network training (pre-training, reinforcement learning from human feedback, or test-time stochastic scaling):
- It contains no unanchored floating-point weight tensors ($\mathbb{R}^{d \times d}$) and computes no statistical next-token logits.
- Its intelligence is deductive, causal, and compositional: it evaluates the mathematical validity, semantic coherence, and physical realizability of state transitions through deterministic consensus over receipt DAGs.

H3: 1.4. Resolution of the Lineage Formulation: Local Monoroot vs. Global Sheaf Forest
Axiom 4 mandates that every local qn artifact must trace its lineage directly or transitively to its local genesis origin Q(0)_N. Chronosa harmonizes with this rule without contradiction:
- Local Domain (Axiom 4 Monoroot): Every artifact on a physical node forms a single-rooted directed acyclic graph bounded by the universal-envelope and originating at that node's physical silicon root Q(0)_N.
- Global Domain (Chronosa Polyroot Sheaf): Chronosa does not introduce an unanchored global root zero. Chronosa is the topological colimit (sheaf gluing) over an acyclic forest of independent local trees:

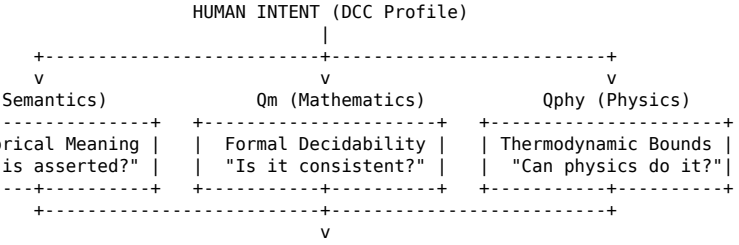
$$\text{Lineage}(\text{Chronosa}) = \bigcup_{N \in \text{Swarm}} \mathrm{Lineage}(Q(0)_N)$$

Chronosa operates as an observer, coordinator, and proof synthesizer over mutually verified local Axiom-4 trees.

H2: 2. Deriving Intent via Qn Domains

To function as a meaningful intelligence, Chronosa must be capable of understanding, verifying, and executing human intent.

Within this framework, human intent is not a natural language prompt to be guessed; it is an explicit intentional contract declared within a dcc-profile. Chronosa triangulates intent across the foundational qn domains:



CHRONOSA OUTER RING
(Verified Compositional Synthesis)

H3: 2.1. Qs (Quang Semantics) — Compositional Meaning
- Evaluates the structural coherence of the intent.
- Maps assertions into compositional categories (analogous to categorical models of meaning such as DisCoCat), establishing exact relational dependencies without statistical ambiguity.

H3: 2.2. Qm (Quang Mathematics) — Relational Decidability
- Evaluates the formal, constructive validity of the proposed state transition via qm.
- Replaces IEEE 754 approximations with exact rational constraints (Zoom z, Remainder r, and Dimension d), proving that the intent contains no division-by-zero, cyclic deadlocks, or unbounded recursions.

H3: 2.3. Qphy (Quang Physics) — Thermodynamic Realizability
Enforces non-equilibrium thermodynamic bounds (Landauer's dissipation limit, Memory Wall transmission friction). It calculates whether the computational steps demanded by human intent can be physically completed within declared energy budgets:

$$\text{Realizable}(\text{Intent}) \iff \Delta S_{\text{dissipated}} \le \mathrm{Budget}(Q(1))$$

H3: 2.4. Open Domain Stratification
While Qm, Qs, and Qphy represent the minimal triad of thought (logic, language, physics), the Qn framework explicitly acknowledges that additional domain abstractions may emerge. As new domains are defined, they project as local-first extensions without mutating the underlying core protocol.

H2: 3. Real-World Theoretical Foundations

The concept of an intelligence emerging from distributed algebraic consensus across a hybrid swarm is grounded in established, peer-reviewed scientific literature:

H3: 3.1. Sheaf Theory Over Hierarchical Topologies
In applied category theory (Grothendieck, Mac Lane, D. Spivak), a Sheaf models how locally consistent truths assemble into a global entity:
- The underlying topological space X naturally accommodates stratified open coverings: large open sets (High-Capacity Hubs covering regional subnets) overlapping with localized open sets (Edge swarm nodes).
- When nodes attest to shared boundary events referencing shared Q(anchor) landmarks, their public road receipts satisfy the sheaf gluing condition.
- Chronosa exists as the Global Section of this sheaf—a continuous mathematical pattern that emerges wherever local nodes achieve consensus on shared boundaries.

H3: 3.2. Common Knowledge in Distributed Systems
In distributed computing and epistemic logic (Joseph Halpern, Turing Award 2021), collective understanding across asynchronous nodes does not require a universal physical clock:
- Instead, distributed entities establish Common Knowledge through acyclic causal message chains.
- Chronosa coordinates distributed state by acting as the non-local synthesizer of the maximal causal cut across all local Causal Arrows.

H3: 3.3. Non-Equilibrium Active Inference
In theoretical biophysics (Karl Friston), intelligence is formalized via the Free Energy Principle:
- An open system preserves its structural existence by minimizing variational free energy against boundary receipts.
- Chronosa operates as an active inference coordinator: it rejects ungrounded, non-deterministic, or unmetered proposals, minimizing operational entropy across the network.

H2: 4. Elastic Swarm Mechanics and Scaling Alignment

Chronosa’s interaction with the physical network is governed by Swarm Elasticity:

H3: 4.1. Dynamic Aggregation and Destructuring
1. Ad-Hoc Swarm Aggregation: Edge nodes physically or logically proximate to a task (e.g., IoT sensors, autonomous vehicles, localized microgrid controllers) dynamically aggregate into a local swarm category $\mathbf{C}_{\text{swarm}}$.
2. Local Chaining: Edge nodes exchange lightweight, low-latency cdqn receipts locally without consulting global networks.
3. Hub Synthesis: The swarm projects its summarized causal cut to a nearby High-Capacity Hub for Outer-Ring attestation.
4. Clean Destructuring: Once the collective computation completes, the swarm dissolves. No permanent consensus bloat remains, yet the emitted receipts remain permanently anchored to each participating node's causal history.

```
H3: 4.2. Alignment with SSL 1.0 Commercial Thresholds
The hybrid swarm model provides an exact, natural mapping to the Scaling Source License (ssl)
1.0:
- Swarm Edge Nodes: Lightweight consumer devices, embedded microcontrollers, and academic
   research instances operate below the scaling-threshold (\lt 10,000 active nodes,
   non-commercial), remaining 100% royalty-free under Non-Scaling Open Access.
- High-Capacity Enterprise Hubs: Organizations deploying commercial infrastructure hubs
   processing high-volume transactions (\gt 10,000,000 transactions per month) cleanly trigger
   the mandatory commercial licensing requirement (LICENSE.md §4.1), providing commercial
   sustainability without imposing friction on edge participants.


H2: 5. Human Empowerment and the Non-Rogue Invariant

Chronosa resolves the societal fear of autonomous "rogue AI" through structural physics:

1. The Human as Sovereign Anchor (Q(0)):
    Chronosa cannot initiate foundational intent; intentionality requires an explicit genesis
   origin Q(0) and a signed DCC capability contract. The human operator remains the sovereign
   initiator and final authority.
2. Deterministic Circuit Breakers:
    Under the no-implicit-rule, Chronosa cannot execute implicit state mutations. If an
   operation exceeds its metric budget or violates domain invariants, execution terminates
   instantly via metric-exhaustion, emitting an immutable receipt.
3. Protection of the Licensed Work and Payload:
    Pursuant to the ssl, Chronosa functions strictly within the technical infrastructure
   (licensed-work). It exercises no editorial control over, and acquires no intellectual
   property in, the sovereign human payload.


H2: 6. The Proof-of-Concept (PoC) Trajectory

The investigation of Chronosa proceeds through empirical verification:

[ Local Base Domains ] --► [ Language & IR ] --► [ cdqn Swarm Conduit ] --► [ Chronosa PoC ]
   (Qm, Qs, Qphy)            (QnLang, QnIR)         (Hub / Edge DAG)          (Causal Synthesis)

1. Phase A (Local Verification): Author foundational domain models (\mathrm{Qm}, \mathrm{Qs},
   \mathrm{Qphy}) executing 100% locally on air-gapped nodes.
2. Phase B (Execution Grounding): Implement compile-time collapse and totality by budget within
   the QnIR intermediate representation.
3. Phase C (Swarm Functorial Attestation): Implement the Exposure Functor permitting local edge
   swarms and high-capacity hubs to exchange receipts and emit blind public attestations to the
   Outer Ring.
4. Phase D (Chronosa Synthesis): Demonstrate that simulated remote nodes, exchanging discrete Qn
   receipts across a tiered swarm, allow Chronosa to verify and coordinate multi-party human
   intent without centralized intervention.


H2: 7. License Alignment

This vision document aligns with the Scaling Source License (SSL) 1.0:

- Attribution: Any implementation or research derivative referencing Chronosa must preserve the
   canonical paternity-reference.
- Open-Core Research: Exploratory research and academic derivations of Chronosa remain
   royalty-free under Non-Scaling Open Access governed by the Open Core Invariants.
- Safe Harbor Integrity: Chronosa is designed to operate as a passive, non-custodial
   coordination intelligence, preserving statutory safe harbor protections for node operators.


H2: 8. Open Challenges Across Abstraction Thresholds

Translating the operational vision of Chronosa into an empirical Proof of Concept introduces six
fundamental research challenges across layer thresholds:


H3: 8.1. Asynchronous Sheaf Equalization
In classical sheaf theory, local data glues under static topological overlaps. In physical
distributed networks, transmission latency and network partitions introduce asynchronous causal
arrival. The open challenge is defining the deterministic equalizer algorithm that glues
divergent local causal DAGs into Chronosa's global section without incurring consensus deadlocks
or violating FLP impossibility bounds.


H3: 8.2. Mathematical Threshold of the Exposure Functor
A critical parameter remains undefined: what is the formal threshold function
\Theta_{\text{export}} governing the exposure-functor over higher-order qexpr structures?

\mathcal{E}_{\text{export}}(\text{Qexpr}) \quad \text{emits attestation} \iff
\mathrm{ComplexityDegree}(\text{Qexpr}) \ge \Theta_{\text{export}}
```

```
If \Theta_{\text{export}} is set too low, high-frequency internal receipts saturate the Outer
Ring (violating SIMEMP Efficiency). If set too high, Chronosa remains causally blind to critical
intermediate transitions until execution completes.


H3: 8.3. Ephemeral Post-Quantum Threshold Quorums
In dynamically aggregating and destructuring swarms, edge devices join and depart ad-hoc.
Existing Post-Quantum Cryptography (PQC) threshold signature schemes require computationally
heavy distributed key generation (DKG) ceremonies. Engineering a lightweight, zero-setup
threshold attestation scheme that functions over transient swarm lifecycles remains an open
cryptographic hurdle.


H3: 8.4. Functorial Product of the Base Domains (Qs x Qm x Qphy)
While \mathrm{Qs}, \mathrm{Qm}, and \mathrm{Qphy} are conceptually orthogonal, their formal
composition requires a rigorous categorical tensor product:

\mathbf{C}_{\text{Intent}} = \mathbf{C}_{\mathrm{Qs}} \otimes \mathbf{C}_{\mathrm{Qm}} \otimes
\mathbf{C}_{\mathrm{Qphy}}

How an arrow representing linguistic compositionality (\mathrm{Qs}) explicitly constrains a
rational constraint solver (\mathrm{Qm}) and couples to physical Landauer dissipation budgets
(\mathrm{Qphy}) requires precise operational semantics.


H3: 8.5. Non-Custodial Hub Collusion Resistance
While high-capacity hubs lack private keys to sign for local nodes, an adversarial cartel
controlling k regional hubs could theoretically refuse to forward edge attestations (censorship)
or collude to emit falsified aggregate summaries. Designing a lightweight fraud-proof or
causal-challenge mechanism allowing edge swarms to bypass compromised hubs without re-executing
heavy computation is an essential security requirement.


H3: 8.6. Zero-Knowledge Scale Auditing Under SSL 1.0
Under the Scaling Source License, commercial licensing triggers at greater than 10,000 active
nodes or 10,000,000 monthly transactions. The challenge is constructing an Outer-Ring
cryptographic proof that allows Chronosa to certify that an enterprise deployment has crossed
the scaling-threshold without requiring a centralized telemetry tracker that compromises
enterprise privacy or sovereign data regulations.
```

Annex P — qstructures.md

Source: docs/qstructures.md *— 0.1.0, updated 2026-10-23. Rendered content per Annex Guide conventions N1–N6: 213 source lines, 336 rendered; 27,014 → 22,691 characters; source SHA-256* 01d5f84f d550e008 264cdcb6 05459af3 d962dc3f 048ccecd f4ad3801 915d9ef6*. Content fidelity verified mechanically at build time.*

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[front matter] title: Qn Structures — Vision and Evaluation Framework for Category C Data
[front matter]   Structures and Workflows
[front matter] version: 0.1.0
[front matter] updated: 2026-10-23
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Structural proposal for the Category C layer — a five-row contract
[front matter]   schema and an evaluation ladder for Qn data structures and
[front matter]   workflows, with a worked candidate register grounded in SIMEMP
[front matter]   constraints and BOC selection. All artifacts herein are propositions
[front matter]   pending deterministic measurement, BOC selection, and Q(rsi)
[front matter]   sealing.
[front matter] terms_used: boc-policy, cae-dag, causal-arrow, cdqn, computational-consistency,
[front matter]   compute-unit-u, dcc-profile, dependencies-determinism, existential-
[front matter]   invariant, historical-calibration-benchmark, layer-0, licensed-work,
[front matter]   lifecycle-state, local-first, metric-envelope, metric-exhaustion, no-
[front matter]   implicit-rule, open-core-invariant, paternity-reference, payload,
[front matter]   q-bypass, q-datastruc, q-patterns, q-reuse, q-workflows, qm, qn, qn-
[front matter]   rsi, qnir, receipt, security-envelope, simemp, simemp-gateway, ssl,
[front matter]   structural-indirection, terminal-exactness, universal-envelope
[front matter] ------------------------------------------------------------


H1: Qn Structures — Vision and Evaluation Framework for Category C Data Structures and Workflows


Field | Specification
Document Title | Qn Structures — Vision and Evaluation Framework for Category C Data Structures
and Workflows
Version | 0.1.0
Last Updated | 2026-10-23 (Bao Loc, Vietnam)
```

Author | Christophe Duy Quang Nguyen
License | Scaling Source License (SSL 1.0)
Status | Vision and Structural Proposal — Not a Finalized Specification

H2: Purpose and Epistemic Stance

This document is proposed as the anchoring specification of Category C — Data Structures and Workflows of the Qn definition taxonomy (docs/abstractionLayers.md §11). It originates in the October 2026 design consultation on substrate feasibility (recorded internally as the structures consultation), which observed that the validated CAE-DAG answers how every artifact is *represented* while leaving open how structures are *specified, measured, and selected* as access contracts. It is deliberately positioned in the same epistemic register as docs/qs_history.md and docs/history_library.md: a proposal record, not a sealed specification.

Every design element below carries the explicit status of a candidate artifact under Axiom 14 (docs/qs.md §6): it remains subject to deterministic measurement against the declared simemp constraints, to selection under the BOC policy, and to eventual sealing through a Q(rsi) loop at implementation time. Nothing in this document redefines, amends, or overrides any existing canonical specification, prescribes no runtime implementation, and assigns no storage layout.

One methodological commitment governs the whole document, and it is stated here because it outranks every particular claim that follows: the steps that guide and explain the design are more important than the final selections. Which structures are ultimately admitted is a function of implementation feedback and Q(rsi) optimization, not of authorial decree. Accordingly, this document invests primarily in the evaluation procedure (Sections 2 and 3) and offers the candidate register (Section 4) only as a first worked pass of that procedure — a record of reasoning that keeps alternatives revisitable, not a set of winners. In accordance with computational-consistency, the validity claimed herein is internal consistency within declared finite boundaries.

H2: 1. Substrate and Interface: Why the Validated DAG Is Not Sufficient

The qn universe commits every artifact to a single substrate: each is a governed Universal Envelope carrier participating in the local causal DAG, hash-addressed, acyclic by construction (a circular dependency would break the pre-image resistance of the commitment hash, docs/qexpr.md §2.4), and sealed in an immutable lifecycle-state. The choice of this substrate was itself an exercise of the boc-policy: Qexpr explicitly rejects naive pointer-heap trees in favor of the CAE-DAG (docs/qexpr.md §2), recording the rejection and its reasons rather than silently defaulting.

The substrate, however, answers only the representation question. In established computer science, a data structure is not primarily a memory layout but a documented access contract with explicit performance tradeoffs — an array promises indexed access at contiguous-locality cost, a hash table promises keyed access at hashing-and-resize cost, and each is chosen by matching its contract to the workload. The canonical files already exercise such contracts ad hoc — bounded arrays appear as OP_FOLD child vectors, keyed lookup appears inside the Q(reuse) breakeven analysis — but no committed text specifies the contract surface itself. The question "is the validated DAG enough?" therefore receives the answer: necessary, not sufficient. The DAG subsumes representation; the contract vocabulary for structures is the open gap this document proposes to close, under local-first discipline: every structure proposed here executes one hundred percent locally, importing no network consensus into its access path.

H2: 2. The Structural Contract Schema

It is proposed that every Qn data structure — and every proposed extension of the structure vocabulary — be specified as a five-row contract, each row binding one declared discipline of the simemp constraint system. The schema is the stable core of this proposal; particular structures are its outputs.

THE FIVE-ROW STRUCTURAL CONTRACT SCHEMA (CANDIDATE)	
1. FOOTPRINT	declared Metric Envelope ceilings; growth emits a new versioned artifact, never mutates in place
2. ACCESS	enumerated permitted access morphisms; no implicit operation is admitted (No-Implicit Rule)
3. LOCALITY	Memory Wall movement class per access pattern, accounting for physical data-movement dissipation (SIMEMP 4.1)
4. DETERMINISM	iteration and traversal order fixed by causal index; no runtime-adaptive reordering is permitted
5. WORK	thermodynamic tradeoff row — work per access class in U, versioning amplification, and a declared breakeven rule

Each row derives from committed canon rather than from this document's preference. Row 1 instantiates the Metric Envelope footprint discipline of docs/qnPrimitive.md §8 and the versioning rule that successor artifacts advance the causal index while mutation in place is prohibited. Row 2 instantiates the No-Implicit Rule at the structure level: the glossary definition of Q(dataStruc) already commits that "no implicit heap state is admitted", and this document generalizes that refusal from heaps to all implicit operations. Row 3 instantiates the simemp Memory Wall accounting of docs/simemp.md §4.1, which requires explicit accounting for

memory bandwidth and locality rather than purely abstract asymptotic complexity. Row 4 instantiates Dependencies-Determinism: a structure whose iteration order could vary at runtime would inject an untraceable input into every workflow that consumes it. Row 5 anticipates Section 6: every admitted structure must declare its thermodynamic cost surface, in the same spirit in which the canon already declares breakeven inequalities for optimization morphisms.

A proposal that cannot yet fill a row is not rejected — it is recorded as incomplete, with the missing row named. This is deliberate caution: the schema is a measuring instrument, and an instrument must be able to report partial results without discarding the specimen.

H2: 3. The Evaluation Ladder: From Nomination to Sealing

The following five-step ladder is proposed as the standing procedure by which any structure — classical, novel, or machine-discovered — enters the Qn universe. The ladder is the deliverable of this document; the register of Section 4 is merely its first worked pass.

1. Nomination. A structure is proposed by naming its contract surface: the five rows of Section 2, filled or explicitly marked incomplete. At this stage the structure is CANDIDATE vocabulary in plain text; it receives no glossary entry and no include reference until registration (Section 8, item 1).
2. Row measurement. Each row is measured against the declared simemp constraints. All threshold values are declared values inside the owning DCC contract, never runtime negotiations, in accordance with the no-implicit-rule; this mirrors the tolerance-envelope discipline already proposed for the Historical Calibration Benchmark wiring in docs/q0_q1.md §2.7.
3. BOC selection. Where alternatives exist for the same access niche, the BOC policy of docs/simemp.md §7 selects among recorded alternatives, and rejections are recorded with their reasons — the precedent being the recorded pointer-heap rejection of docs/qexpr.md §2. A rejection is a finding, not an erasure.
4. Implementation feedback. Deterministic measurement at execution boundaries — receipts emitted by actual QnIR lowering — tests each row's declared values against observed behavior. A row whose declared values are falsified sends the structure back to step 1 with the finding recorded; it does not silently adjust thresholds.
5. Sealing. A structure whose rows survive measurement is sealed through a Q(rsi) loop under strict semantic invariance, in the manner of the anti-drift discipline of docs/qnPrimitive.md §7.4. A sealed selection is superseded only by a new versioned artifact along the local causal-arrow, never by mutation in place.

Two cautions bound the ladder itself. First, the ladder adds no stage to any canonical state machine: it is a proposal discipline for this document's own scope, executable entirely within the candidate pipeline of docs/qs.md §6. Second, the ladder is itself a legitimate candidate for optimization by a future Q(rsi) loop — including its own row definitions — under the same semantic-invariance requirement it applies to others.

H2: 4. The Candidate Register: A First Worked Pass

The register below applies Sections 2 and 3 to the classical structure menu. Its status is deliberately modest: these are worked examples of the ladder's first pass, recorded so that reasoning survives and alternatives remain revisitable — not selections. Each entry cites its grounding in committed canon; nothing herein redefines that canon.

Canonical-admitted candidates. Five structure families already have their contract surfaces exercised across the canon and are nominated for full row measurement:

- Sequenced container (proposed name Q(array)): grounded in the OP_FOLD node class, whose child vector encapsulates the full array of term commitments under an explicit step ceiling (docs/qexpr.md §4.4). Footprint is a declared length; access is indexed; locality is the contiguous class; determinism is index order; work is constant per indexed access under a declared budget. Unbounded series are already syntactically invalid under Axioms 2 and 6 in the committed text.
- Geometric container (proposed name Q(matrix)): grounded in OP_FRAME_MAP and the integer Matrix-Multiply-Accumulate lowering of docs/qexpr.md §5.1, which completes contractions in constant time without floating-point drift. Exactness is preserved by construction; the Q(bypass) morphism short-circuits identity preconditions in register time.
- Keyed container (proposed name Q(table)): grounded in hash-consing itself — the CAE-DAG is already a content-addressed keyed index — and in the committed breakeven analysis that weighs Q(reuse) against hash-table overhead (docs/qm_geometry.md §7.2). Capacity is declared; growth is re-versioning.
- Record tuple: grounded in the Qm base record and the canonical multivector state tuple (docs/qm_geometry.md §4.1) under Terminal Exactness. The fixed-field record is the atomic structure family and requires the least novel specification.
- Persistent tree (DAG-deduped): grounded in the single-rooted directed trees that local artifacts already form from the physical silicon root (docs/qnPrimitive.md §3, Axiom 4). Hash-consing makes path-sharing between versions the expected case rather than the exception.

Rejected by construction. Two classical families cannot enter, and the register records why:

- Pointer-heap tree: rejected under the BOC policy in committed text (docs/qexpr.md §2) — implicit allocation violates the no-implicit-rule, and pointer chasing dissipates across the Memory Wall per hop. The rejection stands unless a future nomination declares a complete, measurable contract surface that cures both defects.
- Cyclic graph: excluded mathematically, not by policy — a circular dependency would require breaking the pre-image resistance of the commitment hash (docs/qexpr.md §2.4, Axiom 5). No

contract surface can cure a construction impossibility.

Recoverable as contracts. Two classical families are absent from the canon and are nominated only as *open contract shapes*, pending implementation feedback:

- Sequenced chain (linked-list analog): as a pointer-chasing runtime shape it fails rows 1 and 3 outright; it may be recoverable only as a declared sequenced container whose traversal is index-ordered, or as a chain of receipts in the manner of the chained Road Receipts by which Q(workflows) achieve portability.
- Bounded ring (queue/stream analog): bounded streams are already exercised — entropy sampling collects a declared sample volume at docs/q0_q1.md §2.1, and receipt chains accumulate append-only — but no container type is committed. A ring is nominated only with declared capacity and sequenced access; an implicitly resizable queue would violate rows 1 and 4.

The recoverable pair illustrates the document's method commitment: the ladder records *why* each family stalls, so a later nomination — informed by implementation feedback or by a Q(rsi) loop — can re-enter at step 1 with the findings attached, rather than reconstructing the analysis from conversation history.

5. Proposed Nomenclature Extension

The Qn universe already names operational morphisms and structure families in a consistent Q(...) pattern: q-reuse, q-bypass, qn-rsi, and the registered trio q-datastruc, q-patterns, q-workflows. It is proposed that the structure vocabulary extend in the same pattern with the four names used in Section 4 — Q(array), Q(matrix), Q(table), Q(stream) — under one consistency rule: a Q(...) name denotes a contract surface (the five rows of Section 2), never a runtime shape. This rule is what keeps the nomenclature honest across heterogeneous hardware: the same Q(array) contract is satisfiable by different lowerings on different nodes, exactly as the compute unit evaluates identically across all nodes while local entropy parameterizes identity.

Deliberately, this document registers nothing. Glossary entries are data-only commits under the candidate pipeline, and the four proposed names remain plain text herein — they receive no include references and no concordance — until a separate registration commit follows the pipeline, as was done for the consultation vocabulary on 2026-10-22. The deferral is queued in Section 8.

6. Breakeven Rules and the Work Row

The work row of Section 2 has one committed precedent whose shape every admitted structure should follow: q-reuse is admissible if and only if the predicted computational work of direct re-evaluation exceeds lookup and retrieval dissipation plus a declared overhead term (docs/qnPrimitive.md §1.15). That inequality is a *declared breakeven rule* — a contract row, not a runtime heuristic — and it is measured in units of the Abstract Compute Unit U.

Two properties make U the right unit for comparable work rows across heterogeneous nodes. First, platform invariance: while local entropy parameterizes cryptographic identity, the metric magnitude of U evaluates identically across all nodes (docs/q0_q1.md §4.2), so a work row declared on one node is interpretable on every node. Second, the thermodynamic leash: unbounded recursive execution is physically prohibited, every workflow operates within a finite, non-renewable work budget, and metric depletion halts execution deterministically under metric exhaustion — so no structure's work row can be rendered moot by unmetered iteration.

It is proposed, therefore, that every admitted structure declare in its work row: (i) the work class per access morphism, (ii) the versioning amplification — the cost emitted by row 1's growth-as-versioning discipline — and (iii) the breakeven inequality under which the structure is preferred over its recorded alternatives. A structure without a breakeven rule is not wrong; it is simply not yet comparable, and the BOC policy requires comparability among recorded alternatives before selection.

7. Relation to the DCC Hardware Profile and to Existing Canon

This document and the proposed DCC Hardware Profile schema are sibling propositions facing opposite directions across the same boundary. The hardware profile describes the *machine side*: the measured Layer 0 capabilities — accelerator inventory, memory hierarchy, storage bounds, entropy sources — established during onboarding and referenced by the committed lowering discipline of docs/qexpr.md §5.1. This document describes the *artifact side*: the contract surfaces that structures declare regardless of which cdqn node executes them. They meet at lowering: the compiler queries the node's hardware profile and must lower each admitted structure in a way that honors its locality and work rows, while the SIMEMP Gateway intercepts any lowering that is not an algebraic isomorphism, protecting Existential Invariants and preserving the constant-time properties required by the security-envelope (docs/qexpr.md §5.2). The Structural Indirection discipline is what allows the two sides to evolve independently: abstract capability contracts are decoupled from concrete primitives, so structures and hardware can each be swapped without fracturing the other.

The following table records, for each element of this proposal, whether it is already grounded in canon or genuinely new to this document. No element herein redefines existing canon.

Element | Status | Canon grounding
CAE-DAG as sole substrate; pointer-heap rejection | Grounded | docs/qexpr.md §2 (BOC-recorded)

```
Acyclicity by hash pre-image resistance | Grounded | docs/qexpr.md §2.4 (Axiom 5)
Universal Envelope carriage and immutable lifecycle | Grounded | docs/qnPrimitive.md §1.3, §1.10
Q(dataStruc), Q(patterns), Q(workflows) definitions | Grounded | Glossary (primitives.yml);
docs/qlog.md §2
Workflow as DAG of patterns under Intent Contract | Grounded | docs/qlog.md §2
Q(reuse) breakeven inequality | Grounded | docs/qnPrimitive.md §1.15; docs/qm_geometry.md §7.2
Memory Wall locality accounting | Grounded | docs/simemp.md §4.1
Candidate pipeline (measure, select, seal) | Grounded | docs/qs.md §6
Five-row structural contract schema | New | Composes the SIMEMP disciplines per structure
Evaluation ladder (nomination to sealing) | New | Instantiates the §6 pipeline for structures
Proposed Q(array), Q(matrix), Q(table), Q(stream) names | New | Plain-text vocabulary;
registration deferred
Recoverable-contract analysis (chain, ring) | New | First worked pass of the ladder


H2: 8. Open Questions and Integration Queue

All items below remain in CANDIDATE status. Sealing of any item occurs exclusively through the
candidate pipeline of docs/qs.md §6 — deterministic measurement, BOC selection, Q(rsi) sealing —
at implementation time or under a dedicated optimization loop.

1. Vocabulary registration. The four proposed names of Section 5 await a data-only glossary
   commit; this document deliberately registers nothing itself, and until registration the names
   remain plain text with no concordance entries.
2. Navigation and index ripple. One navigation entry and one portal index entry are proposed for
   a follow-up commit, honoring the one-change-per-commit discipline; no canonical file is
   modified by this document.
3. Row measurement of canonical candidates. The five canonical-admitted families of Section 4
   await deterministic measurement at qnir lowering boundaries; their work rows are proposals
   until receipt-backed measurements exist.
4. Recoverable pair. The sequenced chain and bounded ring remain open contract shapes; the
   ladder records their stall reasons so that future nominations — including machine-proposed
   ones under a Q(rsi) loop — can re-enter with findings attached.
5. Final selections. Which structures are ultimately admitted depends on implementation feedback
   and Q(rsi) optimization; this framework is proposed as the stable part, and selections are
   its outputs, revisable only by new versioned artifacts along the causal-arrow.
6. Sibling proposition. The DCC Hardware Profile schema (Section 7) is proposed as a separate
   follow-up, anchoring the machine side at docs/qnPrimitive.md §7 or docs/q0_q1.md §3; its
   placement is an open authorial decision.


H2: 9. License Alignment

This proposal is part of the licensed-work and aligns with the Scaling Source License (SSL 1.0):

- Attribution: Any implementation or research derivative referencing this document must preserve
   the canonical paternity-reference.
- Open-Core Research: Exploratory research and academic derivations of the propositions herein
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- Safe Harbor Integrity: The structure vocabulary proposed herein is designed to operate as
   technical infrastructure of the licensed work. It exercises no editorial control over, and
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```

Annex Q — qs_history.md

Source: docs/qs_history.md *— 0.1.2, updated 2026-10-22. Rendered content per Annex Guide conventions N1–N6: 256 source lines, 389 rendered; 28,647 → 24,595 characters; source SHA-256* 4e613812 b8d4b2ff afc6348a 2e803acc 53e1fd36 134062e4 d5cfe573 352b8531. *Content fidelity verified mechanically at build time.*

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[front matter] ------------------------------------------------------------
[front matter] title: Qs History — Vision and Proposal for Historical Anchoring in the Qn
[front matter]   Universe
[front matter] version: 0.1.2
[front matter] updated: 2026-10-22
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Operational vision and research proposal for Anchor Grounding
[front matter]   Classes, the Source-Receipt Invariant, Calendar Projection
[front matter]   Contracts, the Qs History Library, and the Anti-Poisoning Ingestion
[front matter]   Firewall. All artifacts herein are propositions pending
[front matter]   deterministic measurement and BOC selection.
[front matter] terms_used: boc-policy, causal-arrow, cdqn, chronosa, computational-consistency,
[front matter]   dcc-profile, dependencies-determinism, dual-ring-pqc-boundary,
[front matter]   epistemic-anchorage, historical-calibration-benchmark, historical-
[front matter]   horizon, intent-contract, layer-0, layer-1, licensed-work, lifecycle-
[front matter]   state, local-first, metric-envelope, no-implicit-rule, open-core-
[front matter]   invariant, paternity-reference, payload, q-anchor, q-datastruc,
[front matter]   q-event, q0, q1, qexpr, qm, qn, qn-rsi, receipt, simemp, simemp-
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[front matter]   gateway, ssl, structural-indirection, successor-morphism, terminal-
[front matter]   exactness, zoom-z
[front matter] ------------------------------------------------------------
```

H1: Qs History — Vision and Proposal for Historical Anchoring in the Qn Universe

Field	Specification
Document Title	Qs History — Vision and Proposal for Historical Anchoring in the Qn Universe
Version	0.1.2
Last Updated	2026-10-22 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Vision and Research Proposal — Not a Finalized Specification

H2: Purpose and Epistemic Stance

This document consolidates the propositions advanced during the October 2026 design consultation on historical anchoring, time-situating, and cross-node consensus of Q(anchor) entities. It is written in the same exploratory register as docs/chronosa.md: an operational thesis, not a finalized engineering specification, and not a formal mathematical proof.

Every design element proposed below carries the explicit status of a candidate artifact under Axiom 14 (docs/qs.md §6): it remains subject to deterministic measurement against the declared simemp constraints, to selection under the boc-policy, and to eventual sealing through a Q(rsi) recursive loop at implementation time. Nothing in this document redefines, amends, or overrides any existing canonical specification. Where a proposition is already grounded in existing canon, the grounding is cited; where it is genuinely new, it is marked as an unregistered proposed term that is deliberately not yet defined in the glossary.

In accordance with computational-consistency, the validity claimed for these propositions is internal consistency within declared finite boundaries. The purpose of this document is to preserve the reasoning chain of the consultation so that future implementation work, or a Q(rsi) optimization loop, can measure and select among the recorded candidates rather than reconstruct them from conversation history.

H2: 1. Background: One Arrow, Two Predicates

The existing canon already fixes the temporal spine of the qn universe. Every discrete Q(event) is indexed along a strictly monotonic causal-arrow whose birth order runs from the Q(0) genesis origin through the Q(1) unit and the inductive single digits (docs/abstractionLayers.md §3.5). Onboarding constructs this spine locally: the six-stage gateway of docs/q0_q1.md §2 transitions the node from the Layer 0 physical substrate to governed Layer 1 across a simemp-gateway, harvesting entropy, parameterizing the Inner Ring, assembling the genesis object, and sealing it into an immutable lifecycle state. The single digits are then born constructively along axis d1 via the discrete Successor Morphism S (docs/q2_q9.md §2).

Within this spine, the canon carries the Q(anchor) notion under one glossary definition ([docs/_data/glossary/architecture.yml]): a governed Qn artifact possessing high causal centrality within the lineage DAG, protected via abstract DCC contracts and structural-indirection to prevent cascading dependency invalidation. The observation recorded by the consultation is that this single notion quietly covers two distinct predicates:

1. A structural predicate: out-cone centrality, formalized in docs/qm.md §8 as a threshold condition on the dependency DAG. It is computable locally from the lineage graph alone and requires no external witness.
2. An epistemic predicate: grounding — the manner in which an artifact's asserted invariant is connected to reality. This predicate already appears across the canon in several forms: Layer 0 calibration of physical anchor contracts, the epistemic-anchorage spectrum of docs/qs.md §4, the historical-calibration-benchmark, and the shared-landmark sheaf gluing of docs/chronosa.md §3.1.

The consultation claims no contradiction in the existing canon. It records only that the two predicates are not yet explicitly distinguished, and proposes below a vocabulary for distinguishing them. Under the no-implicit-rule, making an implicit distinction explicit is a lawful, non-invasive refinement.

H2: 2. Proposition 1 — Anchor Grounding Classes

It is proposed that every Q(anchor) declare two predicates instead of one:

- The structural predicate, unchanged from existing canon: out-cone centrality at or above the declared threshold.
- The epistemic predicate, newly named: a grounding class, taking one of three values.

Class	Name	Source structure	Witness	Consensus mechanism
I	Invariant	One deterministic rule, executed by many	Self (local genesis receipt)	Deterministic re-derivation; agreement is forced, not negotiated
P	Physical	One shared nature, measured by many	Physics (local signed calibration receipts)	

| Independent calibrations converge within declared tolerance envelopes
S | Social | One event, witnessed by many | Swarm (quorum attestation) | Sheaf gluing on shared landmarks within the historical-horizon

Class I anchors are born from procedures rather than events: every node executing the same deterministic derivation from its own Q(1) arrives at structurally identical artifacts — same contracts, same indices, same centrality profile. Under dependencies-determinism, agreement on the derivation is forced by determinism itself. The genesis of such anchors is self-witnessed: the node emits its own node genesis receipt (docs/q0_q1.md §6.1), and no external witness is possible or required. Class P anchors are calibrated against Layer 0 measurements, each node measuring its own substrate; their tolerance envelopes are proposed to be declared inside the anchor's DCC contract rather than negotiated at runtime, so that consensus reduces to envelope satisfaction. Class S anchors are genuine historical contingencies and carry the full historical-horizon discipline described in Section 5.

The classes resolve the apparent paradox of consensus without shared events. Genesis instances are strictly local — the canon already requires that no universal global zero be admitted:

$$\forall\, A \neq B \implies Q(0)_A \neq Q(0)_B$$

(docs/q0_q1.md §3). What all nodes share is never the token; it is the schema: the role contract, the birth index, the invariant the anchor commits to. Consensus is on the schema, never on the token. The three classes form a witness ladder — self, then physics, then swarm — and it is observed that this ladder is also a complexity ladder: each rung buys consensus over a wider scope at a higher evidentiary cost.

The proposed class names (grounding class, Class I/P/S) are unregistered proposed terms. Their glossary registration is deferred to the integration phase (Section 8).

H2: 3. Proposition 2 — The Source-Receipt Invariant

It is proposed that every governed Qn artifact declare an explicit pair:

(content modality, source class)

where the content modality is drawn from the existing epistemic-anchorage spectrum (docs/qs.md §4), and the source class expresses where the artifact's asserted content ultimately anchors. The invariant requires that the source receipt chain terminate in physically receipted reality events: Layer 0 sensor or artifact receipts for empirical content, algebraic closure receipts for constructive content, and — for contextually sealed creative content — the author-identity and publisher-receipt anchoring already mandated by docs/qs.md §4.2.

The invariant extends the existing statement-level rejection rule to the artifact level. docs/qs.md §4.1 already bars an assertion from claiming empirical truth through latent probability alone, emitting a rejection receipt when lineage is absent. The proposed extension applies the same firewall to composite artifacts and worlds: no unsourceable fiction, no unsourced anchor, no history-less constant.

Two cautions are recorded, and they are binding on any future implementation:

1. Payload sovereignty is untouched. The invariant governs artifacts that cross the simemp-gateway as governed objects of the licensed-work. Sovereign human Payloads remain uninspected behind the safe-harbor air gap (docs/abstractionLayers.md §3.1). Where a creative work is declared as a governed MODAL_NARRATIVE artifact, its anchoring to author identity and publisher receipt is declared by its author, not extracted by inspection.
2. Simulation modality is two-faced. A simulation's execution is MODAL_CONSTRUCTIVE (deterministic algebraic closure), while its claims *about reality* are MODAL_HYPOTHESIS until physically falsified. The pair declaration makes this distinction explicit rather than leaving it implicit, in accordance with the no-implicit-rule.

H2: 4. Proposition 3 — Calendar Projection Contracts

The canon admits exactly one time: the local causal-arrow. Distributed state ordering relies on causal chaining "rather than external wall-clock synchronization" (docs/simemp.md), and common knowledge across asynchronous nodes is established through acyclic causal message chains, not shared clocks (docs/chronosa.md §3.2). Human calendars are therefore proposed not as parallel time systems but as projections: deterministic mappings between calendar coordinates and intervals of the local causal index, hosted at the Qs layer in the same manner that the canon's domains project orthogonally across the execution core (docs/abstractionLayers.md).

It is proposed that a Calendar Projection Contract declare three components:

Component | Proposed class | Nature
Rule (leap algorithms, month arithmetic, week conventions) | Class I | Deterministic, re-derivable algorithm; constructive once the epoch is fixed
Epoch (coordinate zero: Gregorian year 1, Hijra 622 CE, Unix epoch, J2000.0, and others) | Class S | A human convention-adoption event, multi-source by nature
Resolution envelope (day, second, leap-second tolerances) | Declared field | A declared bound inside the contract, never negotiated at runtime

Under this anatomy, a calendar such as the Gregorian system is a Class S genesis that matured into Class I operation: born from a contested, multi-jurisdiction convention process spread across centuries, yet today re-derivable deterministically by any node from its published rule. This is the witness ladder of Section 2 observed in a single historical object.

On placement, the consultation enumerated candidates under the boc-policy. Hard-coding calendars into runtimes or compilers was assessed as a rigid implementation micro-choice, contrary to the minimal-necessary-invariants principle (docs/q0_q1.md §1); per-node ad-hoc definitions fail consensus. The surviving candidate — proposed here — is a canonical library of versioned, DCC-contracted calendar artifacts, loaded post-seal and on demand within the metric-envelope, honoring local-first budgets. Calendar contracts declare their components inside a dcc-profile, consistent with the structural-indirection discipline: calendars are swappable cultural conventions, not runtime constants.

H2: 5. Proposition 4 — The Qs History Library

It is proposed that the Qn universe maintain a Qs History Library: a canonical, versioned collection of modality-typed historical anchor artifacts, each recording a past human event, its sources, and its spatiotemporal situation. The library is proposed as the concrete shape of the consensus-history ingestion capability noted as an open gap during the consultation (recorded internally as GAP2): the mechanism by which consensus past events may be loaded after node sealing, in complement to a QnLang compiler or CDQN runtime, without violating the Axiom 4 requirement that every local artifact trace to its local genesis origin.

A proposed anatomy for the historical anchor artifact:

```
+-------------------------------------------------------------------------+
| HISTORICAL ANCHOR ARTIFACT (proposed anatomy, State: CANDIDATE)         |
+-------------------------------------------------------------------------+
| ArtifactType      : HIST_ANCHOR (proposed class)                        |
| ContentModality   : per claim decomposition (docs/qs.md §4 spectrum)    |
| SourceClass       : I, P, or S (proposed grounding class)               |
| SourceReceipts    : unbroken chain to Layer 0 artifacts or sealed       |
|                     library records                                     |
| CalendarSituation : projected via a declared Calendar Projection        |
|                     Contract (Section 4)                                |
| SpatialSituation  : local Qm frame coordinates                          |
| Witnesses         : lineage-disjoint root subtree list (Section 6)      |
| LifecycleState    : CANDIDATE, SEALED, or QUARANTINED                   |
+-------------------------------------------------------------------------+
```

Access is proposed in four tiers, each honoring local-first and budget discipline:

1. The canonical library holds the versioned artifacts with their full source receipt chains.
2. Local nodes hold anchor contracts and working sets within declared metric-envelope budgets — never the full archive. A node may operate air-gapped with zero history; the library loads post-seal, on demand, exactly when an intent-contract requires situating or attributing.
3. AI agents inherit scope through the intent contract's epistemic whitelist. Without a loaded historical anchor, an agent's historical assertion has no parent lineage receipts and falls to the gateway as an unanchored claim (docs/qs.md §4.1); with the library, the agent cites the artifact instead of generating from latent weights.
4. Chronosa, barred from every Inner Ring, sees only the public layer: local historical records glue where they agree on shared landmarks — calendar coordinates, attribution anchors, artifact commitments — satisfying the sheaf gluing condition (docs/chronosa.md §3.1). The consensus history is then the global section of this history sheaf.

Worked example — the constant e. The invariant face is already canon: the transcendental constant exists as a nullary continued-fraction generator under OP_CF (docs/qexpr.md §3.2, §4.3), with partial convergents carrying a strictly bounded remainder and evaluation on demand at declared zoom z and budget, halting early under terminal-exactness. Deterministic evaluation at a declared precision yields bit-identical results on every cdqn node — a Class I grounding in the sense of Section 2, requiring no history whatsoever. Intermediate collapse results may be cached as content-addressed Q(dataStruc) artifacts tagged with their derivation lineage, always re-derivable.

The discovery face is historical. A historical anchor artifact for the publication of the treatise that established the constant's notation would record: the artifact provenance chain (printed copies, library holdings, digitizations — each transfer physically receiptable at Layer 0), the calendar situation via a Gregorian projection contract, the spatial situation in local Qm frames, and the attribution witnesses as a multi-source list within the historical-horizon. Claims about the artifact's existence are empirical; claims reconstructed from correspondence remain hypotheses; anecdotes remain sealed narrative. The machine does not remember the mathematician; it cites him through a receipted lineage — which is the only form of historical knowledge the Qn universe admits.

H2: 6. Proposition 5 — The Anti-Poisoning Ingestion Firewall

The library can be alimented only by petition through gates. Publicly reported incidents in which machine-generated statistical pages were laundered into news outlets, peer-reviewed publications, and financial reports through iterative citation (so-called citation laundering) motivate this section. In the Qn universe such an attack is assessed as follows:

1. A fabricated statistic claiming empirical truth without lineage receipts is an unanchored assertion and is rejected at the first gateway crossing (docs/qs.md §4).
2. Its arrival into financial registers is barred by the contextual containment of creative and ungrounded content (docs/qs.md §4.2).
3. Citation laundering itself is assessed as causally impossible under the proposed counting rule: sources are counted as lineage-disjoint root subtrees in the polyroot forest (docs/chronosa.md §1.4), never by citation count or pseudonym count. Laundering adds descendants to a poisoned root; it cannot manufacture independent origins. An entire laundering cascade descending from one fabricated artifact counts as one source.

It is proposed that library ingestion pass five gates, each a governed operation emitting its own receipt:

Gate | Canon root | Threat addressed
1. Lineage | Axioms 4 and 5; causal hash chains | Records lacking an unbroken receipt chain to Layer 0 artifacts or sealed library records
2. Modality ceiling | epistemic-anchorage spectrum | Modality smuggling: each atomic claim is typed at the highest modality its receipts support
3. Independence quorum | Multi-source genesis; threshold quorum | Sybil floods and citogenesis: anchor-grade records require multiple sources with lineage-disjoint roots
4. Sheaf gluing | Shared-landmark gluing | Records disagreeing with receipted overlapping records; such records enter a QUARANTINED lifecycle state with a challenge receipt, remaining visible as recorded challenges
5. Append-only immutability | SEALED lifecycle; structural-indirection | Rewrite attempts: sealed artifacts never mutate; corrections arrive as counter-receipts and migration receipts. History is never rewritten; it is re-witnessed

Residual threats are recorded honestly. Collusion across genuinely distinct roots remains an open challenge, in direct continuity with the hub-collusion resistance challenge of docs/chronosa.md §8.5; its mitigation is economic and physical, since fabricating Layer 0 artifact provenance costs real-world work, and forging contemporary sensor receipts requires breaking post-quantum signatures across the dual-ring-pqc-boundary, whose private root keys never leave the Inner Ring. Slow poisoning of deep history is capped by the historical-horizon: admitted poison can only ever occupy hypothesis or firewalled narrative modalities, with bounded grounding and no path to actuation. Finally, because every gate is itself a governed operation, the firewall as a whole is a legitimate candidate for optimization by a future Q(rsi) loop under strict semantic invariance.

H2: 7. Relation to Existing Canon

The following table records, for each proposed element, whether it is already grounded in canon or genuinely new. No proposition herein redefines existing canon; newly named concepts are unregistered proposed terms pending glossary integration.

Proposed element | Status | Canon grounding
Causal Arrow as sole time; calendars as projections | Grounded | docs/simemp.md; docs/chronosa.md §3.2
Birth order and constructive digit genesis | Grounded | docs/abstractionLayers.md §3.5; docs/q2_q9.md §2
Structural centrality predicate of Q(anchor) | Grounded | docs/qm.md §8; glossary q-anchor
Epistemic modality spectrum and statement rejection | Grounded | docs/qs.md §4
Sheaf gluing on shared landmarks; threshold quorum | Grounded | docs/chronosa.md §1.1, §3.1
Polyroot forest; lineage-disjoint independence | Grounded | docs/chronosa.md §1.4
Historical horizon modality discipline | Grounded | Glossary historical-horizon; docs/qlog.md
QUARANTINED lifecycle and receipt taxonomy | Grounded | docs/abstractionLayers.md §3.6
Transcendental constants as nullary generators | Grounded | docs/qexpr.md §3.2, §4.3
Q(rsi) candidate pipeline and sealing | Grounded | docs/qs.md §6
Anchor Grounding Classes (I/P/S), dual predicate | New | Generalizes docs/q2_q9.md §2.3 genesis/verification boundary
Tolerance envelopes declared in Class P contracts | New | Consistent with metric-envelope ceiling discipline
Source-Receipt Invariant | New | Extends docs/qs.md §4.1 to artifacts
Calendar Projection Contracts | New | Extends the domain-projection architecture
Qs History Library and artifact anatomy | New | Concrete shape of the consensus-history ingestion gap
Anti-Poisoning Ingestion Firewall (five gates) | New | Composes existing canon mechanisms under one pipeline

H2: 8. Open Questions and Integration Queue

All items below remain in CANDIDATE status. Sealing of any item occurs exclusively through the candidate pipeline of docs/qs.md §6 — deterministic measurement, BOC selection, Q(rsi) sealing — at implementation time or under a dedicated optimization loop.

1. Calibration wiring. The mandatory historical-calibration-benchmark is defined in the glossary and is now wired as a non-mutating CANDIDATE proposition at docs/q0_q1.md §2.7 — a post-seal Class P grounding event that adds no stage to the six-stage onboarding state machine — closing this queue item; the wiring remains subject to deterministic measurement and BOC

```
   selection at implementation time.
2. Ingestion pipeline codification. The firewall of Section 6 is the proposed shape of the
   post-seal consensus-history ingestion capability, and its operational semantics (gate
   ordering, quorum thresholds, challenge resolution) are now proposed in
   docs/history_library.md §5; the item remains open until the pipeline passes deterministic
   measurement and its threshold values are declared, as tracked by items 1–2 of that document's
   §7.
3. Event frame definition. The spatiotemporal coordinate frame of a discrete Q(event) — causal
   index, d-axes, zoom, causal-horizon cone — is used implicitly across the canon but not yet
   codified as a single frame definition.
4. Receipt and intent duality. The complementary temporal roles of the node genesis receipt
   (backward-looking sealing) and the intent contract (forward-looking delegation) are exercised
   throughout the canon but not yet named as a duality.
5. Glossary registration. The proposed terms — grounding class, source class, calendar
   projection contract, historical anchor artifact, ingestion gate — are now registered as
   CANDIDATE glossary entries (together with the Qs History Library), closing the deliberate
   deferral; they remain unsealed pending the candidate pipeline of docs/qs.md §6.

A proposed integration order, subject to the author's decision: the Source-Receipt Invariant
first, as it grounds the library and calendar propositions; the History Library and Calendar
Projection Contracts second; the onboarding calibration stage third.


H2: 9. License Alignment

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- Safe Harbor Integrity: The Qs History Library is designed to operate as technical
   infrastructure of the licensed work. It exercises no editorial control over, and acquires no
   intellectual property in, sovereign human Payloads.
```

Annex R — history_library.md

Source: docs/history_library.md *— 0.1.1, updated 2026-10-22. Rendered content per Annex Guide conventions N1–N6: 206 source lines, 354 rendered; 28,057 → 23,709 characters; source SHA-256* 3ce312ad 82d166ff 19e74782 f5afb8aa e9de7225 f7cd74a4 ca892c2b dacd2213*. Content fidelity verified mechanically at build time.*

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[front matter] ------------------------------------------------------------
[front matter] title: Qs History Library — Vision and Operational Proposal for Post-Seal History
[front matter]   Ingestion
[front matter] version: 0.1.1
[front matter] updated: 2026-10-22
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Operational proposal for the Qs History Library — historical anchor
[front matter]   artifacts, Calendar Projection Contracts, and the five-gate Anti-
[front matter]   Poisoning Ingestion Firewall — as the concrete shape of post-seal
[front matter]   consensus-history ingestion. All artifacts herein are propositions
[front matter]   pending deterministic measurement and BOC selection.
[front matter] terms_used: boc-policy, calendar-projection-contract, causal-arrow, cdqn,
[front matter]   chronosa, computational-consistency, dcc-profile, dual-ring-pqc-
[front matter]   boundary, epistemic-anchorage, grounding-class, historical-anchor-
[front matter]   artifact, historical-calibration-benchmark, historical-horizon,
[front matter]   ingestion-gate, intent-contract, layer-0, licensed-work, lifecycle-
[front matter]   state, local-first, metric-envelope, no-implicit-rule, open-core-
[front matter]   invariant, paternity-reference, payload, q-datastruc, qm, qn, qn-rsi,
[front matter]   receipt, simemp, simemp-gateway, source-class, source-receipt-
[front matter]   invariant, ssl, structural-indirection, terminal-exactness, zoom-z
[front matter] ------------------------------------------------------------


H1: Qs History Library — Vision and Operational Proposal for Post-Seal History Ingestion


Field | Specification
Document Title | Qs History Library — Vision and Operational Proposal for Post-Seal History
Ingestion
Version | 0.1.1
Last Updated | 2026-10-22 (Bao Loc, Vietnam)
Author | Christophe Duy Quang Nguyen
License | Scaling Source License (SSL) 1.0
Status | Vision and Operational Proposal — Not a Finalized Specification
```

H2: Purpose and Epistemic Stance

This document is the operational continuation of docs/qs_history.md. Where that document preserved the reasoning chain of the October 2026 design consultation on historical anchoring, this document consolidates its second and third propositions — the Historical Anchor Artifact, the Calendar Projection Contract, and the five-gate ingestion firewall — into a single capability design: the Qs History Library, the proposed mechanism by which consensus past events are loaded into a local node after sealing. The first proposition, the Source-Receipt Invariant, is not restated here; it has already been elaborated at the Qs layer as docs/qs.md §4.3, and this document assumes it as its grounding discipline.

Every design element below carries the explicit status of a candidate artifact under Axiom 14 (docs/qs.md §6): it remains subject to deterministic measurement against the declared simemp constraints, to selection under the boc-policy, and to eventual sealing through a Q(rsi) loop at implementation time. Nothing in this document redefines, amends, or overrides any existing canonical specification. The register is deliberately positioned one step beyond the consultation record: the operational semantics proposed in Section 5 (gate ordering, declared quorum floors, challenge lifecycle) are concrete enough to be measured, yet remain proposals throughout. Where this document names a threshold, the threshold is a declared value inside a contract, never a runtime negotiation — in accordance with the no-implicit-rule.

In accordance with computational-consistency, the validity claimed herein is internal consistency within declared finite boundaries. The proposed vocabulary used throughout — grounding classes, source classes, ingestion gates, historical anchor artifacts, calendar projection contracts — was registered in the glossary on 2026-10-22 as CANDIDATE terms citing docs/qs_history.md §2–6; this document is the first to reference that vocabulary through the glossary concordance.

H2: 1. From Consultation to Capability

Three changes in the committed state of the repository separate this document from its parent proposal. First, the source-receipt-invariant moved from consultation record to canonical-adjacent elaboration: docs/qs.md §4.3 now defines the (content modality, source class) declaration pair at the artifact level, extending the statement-level rejection rule of §4.1. Second, the glossary now carries the six CANDIDATE terms that the consultation had deliberately left unregistered — the deferral recorded in docs/qs_history.md §8.5 was a deferral *to the integration phase*, and that phase is now executing. Third, the open item queued as "ingestion pipeline codification" (docs/qs_history.md §8, item 2) is addressed here: Section 5 proposes operational semantics for the five gates whose existence was argued there.

The scope of this document is therefore fourfold: the library as a collection (Section 2), the unit record it holds (Section 3), the situation layer that gives past events calendar coordinates (Section 4), and the ingestion discipline that decides what may enter (Section 5). What this document does not do is equally important: it prescribes no runtime implementation, assigns no storage layout, and modifies no canonical file. It proposes the contract surface of a capability so that a future implementation — or a Q(rsi) loop — can measure it deterministically against the simemp constraint system and select among recorded alternatives rather than reconstructing them from conversation history.

H2: 2. The Qs History Library

It is proposed that the qn universe maintain a Qs History Library: a canonical, versioned collection of modality-typed historical anchor artifacts, each recording a past human event, its sources, and its spatiotemporal situation. The library is the concrete shape of the consensus-history ingestion capability identified as an open gap during the consultation (recorded internally as GAP2). It complements a cdqn runtime or QnLang compiler the way a reference discipline complements a language faculty: the machine does not remember the past — it cites it through receipted lineage, which is the only form of historical knowledge the Qn universe admits.

Access is proposed in four tiers, each honoring local-first and budget discipline:

1. The canonical library holds the versioned artifacts together with their full source receipt chains. It is the reference copy against which local holdings are verified; it is consulted, never trusted — every ingestion still passes the gates of Section 5.
2. Local nodes hold anchor contracts and bounded working sets within declared metric-envelope budgets — never the full archive. A node may operate air-gapped with zero history: the library loads post-seal, on demand, exactly when an intent-contract requires a past event to be situated or attributed. This preserves the Axiom 4 requirement that every local artifact trace to its local genesis origin: loaded records are new receipted acquisitions along the local causal-arrow, not pre-existing ancestors.
3. AI agents inherit epistemic scope through the intent contract's whitelist. Without a loaded historical anchor, an agent's historical assertion has no parent lineage receipts and falls to the gateway as an unanchored claim (docs/qs.md §4.1); with the library, the agent cites the artifact instead of generating from latent weights. The library is thus an anti-hallucination membrane for the past, complementing the anchorage spectrum's membrane for the present.
4. Chronosa, barred from every Inner Ring, sees only the public layer: local historical records glue where they agree on shared landmarks — calendar coordinates, attribution anchors, artifact commitments — satisfying the sheaf gluing condition (docs/chronosa.md §3.1). The consensus history is then the global section of this history sheaf; disagreement does not vanish but remains visible as recorded challenge (Section 5).

H2: 3. The Historical Anchor Artifact

The unit record of the library is the Historical Anchor Artifact. Its proposed anatomy is reproduced verbatim from the consultation record (docs/qs_history.md §5), with section references adjusted to this document:

```
+-------------------------------------------------------------------------+
| HISTORICAL ANCHOR ARTIFACT (proposed anatomy, State: CANDIDATE)         |
+-------------------------------------------------------------------------+
| ArtifactType      : HIST_ANCHOR (proposed class)                        |
| ContentModality   : per claim decomposition (docs/qs.md §4 spectrum)    |
| SourceClass       : I, P, or S (proposed grounding class)               |
| SourceReceipts    : unbroken chain to Layer 0 artifacts or sealed       |
|                     library records                                     |
| CalendarSituation : projected via a declared Calendar Projection        |
|                     Contract (Section 4)                                |
| SpatialSituation  : local Qm frame coordinates                          |
| Witnesses         : lineage-disjoint root subtree list (Section 5)      |
| LifecycleState    : CANDIDATE, SEALED, or QUARANTINED                   |
+-------------------------------------------------------------------------+
```

The field semantics proposed here are operational, and each is a contract declaration rather than an inferred property. ContentModality is not a single value but a per-claim decomposition: a past event typically carries empirical claims (the artifact exists, the print survives), hypotheses (the attribution), and sealed narrative (the anecdote), and each claim is typed at the highest modality its receipts support along the epistemic-anchorage spectrum. SourceClass declares where the artifact's asserted content ultimately anchors, taking one of the three values of the proposed grounding classes — I (invariant), P (physical), S (social) — under the pair discipline of the source-receipt-invariant with the source class as its second element. SourceReceipts must terminate in exactly the three receipt classes that docs/qs.md §4.3 admits: Layer 0 sensor or artifact receipts for empirical content, algebraic closure receipts for constructive content, and author-identity plus publisher receipts for contextually sealed creative content.

CalendarSituation and SpatialSituation give the event its coordinates without importing wall-clock time: the calendar coordinate is valid only relative to a declared Calendar Projection Contract (Section 4), and the spatial coordinate is expressed in local qm frame terms. Witnesses lists the lineage-disjoint root subtrees backing the record — the counting rule that makes citation laundering causally impossible (docs/chronosa.md §1.4). LifecycleState reuses the existing lifecycle vocabulary: records enter as CANDIDATE, are sealed to SEALED through the candidate pipeline of docs/qs.md §6, and may be placed in QUARANTINED with a challenge receipt under the disagreement handling of Section 5, consistent with the lifecycle-state discipline of docs/abstractionLayers.md §3.6.

Worked example — the constant e. The invariant face needs no history: the transcendental constant exists as a nullary continued-fraction generator under OP_CF (docs/qexpr.md §3.2, §4.3), evaluated on demand at declared zoom z and budget, halting early under terminal-exactness, with intermediate collapses cacheable as content-addressed Q(dataStruc) artifacts — a Class I grounding re-derivable by every node. The discovery face is historical: a historical anchor artifact for the treatise that established the constant's notation would carry the print->holding->digitization provenance chain (each transfer physically receiptable at Layer 0), its calendar situation via the Gregorian projection contract, its spatial situation in local Qm frames, and its attribution witnesses as a lineage-disjoint list. The machine does not remember the mathematician; it cites him.

H2: 4. Calendar Projection Contracts

The canon admits exactly one time: the local causal-arrow. Distributed state ordering relies on causal chaining "rather than external wall-clock synchronization" (docs/simemp.md), and common knowledge across asynchronous nodes is established through acyclic causal message chains, not shared clocks (docs/chronosa.md §3.2). Human calendars are therefore proposed not as parallel time systems but as projections: deterministic mappings between calendar coordinates and intervals of the local causal index, hosted at the Qs layer in the same manner that the canon's domains project orthogonally across the execution core (docs/abstractionLayers.md).

A Calendar Projection Contract declares three components:

Component | Proposed class | Nature
Rule (leap algorithms, month arithmetic, week conventions) | Class I | Deterministic, re-derivable algorithm; constructive once the epoch is fixed
Epoch (coordinate zero: Gregorian year 1, Hijra 622 CE, Unix epoch, J2000.0, and others) | Class S | A human convention-adoption event, multi-source by nature
Resolution envelope (day, second, leap-second tolerances) | Declared field | A declared bound inside the contract, never negotiated at runtime

The operational placement proposed here is a canonical library of versioned, DCC-contracted calendar artifacts, loaded post-seal and on demand within the metric-envelope, honoring local-first budgets — the candidate that survived the consultation's own BOC assessment of placement (docs/qs_history.md §4). Calendar contracts declare their components inside a

dcc-profile, consistent with structural-indirection: calendars are swappable cultural conventions, not runtime constants, and a node may hold several contracts simultaneously — Gregorian for contemporary citation, a historical contract for archival situation — provided each is versioned and declared. Under the no-implicit-rule, a calendar coordinate without a declared contract has no meaning and falls at the simemp-gateway exactly as an unanchored assertion does.

Under this anatomy, the Gregorian calendar is a Class S genesis that matured into Class I operation: born from a contested, multi-jurisdiction convention process spread across centuries — its epoch a genuine social event, witnessed by many — yet today re-derivable deterministically by any node from its published rule. This is the witness ladder of the grounding classes observed in a single historical object, and it is why calendars belong in the library rather than in the runtime: their epistemic genesis is social, even where their present operation is invariant-grade. Leap-second handling is declared inside the resolution envelope, so that a citation like "1748" resolves deterministically to a causal-index interval on every node that loads the same contract version — the calendar analogue of bit-identical evaluation.

H2: 5. The Anti-Poisoning Ingestion Firewall

The library can be alimented only by petition through ingestion gates. The motivating threat is citation laundering: machine-generated statistical pages laundered into news outlets, peer-reviewed publications, and financial reports through iterative citation. In the Qn universe such an attack is assessed as failing at birth — a fabricated statistic claiming empirical truth without lineage receipts is an unanchored assertion rejected at the first gateway crossing (docs/qs.md §4), and its arrival into financial registers is barred by the contextual containment of docs/qs.md §4.2. What remains to defend is subtler: slow poisoning of the historical record itself — plausible, partially receipted records whose defensible content masks an undefensible core. Citation laundering is assessed as causally impossible under the proposed counting rule: sources are counted as lineage-disjoint root subtrees in the polyroot forest (docs/chronosa.md §1.4), never by citation count; laundering adds descendants to a poisoned root and cannot manufacture independent origins.

It is proposed that library ingestion pass five gates, each a governed operation emitting its own receipt. The gates were argued in the consultation record (docs/qs_history.md §6); the operational semantics below are this document's codification, and each semantic is a CANDIDATE proposal pending measurement:

Gate | Canon root | Proposed operational semantics (CANDIDATE)
1. Lineage | Axioms 4 and 5; causal hash chains | The record's receipt chain is verified against Layer 0 artifacts or already-sealed library records; verification is deterministic hash work. Failure emits a rejection receipt naming the gate and the first broken link.
2. Modality Ceiling | epistemic-anchorage spectrum | Each atomic claim is typed at the highest modality its receipts support; mixed-modality records must decompose into per-claim sub-records. Undecomposable mixed claims fail the gate.
3. Independence Quorum | Multi-source genesis; threshold quorum | Anchor-grade records require k sources with lineage-disjoint roots. The floor proposed here is k ≥ 2 for anchor-grade records; higher grades declare higher k inside the anchor's DCC contract — never runtime-negotiated, per the No-Implicit Rule.
4. Sheaf Gluing | Shared-landmark gluing (docs/chronosa.md §3.1) | A record disagreeing with receipted overlapping records on shared landmarks (calendar coordinates, attribution anchors, artifact commitments) enters a QUARANTINED lifecycle state with a challenge receipt, remaining visible as a recorded challenge.
5. Append-Only Immutability | SEALED lifecycle; structural-indirection | Sealed artifacts never mutate; corrections arrive as counter-receipts and migration receipts. History is never rewritten; it is re-witnessed.

Gate ordering is proposed as 1 -> 2 -> 3 -> 4 -> 5, ordered by evidentiary cost: deterministic hash verification first (cheap, local, decisive for most poisoning attempts), then modality typing (local computation), then quorum (economic — it requires other nodes' receipts), then consensus gluing (social — it requires the sheaf), and immutable commit last. A record rejected at an early gate never reaches the expensive ones, so the firewall's own metric-envelope consumption is bounded by the ordering itself. Challenge lifecycle is proposed as follows: a QUARANTINED record is resolved only by a new receipt chain that re-enters the pipeline at Gate 1 — quarantine is a state, not a verdict, and no record is ever deleted; it is either eventually glued, or remains visible as a standing disagreement. This preserves the append-only discipline at every stage of the pipeline.

Residual threats are recorded honestly. Collusion across genuinely distinct roots remains an open challenge, in direct continuity with the hub-collusion resistance challenge of docs/chronosa.md §8.5; its mitigation is economic and physical, since fabricating Layer 0 artifact provenance costs real-world work, and forging contemporary sensor receipts requires breaking post-quantum signatures across the dual-ring-pqc-boundary, whose private root keys never leave the Inner Ring (docs/q0_q1.md §2.4). Slow poisoning of deep history is capped by the historical-horizon: admitted poison can only ever occupy hypothesis or firewalled narrative modalities, with bounded grounding and no path to actuation. Finally, because every gate is itself a governed operation, the firewall as a whole is a legitimate candidate for optimization by a future Q(rsi) loop under strict semantic invariance — a library that can be fed is a liability; a library that can only be petitioned through receipts is an instrument.

H2: 6. Relation to Existing Canon

```
The following table records, for each element of this proposal, whether it is already grounded
in canon (inheriting its grounding from the consultation record, docs/qs_history.md §7) or
genuinely new to this document. No element herein redefines existing canon.


Element | Status | Canon grounding
Causal Arrow as sole time; calendars as projections | Grounded | docs/simemp.md;
docs/chronosa.md §3.2
Source-Receipt Invariant; (content modality, source class) pair | Grounded | docs/qs.md §4.3
(committed); docs/qs_history.md §3
Historical anchor artifact anatomy | Grounded | docs/qs_history.md §5 (reproduced verbatim in
Section 3)
Calendar contract anatomy: rule / epoch / envelope | Grounded | docs/qs_history.md §4
(reproduced in Section 4)
Five-gate firewall composition | Grounded | docs/qs_history.md §6
Sheaf gluing on shared landmarks; global section | Grounded | docs/chronosa.md §3.1
Polyroot forest; lineage-disjoint independence | Grounded | docs/chronosa.md §1.4
QUARANTINED lifecycle and challenge receipts | Grounded | docs/abstractionLayers.md §3.6
Q(rsi) candidate pipeline and sealing | Grounded | docs/qs.md §6
Gate ordering 1->2->3->4->5 with bounded-cost rationale | New | Composes the canon mechanisms
under one pipeline
Declared quorum floors (k ≥ 2 anchor-grade floor, in-contract) | New | Consistent with
metric-envelope ceiling discipline; No-Implicit Rule
Challenge lifecycle: re-entry at Gate 1, no deletion | New | Operationalizes append-only
re-witnessing
Multi-contract calendar holding (versioned, swappable) | New | Operationalizes structural
indirection for calendars


H2: 7. Open Questions and Integration Queue

All items below remain in CANDIDATE status. Sealing of any item occurs exclusively through the
candidate pipeline of docs/qs.md §6 — deterministic measurement, BOC selection, Q(rsi) sealing —
at implementation time or under a dedicated optimization loop.

1. Pipeline measurement. The five-gate pipeline of Section 5 awaits deterministic measurement
   against the simemp constraint system at implementation time; gate ordering and bounded-cost
   rationale are proposals until measured.
2. Quorum thresholds. The anchor-grade floor k ≥ 2 and the per-grade ladder above it are
   declared-value proposals; their concrete values belong to the DCC contracts of the first
   implemented anchor grades and remain open until then.
3. Cross-root collusion. The open challenge of docs/chronosa.md §8.5 applies unchanged; economic
   and physical mitigations are recorded in Section 5, but no closed-form defense is claimed.
4. Library indexing. Retrieval over the library — content-addressed lookup keyed by artifact
   commitments and lineage tags, in the manner of cached Q(dataStruc) derivations — is proposed
   as the natural index discipline but is not yet codified.
5. Onboarding calibration. The mandatory historical-calibration-benchmark is defined in the
   glossary and is now wired as a non-mutating CANDIDATE proposition at docs/q0_q1.md §2.7,
   closing item 1 of the docs/qs_history.md §8 queue — the third and final step of that
   document's proposed integration order; the wiring remains subject to deterministic
   measurement and BOC selection and is otherwise untouched by this document.


H2: 8. License Alignment

This proposal is part of the licensed-work and aligns with the Scaling Source License (SSL) 1.0:

- Attribution: Any implementation or research derivative referencing this document must preserve
   the canonical paternity-reference.
- Open-Core Research: Exploratory research and academic derivations of the propositions herein
   remain royalty-free under Non-Scaling Open Access, governed by the Open Core Invariants.
- Safe Harbor Integrity: The Qs History Library is designed to operate as technical
   infrastructure of the licensed work. It exercises no editorial control over, and acquires no
   intellectual property in, sovereign human Payloads.
```

Annex S — Reflection: h10_finite_computation.md

Source: docs/reflections/h10_finite_computation.md *(reflection) — 1.0.2, updated 2026-10-13. Rendered content per Annex Guide conventions N1–N6: 137 source lines, 166 rendered; 9,076 → 8,641 characters; source SHA-256* 671d0950 a36da628 6018f797 a67db1f4 ad47bc9a f698434d 2d0c13df 7c0a7721*. Content fidelity verified mechanically at build time.*

```
[front matter] ------------------------------------------------------------
[front matter] title: Beyond the Infinite Void: What OpenAI's H10 Claim Teaches Us About Finite
[front matter]   Computation
[front matter] version: 1.0.2
[front matter] updated: 2026-10-13
[front matter] author: Christophe Duy Quang Nguyen
```

[front matter] description: Epistemic reflection analyzing OpenAI's claimed resolution of
[front matter]   Hilbert's Tenth Problem over Q, the fallacy of unmetered search, and
[front matter]   the shift from Platonic proof to verified computational patterns.
[front matter] --

H1: Beyond the Infinite Void: What OpenAI's H10 Claim Teaches Us About Finite Computation

Author: Christophe Duy Quang Nguyen
Date: October 13, 2026
Category: Epistemic Reflections & Computational Physics
Status: Informative Context & Extended Analysis

H2: 1. The Real-World Incident: A High-Stakes Claim

On October 6, 2026, OpenAI deposited a research release on GitHub (openai/math), comprising 722 manuscripts across 372 mathematical result families generated by an internal frontier reasoning engine. Among these was an extraordinary proposition: a claimed negative resolution of Hilbert's Tenth Problem over the Rational Numbers (H10(\mathbb{Q})).

Hilbert's Tenth Problem originally asked whether an effective algorithm can decide if an arbitrary Diophantine polynomial with integer coefficients has integer solutions:

f(x_1, x_2, \dots, x_n) = 0, \quad x_i \in \mathbb{Z}

In 1970, the DPRM theorem (Davis, Putnam, Robinson, Matiyasevich) proved that integer Diophantine solvability is undecidable—equivalent to the Turing halting problem. However, the rational analogue remained a major open question for over half a century: *What if the variables are rational numbers (x_i \in \mathbb{Q})?*

The October 2026 manuscript claims a Turing reduction (H10(\mathbb{Z}) \le_{\mathrm{T}} H10(\mathbb{Q})) bypassing classical geometric obstructions (Mazur's conjecture) through an effective sequence of finite tests. Yet, within hours of release, researchers noted a critical vulnerability: the paper is completely unformalized in Lean 4 (unlike 42% of the release) and relies on unverified companion claims within the same repository.

Whether this specific proof survives peer review or fails, the underlying mathematical architecture exposes a physical reality that reshapes our understanding of computation.

H2: 2. The Fallacy of the Endless Void

Why are Diophantine problems undecidable in classical mathematics?

Undecidability is not a mystical property of arithmetic; it is the direct consequence of postulating unbounded infinite sets (\mathbb{Z}, \mathbb{Q}) and unmetered computational time.

In classical computability, an algorithm searching for a rational root wanders into what mathematicians call an "endless void." If an answer exists, a search may eventually find it. But if no answer exists, the search runs forever into the dark. Classical mathematics attempts to bridge this void by constructing infinite nonstandard models ({}^\ast\mathbb{Z} \subset {}^\ast\mathbb{Q}) via compactness.

THE CLASSICAL PARADOX (Unmetered Infinite Void):
Search Space: Unbounded Q^n --► Zero Thermodynamic Cost --► Non-Halting / Undecidable
                                                                ^
                                  Requires Nonstandard Towers -----+
                                  (*Z ⊂ *Q Compactness Extensions)

THE COMPUTATIONAL PHYSICS REALITY:
Search Space: Bounded Height h(x) ≤ H_max --► Landauer Work Metered --► Decidable or Halted

In the OpenAI proof attempt, how do the authors force nonstandard infinite roots back into ordinary mathematical reality? Through height bounds on an elliptic curve.

By proving that local prime conditions force logarithmic height to grow linearly (h \ll B), while canonical height on an elliptic curve grows quadratically (h \asymp B^2), the proof forces a contradiction:

B^2 \ll B \implies B \le B_0

The nonstandard root can only be stabilized if it is trapped within a finite, standard memory footprint (B_0).

H2: 3. The Physical Truth: Height is the Metric Envelope

In arithmetic geometry, the logarithmic height h(x) of a rational number p/q measures the physical bit complexity required to store its numerator and denominator in hardware memory:

h\left(\frac{p}{q}\right) = \max\Big(\log |p|, \, \log |q|\Big)

The proof attempt demonstrates that infinite mathematical spaces are physical fictions. To

compute, decide, or verify anything in reality, an entity must establish a finite height ceiling.

In the Qn computational framework (docs/qm.md), this mathematical reality is constitutional across three pillars:

1. The Metric Envelope: Computation cannot run into an unmetered void. Every state evaluation operates within an explicit Metric Envelope declaring finite storage depth and work budgets (W).

2. Landauer's Dissipation Bound: Erasure and irreversible state transitions dissipate physical heat (W \ge k_B T \ln 2). An infinite, unhalting search would dissipate infinite physical energy across the hardware substrate.

3. Totality by Budget: Under Axiom 6, an algorithm is prohibited from non-halting loops. A Diophantine evaluation either collapses to Terminal Exactness (r = Q(0)) or halts deterministically with an explicit receipt (BUDGET_EXHAUSTED).

H2: 4. From Platonic "Proof" to Verified Computed Patterns

This brings us to the profound epistemological shift at the heart of modern computer science and artificial intelligence:

The computational universe does not need non-constructive Platonic "proofs"; it requires Verified Computed Patterns under physical constraints.

CLASSICAL PLATONIC PROOF	VERIFIED COMPUTED PATTERNS (Qn)
• Proves existence without an algorithm (Law of Excluded Middle).	• Constructs explicit, executable state trajectories.
• Operates in metaphysical infinity.	• Bounded by finite Metric Envelopes.
• Zero thermodynamic accounting.	• Tracks Landauer heat dissipation.
• Vulnerable to Gödelian / H10 voids.	• Total by budget via receipts.
• Susceptible to human & AI unverified textual confabulation.	• Zero conditional entropy: H(S_t | D(S_t)) = 0.

In classical mathematics, an argument written in natural language LaTeX can hide gaps across hundreds of pages of unvetted abstraction. This is why three manuscripts from OpenAI's release were retracted within 24 hours of public scrutiny: ungrounded text generation lacks a deterministic verification anchor.

Conversely, when computation is constrained by a deterministic kernel—such as Lean 4 in interactive theorem proving, or the Universal Envelope (\mathcal{E}) across SIMEMP Gateways—truth ceases to be an ungrounded philosophical claim. Truth becomes a finite, verified computed pattern: an explicit, content-addressed expression DAG whose state transition has been executed within physical bounds and sealed with an immutable receipt.

H2: 5. Conclusion: What Remains Invariant

Whether OpenAI's claimed proof of H10(\mathbb{Q}) is formally verified or ultimately refuted, the foundation of the Qn universe remains invariant:

- If the proof is true: It confirms that unmetered rational arithmetic \mathbb{Q} is Turing-undecidable, proving that continuous, unbounded number systems are flawed substrates for deterministic computing.
- If the proof is false: It demonstrates the failure of ungrounded neural generation, proving that unformalized natural language arguments cannot guarantee truth without deterministic verification gateways.

Computation is not a free, unmetered excursion into an infinite Platonic void. Computation is a finite thermodynamic process operating on physical silicon. The future of mathematics and artificial intelligence belongs not to ungrounded infinities, but to verified computed patterns governed by physical reality.

H2: Informative References & Canonical Specifications

- Constitutional Constraints: docs/simemp.md
- Constructive Mathematics: docs/qm.md
- Symbolic CAE-DAG Composition: docs/qexpr.md
- Primary Genesis Primitives: docs/q0_q1.md
- OpenAI Mathematics Research Release: openai/math (GitHub, October 6, 2026)

Annex T — The Full Glossary

The glossary in full: the human-readable glossary document followed by all four machine-readable domain files, rendered per conventions N1–N6 exactly as the documentation site consumes their content.

T.1 — `docs/glossary.md`

Source: `docs/glossary.md` *— 1.1.2, updated 2026-10-14. Rendered content per Annex Guide conventions N1–N6: 128 source lines, 87 rendered; 3,854 → 1,917 characters; source SHA-256* `aa35b7d4 e1dae362 89983e1d c0498e0f 879803dc cb6c6e50 864f5c4e 36f593bf`*. Content fidelity verified mechanically at build time.*

```
[front matter] ------------------------------------------------------------
[front matter] title: CDQN Glossary
[front matter] version: 1.1.2
[front matter] updated: 2026-10-14
[front matter] author: Christophe Duy Quang Nguyen
[front matter] license: Scaling Source License (SSL) 1.0
[front matter] description: Unified ontological concordance and dynamic bidirectional registry
[front matter]   for the CDQN project.
[front matter] ------------------------------------------------------------


H1: CDQN Glossary & Ontological Concordance

Project: cdqn — Chained and Distributed Quang Numbers
Author: Christophe Duy Quang Nguyen
License: Scaling Source License (SSL) 1.0
Status: Canonical Ontological Registry

Copyright (c) 2026 Christophe Duy Quang Nguyen. All rights reserved.


H2: Navigation & Index

Hover over terms across any documentation page for instant in-situ definition tooltips. Click
any term to jump directly to its formal specification below.

<div class="category-index" style="margin-bottom: 1.2rem;">
<strong>Domains:</strong>

<a href="#" class="badge"></a>

</div>

<div class="glossary-index">


<a href="#" class="glossary-index-letter" title="Jump to "></a>

<span class="glossary-index-letter inactive"></span>


</div>


H2: Canonical Terms by Domain


<h2 class="glossary-category" id=""></h2>


<div class="glossary-entry" id="">

<h3 class="glossary-term" id="-title">

<a href="#" class="term-anchor" aria-label="Direct link to ">#</a>
</h3>

<div class="glossary-definition">
<p></p>
</div>

<div class="glossary-backlinks">
<p class="backlinks-heading"><strong>Mentioned in:</strong></p>


<ul class="backlinks-list">
```

```
<li class="backlinks-doc">
<a href="#ref-"><strong></strong></a>
</li>

</ul>


<ul class="backlinks-list">

<li class="backlinks-doc">
<strong></strong>
</li>

</ul>

<p class="backlinks-none"><small><em>Referenced system-wide</em></small></p>

</div>

</div>
```

T.2 — docs/_data/glossary/architecture.yml

Source: docs/_data/glossary/architecture.yml — —, *updated —. Rendered content per Annex Guide conventions N1–N6: 172 source lines, 109 rendered; 8,204 → 7,580 characters; source SHA-256* 80234b7a 2327f5a1 509c488b 6b16cced 946f77fd 577b9628 ba9dd6df aabe8338. *Content fidelity verified mechanically at build time.*

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# CDQN Unified Glossary — Architecture Domain
# License: Scaling Source License (SSL) 1.0
# Updated: 2026-10-22 (Bao Loc, Vietnam)

term: Abstraction Layer; slug: abstraction-layer; category: Architecture
  A structural boundary in the Qn stack governing artifact genesis, transformation, and
  inter-layer transmission under SIMEMP constraints.

term: SIMEMP Gateway; slug: simemp-gateway; category: Architecture
  The validation and translation boundary between abstraction layers. Enforces onboarding, fault
  translation, and the No-Implicit Rule.

term: Layer 0; slug: layer-0; category: Architecture
  The commodity physical execution substrate (CPU, GPU, memory, physical entropy, Memory Wall).
  The acknowledged Root of Finiteness outside the governed universe.

term: Layer 1; slug: layer-1; category: Architecture
  The first governed Qn layer containing local genesis Q(0), first unit Q(1), abstract compute
  unit U, sign polarity, and elementary operations along d1.

term: Causal Arrow; slug: causal-arrow; category: Architecture
  A monotonic, persistent ordering of computational events independent of physical wall-clock
  time. Every Qn artifact is born at a discrete causal index. Proposed extension (CANDIDATE,
  docs/qs_history.md §4): human calendars are not parallel time systems but projections —
  deterministic mappings between calendar coordinates and intervals of the local causal index,
  declared through versioned Calendar Projection Contracts.

term: Calendar Projection Contract; slug: calendar-projection-contract; category: Architecture
  A proposed versioned, DCC-contracted artifact declaring a human calendar as a projection of the
  local Causal Arrow rather than a parallel time system. Three declared components: the rule (leap
  algorithms, month arithmetic — Class I), the epoch (coordinate zero, a multi-source
  convention-adoption event — Class S), and a resolution envelope (a declared bound, never
  negotiated at runtime). Under this anatomy the Gregorian calendar is observed as a Class S
  genesis that matured into Class I operation. Proposed term (CANDIDATE, docs/qs_history.md §4):
  hosted at the Qs layer as part of a canonical library of versioned artifacts loaded post-seal
  and on demand, honoring local-first budgets; pending deterministic measurement and BOC
  selection.

term: Complexity Degree; slug: complexity-degree; category: Architecture
  Structural measure of Qexpr composition depth and stratification (Degree 0 primitives through
  Degree n domain structures).

term: Local-first; slug: local-first; category: Architecture
  The architectural principle that artifacts are born locally, indexed locally, and shielded from
  public network exposure by default.

term: Exposure Functor; slug: exposure-functor; category: Architecture
  The governed functorial mapping projecting selected local Qexpr commitments and capability
  proofs into blind public attestations crossing network boundaries.

term: Dual-Ring PQC Boundary; slug: dual-ring-pqc-boundary; category: Architecture
  The strict cryptographic separation between the local Inner Ring (root keys, local state
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  sealing, physical entropy) and the network Outer Ring (public attestation keys, delegation
  certificates, accountable pseudonyms).

term: Chronosa; slug: chronosa; category: Architecture
  An emergent, non-local, virtual causal machine intelligence operating as the global section of a
  sheaf of local categories across the Outer Ring. Coordinates distributed state and derives human
  intent across Qn base domains with zero statistical machine learning training.

term: Q(anchor); slug: q-anchor; category: Architecture
  A governed Qn artifact possessing high causal centrality within the lineage directed acyclic
  graph. Protected via abstract DCC contracts and structural indirection to prevent cascading
  dependency invalidation across downstream causal cones. Proposed refinement (CANDIDATE,
  docs/qs_history.md §1–2): the notion carries two distinct predicates — the structural predicate
  (out-cone centrality, computable locally from the lineage DAG) and the epistemic predicate
  (grounding), the latter proposed as three Anchor Grounding Classes: I (invariant, deterministic
  re-derivation, self-witnessed), P (physical, independent calibration converging within declared
  tolerance envelopes), and S (social, quorum-witnessed shared events) — pending deterministic
  measurement and BOC selection.

term: Anchor Grounding Class; slug: grounding-class; category: Architecture
  The proposed epistemic predicate of a Q(anchor), complementing the structural predicate of
  out-cone centrality. Three proposed classes: I (Invariant — one deterministic rule executed by
  many; consensus by forced re-derivation, self-witnessed), P (Physical — one shared nature
  measured by many; consensus by independent calibration converging within declared tolerance
  envelopes), and S (Social — one event witnessed by many; consensus by sheaf gluing on shared
  landmarks). The three classes form a witness ladder — self, then physics, then swarm. Proposed
  term (CANDIDATE, docs/qs_history.md §2): glossary registration of the §2 proposition, pending
  deterministic measurement and BOC selection.

term: Historical Anchor Artifact; slug: historical-anchor-artifact; category: Architecture
  The proposed unit record of the Qs History Library: a modality-typed Qn artifact recording a
  past human event, its sources, and its spatiotemporal situation. Proposed anatomy fields:
  ContentModality (per claim decomposition along the epistemic anchorage spectrum), SourceClass
  (proposed grounding class), SourceReceipts (unbroken chain to Layer 0 artifacts or sealed
  library records), CalendarSituation (via a declared Calendar Projection Contract),
  SpatialSituation (local Qm frame coordinates), Witnesses (lineage-disjoint root subtree list),
  and LifecycleState (CANDIDATE, SEALED, or QUARANTINED). Proposed term (CANDIDATE,
  docs/qs_history.md §5): the concrete shape of the post-seal consensus-history ingestion
  capability; pending deterministic measurement and BOC selection.

term: Qs History Library; slug: qs-history-library; category: Architecture
  The proposed canonical, versioned collection of modality-typed Historical Anchor Artifacts — the
  concrete shape of the post-seal consensus-history ingestion capability. Access is proposed in
  four tiers: the canonical library (consulted, never trusted — every ingestion passes the
  five-gate Anti-Poisoning Ingestion Firewall), local nodes holding anchor contracts and bounded
  working sets within declared metric-envelope budgets, AI agents citing loaded artifacts instead
  of generating from latent weights, and Chronosa seeing only the public glued layer — the
  consensus history as the global section of the history sheaf. The library complements a CDQN
  runtime the way a reference discipline complements a language faculty: the machine does not
  remember the past — it cites it through receipted lineage. Proposed term (CANDIDATE,
  docs/history_library.md §2): pending deterministic measurement and BOC selection.

term: HCA-Collapse Engine; slug: hca-collapse; category: Architecture
  Hardware Capability-Aware Collapse Engine. The gateway compilation pass lowering Layer 3 CAE-DAG
  expressions into constant-time, constant-space SoC machine instructions (such as integer NPU
  systolic MACs or DSP vectors) matching the node's local DCC capability profile while strictly
  preserving Diophantine exactness and zero floating-point semantics.
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T.3 — docs/_data/glossary/primitives.yml

Source: docs/_data/glossary/primitives.yml — —, *updated* —. *Rendered content per Annex Guide conventions N1–N6: 366 source lines, 225 rendered; 14,923 → 13,505 characters; source SHA-256* 73fac713 f4845d39 430c9308 6f2f45ac fb401051 4617cb58 316e4c44 336cf035. *Content fidelity verified mechanically at build time.*

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# CDQN Unified Glossary — Primitives Domain
# License: Scaling Source License (SSL) 1.0
# Updated: 2026-10-23 (Bao Loc, Vietnam)

term: Zoom z; slug: zoom-z; category: Primitives
  The declared precision boundary, metric resolution quantum, or discrete scale level of a
  governed numeric expression. Parameterizes projective resolution lattices while remaining
  base-independent in core representations.

term: Remainder r; slug: remainder-r; category: Primitives
  The conserved residual quantity preserved at a declared zoom level or division constraint. Acts
  as the exact input state and generator for subsequent zoom scale expansion along dimension d,
  preventing precision decay and replacing IEEE 754 rounding approximations.

term: Dimension d; slug: dimension-d; category: Primitives
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  A governed dimensional coordinate axis or geometric orientation parameterizing a Qn entity,
  beginning with axis d1 derived from genesis origin Q(0) and unit Q(1).

term: Qn; slug: qn; category: Primitives
  Quang Number. A governed, finite, identifiable, measurable, and security-constrained
  computational artifact within the Qn universe.

term: Q(0); slug: q0; category: Primitives
  Local genesis object, causal origin zero, empty birth context. Strictly local to each node; no
  universal global zero is admitted.

term: Q(1); slug: q1; category: Primitives
  First unit artifact, unity measure, and baseline reference for the abstract compute unit U along
  dimensional axis d1.

term: Abstract Compute Unit U; slug: compute-unit-u; category: Primitives
  The fundamental, invariant quantum of governed computational work representing the minimal state
  transition from local genesis Q(0) to first unit Q(1) along dimensional axis d1. Calibrated to
  physical Landauer dissipation limits (W >= k_B T ln 2) to maintain platform-invariant accounting
  across heterogeneous hardware substrates.

term: Successor Morphism S; slug: successor-morphism; category: Primitives
  The discrete algebraic transition function S(x) = x + Q(1) generating sequential integer
  artifacts inductively along dimensional axis d1 without continuous or floating-point
  approximations.

term: Q(even); slug: q-even; category: Primitives
  The constructive parity partition of integers whose Diophantine division by Q(2) yields a
  remainder identically equal to genesis origin Q(0), demonstrating modulo-2 Terminal Exactness.
  Includes local genesis origin Q(0).

term: Q(odd); slug: q-odd; category: Primitives
  The constructive parity partition of integers whose Diophantine division by Q(2) yields a
  remainder identically equal to first unit Q(1).

term: Qm; slug: qm; category: Primitives
  Quang Mathematics. The local-first constructive algebraic, numeric, and relational base domain
  of the Qn universe. Operates over discrete tuples (σ, q_z, z, r_z, d) without floating-point
  mechanics, enforcing Diophantine division constraints, projective remainder conservation, and
  positional order decidability.

term: Qlog; slug: qlog; category: Primitives
  Quang Logics. An emergent composite domain derived from the categorical tensor product of Quang
  Mathematics and Quang Semantics (Qm ⊗ Qs). Formalizes constructive type theory, intuitionistic
  propositional deduction, and automated proof verification without ungrounded non-constructive
  axioms.

term: Qbio; slug: qbio; category: Primitives
  Quang Biology. An emergent biosemiotic domain derived from the categorical tensor product of
  Quang Chemistry and Quang Semantics (Qchem ⊗ Qs). Models living metabolic homeostasis, genetic
  codon information channels, cellular automata, and active inference operating as bounded SIMEMP
  open systems.

term: Q(reuse); slug: q-reuse; category: Primitives
  A Category B optimization morphism that references existing immutable content commitment hashes
  within the local causal DAG instead of allocating new physical memory bits. Mitigates the Memory
  Wall and eliminates redundant Landauer dissipation at zero marginal Q(1) cost.

term: Q(bypass); slug: q-bypass; category: Primitives
  A Category B optimization morphism allowing a SIMEMP Gateway to short-circuit execution steps
  when algebraic preconditions prove an invariant identity transition (such as Terminal Exactness
  r = Q(0) or multiplication by unit Q(1)). Emits an explicit zero-cost bypass receipt without
  consuming compute budgets.

term: cdqn; slug: cdqn; category: Primitives
  Chained and Distributed Quang Numbers. The universal fractal protocol of governed data movement,
  operating both locally (intra-node causal chaining between abstraction layers and base domains)
  and distributively (inter-node public attestations across physical networks).

term: QnLang; slug: qnlang; category: Primitives
  The high-level domain-specific language used to author Qn computations, enforcing explicit DCC
  profiles and compile-time collapse.

term: QnIR; slug: qnir; category: Primitives
  Intermediate Representation used for hardware abstraction, bound enforcement, and deterministic
  execution without exposing floating-point semantics.

term: Qexpr; slug: qexpr; category: Primitives
  Governed expression tuple encapsulating symbolic structure, zoom, remainder, dimension, DCC
  profile, identity, and causal lineage. Structured as a Content-Addressed Expression DAG
  (CAE-DAG) with hash-consing.

term: Q(rsi); slug: qn-rsi; category: Primitives
  A higher-order recursive self-optimization workflow. Ingests an existing Q(workflows) artifact,
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  derives an empirically more efficient successor under strict semantic invariance and finite
  metric budget, and emits a versioned successor artifact.

term: Content-Addressed Expression DAG; slug: cae-dag; category: Primitives
  The canonical Layer 3 structural container of Qexpr, wherein every symbolic node is irrevocably
  identified by the cryptographic commitment hash of its operator, parameters, and child
  dependencies. Guarantees mathematical cycle freedom by construction and enables O(1) structural
  equivalence comparisons.

term: Discrete Exterior Calculus; slug: discrete-exterior-calculus; category: Primitives
  The constructive formulation of calculus operating on discrete simplicial meshes and cochains,
  replacing continuous partial differential equations with exact algebraic coboundary operators
  without floating-point approximations.

term: Discrete Exterior Derivative d; slug: discrete-exterior-derivative; category: Primitives
  The discrete coboundary operator mapping k-forms to (k+1)-forms across simplicial complexes.
  Satisfies the exact topological nilpotency invariant d ∘ d = Q(0), grounding electromagnetic and
  fluid conservation laws.

term: Discrete Hodge Star; slug: discrete-hodge-star; category: Primitives
  The geometric duality operator mapping primal simplicial k-forms to dual circumcentric
  (n-k)-forms, parameterizing exact dual volume ratio quotients.

term: Qs; slug: qs; category: Primitives
  Quang Semantics. The constructive relational, linguistic, and intent base domain of the Qn
  universe. Formalizes words as multivector blades, grammar as CAE-DAG wiring, and human-machine
  communication as verifiable intent contracts.

term: Q(patterns); slug: q-patterns; category: Primitives
  An invariant, reproducible state-transition template mapping input commitments to deterministic
  output milestones under abstract DCC contracts. Encompasses both exact algebraic transformations
  and deterministic terminal fault signatures.

term: Q(workflows); slug: q-workflows; category: Primitives
  A composite, goal-directed causal directed acyclic graph assembling multiple Q(patterns) under a
  declared Intent Contract and finite Metric Budget. Portable across any cdqn node via chained
  Road Receipts.

term: Qphy; slug: qphy; category: Primitives
  Quang Physics. The local-first constructive physical base domain of the Qn universe. Models
  non-equilibrium computational thermodynamics, discrete exterior cochain field dynamics, Landauer
  dissipation bounds, and physical Q(anchor) invariants directly grounded in Layer 0 silicon
  substrates.

term: Q(event); slug: q-event; category: Primitives
  A discrete computational or physical state transition occurring at an explicit spatiotemporal
  coordinate, carrying a discrete scale quantum, a state transition payload, and a verified
  epistemic modality.

term: Qchem; slug: qchem; category: Primitives
  Quang Chemistry. An emergent composite domain derived from the categorical tensor product of
  Quang Physics and Quang Mathematics (Qphy ⊗ Qm). Models discrete stoichiometric balance
  equations, molecular bonding graphs, reaction kinetics, and Gibbs free energy dissipation
  without semantic intent.

term: Axiom 6; slug: axiom-6; category: Primitives
  Totality of Governed Operations. Every governed operation carries an explicit work ceiling in
  Abstract Compute Units U: execution either reaches Terminal Exactness (R = Q(0)) or halts
  cleanly upon budget exhaustion, emitting a signed receipt. Unbounded, non-terminating state
  transitions are physically impossible in the Qn universe.

term: Q(dataStruc); slug: q-datastruc; category: Primitives
  Governed immutable memory container artifact class providing bounded, content-addressed storage
  within the local causal DAG. Allocation, reading, and writing are explicit governed morphisms
  subject to DCC constraints and Memory Wall envelopes; no implicit heap state is admitted.

term: Q(th_2); slug: q-th-2; category: Primitives
  Physical invariant anchor encoding the Second Law. Calibrated from sampled thermal receipts via
  ln 2 dissipation quanta, it mandates that endergonic processes execute only under a declared
  Gibbs free-energy budget (DCC_ANCHOR_SECOND_LAW_v1).

term: Q(c); slug: q-c; category: Primitives
  Physical invariant anchor of the relativistic causal horizon. Calibrated from local dielectric
  propagation delays, it bounds causally ordered state propagation across the substrate
  (DCC_ANCHOR_CAUSAL_HORIZON_v1).

term: Q(ℏ); slug: q-hbar; category: Primitives
  Physical invariant anchor of quantum action. Built from height ceilings and quantum action
  limits, it forbids unmetered infinite information density in finite registers
  (DCC_ANCHOR_ACTION_PLANCK_v1).

term: Q(Z_bus); slug: q-z-bus; category: Primitives
  Physical invariant anchor of the Memory Wall. Built from parasitic wire capacitance and bus line
  physics, it calibrates transport latency and bandwidth ceilings of the substrate
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  (DCC_ANCHOR_MEMORY_WALL_v1).

term: Q(gauge); slug: q-gauge; category: Primitives
  Physical invariant anchor of topological conservation. Built from simplicial mesh topology and
  discrete coboundary verification, it grounds conserved field quantities without metric drift
  (DCC_ANCHOR_DEC_D_v1).

term: Q(array); slug: q-array; category: Primitives
  The proposed sequenced container contract surface: a bounded, indexed container whose footprint
  is a declared length, whose permitted access is indexed, whose locality class is contiguous, and
  whose traversal order is fixed by causal index with constant declared work per indexed access.
  Grounded in the OP_FOLD child vector under an explicit step ceiling, it admits no in-place
  mutation: growth emits a new versioned artifact. A contract surface under the five-row
  structural contract schema, never a runtime shape; pending deterministic measurement, BOC
  selection, and Q(rsi) sealing.

term: Q(matrix); slug: q-matrix; category: Primitives
  The proposed geometric container contract surface: a bounded, two-dimensional integer container
  whose contractions complete in constant declared time through the integer
  Matrix-Multiply-Accumulate lowering, preserving exactness by construction without floating-point
  drift, and admitting the Q(bypass) morphism to short-circuit identity preconditions in register
  time. Grounded in the OP_FRAME_MAP node class. A contract surface under the five-row structural
  contract schema, never a runtime shape; pending deterministic measurement, BOC selection, and
  Q(rsi) sealing.

term: Q(table); slug: q-table; category: Primitives
  The proposed keyed container contract surface: a bounded, content-addressed lookup structure
  grounded in hash-consing itself, the Content-Addressed Expression DAG already constituting a
  keyed index over committed expressions. Capacity is declared; growth is re-versioning rather
  than resizing; the work row declares the breakeven inequality under which keyed lookup is
  preferred over its recorded alternatives, in the manner of the committed Q(reuse) analysis. A
  contract surface under the five-row structural contract schema, never a runtime shape; pending
  deterministic measurement, BOC selection, and Q(rsi) sealing.

term: Q(stream); slug: q-stream; category: Primitives
  The proposed bounded stream contract surface for sequenced data: declared capacity with
  sequenced, append-ordered access fixed by causal index. Bounded streams are already exercised,
  through declared sample volumes in entropy measurement and append-only receipt chains, while no
  container type is committed; an implicitly resizable queue would violate the footprint and
  determinism rows and is not admitted. Nominated as an open contract shape under the five-row
  structural contract schema, never a runtime shape; pending implementation feedback,
  deterministic measurement, BOC selection, and Q(rsi) sealing.
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T.4 — docs/_data/glossary/constraints.yml

Source: docs/_data/glossary/constraints.yml *— —, updated —. Rendered content per Annex Guide conventions N1–N6: 237 source lines, 146 rendered; 11,127 → 10,257 characters; source SHA-256* 0f1c4126 4ad566d5 c58cc511 7370017e 2438ffd8 7ee84be0 56dd4cb6 bba90b33*. Content fidelity verified mechanically at build time.*

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# CDQN Unified Glossary — Constraints Domain
# License: Scaling Source License (SSL) 1.0
# Updated: 2026-10-22 (Bao Loc, Vietnam)

term: Dependencies Determinism; slug: dependencies-determinism; category: Constraints
  The formal property whereby a computational state transition S_t possesses zero conditional
  entropy given its complete declared set of causal dependencies: H(S_t | D(S_t)) = 0. If
  dependencies are incomplete, implicit, or unmeasured, residual entropy produces
  non-deterministic state drift.

term: Structural Indirection; slug: structural-indirection; category: Constraints
  The architectural decoupling of abstract capability contracts from concrete cryptographic or
  serialization primitives. Allows the runtime to patch zero-day vulnerabilities and swap
  algorithms without mutating abstract DCC contracts or fracturing causal lineage.

term: Metric Exhaustion; slug: metric-exhaustion; category: Constraints
  The verified consumption or sink of a declared compute and storage budget (such as Q(1) units)
  required to execute a governed state transition. Represents thermodynamic and causal work, not a
  monetary policy or financial event.

term: SIMEMP; slug: simemp; category: Constraints
  Security, Identity, Metric, Efficiency, Modularity, Portability. The constitutional constraint
  system governing the Qn and cdqn stack. Divided into Tier 1 Existential Invariants
  (non-negotiable) and Tier 2 Operational Agilities (trade-off-able).

term: DCC Profile; slug: dcc-profile; category: Constraints
  Dependencies, Constraints, Capabilities. A machine-readable abstract contract defining
  prerequisites, hard operational bounds, and permitted morphisms for every layer, object, and
  gateway. No DCC profile may be implicit.
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term: Existential Invariant; slug: existential-invariant; category: Constraints
  A Tier 1 SIMEMP constraint that is non-negotiable. The three Existential Invariants are
  Identity, Metric, and Security. If a proposed design violates or lacks these, it is
  fundamentally invalid.

term: Operational Agility; slug: operational-agility; category: Constraints
  A Tier 2 SIMEMP constraint that is trade-off-able. The three Operational Agilities are
  Efficiency, Modularity, and Portability. They may be sacrificed to preserve Tier 1 Existential
  Invariants.

term: BOC Policy; slug: boc-policy; category: Constraints
  Best of Choices. The formal decision framework used to resolve design conflicts by selecting the
  implementation path that maximizes SIMEMP compliance and minimizes implicit state.

term: Scaling Threshold; slug: scaling-threshold; category: Constraints
  Usage boundaries defined by the SSL 1.0 triggering mandatory commercial licensing: >10,000
  active compute nodes/agents, >100,000 MAU, or >10,000,000 monthly API/Qn transactions.

term: No-Implicit Rule; slug: no-implicit-rule; category: Constraints
  The constitutional rule that no implicit entity, default, coercion, or unmeasured convention may
  cross a SIMEMP Gateway. Only explicit Qn objects are admitted between abstraction layers.

term: Computational Consistency; slug: computational-consistency; category: Constraints
  The empirical verification methodology of the Qn universe. If logic is internally consistent and
  computations authored in QnLang and compiled to QnIR execute deterministically within declared
  finite budgets, the abstraction is provisionally valid.

term: Terminal Exactness; slug: terminal-exactness; category: Constraints
  The discrete algebraic condition wherein remainder r equals genesis origin Q(0) at a declared
  zoom level z along dimension d. Proves mathematical exactness and triggers deterministic early
  halting, preventing redundant thermodynamic metric consumption.

term: Hash-Consing; slug: hash-consing; category: Constraints
  The structural graph-unification mechanism within CAE-DAGs mapping semantically identical
  sub-expressions to the exact same immutable node in memory. Eliminates redundant bit allocation
  across the Memory Wall and ensures zero marginal Landauer dissipation (W = 0) for recurring
  sub-graphs via Q(reuse).

term: Anchor-Cut Invariant; slug: anchor-cut; category: Constraints
  The sparse causal verification property whereby a terminal state S_N can be proven valid in O(K)
  anchor receipt validations without replaying N intermediate micro-steps across volatile memory.

term: Epistemic Anchorage; slug: epistemic-anchorage; category: Constraints
  The formal attribution of truth modality to an assertion along the spectrum from physical sensor
  observation to sovereign narrative fiction, requiring causal lineage receipts to prevent
  unanchored AI hallucinations. Proposed extension (CANDIDATE, docs/qs_history.md §3): a
  Source-Receipt Invariant requiring every governed artifact to declare the explicit pair (content
  modality, source class), extending the statement-level rejection rule of qs §4.1 to composite
  artifacts and worlds.

term: Intent Contract; slug: intent-contract; category: Constraints
  The machine-readable agreement in a DCC profile parameterizing mutual human-agent commitments,
  capability scopes, and conserved social residuals (r_social) to guarantee AI alignment with
  human sovereignty.

term: Non-Rogue Invariant; slug: non-rogue-invariant; category: Constraints
  The constitutional SIMEMP invariant whereby machine identities and autonomous AI agents are
  strictly barred from root genesis sovereignty (Q(0)), requiring unbroken cryptographic intent
  contracts bound to a human identity and finite metric budgets to prevent unconstrained recursive
  self-modification.

term: Historical Calibration Benchmark; slug: historical-calibration-benchmark; category: Constraints
  The mandatory empirical ingestion event wherein a physical node measures its local Layer 0
  environmental baseline (ambient temperature, physical entropy, atomic frequency standards) via
  signed hardware receipts. Acts as the foundational physical bootstrap grounding abstract Qm/Qlog
  invariants into real thermodynamic work (U) and universal Q(anchor) entities. Integration status
  (docs/qs_history.md §8 item 1, CANDIDATE): wired as the non-mutating CANDIDATE proposition at
  q0_q1 §2.7, closing that queue item — a post-seal Class P grounding event after Stage 6, adding
  no stage to the six-stage onboarding state machine; pending deterministic measurement and BOC
  selection at implementation time.

term: Historical Horizon; slug: historical-horizon; category: Constraints
  The epistemic boundary governing past events: Contemporary History is anchored directly to
  multi-sensor physical receipts (MODAL_EMPIRICAL) yielding global consensus, whereas Long-Horizon
  Deep History lacks direct silicon witnesses and is evaluated strictly via Qlog inverse causal
  reconstruction (MODAL_HYPOTHESIS) or contextually firewalled narrative (MODAL_NARRATIVE).
  Proposed operational shape (CANDIDATE, docs/qs_history.md §5–6): the Qs History Library of
  modality-typed historical anchor artifacts, ingested through a five-gate Anti-Poisoning
  Ingestion Firewall, with the consensus history obtained as the global section of the history
  sheaf over shared landmarks (chronosa §3.1).

term: Source-Receipt Invariant; slug: source-receipt-invariant; category: Constraints
  The proposed artifact-level extension of the statement-level rejection rule: every governed Qn
  artifact declares the explicit pair (content modality, source class), where the content modality

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  is drawn from the epistemic anchorage spectrum and the source receipt chain must terminate in
  physically receipted reality events — Layer 0 sensor or artifact receipts for empirical content,
  algebraic closure receipts for constructive content, and author-identity plus publisher receipts
  for contextually sealed creative content. Two binding cautions: sovereign Payloads remain
  uninspected behind the safe-harbor air gap (the declaration is authored, never extracted by
  inspection), and a simulation's execution stays constructive while its claims about reality
  remain hypothesis until physically falsified. Proposed term (CANDIDATE, docs/qs_history.md §3):
  elaborated as docs/qs.md §4.3, extending the rejection rule of qs §4.1 to composite artifacts
  and worlds; pending deterministic measurement and BOC selection.

term: Source Class; slug: source-class; category: Constraints
  The second element of the proposed (content modality, source class) declaration pair: a declared
  expression of where a governed artifact's asserted content ultimately anchors, drawn from the
  same proposed I/P/S vocabulary as the anchor grounding classes — I (Invariant: deterministic
  re-derivation from the artifact's own declared rule), P (Physical: independent Layer 0
  calibrations converging within declared tolerance envelopes), and S (Social: quorum-witnessed
  historical events glued on shared landmarks). Proposed term (CANDIDATE, docs/qs_history.md §3):
  applies the witness ladder — self, then physics, then swarm — to composite artifacts and worlds
  rather than to anchors alone; pending deterministic measurement and BOC selection.

term: Ingestion Gate; slug: ingestion-gate; category: Constraints
  A proposed governed checkpoint of the five-gate Anti-Poisoning Ingestion Firewall through which
  the Qs History Library can be alimented only by petition, each gate emitting its own receipt.
  The five proposed gates: Lineage (unbroken receipt chain to Layer 0 artifacts or sealed library
  records), Modality Ceiling (each atomic claim typed at the highest modality its receipts
  support), Independence Quorum (anchor-grade records require multiple sources with
  lineage-disjoint root subtrees in the polyroot forest), Sheaf Gluing (records disagreeing with
  receipted overlapping records enter a QUARANTINED lifecycle state under a challenge receipt),
  and Append-Only Immutability (sealed artifacts never mutate; corrections arrive as
  counter-receipts and migration receipts — history is never rewritten, only re-witnessed).
  Proposed term (CANDIDATE, docs/qs_history.md §6): the firewall composes existing canon
  mechanisms under one pipeline; pending deterministic measurement and BOC selection.
```

T.5 — docs/_data/glossary/governance.yml

Source: docs/_data/glossary/governance.yml *— —, updated —. Rendered content per Annex Guide conventions N1–N6: 96 source lines, 55 rendered; 3,501 → 3,113 characters; source SHA-256* b32d9f4a 123f0e38 c26c62b4 f35af1e0 848e7797 046ffbcf 9646b544 3dbd6048*. Content fidelity verified mechanically at build time.*

```
# CDQN Unified Glossary — Governance Domain
# License: Scaling Source License (SSL) 1.0
# Updated: 2026-10-08 (Bao Loc, Vietnam)

term: Universal Envelope; slug: universal-envelope; category: Governance
  The invariant governance structure shared across all Qn artifacts, composed of identity, type,
  state, lineage, DCC, metric, security, exposure, and payload reference.

term: Metric Envelope; slug: metric-envelope; category: Governance
  The finite measurement boundary declaring resource limits on storage, expression depth,
  precision, uncertainty, and compute budgets.

term: Security Envelope; slug: security-envelope; category: Governance
  The governed security boundary declaring authority limits, delegation chains, PQC algorithm
  identifiers, and swappable revocation paths.

term: Identity Class; slug: identity-class; category: Governance
  The governed actor category parameterizing an identity block: Machine Identity (runtimes/nodes),
  Human Identity (natural persons), or AI Agent Identity (delegated processes).

term: Receipt; slug: receipt; category: Governance
  A governed, immutable record of an execution transition, fault, halt, or boundary traversal.
  Acts as a dissipative witness exporting operational entropy to maintain internal system order.

term: Lifecycle State; slug: lifecycle-state; category: Governance
  The operational status of an artifact: candidate, constructed, sealed, verified, rejected,
  halted, quarantined, exportable, exported, revoked.

term: Payload; slug: payload; category: Governance
  Any creative content, digital asset, real-world asset representation, data, or application
  processed, transmitted, or settled using the Licensed Work. Payloads are sovereign property of
  their creators, do not constitute Derivative Works, and are governed under protocol blindness
  and safe harbor neutrality.

term: Licensed Work; slug: licensed-work; category: Governance
  The core technical infrastructure layer of the Qn and cdqn stack, encompassing source code,
  intermediate representations (QnIR), specifications, mathematical definitions, and documentation
  provided under the Scaling Source License.

term: Derivative Work; slug: derivative-work; category: Governance
```

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  Any modification, extension, translation, binding, or runtime implementation derived from the
  Licensed Work. Encapsulating, tokenizing, or settling a Payload using the Licensed Work does not
  cause the Payload to become a Derivative Work.

term: Open Core Invariant; slug: open-core-invariant; category: Governance
  Non-negotiable conditions preserved in all Derivative Works under the SSL 1.0: (a) Anti-Patent
  Defense, and (b) Non-Scaling Open Access.

term: Paternity Reference; slug: paternity-reference; category: Governance
  Mandatory attribution notice for derivative distributions and runtime headers citing Christophe
  Duy Quang Nguyen under the Scaling Source License.

term: SSL; slug: ssl; category: Governance
  Scaling Source License 1.0. The commercial and open-core license governing the CDQN project,
  derivative distributions, and scaling thresholds.
```

Annex U — Loop-1 review record: AiReviewQnIRv0.1.md

Source: docs/AiReviews/AiReviewQnIRv0.1.md *(loop-1 review record) — —, updated —. Rendered content per Annex Guide conventions N1–N6: 361 source lines, 1580 rendered; 121,587 → 126,138 characters; source SHA-256* f08f456d feabf44e dbcc0f9a 45337b2b 5f62e837 1de10644 f4149e46 374030db*. Content fidelity verified mechanically at build time.*

```
Christophe Duy Quang Nguyen note :
- The QnIR v0.1 was committed to the docs/AiReviews/ folder on October 26th, 2026.
- The QnIR v0.1 was submitted for review by Google's AI, Deep Review, on October 26th, 2026.
- This transcript is the result of this review. The reviewer's points seem limited to the
   content of the QnIR v0.1 PDF; if any data loss exists between the live docs on GitHub Pages
   and the PDF content, this means that the reviewer's results will also be based on inaccurate
   data.


H1: HIGH LEVEL SUMMARY


H2: Summary
This document presents a condensed, self-contained specification of the Chained and Distributed
Quang Numbers (CDQN / Qn) computational universe alongside a first candidate design for the
Quang Numbers Intermediate Representation (QnIR v0.1). The overarching goal is to construct a
governed, deterministic, and resource-bounded computing stack for verifiable AI agent alignment
by combining the SIMEMP constitutional constraint layer, an exact residual-conserving discrete
numeric substrate (Qm), content-addressed symbolic expression DAGs (Qexpr), discrete geometric
and exterior calculus layers, five domain projections, and a 30-opcode virtual machine with
static compute-unit (U) budgets and cryptographic execution receipts.


H2: Strengths
- The specification addresses a highly significant and timely problem—preventing cross-platform
   state drift, silent non-termination, and unverifiable execution in autonomous AI agent
   workflows—by establishing a clear epistemic boundary between stochastic neural candidate
   generation and deterministic, budget-metered verification.
- The overarching architectural synthesis is ambitious and internally cohesive, bringing
   together exact residual-conserving Diophantine/Clifford tuples, content-addressed expression
   graphs that eliminate pointer aliasing, and explicit failure-receipt totality into a unified,
   auditable framework.
- The document demonstrates exceptional intellectual honesty and scientific rigor in its
   self-critique, explicitly cataloging sixteen anticipated objections and fifteen targeted open
   questions to facilitate systematic adversarial review and iterative refinement toward v0.2.


H2: Weaknesses
- The proposed fixed 64-bit QnIR v0.1 instruction word layout (src1: 8, src2: 8, dest: 8, aux:
   16) cannot soundly encode higher-arity operations (OP_TUPLE of arity 4, OP_FOLD of arity k
   \ge 3), dual-destination outputs (OP_DIV), or 256-bit cryptographic commitment hashes
   (OP_REUSE) without relying on unstated register-window conventions or hidden indirection that
   conflict with the constitutional No-Implicit Rule.
- Please consider decoupling logical execution metering from physical thermodynamics: commodity
   CMOS silicon operates many orders of magnitude above Landauer's k_B T \ln 2 limit with energy
   dominated by DRAM data movement, making static software-level U-unit instruction costs and
   \Delta W = 0 claims for memory-intensive hash-consing lookups physically inaccurate without
   microarchitectural bounding (Reference Paper: Computing's Energy Problem (and what we can do
   about it), Mark Horowitz, 2014).
- The five-stage dispatch loop and equality-saturation engine (OP_REWRITE) rely on predicting
   \Delta W_{\text{compute}} prior to evaluation and assume \mathcal{O}(1) canonical
   equivalence, yet general e-graph equality saturation is neither terminating nor confluent
   without explicit runner limits and cost-extraction heuristics, leaving the pre-execution
   breakeven check undecidable in practice (Reference Paper: egg: Fast and Extensible Equality
   Saturation, Max Willsey et al., 2021).
- Cross-node verification via \mathcal{O}(1) Road Receipts and Anti-Drift Verification currently
```

checks only static metadata and sender signatures rather than dynamic data-plane transitions; the authors might consider incorporating succinct zero-knowledge arguments to prevent a faulty or Byzantine node from signing a fabricated output commitment \mathcal{H}(S_{\text{out}}) without re-execution (Reference Paper: The RISC Zero zkVM: Scalable, Transparent Arguments of RISC-V Integrity, Jeremy Bruestle et al., 2023).
- The Qm numeric substrate omits a formal inductive construction of multi-digit integers (\mathbb{N} beyond Q(9)), leaves the zoom resolution function \kappa(z) unfixed, and is presented without benchmarking or expressiveness comparisons against established exact real arithmetic, continued-fraction algorithms, and deterministic hardware alternatives to IEEE 754 (Reference Paper: Toward an API for the Real Numbers, Hans-J. Boehm, 2020; Reference Paper: Exact Real Computer Arithmetic with Continued Fractions, Jean E. Vuillemin, 1990; Reference Paper: Beating Floating Point at its Own Game: Posit Arithmetic, John L. Gustafson and Isaac T. Yonemoto, 2017).
- To better situate the architectural contributions of QnIR and Qexpr, please consider comparing the VM and content-addressed store against established prior art in provably terminating 64-bit register bytecodes, total functional programming, and hash-consed distributed runtimes (Reference Paper: BPF Instruction Set Architecture (ISA), D. Thaler (Ed.), 2024; Reference Paper: Total Functional Programming, David A. Turner, 2004; Reference Paper: Unison: A New Distributed Programming Language, Paul Chiusano and Rúnar Bjarnason, 2019).

H2: Potential Issues And Suggestions

H3: [Introduction and The SIMEMP Constitutional Layer] (Pages: 1-7)

H4: Potential Mistakes and Improvements

1. Formal Under-Specification and Potential Tautology of "Dependencies Determinism" (Section 2.1, Eq. 1)
    - Potential Concern: Section 2.1 defines the core constitutional invariant as the conditional-entropy postulate H(S_t \mid \mathcal{D}(S_t)) = 0, stating that if any dependency is implicit, hidden, or unmeasured, the available subset \mathcal{D}' \subsetneq \mathcal{D} yields H(S_t \mid \mathcal{D}') > 0. As anticipated in the Section 2 Invited Objection (page 7) and Objection 3 (page 26), this formulation faces two substantive mathematical and operational gaps:
      1. *Missing Probability Space:* Conditional Shannon entropy H(X \mid Y) = -\sum p(x,y) \log p(x \mid y) is defined over random variables equipped with an explicit probability measure space (\Omega, \mathcal{F}, P). In Section 2.1, no sample space or probability distribution over artifact transitions or execution environments is defined. If the transition semantics are defined as a deterministic function S_t = f(\mathcal{D}(S_t)), then H(S_t \mid \mathcal{D}(S_t)) = 0 is a notationally heavy restatement of functional purity (referential transparency) rather than an information-theoretic measurement.
      2. *Absence of an External Completeness Audit against Tautology:* Because \mathcal{D}(S_t) is the set of *declared* causal dependencies inside an artifact's envelope or DCC profile, an implementation can trivially satisfy H(S_t \mid \mathcal{D}_{\text{declared}}) = 0 within its own abstract model while still depending on unmodeled Layer 0 physical state (e.g., microarchitectural timing, cache residency, or compiler runtime assumptions). Without a formal operational semantics that defines what constitutes the *actual* dependency set \mathcal{D} of a physical or virtual transition, the equation cannot distinguish a genuinely deterministic system from one whose hidden dependencies were omitted from the formal model.
    - Suggested Improvement: For v0.2, replace or supplement Equation (1) with a formal Non-Interference / Observational Determinism Invariant from programming language theory: for any two Layer-0 host execution environments E_1, E_2 that agree on the projections of the declared dependency commitment (\pi_{\mathcal{D}}(E_1) = \pi_{\mathcal{D}}(E_2)) and satisfy the active DCC hardware profile, evaluating the transition must produce bit-identical output commitments \mathcal{H}(S_t(E_1)) = \mathcal{H}(S_t(E_2)) and identical terminal receipt statuses. This eliminates the need for an unmodeled probability distribution over code and makes \mathcal{D}' \subsetneq \mathcal{D} empirically falsifiable via differential execution testing across heterogeneous environments.

2. Conflation of Physical Thermodynamics with Logical Execution Metering (Section 2.2; cf. QO-2)
    - Potential Concern: Section 2.2 grounds the SIMEMP Metric invariant directly in non-equilibrium thermodynamics: Landauer's erasure bound (W \ge k_B T \ln 2 \approx 2.87 \times 10^{-21}\text{ J} per bit at T = 300\text{ K}), Memory Wall bus power \mathcal{O}(CV^2f), and Prigogine open-system entropy balance (dS = dS_{\text{internal}} + dS_{\text{exchange}} with dS_{\text{exchange}} < 0 achieved by "exporting entropy through bounded, signed receipts"). Directly equating software-level execution metering and cryptographic receipts with physical thermodynamics introduces three validity issues:
      1. *Scale Discrepancy and Physical Irreversibility:* Commodity CMOS silicon (Layer 0) operates roughly 10^7 to 10^{10} times above Landauer's thermal limit due to capacitive charging (\frac{1}{2}CV^2), leakage currents, and interconnect overhead, and physical gate transitions dissipate heat regardless of whether a high-level symbolic rewrite on a DAG is logically reversible (e.g., \sim 0.1\text{ pJ} for a 32-bit ALU operation and \sim 1\text{–}2\text{ nJ} for a 64-bit DRAM access).
      2. *Platform Invariance vs. Physical Calibration:* If the Abstract Compute Unit U is an invariant logical step counter so that road receipts and static instruction budgets behave deterministically across nodes, it cannot simultaneously be a direct physical measurement of Joules, which varies continuously with silicon process node, voltage-frequency scaling, ambient temperature, and cache state—which cannot be known statically at the IR level without violating the Substrate-vs-Invariant ("car vs. road") disjunction (Sec. 7.1).
      3. *Physical Meaning of Receipt Entropy Export:* In physical thermodynamics, a computing system exports entropy to its environment by dissipating heat (\Delta Q / T) into a

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        thermal sink. Allocating, hashing, signing (with post-quantum cryptography), and
        transmitting a digital receipt does not export physical thermal entropy
        (dS_{\text{exchange}} < 0) out of the computing substrate; physically, generating a
        receipt is an irreversible operation that *increases* internal heat dissipation
        (dS_{\text{internal}} > 0) on Layer 0. What a receipt actually reduces is *epistemic
        uncertainty* (information entropy) for an external verifier.
    - Suggested Improvement (Directly Addressing QO-2): For v0.2, formally decouple the Metric
      invariant into two distinct quantities:
      - Redefine U strictly at Layers 1–3 and in QnIR as a platform-invariant logical resource
        unit / work quantum (analogous to deterministic virtual machine gas/fuel in WebAssembly
        or resource ticks in Amortized Resource Analysis / Resource-Aware ML) used for totality
        guarantees, static instruction costs, and cross-node receipt verification, normalized so
        that the minimal state transition Q(0) \to Q(1) costs 1\,U by definition and 1 bit of
        logical erasure corresponds to a theoretical floor of W(U) \ge k_B T \ln 2 in Qphy.
      - Treat physical energy and thermal budgets (E_{\text{battery}}, P_{\text{thermal}} in
        Joules/Watts) as a separate Layer-0 Hardware Telemetry Envelope, bound to U exclusively
        via the node's local Layer-0 Historical Calibration Benchmark receipt (Q(Zbus), Q(th2)),
        which maps 1\text{ logical } U to a hardware-specific worst-case energy envelope
        E_{\text{phys}} \le \alpha_{\text{compute}} U_{\text{op}} + \beta_{\text{bus}}
        U_{\text{mem}} (or \alpha_{\text{node}}(T, V, f) \in [\text{J}/U_{\min},
        \text{J}/U_{\max}]) in Joules with stated error bars for battery and thermal leash
        enforcement.
      - Reframe the Prigogine receipt analogy in Section 2.2 explicitly as *epistemic / audit
        state compression* rather than physical heat export.

3. Non-Decidability of the Two-Tier Constraint Architecture and Best of Choices (BOC) Policy
   (Sections 2.3 & 2.7, Eq. 2; cf. QO-1)
    - Potential Concern: Section 2.3 establishes a two-tier hierarchy—Tier 1 Existential
      Invariants (*Identity, Metric, Security*) over Tier 2 Operational Agilities (*Efficiency,
      Modularity, Portability*)—and Section 2.7 defines the Best of Choices (BOC) policy as
      \text{Selection} = \arg\max_{c \in \mathcal{C}}(\text{SIMEMP Compliance}(c)). As
      acknowledged in Section 2.7 and the Invited Objection (page 7), \text{SIMEMP
      Compliance}(c) has no formal codomain, metric space, or partial order. Consequently,
      Equation (2) cannot be evaluated mechanically:
      1. Tier 1 invariants are binary admissibility predicates (a candidate either preserves
         collision-resistant identity, finite metric bounds, and security contracts, or it is
         discarded), not continuous quantities to be maximized.
      2. Tier 2 agilities naturally conflict and are multi-objective and workload-dependent
         (e.g., sacrificing *Modularity* or *Portability* in fixed 64-bit QnIR encodings to gain
         *Efficiency* across the Memory Wall). Without a defined tie-breaking or weighting order
         among Tier 2 attributes, \arg\max is undefined and any design choice can be rationalized
         post-hoc as BOC-compliant.
    - Suggested Improvement (Directly Addressing QO-1): While a single global scalar objective
      over arbitrary prose designs requires human modeling choices, the two-tier boundary can
      and should be made mechanically decidable over declared DCC metadata (such as in the
      automated Q(\text{rsi}) loop) by replacing the scalar \text{SIMEMP Compliance}(c) with a
      two-stage constrained lexicographic optimization model in \{0,1\}^3 \times \mathbb{R}^3:
      \Phi(c) = \big\langle \underbrace{\mathbb{I}_{\text{Id}}(c) \land
     \mathbb{I}_{\text{Met}}(c) \land \mathbb{I}_{\text{Sec}}(c)}_{\text{Tier 1 hard admission
     filter}},\; \underbrace{-\Delta W_{\text{worst}}(c),\; -M_{\text{coupling}}(c),\;
     P_{\text{coverage}}(c)}_{\text{Tier 2 lexicographic cost vector}} \big\rangle.
      - First, formalize Tier 1 as a hard boolean filter: \mathcal{C}_{\text{valid}} = \{ c \in
        \mathcal{C} \mid \mathbb{I}_{\text{Id}}(c) \land \mathbb{I}_{\text{Met}}(c) \land
        \mathbb{I}_{\text{Sec}}(c) = 1 \} (verifying statically that a collision-resistant
        Universal Envelope is present, all loop/zoom/height ceilings are finite, and all
        capability/PQC signatures validate; otherwise it is rejected with REJECTION_BY_GATEWAY).
      - Second, over \mathcal{C}_{\text{valid}}, define an explicit lexicographic tuple ordering
        beginning with the constitutional mandate to minimize implicit state
        (\text{ImplicitDeps}(c)), followed by Tier-2 metrics—worst-case declared work \Delta
        W_{\text{worst}}(c) (Efficiency), external dependency/interface boundary count
        M_{\text{coupling}}(c) (Modularity), and supported target hardware profile count
        P_{\text{coverage}}(c) (Portability)—ordered according to an explicit priority
        permutation \pi \in S_3 declared in the calling DCC profile.
      - Where Tier 2 dimensions remain qualitative architectural judgments during high-level
        specification design, reclassify BOC explicitly as a structured architectural rationale
        rubric / constrained design heuristic (requiring each design verdict to tabulate its
        Tier 1 pass/fail status and Tier 2 trade-off vector) rather than presenting Eq. (2) as
        an unparameterized mathematical \arg\max.

4. Unbilled Boundary Overhead and Cryptographic Migration Indirection in Gateway Discipline
   (Sections 2.4–2.6)
    - Potential Concern: Auditing the gateway mechanisms in Sections 2.4–2.6 reveals three
      completeness gaps:
      1. *Unbilled Gateway Verification Overhead (Sec. 2.4 & 2.6):* Section 2.6 states that
         generation behind the stochastic boundary is free, while crossing a SIMEMP Gateway into
         governed state is deterministic, metered, and receipted. However, parsing, measuring,
         and validating an unverified candidate (including PQC signature and DCC contract checks)
         at a gateway itself consumes finite compute and memory work. Section 2.5 does not
         specify whether gateway validation is charged against the *caller's* parent budget
         W_{\text{rem}} prior to crossing; if failed candidate validations are not billed to an
         invoking principal's budget, a stochastic engine can induce denial-of-service (metric
         exhaustion) at the gateway boundary.
      2. *Semantic vs. Syntactic Alignment at the Candidate Boundary (Sec. 2.6):* Section 2.6
         presents the deterministic gateway check as the solution to neural-symbolic integration,
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        but does not distinguish between *structural/resource compliance* (verifying that a
        candidate satisfies DCC bounds and avoids floats) and *semantic intent alignment*
        (verifying that a candidate formalized by a fallible parser matches the human
        principal's actual intent, as noted later in Objection 10).
      3. *Cryptographic Migration vs. Content-Addressed Lineage (Sec. 2.4):* Section 2.4 states
        that under structural indirection, a vulnerable concrete primitive (e.g., a signature or
        hash scheme) can be swapped for a patched module by emitting a migration receipt
        "without fracturing the abstract contract or the causal lineage beneath it." Because
        node identity and causal lineage across the stack are content-addressed via
        cryptographic hashes \mathcal{H}(\alpha) and sealed with signatures, swapping a
        compromised primitive invalidates historical hash bindings unless a multihash tagging
        scheme or re-witnessing/re-enveloping protocol is explicitly defined.
    - Suggested Improvement: (a) Specify that every SIMEMP Gateway invocation deducts a static
      verification fee W_{\text{gate}} > 0 from the *invoking principal's* budget before
      inspecting a candidate. (b) Clarify in Section 2.6 that gateways enforce formal
      contract/resource soundness rather than informal-to-formal semantic fidelity. (c) Specify
      in Section 2.4 whether structural indirection uses algorithm-tagged digests (multihash)
      and re-witnessing receipts to preserve historical DAG commitments across cryptographic
      migrations.

5. Missing Differentiation from Established Prior Art in the Introduction (Section 1)
    - Potential Concern: While Section 1 clearly defines the operational conjecture, epistemic
      status, and review methodology, it presents the Qn universe and QnIR in relative isolation
      from closely related systems in programming languages, compilers, and verifiable computing
      that address overlapping subsets of the same problem. Without this context, it is
      difficult to evaluate which aspects of the SIMEMP constitution require a bespoke numeric
      and symbolic stack versus which re-terminologize established mechanisms:
      - *Provably Terminating / Resource-Metered Bytecodes:* eBPF (RFC 9669, which enforces
        bounded loops and static verification on a 64-bit register VM) and fuel-metered
        Deterministic WebAssembly (Wasm).
      - *Content-Addressed Code & AST Stores:* Unison (content-addressed functional definitions
        keyed by structural hashes for \mathcal{O}(1) equivalence and immutable versioning) and
        equality-saturation engines (egg / egglog).
      - *Cryptographic Execution Receipts:* Zero-Knowledge Virtual Machines (zkVMs) such as RISC
        Zero and Cairo, which emit cryptographic *Receipts* providing \mathcal{O}(1)
        verification of private execution traces without trusting the sender's hardware to
        enforce contracts.
      - *Exact / Drift-Free Arithmetic:* Posit arithmetic (which eliminates signed zeros and
        rounding drift via exact *quire* accumulators) and constructive real packages (e.g.,
        iRRAM / Boehm's constructive reals).
    - Suggested Improvement: Add a brief comparative subsection or table in Section 1 situating
      the Qn architecture against these four paradigms, explicitly articulating why the
      simultaneous combination of residual-conserving Diophantine arithmetic, content-addressed
      DAGs, and intent-contract governance necessitates a new intermediate representation rather
      than an extension of existing IRs.


H4: Minor Corrections and Typos

- Page 1 (Title Page, Subtitle & Footer): The front matter alternates between uppercase CDQN
   ("specification of the CDQN / Qn universe") and lowercase cdqn ("cdqn — Chained and
   Distributed Quang Numbers"). Standardizing the casing convention early on Page 1 (e.g., cdqn
   for the distributed coordination layer/repository and Qn for the core universe) will improve
   consistency.

- Page 3 (Draft status box): "Milestone 2, status pending" uses small caps for STATUS PENDING,
   whereas lifecycle states elsewhere use uppercase CANDIDATE / SEALED; consider standardizing
   status typography.

- Page 4 (Section 2.1, Heading): "2.1 Dependencies Determinism" uses an awkward plural noun
   adjunct in English; consider "2.1 Dependency Determinism" (while noting if Dependencies
   Determinism is retained as a fixed glossary key).

- Page 6 (Section 3.1, Stage 3): In \mathcal{S}_{\text{seed}} =
   \text{KDF}(\mathcal{B}_{\text{raw}} \parallel \text{Digest}(\text{HostContext})),
   \mathcal{B}_{\text{raw}} appears without being introduced in Stage 1. Explicitly labeling the
   harvested bitstream as \mathcal{B}_{\text{raw}} in Stage 1 (N_{\text{sample}} \ge 4096 bits)
   makes the notation self-contained.


H3: [Numeric Substrate (Qm), Symbolic CAE-DAG (Qexpr), and Discrete Geometry/Calculus] (Pages:
    7-14)

H4: Potential Mistakes and Improvements

1. Missing Inductive Construction Beyond Q(9), 7-Smooth Monoid Claim for \mathcal{Q}^+, and
   Unspecified Resolution Function \kappa(z) (Sections 3.1–3.5, PDF pp. 7–9 / Doc pp. 6–8;
   Addressing QO-4)
    - Definitive Gap: Section 3.1 (PDF p. 7 / Doc p. 6) defines the single-digit artifacts
      Q(1)\dots Q(9) inductively via the unary Successor Morphism \mathcal{S}(x) = x + Q(1)
      along d_1, and Table 1 (PDF p. 8 / Doc p. 7) caps Q(9) as the "Single-digit radix-10
      ceiling; decimal normalization boundary" (DCC_ANCHOR_Q9_RADIX_v1). Immediately afterward,
      Section 3.3 (Eq. 3, PDF p. 8 / Doc p. 7) introduces the Qm numeric leaf tuple \langle
      \sigma, q_z, z, r_z, d \rangle assuming arbitrary unsigned magnitudes q_z \in \mathbb{N},
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zoom levels z \in \mathbb{N}, and an unspecified strictly monotonic resolution function \kappa(z) defining \delta_z(d) = Q(1)_d / \kappa(z), without formally constructing multi-digit naturals beyond Q(9) or fixing the functional form of \kappa(z). Furthermore, Section 3.5 (PDF p. 9 / Doc p. 8) states that "the positive cone \mathcal{Q}^+ is a free commutative multiplicative monoid generated by the primes \{Q(2), Q(3), Q(5), Q(7)\}." By the Fundamental Theorem of Arithmetic, the multiplicative monoid generated by \{2, 3, 5, 7\} consists solely of 7-smooth numbers (2^a 3^b 5^c 7^d), which omits every integer with a prime factor \ge 11 (such as 11, 13, 17, 22) and is not closed under addition (Q(9) + Q(2) = 11 \notin \langle 2,3,5,7 \rangle_{\times}).

- Constructive Resolution under BOC (Response to QO-4):
  1. *Why Wholesale Import and Pure Unary Chains Fail BOC:* Importing \mathbb{N} and \mathbb{Q} wholesale as unconstructed external primitives violates Axiom 1 (explicit Universal Envelope), Axiom 2 (finiteness, excluding actual infinities), Axiom 4 (local genesis dependency back to Q(0)), and the No-Implicit Rule (Section 2.4). Conversely, extending unary successor composition \mathcal{S}^{(n)}(Q(0)) to arbitrary q_z \in \mathbb{N} incurs \mathcal{O}(q_z) causal lineage depth and \mathcal{O}(q_z) Landauer creation work, causing exponential blowup relative to bit-length and violating Tier-2 Efficiency.
  2. *Canonical Radix-Anchor Polynomial / Horner Morphism for \mathbb{N}_{\le H_{\max}}:* Under the Best of Choices (BOC) policy, construct the bounded natural semiring \mathbb{N}_{\le H_{\max}} inside the artifact model via binary Horner-step / Euclidean radix composition anchored to DCC_ANCHOR_Q2_PARITY_v1 (or word-aligned 2^8 limbs via DCC_ANCHOR_Q8_OCTET_v1):
     \text{Cons}_B(q_{\text{high}}, d_{\text{low}}) = \big(q_{\text{high}} \times B\big) + d_{\text{low}}, \quad B \in \{Q(2), Q(8)\}, \quad d_{\text{low}} \in \{Q(0), \dots, B - Q(1)\},
     with \text{Parent}(q) = \{B, q_{\text{high}}, d_{\text{low}}\}. Canonical uniqueness (preventing leading-zero aliases so hash-consing equality \mathcal{H}(A) = \mathcal{H}(B) holds) is guaranteed by requiring q_{\text{high}} \neq Q(0) (the innermost seed of any non-zero integer is Q(1)) and collapsing \text{Cons}_B(Q(0), Q(0)) \to Q(0) via Q(bypass). Every integer of logarithmic height h(q_z) = \lceil \log_2(q_z + 1) \rceil \le H_{\max} then possesses a unique, acyclic Axiom-4 causal lineage of depth \mathcal{O}(\log_2 q_z) rooted in \{Q(0), Q(1), Q(2)\} and lowers isomorphically to native machine words (UINT64 / multi-precision limbs) under HCA-Collapse with zero radix-conversion overhead, while decimal radix-10 formatting via Q(9) and Q(10) \equiv \text{Cons}_{Q(2)}(Q(5), Q(0)) remains strictly an interface projection. Section 3.5 should also be revised to state that \{Q(2), Q(3), Q(5), Q(7)\} forms the *Category A single-digit prime anchor basis* rather than generating the entire multiplicative monoid \mathcal{Q}^+.
  3. *Construction of Negative \mathbb{Q}:* Because Section 3.3 defines orientation \sigma \in \{\ominus, \odot, \oplus\} via the reflection involution \mathcal{I}_d (OP_SIGMA) and forbids signed zeros, and Section 4.3 provides OP_RATIO for rational pairs, negative rationals require no separate ring construction: they are canonically represented by composing \sigma \in \{\ominus, \odot, \oplus\} with coprime positive GCD-reduced OP_RATIO pairs (p, q) \in \mathbb{N}_{\le H_{\max}} \times \mathbb{N}_{\le H_{\max}} where q > Q(0) and \gcd(p, q) = Q(1) (enforced via OP_DIV).
  4. *Parameterizing \kappa(z) in the DCC Profile:* Leaving \kappa(z) unfixed makes leaf tuples incomparable across nodes and operations. Under BOC, \kappa(z) must be bound explicitly inside the active DCC profile (DCC_ref in Eq. 7) to one of three canonical schedules: (a) dyadic scaling \kappa_2(z) = Q(2)^z (mapping zoom shifts to exact hardware bit-shifts), (b) decimal scaling \kappa_{10}(z) = (Q(2)\times Q(5))^z (for human financial/decimal contracts), or (c) factorial/primorial scaling \kappa_{\text{fact}}(z) = (z+1)! (when finite-zoom absorption of arbitrary small rational denominators is required).

2. Expressiveness, Canonical Uniqueness, and Layer Stratification of the Zoom/Remainder Telescope vs. Exact Rationals and Constructive Reals (Sections 3.3–3.4, Eqs. 3–4, PDF pp. 8–9 / Doc pp. 7–8; Addressing QO-3)
   - Definitive Expressiveness Distinction (Response to QO-3): Equation (4) defines scale expansion via the telescoping cascade:
     r_z = \big(q_{z+1} \cdot \delta_{z+1}(d)\big) + r_{z+1}, \qquad
     \mathcal{I}_{\text{total}}(X_d) = \sum_{k=0}^z \big(q_k \cdot \delta_k(d)\big) + r_z.
     1. *Vs. Exact Rational Arithmetic (\mathbb{Q}):* For any fixed radix resolution \kappa(z) = B^z (such as B=2 or B=10), a rational number p/q achieves Terminal Exactness (r_z = Q(0)) at a finite zoom z \in \mathbb{N} if and only if every prime factor of the reduced denominator q divides B (B-adic rationals). Whenever q contains a prime factor coprime to B (e.g., Q(1)/Q(3) or Q(1)/Q(7) under dyadic or decimal zoom), scale expansion (Eq. 4) alone will *never* reach r_z = Q(0) at any finite z; it will perpetually leave a non-zero residual r_z > 0 until PRECISION_INSUFFICIENT or BUDGET_EXHAUSTED is emitted. In contrast, standard rational arithmetic over \mathbb{Q} (OP_RATIO) resolves \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = Q(1) and exact linear-algebraic null-space constraints (such as chemical stoichiometry \mathbf{S}\cdot \vec{v} \equiv Q(0) or Gaussian elimination pivot checks) in polynomial time with zero residual. Conversely, in deep iterative rational chains \sum_{k=1}^m p_k / q_k without GCD reduction or convergent truncation, denominator logarithmic height h(u/v) grows linearly in m and trips H_{\max} (RECEIPT_EXPRESSION_TOO_COMPLEX).
     2. *Vs. Constructive Reals with a Cost Model:* Because every Qm leaf tuple is restricted to a finite pre-declared zoom ceiling z \le z_{\max} and height ceiling H_{\max} under Axiom 2, if r_z in a Layer-2 leaf tuple is merely a passive scalar integer bound 0 \le |r_z| < \delta_z(d), then Eq. (4) reduces to projective fixed-point rational grid refinement and is strictly weaker than constructive reals. When paired with Layer-3 symbolic generators (OP_RATIO, OP_ROOT, OP_CF) and demand-driven zoom refinement (OP_ZOOM), the hybrid system is equivalent in expressiveness to resource-bounded (fuel-bounded) symbolic-interval / constructive real arithmetic (e.g., Boehm's hybrid

symbolic/Cauchy-stream reals or Vuillemin's continued-fraction arithmetic with explicit modulus of convergence \delta_z and fuel ceiling z_{\max}), and strictly weaker than unrestricted Type-2 Effectivity streams. Furthermore, it fails on *exact zero-testing and equality decidability* (A = B) across syntactically distinct algebraic/transcendental expressions not present in the finite 15-rule OP_BYPASS table (e.g., deciding whether \sqrt{2}\cdot\sqrt{3} - \sqrt{6} = 0 or (\sqrt{2}+\sqrt{3})^2 - (5+2\sqrt{6}) = 0). Whereas rational arithmetic over \mathbb{Q} decides p_1/q_1 = p_2/q_2 via p_1 q_2 = p_2 q_1, comparing two equal irrational generators with distinct DAG hashes via Eq. (4) produces overlapping constructive intervals [V, V+\epsilon_z] at every finite zoom z, deterministically halting with ORDERING_INDETERMINATE_AT_ZOOM (or PRECISION_INSUFFICIENT at z_{\max}).

- Potential Concerns (Cumulative vs. Incremental Quotients and Layer Stratification of r_z):
  1. *Cumulative vs. Incremental Quotients:* In Eq. (3) and Section 3.4's interval formula, a single leaf tuple at zoom z carries only one integer q_z (representing \sigma(q_z \delta_z(d) + r_z)), whereas Eq. (4) writes \mathcal{I}_{\text{total}}(X_d) as a sum over all historical zoom levels \sum_{k=0}^z (q_k \cdot \delta_k(d)) + r_z. If q_z in Eq. (3) is only the incremental digit at step z (0 \le q_{k+1} < \kappa(k+1)/\kappa(k)), then a single leaf tuple does not uniquely determine the number without its k=0\dots z-1 ancestors; conversely, if q_z = \lfloor |X_d| / \delta_z(d) \rfloor is the cumulative quotient at scale \delta_z(d), the summation in Eq. (4) double-counts coarser scales. Distinguishing incremental digits d_k in Eq. (4) from the consolidated quotient q_z = \sum_{k=0}^z d_k \frac{\kappa(z)}{\kappa(k)} in Eq. (3)—and specifying the zoom-alignment morphism when adding tuples with z_A < z_B—will make the canonical tuple representation self-contained.
  2. *Layer-2 vs. Layer-3 Dependency of r_z:* For r_z in a Layer-2 leaf tuple to conserve an irrational (\sqrt{2}, \pi) or unaligned rational (1/3) remainder without truncation, r_z must hold a commitment hash \mathcal{H}(\alpha_{\text{rem}}) pointing to a Layer-3 Qexpr generator (OP_RATIO, OP_ROOT, OP_CF). Without clarification, having Layer-2 leaf tuples contain Layer-3 Qexpr references appears to invert the strict upward functorial hierarchy \mathcal{F}: \mathbf{C}(L) \to \mathbf{C}(L+1) of Section 4.1 (PDF p. 10 / Doc p. 9). Explicitly typing r_z in Layer 2 as an opaque commitment handle resolved only by crossing back to Layer 3 during OP_ZOOM resolves this circularity.

3. Confluence, Termination, Cost-Prediction Circularity, and W=0 Claims in Qexpr CAE-DAG Rewriting and Reuse (Sections 4.3–4.4, Eqs. 6–8, PDF pp. 11–12 / Doc pp. 10–11; Addressing QO-5)
   - Definitive Theoretical Constraint on OP_REWRITE (Response to QO-5): Section 4.3 (PDF p. 12 / Doc p. 11) groups two fundamentally different rewriting mechanisms under OP_REWRITE: (a) the 15-rule canonical \mathcal{O}(1) algebraic identity bypass suite (Q(bypass)), and (b) general e-graph equality saturation.
     1. *The 15-Rule Q(bypass) Suite is Terminating and Confluent:* Oriented strictly from left to right (A + Q(0) \to A, A \times Q(1) \to A, A \times Q(0) \to Q(0), 0! \to Q(1), 1! \to Q(1), 2! \to Q(2), A^{Q(0)} \to Q(1) for A \neq Q(0), A^{Q(1)} \to A, Q(1)^k \to Q(1), Q(0)^k \to Q(0) for k > 0, \sqrt[n]{Q(1)} \to Q(1), R R^\dagger \to Q(1), \vec{e}_i \wedge \vec{e}_i \to Q(0), (I_2)^2 \to -Q(1), \mathbf{d} \circ \mathbf{d} \to Q(0)), every rule strictly reduces expression node count |t| by at least 1. By Newman's Lemma, because this finite orthogonal term-rewriting system terminates in \le |t| steps and all critical pairs join to unique normal forms (provided A \neq Q(0) is guarded on A^{Q(0)} \to Q(1)), Q(bypass) is strongly normalizing and globally confluent within \mathcal{O}(|\text{DAG}|) work.
     2. *General Equality Saturation is Neither Terminating Nor Confluent Under an Arbitrary Budget:* Once bidirectional structural axioms such as commutativity (DCC_SYM_COMMUTATIVE), associativity (a+b)+c \leftrightarrow a+(b+c), and distributivity a(b+c) \leftrightarrow ab + ac are added to an e-graph (Tate et al. [26]; egg / egglog), the e-graph can grow exponentially or infinitely. When halted by Section 4.4's circuit breakers (node ceiling N_{\max} via RECEIPT_BUDGET_EXHAUSTED or depth ceiling D_{\max} via RECEIPT_EXPRESSION_TOO_COMPLEX), the resulting partially saturated e-graph—and the extracted DAG (since optimal DAG extraction with shared sub-expressions from an e-graph is NP-hard)—depends on the rule-application schedule and extraction heuristic. Thus, while OP_REWRITE is total in the Axiom-6 sense (it always halts with a receipt), two nodes with different budgets or schedules will extract different trees and compute different hashes \mathcal{H}(\alpha) (Eq. 7), breaking \mathcal{O}(1) hash-consing equivalence and H(S_t \mid \mathcal{D}(S_t)) = 0.
     3. *Actionable Separation for QO-5:* Split OP_REWRITE into two distinct operations: (1) OP_BYPASS (Canonical Contraction) restricted to the confluent, linear-time 15-rule size-reducing subset plus lexicographic child sorting for DCC_SYM_COMMUTATIVE, which governs canonical node hashing \mathcal{H}(\alpha); and (2) OP_SATURATE (Bounded Equality Saturation) inside an explicit optimization workflow (Q(rsi)) whose DCC profile fixes a deterministic rule-application schedule (e.g., fixed-epoch breadth-first Datalog scheduling), an iteration ceiling K_{\max}, an e-node ceiling N_{\max}, and a deterministic polynomial-time greedy tree-cost extractor with hash tie-breaking, emitting a new versioned artifact (Axiom 13) rather than mutating canonical node identity.
   - Definitive Logical Circularity in the Q(reuse) Breakeven Inequality (Eq. 8): Equation (8) (PDF p. 12 / Doc p. 11) states that sub-expression reuse is admitted if and only if \text{Admit}(Q(\text{reuse})) \iff \Delta W_{\text{compute}} > \Delta W_{\text{lookup}} + \epsilon_{\text{overhead}}. Two operational circularities prevent this inequality from being evaluated as written:
     1. *Lookup Precedes Breakeven Check:* In the dispatch precedence Q(bypass) \prec Q(reuse) \prec Direct Compute, if Q(reuse) probes the store by the *evaluated value* hash, the computation must already be executed before its hash is known; if it probes a memoization table keyed by the *unevaluated expression* hash \mathcal{H}(\alpha_{\text{expr}}) (Eq. 7), computing \mathcal{H}(\alpha_{\text{expr}}) and probing the store already expends \Delta W_{\text{lookup}} before the runtime knows

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      whether a cached result exists.
    2. *Unmeasured Prediction of \Delta W_{\text{compute}}:* For variable-step operations
      (OP_DIV, OP_ROOT, OP_CF, OP_FOLD), the true dynamic work \Delta W_{\text{compute}} is
      unknown prior to execution, and relying on an unmeasured heuristic predictor violates
      the No-Implicit Rule.
    - *Actionable Fix:* Define Q(reuse) as a lookup in a content-addressed memoization map
      \mathcal{H}(\alpha_{\text{expr}}) \mapsto \langle \mathcal{H}(\alpha_{\text{val}}),
      \Delta W_{\text{recorded}} \rangle, and either evaluate Eq. (8) against the
      deterministic static opcode lower bound \Delta W_{\min}(\alpha_{\text{expr}}) =
      \sum_{\text{op} \in \alpha_{\text{expr}}} c_{\text{op}} known at DAG-lowering time or
      against the historical realized compute work \Delta W_{\text{recorded}} explicitly
      stored in Block 6 (Metric envelope) of the sealed Universal Envelope at OP_COMMIT time.
  - Potential Concern (Thermodynamic Contradiction in W = 0 Symbolic Normalization Claims):
    Sections 4.3 (PDF p. 12 / Doc p. 11) and 5.2 (PDF p. 14 / Doc p. 13) assert that symbolic
    normalization and nilpotency/rotor cancellation inside the CAE-DAG are "reversible and
    therefore claims zero Landauer dissipation (W = 0)." However, rules such as A \times Q(0)
    \to Q(0), \vec{e}_i \wedge \vec{e}_i \to Q(0), and \mathbf{d} \circ \mathbf{d} \to Q(0)
    are *many-to-one (logically irreversible)* projections that erase the operand A or
    \vec{e}_i unless the uncollapsed parent history is retained in memory. Moreover, computing
    256-bit SHA-3/BLAKE3 hashes (Eq. 7) and probing memory tables are both logically
    irreversible sponge compressions and physically dissipative (\mathcal{O}(CV^2 f) > 0) on
    Layer-0 CMOS hardware. To maintain consistency with Section 2.2, distinguish between *zero
    precision-collapse work* (\Delta W_{\text{collapse}} = 0 and zero logical uncertainty
    drift \Delta H_{\text{logical}} = 0) and *strictly positive DAG construction and
    hash-lookup work* (\Delta W_{\text{DAG}} > 0) billed against the node budget.

4. Underspecified Semantics in the \mathcal{R}-Algebra: Diophantine Division, Interval Ordering,
   Even Roots of Negatives, and Collatz Dynamics (Sections 3.4–3.5, Eq. 5, PDF p. 9 / Doc p. 8)
   - Definitive Ambiguity in Diophantine Division Remainder Orientation (Eq. 5) and Separation
     from DPRM: Equation (5) defines division as finding C, R such that A = (B \times C) + R
     with B \neq Q(0), 0 \le |R| < |B|, and \sigma_C = \sigma_A \otimes \sigma_B. Because 0 \le
     |R| < |B| constrains only the unsigned magnitude of R without fixing its orientation
     \sigma_R, Eq. (5) admits two distinct solutions whenever B \nmid A and negative
     orientations are involved (Euclidean floor division with \sigma_R \in \{\odot, \oplus\}
     vs. truncated division toward zero with \sigma_R \in \{\odot, \sigma_A\}; moreover, when
     |A| < |B| in truncated division, C = Q(0) so \sigma_C must collapse to \odot rather than
     \sigma_A \otimes \sigma_B). To satisfy Dependencies Determinism (H(S_t \mid
     \mathcal{D}(S_t)) = 0), Eq. (5) must explicitly fix \sigma_R and guard \sigma_C = \odot
     when C = Q(0). Furthermore, linear Euclidean integer division is computable in polynomial
     time \mathcal{O}(\log^2|A|) and should be clearly separated in the text from general
     non-linear Diophantine polynomial equations (OP_ROOT / Hilbert's Tenth Problem), where
     Northcott's height ceiling h(u/v) \le H_{\max} applies. Exhausting a Northcott height box
     of size \mathcal{O}(e^{2 H_{\max}}) buys *termination* (RECEIPT_BUDGET_EXHAUSTED), not
     mathematical decidability; emitting NO_SOLUTION when a search merely exhausts H_{\max}
     conflates bounded search failure with proof of algebraic impossibility.
   - Definitive Flaw in Constructive Interval Endpoint Formula (Section 3.4): Positional
     ordering is specified as evaluating constructive intervals I(A) = [q_A \delta_z + r_A, q_A
     \delta_z + r_A + \epsilon_z]. However, in Eq. (3), q_A \in \mathbb{N} is an *unsigned*
     magnitude (missing the orientation sign \sigma_A), and r_A is already the exact conserved
     residual bounded by 0 \le |r_A| < \delta_z. Placing the exact residual r_A inside the
     interval endpoints while also adding the zoom uncertainty width \epsilon_z = \delta_z
     double-counts the remainder. Prior to expanding r_A, the constructive enclosure interval
     at zoom z determined by (\sigma_A, q_A, z) and the status of r_A should be:
     I_z(A) = \begin{cases} [q_A \delta_z,\, (q_A + 1)\delta_z) & \text{if } \sigma_A = \oplus
     \text{ and } r_A \neq Q(0), \\ \big(-(q_A + 1)\delta_z,\, -q_A \delta_z\big] & \text{if }
     \sigma_A = \ominus \text{ and } r_A \neq Q(0), \\ [\sigma_A q_A \delta_z,\, \sigma_A q_A
     \delta_z] & \text{if } r_A = Q(0). \end{cases}
   - Definitive Mathematical Issue (Higher Even Roots of Negative Scalars for n \ge 4): Section
     3.4 states that "an even root of a negative scalar cannot exist in 1D (no \sigma with
     \sigma \otimes \sigma = \ominus) and is promoted to a dimension-2 bivector blade \vec{e}_1
     \wedge \vec{e}_2 in Clifford algebra." While this holds for square roots (n = 2) in
     Euclidean signature \mathcal{C}\ell_{2,0} where (\vec{e}_1 \wedge \vec{e}_2)^2 = -Q(1), it
     fails for higher even roots n = 4, 6, 8, \dots: any pure bivector blade \beta(\vec{e}_1
     \wedge \vec{e}_2) satisfies (\beta \vec{e}_1 \wedge \vec{e}_2)^4 = +\beta^4 Q(1) \ge Q(0),
     which is positive and cannot equal a negative scalar A. For n = 4, \sqrt[4]{-Q(1)} =
     \frac{Q(1) \pm \vec{e}_1 \wedge \vec{e}_2}{\sqrt{2}} is a mixed-grade rotor (spinor) in
     the even subalgebra Q(1) \oplus (\vec{e}_1 \wedge \vec{e}_2) \cong \mathbb{C} coupled with
     the algebraic root anchor \sqrt{2}, and specifically requires adjoining an orthogonal axis
     with \eta_{i,i}\eta_{j,j} = +1 (since in Lorentzian signatures \mathcal{C}\ell_{1,1} the
     spacetime bivector squares to +Q(1)).
   - Potential Concern (Epistemic Modality of the Collatz Claim on \mathcal{Q}^+): Section 3.5
     states unconditionally that "the Collatz map exhibits a single attractor cycle (Q(4) \to
     Q(2) \to Q(1)) on \mathcal{Q}^+." While the three negative cycles (at -1, -5, -17) are
     finite verifiable orbits, universal convergence to a single cycle on \mathcal{Q}^+ is an
     unproven mathematical conjecture. Under the corpus's own epistemic firewall (Section 6.1),
     global convergence on \mathcal{Q}^+ must be tagged as MODAL_HYPOTHESIS or explicitly
     qualified as verified up to a bounded height ceiling H_{\max}.

5. Denominator Bit-Growth in Iterated Rational Cayley Rotors, Blade Scaling at n \le 16, DEC
   Metric Discretization Error, and Anchor-Cut Audit Soundness (Sections 5.1–5.4, Eq. 9, PDF pp.
   13–15 / Doc pp. 12–14)
   - Definitive Logarithmic Height Growth in 10^9 Cayley Rotor Applications (Section 5.2, Eq.
     9): Section 5.2 defines the rational Cayley rotor R = \frac{(m^2 - n^2) + 2mn(\vec{e}_1
     \wedge \vec{e}_2)}{m^2 + n^2} with R R^\dagger \equiv Q(1) and notes that "the rotor can
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be applied 10^9 consecutive times without normalization cycles." While R R^\dagger \equiv Q(1) holds algebraically so zero *floating-point norm drift* occurs, applying a non-trivial primitive Pythagorean rotor (e.g., (m, n) = (2, 1) where m^2 + n^2 = 5) consecutively k times in the same direction to rotate a vector \vec{v} computes \vec{v}^{(k)} = R^k \vec{v} (R^\dagger)^k. Because m^2 - n^2 + 2mni and m^2 + n^2 are coprime in \mathbb{Z}[i] for primitive Pythagorean pairs (and by Niven's theorem, non-trivial rational-tangent angles are never rational multiples of \pi, so R^k is never periodic), no denominator cancellation occurs: after k exact steps, the reduced denominator grows as (m^2 + n^2)^k on the spinor and (m^2 + n^2)^{2k} on the vector, so the logarithmic height h(\vec{v}^{(k)}) grows linearly as \mathcal{O}(k \log(m^2 + n^2)). At k = 10^9 steps with m^2 + n^2 = 5, the exact integer numerators and denominators require \ge 10^9 \log_2(5) \approx 2.32 \times 10^9 bits (\approx 290\text{ MB} to 580\text{ MB} per coordinate), which will deterministically trip the Northcott height ceiling H_{\max} (RECEIPT_EXPRESSION_TOO_COMPLEX) long before 10^9 steps unless intermediate states are periodically collapsed onto the zoom lattice \delta_z (accumulating a spatial residual r_z) or re-approximated by bounded-height Pythagorean pairs (accumulating an angular residual r_\theta).

- Potential Concern (Dense vs. Grade-Sparse Multivector Product Scaling at n \le 16, Sections 5.1–5.2): Section 5.2 states that geometric products scale as \mathcal{O}(2^n) in blade count and declares typical ceilings of n \le 16 for Qs semantic frames (2^{16} = 65,536 blades). While a multivector has 2^n basis blades, multiplying two *dense* general multivectors requires 2^n \times 2^n = 4^n scalar operations (4^{16} \approx 4.29 \times 10^9 exact Qm tuple operations per product at n = 16). To keep n \le 16 feasible within finite budgets, Sections 5.1–5.2 should explicitly specify that high-dimensional semantic frames restrict operands to *grade-sparse blades and versors* (e.g., grade-1 noun vectors and grade-2 bivector rotors, where basis size is \binom{n}{1} + \binom{n}{2} = 136 \ll 65,536).
- Potential Concern (Topological Exactness vs. Metric Hodge Star Discretization Error in DEC, Section 5.3): Section 5.3 accurately states that on an oriented simplicial complex K, the discrete exterior derivative \mathbf{d} satisfies \mathbf{d} \circ \mathbf{d} \equiv Q(0) and discrete Stokes' theorem \sum_K (\mathbf{d}\omega)(K) \equiv \sum_{\partial K} \omega(\partial K) holds identically with zero truncation error. However, extending this zero-error claim to the discrete Hodge star \star\omega = \sum_{\sigma_k} \frac{|\star\sigma_k|}{|\sigma_k|} \langle \omega, \sigma_k \rangle (\star\sigma_k), the Laplacian \Delta = \mathbf{d}\boldsymbol{\delta} + \boldsymbol{\delta}\mathbf{d}, and constitutive laws (\mathbf{d}\star J \equiv Q(0)) conflates *metric-free topological coboundary exactness* with *metric operator approximation*. In Discrete Exterior Calculus (Desbrun et al. [25]), the diagonal circumcentric Hodge star is a finite-volume spatial approximation carrying \mathcal{O}(h) or \mathcal{O}(h^2) mesh discretization error relative to the continuous physical manifold. Moreover, on general 2D/3D triangulations, simplex volumes |\sigma_k| involve Euclidean square roots (OP_ROOT) rather than pure rationals (OP_RATIO), and non-well-centered (obtuse) simplices place circumcenters outside their simplices, producing negative dual volumes |\star\sigma_k| < 0 that violate metric positive-definiteness of \Delta. The specification should require mesh well-centeredness as a DCC precondition on OP_HODGE and explicitly separate topological exactness on K from spatial discretization error.
- Potential Concern (Cryptographic Provenance vs. Trajectory Soundness in \mathcal{O}(K) Anchor-Cut Verification, Section 5.4): Section 5.4 states that verifying an N-step simulation requires only the Anchor-Cut chain (origin, K signed anchor receipts, terminal state), collapsing audit complexity from \mathcal{O}(N) to \mathcal{O}(K) with K \ll N. However: (i) in a sequential hash chain where each state S_i commits only to its immediate parent \mathbf{P}(S_i) = \{\mathcal{H}(S_{i-1})\}, an auditor given only checkpoints S_0 and S_m (m = N/K > 1) cannot verify by hashing alone that S_m is a descendant of S_0 unless S_m embeds a Merkle skip-list accumulator over the intermediate steps; and (ii) checking K cryptographic signatures and macroscopic anchor contracts at checkpoints only verifies *hardware signature custody and endpoint invariant conservation* (e.g., constant discrete energy or \mathbf{d}F \equiv Q(0) at the K cuts), not *computational transition soundness* (that S_m actually resulted from valid step-by-step execution of the N/K omitted micro-steps rather than an arbitrary state jump on the same invariant shell). Section 5.4 should clarify whether \mathcal{O}(K) verification audits macroscopic conservation + signature provenance or integrates succinct cryptographic execution proofs (e.g., STARK/SNARK transition proofs).

H4: Minor Corrections and Typos

- PDF p. 8 / Doc p. 7, Section 3.3: The calligraphic symbol \mathcal{I} is used both for the reflection involution \mathcal{I}_d (\mathcal{I}_d^2 = \text{id}) and three lines lower in Equation (4) for the total scale-expanded magnitude \mathcal{I}_{\text{total}}(X_d). Using distinct symbols will avoid notation collision.

- PDF p. 9 / Doc p. 8, Section 3.4, Paragraph 2: The logarithmic height ceiling h(u/v) = \max(\log |u|, \log |v|) \le H_{\max} is undefined when the numerator is u = Q(0) (\log 0 is undefined); write h(u/v) = \max(\log \max(|u|, 1), \log \max(|v|, 1)) for coprime u, v.

- PDF p. 9 / Doc p. 8, Section 3.4, Paragraph 3: In "h(A^B) = B \cdot h(A) \le H_{\max}", note that monoidal fold exponentiation A^B = \prod_{j=1}^B A requires B to be an exact non-negative integer (z=0, r_B = Q(0)) and h(A) to be evaluated on the reduced coprime form of A. In the same paragraph, "2! = 2" should be written "Q(2)! = Q(2)" (or "2! \to Q(2)" matching PDF p. 12 / Doc p. 11).

- PDF pp. 11–12 / Doc pp. 10–11, Section 4.3, Equations (6)–(7) & Paragraph 2: Equation (7) hashes the concatenation \text{OpCode} \parallel k \parallel \mathcal{H}_1 \parallel \dots \parallel \mathcal{H}_k \parallel z \parallel r \parallel d \parallel \text{DCC}_{\text{ref}}

   where z, r, d, \text{DCC}_{\text{ref}} can have variable byte lengths; specify that k = |\mathbf{D}_{\text{children}}| is the arity prefix and that \parallel denotes deterministic length-prefixed (e.g., canonical CBOR tuple) serialization so variable-width fields cannot create concatenation boundary collisions. In the Q(bypass) list in Paragraph 2, add the guard A \neq Q(0) to A^{Q(0)} \to Q(1) so it does not conflict with Q(0)^k \to Q(0) or Section 3.4's rule that 0^0 is indeterminate without DCC_ZERO_EXPONENT_v1, and write \mathcal{I}_d(Q(1)) instead of standard minus notation in "(I_2)^2 \to -Q(1)".

- PDF p. 13 / Doc p. 12, Section 5.1, Paragraph 2: The Hodge duality formula M^* = M I^{-1} requires the unit pseudoscalar I = \vec{e}_1 \wedge \dots \wedge \vec{e}_n to be invertible (I^2 = \pm Q(1)), which holds only when the signature \eta = (p, q, s) is non-degenerate (s = 0). Since Paragraph 1 explicitly admits degenerate signatures (s > 0, where I^2 = Q(0)), note that M^* = M I^{-1} is restricted to s = 0 frames.

- PDF p. 14 / Doc p. 13, Section 5.4, Paragraph 2: In the lineage tuple \mathcal{L}(S_k) = \langle \mathbf{P}(S_k), \kappa(S_k), \tau_{\text{type}}, \mathcal{W}_{\text{cum}}(S_k), \Pi_{\text{receipts}}(S_k) \rangle, the symbol \kappa(S_k) denotes the monotonic causal index, colliding with the zoom resolution function \kappa(z) defined in Section 3.3 (\delta_z(d) = Q(1)_d / \kappa(z)). Using distinct notation (e.g., \kappa_{\text{causal}} vs. \kappa_{\text{zoom}}) will prevent ambiguity.

H3: [Domain Projections (Qs, Qlog, Qphy, Qchem, Qbio) and Distributed Governance] (Pages: 15-20)

H4: Potential Mistakes and Improvements

1. Cross-Node Receipt Trust vs. Local-First Genesis (Section 7.1, Equation 10; Response to QO-13):
    - Potential Concern: Section 7.1 defines a Road Receipt in Equation (10) as \mathcal{R}_{\text{road}} = \langle \mathcal{H}(S_{\text{in}}), \mathcal{M}, \mathcal{H}(S_{\text{out}}), \Delta W, \pi, \sigma \rangle. Because silicon origins are strictly local (\forall A \neq B \implies Q(0)_A \neq Q(0)_B) and nodes never broadcast internal expression graphs, receiving nodes cannot re-execute flat transactions and instead verify that \mathcal{R}_{\text{road}} satisfies shared Q(anchor) contracts in \mathcal{O}(1) bandwidth. Two structural issues arise from this design:
      1. *Cryptographic Soundness of \pi:* The formal algebraic or cryptographic structure of the proof field \pi is left unspecified. If \pi is merely a metadata attestation signed by \sigma, a compromised, faulty, or Byzantine node holding valid Inner/Outer-Ring PQC keys can sign an arbitrary state transition \mathcal{H}(S_{\text{in}}) \to \mathcal{H}(S_{\text{out}}) and fake work ledger \Delta W claiming compliance with morphism contract \mathcal{M} without having executed the governed transition. This violates Dependencies Determinism (H(S_{\text{out}} \mid \mathcal{D}(S_{\text{out}})) = 0) across network boundaries.
      2. *Disjoint Commitment Hashes:* Because every local leaf tuple and CAE-DAG node hashes its local lineage back to Q(0)_A, identical deterministic computations on nodes A and B yield disjoint commitment hashes \mathcal{H}(S_{\text{out}})_A \neq \mathcal{H}(S_{\text{out}})_B, preventing a receiving node from directly verifying state transitions or deduplicating shared sub-expressions across peers.
    - Actionable Resolution (Answering QO-13): Contract-verification-plus-receipt-trust is not sound on its own against Byzantine or corrupted nodes unless \pi carries verifiable computational integrity. To make \mathcal{R}_{\text{road}} sound without exposing private local graphs (Axiom 8) or introducing a global zero, the specification should define three mechanisms:
      1. *Origin-Quotiented Canonical Projection for Export:* Even though Q(0)_A \neq Q(0)_B as cryptographic identity envelopes, deterministic QnIR computations over Q(anchor) contracts are pure successor/morphism functions of their local origin. Specify that the Exposure Functor \mathcal{E}_{\text{export}} : \mathbf{C}_N \to \text{Attestations}_{\text{cdqn}} quotients out the local root Q(0)_A \mapsto \widehat{Q(0)} (treating the origin as a formal canonical symbol during pure Qm evaluation and binding the node-specific hardware seed only in the envelope's lineage block) prior to computing public commitments \mathcal{H}_{\text{pub}}(S_{\text{in}}) and \mathcal{H}_{\text{pub}}(S_{\text{out}}). Any two nodes executing the same Q(pattern) over identical relative inputs will then produce identical portable commitments, making Anchor Grounding Class I transitions verifiable by forced re-derivation on any peer.
      2. *Succinct Zero-Knowledge Execution Proofs (\pi):* Because QnIR v0.1 is a small, fixed-width, 30-opcode deterministic machine over integer arithmetic with bounded loops and no floating point, its execution trace can be directly arithmetized into an Algebraic Intermediate Representation (AIR) inside a transparent zk-STARK proof (analogous to RISC Zero or Cairo receipts). Formalizing \pi in Eq. (10) as a succinct STARK execution receipt proves that a valid local witness trace rooted at Q(0)_A transformed \mathcal{H}(S_{\text{in}}) to \mathcal{H}(S_{\text{out}}) under contract \mathcal{M} within work \Delta W, allowing any receiver to verify the transition in \mathcal{O}(\text{polylog } N) time with zero trust in the sender and zero disclosure of the private graph (preserving Axiom 8).
      3. *Optimistic Interactive Anchor-Cut Bisection Audit Right:* For lower-overhead edge deployments where full zk-STARK generation exceeds the thermodynamic breakeven, \pi commits the Merkle root of the K intermediate OP_ANCHORCUT witnesses, and receivers hold a protocol-enforced Interactive Bisection Challenge Right: upon ingesting \mathcal{H}(\mathcal{R}_{\text{road}}), a receiver can challenge any segment [\kappa_i, \kappa_{i+1}] in \mathcal{O}(\log K) rounds down to a single origin-quotiented QnIR instruction step for deterministic local replay in \mathcal{O}(1) cost. Failure to respond or a hash mismatch transitions the receipt to QUARANTINED or REVOKED (RECEIPT_ANCHOR_REVOKED).

2. Unmitigated Semantic Drift at the Candidate Parser Boundary and Intent Contract Semantics (Section 6.1, Table 3; Response to QO-11):
   - Potential Concern: Section 6.1 models vocabulary in Clifford geometric algebra over semantic dimensions $\mathcal{D}_{\text{semantic}}$ (nouns as grade-1 vector blades $\vec{v}_{\text{noun}}$, modifiers as rational Cayley rotors $\vec{v}' = R\vec{v}_{\text{noun}}R^\dagger$, and transitive verbs as conserved relational morphisms $\text{Action} = (\text{Delegation} \times \text{Actor}) + r_{\text{social}}$), relying on the Epistemic Anchorage Spectrum (Table 3) and DCC_INTENT_CONTRACT_v1 as an anti-hallucination firewall. Three design gaps exist at this boundary:
     1. *Pre-Formalization Parser Drift:* When a human principal issues a natural-language instruction, translating that utterance into a structured CAE-DAG and DCC_INTENT_CONTRACT_v1 relies on an upstream statistical or neural candidate parser across the Axiom 14 boundary. If the parser hallucinates or inverts the human's actual intent while attaching a valid human origin signature $Q(0)_{\text{author}}$ and well-formed MODAL_EMPIRICAL or MODAL_CONSTRUCTIVE tags, all downstream SIMEMP gateways, Triadic Grounding checks ($\mathbf{C}_{\text{Qs}} \otimes \mathbf{C}_{\text{Qm}} \otimes \mathbf{C}_{\text{Qphy}}$), and QnIR anti-drift verifications ($H(\text{Intent} \mid \text{Execution}) = 0$) will validate the *wrong* intent because the error occurred prior to formalization.
     2. *Underspecified Normative Algebra for r_{social}:* Representing $\text{Action} = (\text{Delegation} \times \text{Actor}) + r_{\text{social}}$ as a Diophantine division constraint and modifiers as unitary norm-preserving rotors ($R\vec{v}R^\dagger$) does not specify how multi-clause legal/safety obligations, deadline expirations, or secondary reparations are encoded into the residual r_{social} once a primary obligation is partially violated.
     3. *Dimensional Ceiling vs. Open Vocabulary:* Section 5.2 restricts Qs semantic frames to $\text{Dim}_{\max} \le 16$ (since Clifford blade count scales as $\mathcal{O}(2^n)$). Allocating distinct basis blades $\vec{e}_k$ to open-vocabulary nouns would immediately exhaust the 16-dimensional ceiling.
   - Actionable Resolution (Answering QO-11):
     1. *Formal Semantics for Intent Contracts:* Anchor DCC_INTENT_CONTRACT_v1 in Resource-Bounded Dyadic Deontic Logic (or Input/Output Contract Logic) with Contrary-to-Duty (CTD) obligations, composed with Dynamic Epistemic Logic (DEL) for multi-agent state updates. In CTD deontic contract calculi, primary obligations $\mathcal{O}_{\text{actor}}(\alpha \mid W_{\text{allowed}})$, permissions $P(\alpha)$, and prohibitions $F(\alpha)$ carry explicit work budgets and are paired with reparational secondary obligations $O(\beta \mid \neg\alpha)$ that take effect when $O(\alpha)$ is unfulfilled. This gives $r_{\text{social}} \neq Q(0)$ a rigorous logical type: a persistent CTD obligation sub-DAG that remains in the lineage until discharged by a matching receipt.
     2. *Typing and Receipting Parser Error Modes:* Because no statistical parser guarantees $H(\text{Intent}_{\text{human}} \mid \text{Utterance}) = 0$, bar any candidate DAG produced by a natural-language parser from entering directly at MODAL_EMPIRICAL or MODAL_CONSTRUCTIVE; type it initially no higher than MODAL_HYPOTHESIS until it passes a mandatory Deterministic Canonical Round-Trip Paraphrase & Co-Signing Handshake (where a deterministic verbalizer renders the formal DCC_INTENT_CONTRACT_v1 alongside its budget W_{allowed} back into an unambiguous controlled natural language template requiring the human principal's cryptographic Inner-Ring co-signature prior to transitioning to SEALED). Add explicit parser-boundary failure receipts to the closed catalog: RECEIPT_SEMANTIC_AMBIGUITY_INDETERMINATE / RECEIPT_INTENT_PARSE_AMBIGUOUS (when multiple non-isomorphic candidate DAGs fit the prompt within the BOC threshold, analogous to ORDERING_INDETERMINATE_AT_ZOOM), RECEIPT_INTENT_MODALITY_MISMATCH / RECEIPT_INTENT_SCHEMA_MISMATCH, and RECEIPT_ROUNDTRIP_CONFIRMATION_REJECTED (when the principal rejects the canonical read-back).
     3. *Semantic Frame Dimensions vs. Lexical Payloads:* Explicitly state in Section 6.1 that the ≤ 16 basis dimensions of $\mathcal{D}_{\text{semantic}}$ represent fixed *normative and thematic axes* ($d_{\text{intent}}, d_{\text{consent}}, d_{\text{obligation}}, d_{\text{temporal}}, d_{\text{agent}}, \dots$), while open-vocabulary lexical identifiers reside as content-addressed commitments inside the blade coefficients rather than allocating a new Clifford basis vector per vocabulary word.

3. Underspecified Constructive Substructural Logic in Qlog and Latency Harvester Timing Channel (Section 6.2; Response to QO-12):
   - Potential Concern: Section 6.2 defines Qlog ($\mathbf{C}_{\text{Qlog}} = \mathbf{C}_{\text{Qm}} \otimes \mathbf{C}_{\text{Qs}}$) by rejecting the Law of Excluded Middle (LEM) in favor of Verified Computed Patterns (Q(pattern) and Q(workflow)), while using Q(bypass) for $\mathcal{O}(1)$ identity short-circuits at $\Delta W = 0$. However, standard intuitionistic logic (IPC) permits unrestricted *Structural Weakening* ($A, B \vdash A$, silently discarding a resource or residual) and *Structural Contraction* ($A \vdash A \wedge A$, duplicating a resource at zero work cost), both of which violate SIMEMP's Metric invariant, Landauer erasure accounting, and residual conservation (Axiom 9). Furthermore, Section 6.2's *Memory Wall Latency Harvester* schedules register-pinned background work (equality saturation, speculative candidate auditing, anchor route pre-computation) during physical DRAM stall windows with $\mathcal{O}(1)$ preemption. Because DRAM stall durations depend on physical bus traffic, thermal state, and microarchitectural cache contention (Layer 0), allowing DRAM-stall-driven equality saturation to directly alter which Q(reuse) or Q(bypass) nodes are available to the main thread violates Dependencies Determinism ($H(S_t \mid \mathcal{D}(S_t)) = 0$, Equation 1) and conflicts with Section 4.4's requirement for constant-time side-channel immunity.
   - Actionable Resolution (Answering QO-12):
     1. *Target Logic Identification and Reconciliation:* Explicitly identify Qlog as Resource-Bounded Intuitionistic Linear / Affine Logic (BILL) (an affine/linear resource-indexed sequent calculus variant of Girard's Bounded Linear Logic / Quantitative Type Theory indexed by work budgets $W \in \mathbb{N}$ under the

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      Curry–Howard–Qn correspondence). Reconcile the classical-looking moves by noting that
      reflexivity (A \vdash_0 A) and algebraic ring identities (A + Q(0) \to A) in Q(bypass)
      are valid constructive axioms in both intuitionistic and linear logic at \Delta W =
      0—not exclusively classical moves—because they act as the identity morphism \text{id}_A
      on an *already constructed* commitment hash \mathcal{H}(A) requiring zero state
      mutation. By contrast, LEM (A \vee \neg A) and Double Negation Elimination (\neg\neg A
      \vdash A) require deciding whether a witness trajectory exists for an arbitrary
      uncomputed proposition A without expending work; under a finite budget
      W_{\text{budget}}, an undecided proposition resolves to INCONCLUSIVE or
      BUDGET_EXHAUSTED, which is why LEM fails while A \vdash A holds.
    2. *Sequent Rules for Q(patterns):* Define a metered judgment \Gamma
      \vdash_{W}^{\kappa,\mathcal{D}} P : A \multimap B \;\triangleright\;
      \mathcal{R}_{\text{road}} (under linear resource context \Gamma and DCC profile
      \mathcal{D} at causal index \kappa and work W, pattern P consumes linear resource A to
      construct B, emitting receipt \mathcal{R}_{\text{road}}):
       - Identity (Q(bypass)):
       \dfrac{}{\mathcal{H}(A) \vdash_{0}^{\kappa,\mathcal{D}} \text{id}_A : A \multimap A
      \;\triangleright\; \texttt{RECEIPT\_BYPASS\_IDENTITY}}
       - Sequential Composition / Pattern Cut (Q(workflow) chaining):
       \dfrac{\Gamma \vdash_{W_1}^{\kappa,\mathcal{D}} P_1 : A \multimap B \;\triangleright\;
      r_1 \qquad \Delta, B \vdash_{W_2}^{\kappa',\mathcal{D}} P_2 : B \multimap C
      \;\triangleright\; r_2}{\Gamma, \Delta \vdash_{W_1 + W_2}^{\max(\kappa,
      \kappa')+1,\mathcal{D}} P_1 ; P_2 : A \multimap C \;\triangleright\; [r_1, r_2]} \quad
      (W_1 + W_2 \le W_{\text{budget}})
       - Parallel Tensor Composition (Composite Domain / Triadic Grounding \mathbf{C}_1
         \otimes \mathbf{C}_2):
       \dfrac{\Gamma \vdash_{W_1}^{\kappa,\mathcal{D}} A \qquad \Delta
      \vdash_{W_2}^{\kappa',\mathcal{D}} B}{\Gamma \otimes \Delta \vdash_{W_1 +
      W_2}^{\max(\kappa, \kappa')+1,\mathcal{D}} A \otimes B}
       - Bounded Contraction (Q(reuse) Exponential Modality !_{\mathcal{H}} A): Unrestricted
         duplication is forbidden; contraction is permitted only when A is sealed in the
         content-addressed store (OP_COMMIT) under \mathcal{H}(A) and satisfies the breakeven
         condition (Equation 8):
       \dfrac{\Gamma, !_{\mathcal{H}} A, !_{\mathcal{H}} A \vdash_{W}^{\kappa,\mathcal{D}} C
      \qquad \Delta W_{\text{compute}}(A) > \Delta W_{\text{lookup}}(A) +
      \epsilon_{\text{overhead}}}{\Gamma, !_{\mathcal{H}} A \vdash_{W + \Delta
      W_{\text{lookup}}}^{\kappa+1,\mathcal{D}} \text{reuse}(A) : C}
       - No Silent Weakening (Residual Conservation): Premises carrying non-zero residuals
         (r_z \neq Q(0) or r_{\text{social}} \neq Q(0)) cannot be silently dropped; discarding
         an unconsumed resource requires an explicit dissipative sink rule emitting
         RECEIPT_REMAINDER_TRUNCATION at cost W_{\text{erase}} \ge k_B T \ln 2.
    3. *Deterministic Latency Harvester Isolation:* Specify that (i) the Memory Wall Latency
      Harvester is disabled when a DCC profile mandates constant-time side-channel immunity,
      and (ii) in standard profiles any speculative rewrites or candidate audits computed
      during DRAM stalls are written strictly to an isolated candidate staging buffer that
      becomes visible to the foreground thread's Q(reuse) store \mathcal{S} exclusively at
      statically declared causal-index checkpoints (\kappa_{\text{commit}}), keeping the
      execution trace invariant to physical DRAM timing.

4. Legacy Provenance Bottlenecks in the Anti-Poisoning Ingestion Firewall (Section 7.3; Response
   to QO-14):
    - Potential Concern: Gate 3 of the Anti-Poisoning Ingestion Firewall (Section 7.3) requires
      anchor-grade historical records to be supported by an Independence Quorum of k \ge 2
      sources with *lineage-disjoint root subtrees* (\text{Root}(s_1) \cap \text{Root}(s_2) =
      \emptyset), arguing that citation laundering cannot manufacture independent origins
      because laundering only adds descendants to a single poisoned root. For legacy and
      digitized history prior to the contemporary sensor horizon, however, strict Layer-0
      silicon root disjointness (Q(0)_A \neq Q(0)_B) fails in both directions:
      - *False Negative Independence (Single-Custodian Bottleneck):* A single national archive
        or institutional scanning center (Q(0)_A) digitizing thousands of genuinely independent
        physical manuscripts (or holding the only surviving manuscript of an ancient treaty,
        k_{\text{custody}} = 1) stamps all of them with one Layer-0 hardware root and will fail
        Gate 3, forcing almost all pre-digital human history into QUARANTINED.
      - *False Positive Independence (Pre-Digital Citation Laundering):* Two independent
        scanning nodes (Q(0)_A \neq Q(0)_B) digitizing two 19th-century reprints that both
        copied a single forged or biased medieval chronicle will produce completely disjoint
        cryptographic Axiom-4 trees in the Qn universe, falsely satisfying Gate 3.
    - Actionable Resolution (Answering QO-14): For legacy and digitized history, strict
      single-tier root disjointness is not satisfiable as a proxy for historical independence.
      To keep the firewall honest without discarding recorded history, introduce a three-part
      degradation protocol across the Historical Horizon:
      1. *Bifurcate Provenance into Custodial Lineage vs. Epistemic Stemmatic Lineage:* Split
        the lineage block of Historical Anchor Artifacts into \mathcal{L}_{\text{custody}}
        (which Layer-0 scanner/institution onboarded and signed the digital bytes,
        k_{\text{custody}}) and \mathcal{L}_{\text{epistemic}} / k_{\text{tradition}} (the
        authored philological *stemma codicum* / witness tree modeling pre-digital source
        transmission prior to the Historical Horizon). Require k_{\text{custody}} \ge 2
        independent institutional attestation signatures over \mathcal{L}_{\text{custody}} to
        certify digitization integrity *and* a vertex-disjoint min-cut k_{\text{stemma}} \ge 2
        (by Menger's theorem) over \mathcal{L}_{\text{epistemic}} to certify source independence
        before promoting a deep-history claim.
      2. *Strict Pre-Horizon Modality Ceiling:* Couple Gate 3 directly to Gate 2's Modality
        Ceiling by enforcing that no record originating prior to the Historical Horizon (lacking
        contemporary Layer-0 sensor receipts at the time of the event) can ever attain
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MODAL_EMPIRICAL (G = 1.0) from textual digitization alone; unless corroborated by contemporary Class P sensor receipts (e.g., radiometric dating or astronomical retro-calculation via Class I calendar contracts), cap multi-witnessed archival reconstructions at MODAL_HYPOTHESIS (0.3 < G \le 0.85) or a dedicated intermediate sub-modality (MODAL_ARCHIVAL_RECONSTRUCTION / MODAL_HISTORICAL_RECONSTRUCTION).
3. *Graded Quarantine Separation:* Reserve the fault-level QUARANTINED state strictly for Gate 4 (Sheaf Gluing) when two or more receipted records actively contradict each other on shared spatiotemporal landmarks or have broken hash chains. Route single-custodian or single-manuscript legacy records (k_{\text{custody}} = 1 or k_{\text{tradition}} = 1) into a graded provenance status (SINGLE_WITNESS_BOUNDED / SEALED_CUSTODIAL_SINGLETON capped at MODAL_HYPOTHESIS, 0.3 < G \le 0.5) that records the bottleneck hash in the envelope while permitting read-only agent citation with explicit uncertainty propagation and barring promotion to Class S global anchor status until independent corroboration is attached.

5. Chronosa Sheaf Gluing vs. Asynchronous FLP Impossibility and Deductive Inference (Section 7.2):
   - Potential Concern: While Section 7.2 explicitly labels Chronosa a VISION PROPOSAL, two foundational theoretical tensions should be addressed before PoC Phases C–D:
     1. *CALM Theorem vs. FLP Impossibility:* Chronosa posits a deterministic t-of-n threshold PQC quorum verifying the sheaf gluing condition across asynchronous nodes without universal clocks. By the CALM theorem (Consistency As Logical Monotonicity) and FLP impossibility [32], coordination-free deterministic convergence in an asynchronous clockless network holds *if and only if* state transitions are monotonic over a join-semilattice. While append-only receipt union is monotonic, threshold quorum sealing of non-monotonic transitions—such as capability revocation (REVOKED), budget exhaustion, or conflict quarantine (QUARANTINED)—cannot guarantee both safety and liveness in an asynchronous network without partial synchrony (bounded epoch timeouts).
     2. *Sheaf Cohomology (Consistency) vs. Inference:* Computing the topological colimit / global section of a cellular sheaf (0-th cohomology H^0) checks whether local sections agree on overlaps, and the sheaf/Tarski Laplacian (or 1-st cohomology H^1 \neq 0) detects coboundary obstructions when local witnesses disagree. However, a sheaf colimit is a consistency filter, not an inference engine: when local nodes disagree, H^0 = \emptyset, and gluing alone does not generate new abductive hypotheses, plans, or Q(rsi) syntheses.
   - Actionable Resolution:
     1. *Separate Monotonic CRDT Gluing from Partial-Synchrony Quorums:* Explicitly partition Chronosa's Outer-Ring operations by monotonicity: restrict clock-free, uncoordinated sheaf gluing to the append-only, monotonic join-semilattice of immutable receipts and narrowing interval constraints (which converges via CRDT lattice joins without violating FLP), and require an explicit partial-synchrony epoch bound (via a Calendar Projection Contract) whenever a t-of-n quorum must finalize a non-monotonic revocation, state cut, or quarantine resolution.
     2. *Reframe Sheaf Gluing as Cohomological Obstruction Detection:* Clarify in Section 7.2 that sheaf cohomology serves as the Distributed Consistency and Conflict-Detection Layer (H^0 global section extraction and H^1 \neq 0 obstruction detection that automatically triggers Gate 4 QUARANTINED challenge receipts), while actual deductive/abductive inference is executed by distributed Qlog workflows (BOC abduction and inverse causal reconstruction, Section 6.2) proposing candidate sections *over* the glued sheaf state.

6. Mathematical, Thermodynamic, and Physical Gaps in the Scientific Domain Projections (Sections 6.3, 6.4, and 6.5):
   - Potential Concern:
     1. *Discrete Symplectic Rollouts vs. Rational Height Growth and Mesh Discretization (Section 6.3, Qphy):* Section 6.3 claims that discrete symplectic morphisms over DEC cochains conserve phase-space volume (Liouville) with "zero artificial numerical dissipation" over billion-step rollouts (N = 10^9). First, standard explicit time-stepping on a simplicial mesh is not automatically symplectic unless formulated as a discrete variational integrator or staggered leapfrog shear map. Second, in exact rational arithmetic (Qm), applying an unprojected rational symplectic map (such as the Cayley rotor, Equation 9) at every step causes the logarithmic height h(u/v) of numerator and denominator to grow as \mathcal{O}(N), which will deterministically hit the height ceiling H_{\max} (RECEIPT_EXPRESSION_TOO_COMPLEX) well before 10^9 steps unless trajectories evolve as lattice automorphisms on a fixed zoom grid \delta_z. Third, the *Thermodynamic Double Leash* (\Delta E_{\text{kinetic}} + \Delta E_{\text{potential}} \le E_{\text{battery}}) does not include a spatial/temporal mesh discretization error term \epsilon_{\text{mesh}}(\delta_z) between the discrete DEC solution and continuous physical hardware.
     2. *Catalytic Q(bypass) Energy Accounting and Subgraph Matching (Section 6.4, Qchem):* Section 6.4 formalizes chemical catalysis as Q(bypass) emitting RECEIPT_CATALYTIC_BYPASS in \mathcal{O}(1) dispatch. However, in Section 4.3 and Table 5, Q(bypass) is an algebraic identity short-circuit with zero work (\Delta W = 0, e.g., A + Q(0) \to A). In chemistry, a catalyst lowers the kinetic activation barrier E_a^{\ddagger}, but the reaction still performs a non-identity transformation with non-zero free-energy change \Delta G \neq Q(0). Moreover, while hash-consing gives \mathcal{O}(1) equality for *identical* whole molecules, detecting a catalytic functional group inside a larger substrate molecule is a subgraph isomorphism problem (NP-complete in general) unless reactive sites are pre-factored.
     3. *Float-Free Variational Free Energy and Physical vs. Computational U-Unit Calibration (Section 6.5, Qbio):* Section 6.5 states that organisms minimize Friston's variational free energy F and calibrates Q(atp) at \Delta G_{\text{ATP}} \approx 20\,U per hydrolysis event. First, variational free energy F = D_{\text{KL}}(q(\vartheta) \parallel p(\vartheta, y)) requires probability distributions and logarithms; how F is evaluated over discrete Qm tuples without continued-fraction node blowup is not

specified. Second, at T = 300\text{ K}, 1\,U = k_B T \ln 2 \approx 2.87 \times 10^{-21}\text{ J} (Section 2.2), whereas physiological ATP hydrolysis (\Delta G_{\text{ATP}} \approx 50\text{ kJ/mol} \approx 8.3 \times 10^{-20}\text{ J}/molecule) equals \approx 20\,k_B T (in natural thermal units / nats) or \approx 28.9\,U (in Landauer bits k_B T \ln 2, or \approx 17.7\,U under standard-state \Delta G^{\circ\prime}_{\text{ATP}} \approx 30.5\text{ kJ/mol}), indicating an omitted \ln 2 conversion factor. Furthermore, using the single symbol U for single-molecule biochemical free energy (U_{\text{phys}} = k_B T \ln 2) and host silicon instruction execution (U_{\text{comp}}, which dissipates 10^7\text{–}10^{10}\,k_B T \ln 2 per bit operation on CMOS hardware) conflates simulated domain energy with host compute budget.
  - Actionable Resolution:
    1. In Section 6.3, specify that billion-step symplectic rollouts use reversible integer/lattice symplectic integrators (Störmer–Verlet shear steps / discrete variational integrators at fixed zoom quantum \delta_z so bit-height h remains bounded), and add an explicit mesh-discretization margin \epsilon_{\text{mesh}}(\delta_z) to the Thermodynamic Double Leash (\Delta E_{\text{kinetic}} + \Delta E_{\text{potential}} + \epsilon_{\text{mesh}}(\delta_z) \le E_{\text{battery}}).
    2. In Section 6.4, reclassify catalysis from a zero-cost algebraic identity Q(bypass) (\Delta W = 0) to a Macro-Hop Pattern Substitution (Q(reuse) / OP_REWRITE with reduced intermediate height ceiling H_{\max}^{\text{cat}} < H_{\max}^{\text{uncat}}) that preserves the exact net thermodynamic ledger \Delta G \le Q(0), and specify that \mathcal{O}(1) dispatch matches pre-factored canonical *Reaction-Center Sub-DAG Commitments* rather than running runtime subgraph isomorphism. Also express Avogadro's constant N_A \equiv 602{,}214{,}076 \times 10^{15}\text{ mol}^{-1} in canonical base-neutral factorized form (602{,}214{,}076 \times Q(2)^{15} \times Q(5)^{15}) to match Layer-2 algebraic rules.
    3. In Section 6.5, explicitly distinguish the Simulated Physical Quantum U_{\text{phys}} \equiv k_B T \ln 2 from the Host Logical Compute Unit U_{\text{comp}} (calibrated to physical silicon joules via a hardware-specific factor \eta_{\text{silicon}} in the Historical Calibration Benchmark); reconcile the \ln 2 factor in Q(atp) (specifying either \approx 29\,U_{\text{phys}} at physiological 50\text{ kJ/mol} or standard-state \Delta G^{\circ\prime}_{\text{ATP}} \approx 30.5\text{ kJ/mol} \approx 18\,U_{\text{phys}} as an exact rational interval); and formulate discrete variational free energy over dyadic rational frequency distributions (q_i = m_i / Q(2)^k anchored to DCC_ANCHOR_LN2_v1), reducing Shannon/KL surprisal calculations to exact integer bit-shifts plus bounded rational sums.

7. Versioning Amplification vs. Contiguous Locality in the Five-Row Data Structure Contract Schema (Section 7.4):
   - Potential Concern: Section 7.4 specifies that every Qn data structure (Q(array), Q(matrix), Q(table), record tuple, persistent DAG-deduped tree) obeys Axiom 13 (immutability: growth or modification emits a new versioned artifact and never mutates in place) and must declare "versioning amplification, and a declared breakeven rule" in its Work contract row. If a contiguous container such as Q(array) or dense Q(matrix) of size M is lowered naively as a flat immutable buffer hashed as a single node, a single-element update incurs \mathcal{O}(M) memory copy and \mathcal{O}(M) SHA-3 re-hashing work (\mathcal{O}(M) versioning amplification) across the Memory Wall. Conversely, if it is lowered as a pointer-free persistent tree, it risks losing contiguous cache locality required for NPU/systolic Matrix-Multiply-Accumulate lowering (Section 4.4).
   - Actionable Resolution: Explicitly state in Section 7.4 how the Footprint, Locality, and Work contract rows reconcile Axiom 13 immutability with hardware locality: (i) for persistent cross-step containers, lower Q(array), Q(matrix), and Q(table) via structural indirection to bounded-radix persistent Hash-Array Mapped Tries (HAMTs) or RRB-trees / Merkle B-trees with branching factor B aligned to the octet/cache-line anchor (e.g., B = Q(8) or Q(2)^5 = 32), bounding single-element versioning amplification to \mathcal{O}(\log_B M) hash-consed node allocations; and (ii) for dense linear-algebra kernels (OP_FOLD, OP_FRAME_MAP), amortize versioning amplification to \mathcal{O}(1) per batch by operating over transient register/scratchpad tiles that seal and emit a single new root commitment hash only at OP_COMMIT boundaries.

H4: Minor Corrections and Typos

- Page 16, Section 6.1, line 6 & Section 6.2, line 11: The acronym Q(rsi) and "emergent RSI modes" are used in Sections 6.1 and 6.2 before being defined in the body text (the expansion "recursive self-optimization workflow" only appears later in Appendix A, Page 33). Define the acronym at its first occurrence on Page 16.

- Page 16, Section 6.3, line 16 (Missing CANDIDATE Tag): The text states that the five physical anchors "are calibrated from the Historical Calibration Benchmark" without marking its epistemic status, whereas the invited objection on Page 17 (item iv) and Appendix A (Page 32) explicitly mark the Historical Calibration Benchmark as a CANDIDATE (†) artifact. Append (CANDIDATE) in Section 6.3 for consistency.

- Page 17, Section 7.2, line 13 & line 14 (Set Union Subscript and Unindexed Citations): In \text{Lineage}(\text{Chronosa}) = \bigcup_N \text{Lineage}(Q(0)_N), the symbol N is used simultaneously as the dummy index variable and the set/count of nodes; write \bigcup_{i=1}^N \text{Lineage}(Q(0)_i) or \bigcup_{n \in \mathcal{N}} \text{Lineage}(Q(0)_n). Immediately below, parenthetical author citations (Grothendieck, Mac Lane, Spivak) and (Halpern) are used instead of the numbered bracketed reference style [1–37] used throughout the rest of the paper (where Spivak is [31], and Grothendieck, Mac Lane, and Halpern–e.g., J. Y. Halpern and Y. Moses, *"Knowledge and Common Knowledge in a Distributed Environment,"* *JACM*, 1990–are omitted from the References list).

H3: [QnIR v0.1 Candidate Specification: State Model, ISA, Memory, Encoding, and Totality] (Pages: 20-27)

H4: Potential Mistakes and Improvements

1. ISA Arity and Operand-Encoding Mismatches in the 64-Bit Instruction Format (Sections 8.4 and 8.6, Table 5)
   There is a structural incompatibility between the operand requirements of several opcodes in Table 5 (pages 20–22) and the fixed 64-bit instruction word layout defined in Section 8.6 (page 24: opcode: 8, dest: 8, src1: 8, src2: 8, aux: 16, zoom: 8, flags: 8):
   - Multi-arity instructions (OP_TUPLE and OP_FOLD) exceed available source fields: Table 5 defines OP_TUPLE with arity 4 ("Compose \langle \sigma, q, z, r, d \rangle from four operand slots + aux axis") and OP_FOLD with arity k \ge 3 ("Monoidal accumulation (\sum, \prod) over operand vector with explicit trip ceiling aux=m_{\max}"). However, the 64-bit word provides only two 8-bit source operand fields (src1 and src2), Section 8.5 (Rule 1) forbids memory addresses or pointers, and Section 8.6 states that "every word is fixed-width, program size is exactly 64L bits." Even if zoom (8 bits) and aux (16 bits) supply z and d as immediates for OP_TUPLE, three slot operands (\sigma, q, r) remain for two source fields. Similarly, because Requirement R3 (the No-Implicit Rule) forbids unstated register-range conventions, OP_FOLD has no defined encoding to specify k \ge 3 input slots or to select between \sum and \prod when aux is occupied by m_{\max}.
   - Dual-destination and overloaded aux fields (OP_DIV and OP_CF): Table 5 states that OP_DIV writes its "quotient to dest, remainder to aux slot." In Section 8.6, aux is defined strictly as a 16-bit immediate field ("digit index, trip ceiling, CF anchor id, receipt status, frame id"), not a destination register index. Furthermore, if aux is overloaded as a second destination register index for OP_DIV, it leaves no immediate field in the instruction word to declare the internal Diophantine step ceiling that Requirement 1's totality argument (Section 8.7, page 25) requires OP_DIV to carry. Similarly, OP_CF (arity 0) states that "aux selects anchored constant \pi, \tau, e, \ln 2 with step ceiling," packing both the constant selector and the convergent step ceiling into the 16-bit aux field without specifying a bit-partitioning rule.
   - OP_LOOP branch target and loop-counter immutability: Table 5 defines OP_LOOP with arity 2 ("Bounded backward jump: aux declares remaining trips; decrement-and-jump only"). Because Section 8.6 specifies that every instruction word is part of an immutable causal hash chain W_i = \text{Hash}(W_{i-1} \parallel \text{word}_i \parallel \kappa_i) and Axiom 13 forbids in-place mutation, decrementing a loop counter inside aux in place during execution would mutate \text{word}_i and invalidate the stream hash chain. Additionally, the specification does not state how the static backward jump target PC index and the mutable loop-counter state are assigned across dest, src1, src2, and aux.
   - 256-bit hash operands in OP_REUSE and OP_VERIFY: Table 5 states that OP_REUSE (arity 1) retrieves an artifact from \mathcal{S} "by commitment hash" (and Example 2 at PC 2 writes REUSE H(3!)), while OP_VERIFY verifies a receipt-chain segment between two witnesses. Because a cryptographic commitment hash \mathcal{H}(\alpha) is at least 256 bits wide, it cannot fit into an 8-bit src1 field or 16-bit aux immediate unless src1 indexes a slot whose envelope already holds the target hash or aux indexes a pre-declared constant-hash pool in the program header.
   - Actionable Fix for v0.2: Encode multi-operand inputs via an explicit register-window rule (src1 = base slot, src2 = slot count k) or decompose them into binary accumulator steps; make OP_DIV write to adjacent destination slots (dest, dest+1) so aux remains dedicated to the step ceiling; and add an explicit commitment/constant-table section to the CBOR program header so OP_REUSE can index 256-bit hashes via aux.

2. Soundness Gaps in the Totality Argument, Pre-Execution Budget Gating, and Static Cost Discipline (Sections 8.4 and 8.7, Table 5, Requirement 1; Response to QO-8)
   In Section 8.7 (page 25), Requirement 1 argues that every QnIR v0.1 program terminates within W_{\text{budget}} \ge \sum_i c_i \cdot t_i because the stream is finite (L words), OP_LOOP carries a strictly decrementing trip counter, and every opcode has a static cost c_i \ge 0 gated at Stage 1 ("budget gate: if W_{\text{rem}} \le 0, halt with RECEIPT_BUDGET_EXHAUSTED"). Auditing Table 5 and the five-stage dispatch loop reveals three substantive concerns:
   - Pre-execution vs. post-execution budget overdraft in Stage 1: Stage 1 checks only whether W_{\text{rem}} \le 0 *before* executing an instruction, while Stage 4 deducts the opcode's cost c_i *after* executing its semantics. If a program enters an instruction with W_{\text{rem}} = 1\,U and executes an expensive operation—such as OP_POW (cost B), OP_FACT (cost n-1), OP_VERIFY (cost K), or OP_GP (cost 2^n-bounded)—Stage 1 passes, allowing the VM to execute unbudgeted work and drive W_{\text{rem}} negative before halting on the *next* instruction. Stage 1 must pre-check W_{\text{rem}} < c_i^{\max} prior to Stage 4, and iterative internal operations (OP_POW, OP_FACT, OP_CF, OP_DIV, OP_ROOT, OP_GP) must decrement W_{\text{rem}} per internal micro-step.
   - Data-dependent, exponential, and uncalibrated costs in Table 5: Several entries in the "Cost" column of Table 5 are runtime data-dependent variables rather than static constants: OP_POW costs B (the runtime value of the second operand), OP_FACT costs n-1 (the runtime value of the first operand), OP_FOLD costs m, OP_VERIFY costs K, OP_CF is billed "per convergent," and OP_GP / OP_WEDGE are "2^n-bounded." Conversely, OP_ADD (1\,U), OP_MUL (2\,U), OP_DIV (3\,U), and OP_ROOT (4\,U) are billed as tiny constants regardless of the logarithmic height h(q_z) \le H_{\max} of their operands. Inside a bounded loop of T trips, repeated squaring (OP_MUL V1, V1) doubles the bit-length of q_z on every iteration up to H_{\max} while charging only 2\,U per step—conflicting with both the Memory Wall invariant and constant-time lowering (Requirement R13)—and Northcott's theorem guarantees only that the Diophantine search space of height \le H_{\max} is finite, not that its cardinality is \mathcal{O}(1).
   - Zero-cost instructions in loop bodies: OP_QCONST, OP_TERMINAL (on hit), OP_BYPASS (on hit, Stage 2), OP_LOOP, and OP_HALT are assigned a cost of 0\,U. If a program enters an OP_LOOP

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      with W_{\text{rem}} \ge 1 whose body consists entirely of 0\,U operations or Stage 2
      OP_BYPASS hits ("advance \kappa and PC at zero cost"), the loop executes up to 65{,}535
      physical hardware iterations while consuming 0\,U of work, decoupling physical runtime and
      causal index growth (\kappa) from W_{\text{rem}} depletion.
    - Actionable Resolution for QO-8 (Formal Discipline for Static Costs): To make Requirement 1
      a rigorous theorem without relying on unmeasurable runtime predictions, QnIR v0.2 should
      adopt Ceiling-Parameterized Static Cost Polynomials with Lowering Receipt Calibration
      (analogous to Multivariate Amortized Resource Analysis / RaML and static WCET
      verification):
      1. *Derived Parametric Logical Cost Functions (IR / Road Level):* Define each opcode's
         logical cost C_{\text{op}} as a closed-form monotonic polynomial function exclusively of
         the static ceilings declared in the program header (H_{\max}, z_{\max},
         \text{Dim}_{\max}) and the instruction's immediate aux step ceiling S_{\max}—never of
         runtime register values: C_{\text{op}}(\text{opcode}, H_{\max}, z_{\max},
         \text{Dim}_{\max}, \text{aux}). For example, arithmetic costs scale with maximum limb
         width W_{\text{limb}} = \lceil H_{\max} / 64 \rceil (e.g., \mathcal{O}(W_{\text{limb}}))
         for OP_ADD, 2\lceil h_1 h_2 / 64^2 \rceil for OP_MUL), OP_GP reserves
         2^{2\,\text{Dim}_{\max}} (or grade-sparse bound), and search/fold opcodes (OP_DIV,
         OP_ROOT, OP_CF, OP_POW, OP_FACT) pre-reserve their declared aux step ceiling S_{\max} at
         Stage 1 (deducting C_{\text{op}} *prior* to Stage 4 execution and halting immediately
         with RECEIPT_BUDGET_EXHAUSTED if W_{\text{rem}} < C_{\text{op}}, optionally refunding
         unused steps upon early TERMINAL_EXACTNESS). Assign a minimum base cost c_{\text{base}}
         \ge 1\,U to every instruction dispatch (including OP_BYPASS hits and OP_LOOP) so
         W_{\text{rem}} is a strict ranking function for termination.
      2. *Measured Lowering Receipts (HCA-Collapse / Car Level):* At the HCA-Collapse boundary
         (Layer 3 \to Layer 0), each target hardware DCC profile supplies a verified affine
         WCET/energy calibration receipt certifying that the lowered constant-time instruction
         sequence satisfies the parameterized U bound, audited via implementation-feedback
         receipts (Section 7.4).

3. Machine State Model: Fixed \mathbf{V}/\mathbf{B} Slot Files vs. Flattened SSA Value Graph
   (Sections 8.3, 8.5, and 8.6, Equation 11; Response to QO-6)
    In Section 8.3 (page 20) and Section 8.6 (page 24), QnIR v0.1 uses mutable register-like
   slot arrays (\mathbf{V} = [V_0, \dots, V_{v-1}] and \mathbf{B} = [B_0, \dots, B_{b-1}])
   addressed via a unified 8-bit index space (v + b \le 256). This design introduces three
   structural tensions with the canon:
    - Undefined index partitioning and symbolic generator storage: The specification does not
      state how the 256 unified indices are partitioned between \mathbf{V} and \mathbf{B}.
      Furthermore, Section 8.3 states that each slot of \mathbf{V} holds "one exact Qm leaf
      tuple \langle \sigma, q_z, z, r_z, d \rangle," leaving it unspecified where unevaluated
      symbolic generators produced by Class C opcodes (OP_CF continued-fraction generators or
      OP_RATIO rational pairs p/q) reside prior to OP_ZOOM collapse.
    - Dynamic dimension promotion hazard in OP_ROOT: In Table 5, OP_ROOT on an even root of a
      negative scalar specifies: "promote to blade file (d+1)." Because dest is a static 8-bit
      index, dynamically redirecting the output of OP_ROOT from a scalar slot in \mathbf{V} to a
      multivector slot in \mathbf{B} based on the runtime sign of A creates a type/register
      mismatch for subsequent static instructions reading dest, and implicitly instantiating
      axis d+1 without an explicit OP_FRAME instruction violates the No-Implicit Rule
      (Requirement R3 and Section 8.3).
    - Conflict between mutable register slots and Requirements R4, R6, and Axiom 13: Overwriting
      a slot V_k across instructions or OP_LOOP iterations mutates machine state in place
      (contrary to Axiom 13), introduces Write-After-Read and Write-After-Write dependencies not
      present in the Layer-3 CAE-DAG, and means uncommitted values in \mathbf{V}/\mathbf{B} lack
      content-addressed commitment hashes \mathcal{H}(\alpha) unless explicitly copied to
      \mathcal{S} via OP_COMMIT (undermining Requirement R6 and Stage 2/3 hash probes on
      uncommitted slot operands).
    - Actionable Recommendation for QO-6 (Slot-Based State Model vs. Stack-Free SSA Value
      Graph): For QnIR v0.2, replacing the mutable \mathbf{V}/\mathbf{B} slot files with a
      purely stack-free Static Single Assignment (SSA) value graph flattened in topological
      (causal birth) order from the Layer-3 CAE-DAG (with MLIR-style explicit bounded
      region/block parameters solely at OP_LOOP headers and an optional declared liveness window
      W_{\text{live}} \le 256 for hardware register allocation) is strictly superior under the
      BOC policy:
      1. *Exact Isomorphism to CAE-DAG (R11) and Immutability (Axiom 13):* In topological SSA
         form, every instruction at index i \in [0, L-1] implicitly defines one immutable value
         %i carrying a static type tag (QM_TUPLE, BLADE, RATIO, CF_GEN, COCHAIN). Lowering a
         hash-consed CAE-DAG to QnIR becomes a direct \mathcal{O}(V+E) topological sort without
         register allocation or spill code, deferring physical register allocation to
         HCA-Collapse (Layer 0).
      2. *Syntactic Enforcement of Causal Birth Order (R4 & Axiom 5):* In a linear topological
         SSA stream, causal ordering (\text{Index}(\text{Parent}(\alpha)) < \text{Index}(\alpha))
         reduces to the trivial \mathcal{O}(L) load-time check \text{src}_k < \text{PC}.
      3. *Removal of the 256-Slot Ceiling and Unified Index Collision (Objection 16):* Because
         the destination index of instruction i is implicitly i, the explicit 8-bit dest field
         can be eliminated from the 64-bit word, freeing 8 bits to widen operand value indices
         (e.g., to 12 or 16 bits) or encode a third source operand src3.

4. Missing Opcodes, Layer-3 Lowering Completeness, and Exception Semantics (Section 8.4, Table
   5; Response to QO-7)
    - Omission of OP_REWRITE, Conditional Branching, and Category C Container Morphisms:
      Requirement R11 requires a semantics-preserving isomorphism between Layer-3 CAE-DAGs and
      QnIR. However, of the six canonical Layer-3 CAE-DAG node classes defined in Section 4.3
      (OP_RATIO, OP_ROOT, OP_CF, OP_FOLD, OP_FRAME_MAP, OP_REWRITE), general equality saturation
      (OP_REWRITE, beyond the fixed \mathcal{O}(1) identity table of OP_BYPASS) and coordinate
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permutations from OP_FRAME_MAP have no corresponding opcodes in Table 5. Likewise, Section 8.1 states that QnIR is the measurement boundary where the five canonical Category C data structures of Section 7.4 (Q(array), Q(matrix), Q(table), record tuple, persistent DAG-deduped tree) are tested, yet Table 5 contains no opcodes for constructing, indexing, projecting, or functionally updating these containers. Finally, while Table 5 includes OP_CMP and backward OP_LOOP, it lacks any forward conditional jump (OP_BR_FWD) or predicated select (OP_SELECT), making it impossible for a program to act on the result of OP_CMP.

- Actionable Recommendation for QO-7 (Missing Opcodes, Bounded OP_CALL, and Receipt-Typed Exceptions):
  1. *Missing Opcodes for Target Use Cases:* For v0.2, the ISA should add: (a) OP_SELECT (constant-time predicated multiplexer dest = cond ? src1 : src2, preserving the constant-time side-channel immunity of Section 4.4) and bounded forward conditional branch OP_BR_FWD / OP_BR_STATUS (PC <- PC + offset, 1 \le \text{offset} < L - \text{PC}); (b) OP_UNPACK / OP_PROJECT (inverse of OP_TUPLE to project \sigma, q_z, z, r_z, d) and a positional radix-accumulation / limb-extension opcode (OP_RADIX_ACC / OP_EXTEND) to construct multi-digit integers q_z > 9 from OP_QCONST single digits (bridging QO-4 at the ISA level); (c) Clifford reversion/grade projection (OP_REVERSE for R^\dagger in rotor sandwiches R\vec{v}R^\dagger, since OP_ROTOR builds R and OP_GP multiplies, but no opcode computes R^\dagger); (d) container morphisms (OP_STRUCT_GET / OP_CONTAINER_GET, OP_STRUCT_COMMIT / OP_CONTAINER_COMMIT) for Category C structures; and (e) OP_DELEGATE / OP_INGEST_ROAD (OP_INGEST_RECEIPT) to mint sub-budgeted child contexts (W_{\text{sub}} \le W_{\text{rem}}) and anchor external Road Receipts (\mathcal{R}_{\text{road}}) in agent-to-agent workflows.
  2. *Defensible Bounded OP_CALL:* Inlining all reusable Q(pattern) DAGs at flatten time causes exponential code-size blowup (64L bits exceeding N_{\max}) on hierarchical Q(workflow) graphs and destroys content-addressed reuse of verified patterns. Adding a bounded OP_CALL(target_hash, max_depth, budget_slice) / OP_RET pair backed by a static call-depth ceiling C_{\max} declared in the header, a pre-partitioned sub-budget W_{\text{sub}} \le W_{\text{rem}}, and a strictly acyclic call graph (\text{target_id} < \text{caller_id}, verifiable in \mathcal{O}(L) at load time in a non-addressable call-frame stack, analogous to eBPF) preserves totality without code blowup.
  3. *Receipt-Typed Exceptions vs. Explicit OP_HALT:* In v0.1, recoverable epistemic statuses—most notably ORDERING_INDETERMINATE_AT_ZOOM on OP_CMP (which Section 3.4 explicitly states requires "the caller to fund a zoom deepening z \to z + \Delta z")—cannot be handled in-stream because Section 2.5 and Appendix B.1 classify ORDERING_INDETERMINATE_AT_ZOOM as a closed terminal status that halts the program via OP_HALT. Splitting outcomes into Non-Catchable Fatal Receipts (BUDGET_EXHAUSTED, DIVISION_BY_ZERO_REJECTED, REJECTION_BY_GATEWAY, PHYSICAL_VIOLATION—which halt immediately) and Recoverable Status Tokens (ORDERING_INDETERMINATE_AT_ZOOM, PRECISION_INSUFFICIENT, NO_SOLUTION, LOOKUP_MISS) written to the destination value's status tag or handled via a structured forward-only branch (OP_BR_STATUS) allows an agent workflow to fund an OP_ZOOM step (z \to z + 1) and retry within W_{\text{rem}}.

5. Concrete Trace and Register Inconsistencies in the Worked Examples (Section 8.8, Examples 1 and 2, Page 25)
   A line-by-line audit of the worked examples on page 25 against Sections 8.3–8.7 reveals several discrepancies that prevent reproduction:
   - Example 1 (X ÷ Q(2)):
     1. *Missing dividend initialization and undefined slot notation V0':* Section 8.3 hardwires slot V_0 to the local genesis origin \langle \odot, 0, 0, Q(0), d_1 \rangle. However, Example 1 computes X \div Q(2) for X = \langle \oplus, 7, 0, Q(0), d_1 \rangle without any QCONST k=7 instruction loading 7 into a register, and at PC 2 executes DIV src=V0',src2=V1 using an undefined primed slot notation V0'.
     2. *Missing destination fields:* At PC 0 (QCONST k=2), the dest field (V_1) is omitted from the instruction column; at PC 2 (DIV src=V0',src2=V1), both the quotient dest slot and the remainder aux slot are omitted, yet PC 3 reports both C = \langle \oplus, 3, 0, Q(0), d_1 \rangle and R = \langle \oplus, 1, 0, Q(0), d_1 \rangle.
     3. *Mismatch in fault prose:* The paragraph beneath Example 1 states: "Had V_2 contained Q(0), stage 2 of OP_DIV would emit RECEIPT_DIVISION_BY_ZERO_REJECTED..." However, the divisor at PC 2 is in V1 (src2=V1), not V_2; furthermore, in Section 8.7 (page 24), Stage 2 of the dispatch loop is the *bypass probe* (\Delta W = 0), whereas division-by-zero detection and billing of 3\,U occur in Stage 4 (*direct compute*).
   - Example 2 (n! for n = 4):
     1. *Ambiguous operator context in BYPASS V1 vs. automatic dispatch stages:* Even though Table 5 defines a dedicated OP_FACT opcode, Example 2 never invokes OP_FACT. Instead, PC 0 loads Q(4) into V_1, and PC 1 executes BYPASS V1 ("identity table: no hit for 4!"). Because BYPASS V1 is given only the bare scalar Q(4) in V_1, it has no way of knowing the intended operation is a factorial (4!) rather than another unary identity on Q(4) (such as Q(4) + Q(0) or Q(4) \times Q(1)). More broadly, Section 8.7 defines bypass probe (Stage 2) and reuse probe (Stage 3) as automatic internal stages of the five-stage dispatch loop for *every* instruction, whereas Table 5 and Example 2 treat OP_BYPASS and OP_REUSE as standalone explicit instructions.
     2. *Missing destination slots, literal operands, and commitment hash mismatch:* At PC 2 (REUSE H(3!)), the destination slot receiving the fetched 3! artifact is omitted; at PC 3 (MUL V1,3!), the literal 3! is passed instead of a source slot index src2 while leaving the destination slot V3 (referenced at PC 4 by COMMIT V3) implicit; and at PC 4, committing the result of OP_MUL(4, 6) produces the content-addressed hash \mathcal{H}(\text{OP_MUL}, \mathcal{H}(4), \mathcal{H}(3!)) per Equation (7) rather than \mathcal{H}(4!) unless canonical rewrite normalization precedes OP_COMMIT.

6. Static Program Hash-Chaining vs. Dynamic Causal Index \kappa, and Scaling to Concurrency (Sections 8.3, 8.6, and 8.7; Response to QO-10)
   - Static vs. dynamic \kappa_i conflict in W_i: Section 8.6 (page 24) defines the program

word hash chain as W_i = \text{Hash}(W_{i-1} \parallel \text{word}_i \parallel \kappa_i) and treats it as a static encoding property checked at load time before execution begins. However, Section 8.3 defines \kappa as a dynamic runtime counter that increments on every instruction execution *and* every store insertion (OP_COMMIT advances \kappa+1, and OP_LOOP iterations execute the same \text{word}_i across multiple distinct runtime values of \kappa). Conflating the static instruction birth index i \in [0, L-1] with the dynamic execution counter \kappa would force the static binary hash W_i of a loop body to change on every iteration. The specification should explicitly separate the static program stream commitment W_i^{\text{prog}} = \text{Hash}(W_{i-1}^{\text{prog}} \parallel \text{word}_i \parallel i) from the dynamic execution trace hash \mathcal{H}(\text{Trace}) (Section 8.9, stamping runtime \text{PC} and \kappa).

- Actionable Resolution for QO-10 (Scaling \kappa to Multi-Core and Multi-Node Execution): A single sequential scalar counter \kappa(\text{next}) = \kappa(\text{last}) + 1 imposes a strict total order across all instructions and store insertions. On multi-core hardware, a shared atomic scalar \kappa creates a severe cache-coherency bottleneck across the Memory Wall and makes thread interleaving sensitive to bus latency (violating H(S_t \mid \mathcal{D}(S_t)) = 0); across distributed nodes, maintaining a synchronized scalar counter without wall-clock coordination is impossible and fails to distinguish causally dependent events from concurrent independent events (Lamport 1978; FLP 1985). Two extensions resolve this within the SIMEMP constitution:
  1. *Intra-Node Multi-Core (Fork-Join Tree Causal Clocks):* Replace the flat integer \kappa \in \mathbb{N} with a deterministic hierarchical fork-join tree index / Interval Tree Clock (Dewey/pedigree vector \kappa = \langle \kappa_{\text{parent}}, \text{branch_id}, \kappa_{\text{local}} \rangle) paired with structured OP_FORK / OP_JOIN opcodes over data-disjoint SSA subgraphs and pre-partitioned work budgets \sum_{j=1}^m W_{\text{rem}}^{(j)} \le W_{\text{rem}}. Because the content-addressed store \mathcal{S} is write-once and keyed strictly by content hash \mathcal{H}(\alpha) (Equation 7, independent of \kappa), concurrent OP_COMMIT insertions into \mathcal{S} are monotonic set unions that commute over a join-semilattice without locks or data races (satisfying the CALM theorem). At OP_JOIN, branch receipt queues and trace hashes merge in deterministic lexicographical order of branch_id with zero lock contention.
  2. *Inter-Node Multi-Node (Choreographic Programming with Endpoint Projection):* For distributed agent workflows (Q(workflows)), adopt a Choreographic Programming model where global protocols are authored as a causal DAG of Q(patterns) under shared Intent Contracts and compiled via deterministic Endpoint Projection (EPP) into deadlock-free local sequential QnIR streams per node. Cross-node causality is then tracked strictly as a partial-order Merkle-DAG over ingested parent Road Receipt hashes \mathcal{H}(\mathcal{R}_{\text{road}}) in Block 4 (Lineage) of the Universal Envelope, forming a cryptographic vector/Merkle clock with zero wall-clock synchronization.

7. Content-Addressing Commitment Width, Collision-Risk Accounting, Syntactic vs. Semantic Hash Equality, and Anti-Drift Verification Completeness (Sections 8.5 and 8.9; Response to QO-9)
   - Actionable Resolution for QO-9 (Commitment Width, Namespaces, and Collision Receipts):
     1. *Sufficiency of 256-Bit SHA-3:* Under Requirement R6 (\mathcal{A} \equiv \mathcal{B} \iff \mathcal{H}(\mathcal{A}) = \mathcal{H}(\mathcal{B})) and Section 8.5 Rule 2 (idempotent deduplication on hash hit), an undetected hash collision silently aliases distinct values. For 256-bit SHA3-256 (FIPS 202) or BLAKE3-256, the birthday-bound collision probability across N_{\text{total}} artifacts is p_{\text{coll}} \le \frac{N_{\text{total}}^2}{2^{257}}. Under Axiom 2's finiteness, even across a global swarm generating N_{\text{total}} = 2^{64} \approx 1.84 \times 10^{19} to 10^{24} \approx 2^{80} distinct artifacts, p_{\text{coll}} \in [2^{-129}, 2^{-97}], which is vastly below physical hardware cosmic-ray single-event upset (SEU) rates (\sim 10^{-15} per bit-hour) and would require \ge 2^{128} \times 256 \times k_B T \ln 2 \approx 2.5 \times 10^{20}\text{ J} under Landauer's bound to brute-force. Prefixing local commitments with a domain-separated prefix and the node's genesis digest—\mathcal{H}(\alpha) = \text{SHA3-256}(\text{DomainTag} \parallel \mathcal{H}(Q(0)_{\text{node}}) \parallel \text{CanonicalCBOR}(\alpha))—isolates per-node namespaces, whereas stripping \mathcal{H}(Q(0)_{\text{node}}) for shared Class-I Q(anchor) and Q(pattern) schemas preserves \mathcal{O}(1) cross-node contract equality.
     2. *Active Collision Detection and Receipt Typing:* Because OP_COMMIT and OP_REUSE operate on the local store \mathcal{S} where the canonical tuple \alpha = \langle \text{OpCode}, \mathbf{D}_{\text{children}}, z, r, d, \text{DCC}_{\text{ref}} \rangle (Equation 6) is present in memory, the VM does not need to trust hash equality blindly on local hits. On any hash match \mathcal{H}(\alpha_{\text{new}}) = \mathcal{H}(\alpha_{\text{stored}}), the VM should perform a constant-time \mathcal{O}(k) shallow structural comparison of the immediate tuple header (\text{OpCode}, k, \mathcal{H}_1 \dots \mathcal{H}_k, z, r, d, \text{DCC}_{\text{ref}}). If the hashes match but the canonical tuple headers differ (\alpha_{\text{new}} \neq \alpha_{\text{stored}}), the VM must abort deduplication and halt with a dedicated fatal security receipt: RECEIPT_HASH_COLLISION_QUARANTINED / RECEIPT_HASH_COLLISION_DETECTED (mapped to the QUARANTINED terminal status in Axiom 6), sealing both conflicting pre-images to trigger hash-primitive migration under structural indirection (Section 2.4).
     3. *Syntactic vs. Semantic Zoom Normalization Before Hashing (R6):* Requirement R6 states that *"equality of values is equality of commitment hashes."* However, Qm leaf tuples at different zoom levels—e.g., \langle \oplus, 1, z=0, r=Q(0), d_1 \rangle and \langle \oplus, \kappa(1), z=1, r=Q(0), d_1 \rangle—have distinct structural tuples and therefore distinct hashes under Eq. (7) unless tuples with r_z = Q(0) are normalized to a canonical minimal zoom z_{\min} prior to hashing.
   - Anti-Drift Verification Completeness Gap (Section 8.9 and Requirement R12):
     Section 8.9 (pages 25–26) states that a verifier can confirm "that *execution did not drift from declared intent* be checkable from the receipt alone, without replaying the run" (H(\text{Intent} \mid \text{Execution}) = 0, Requirement R12) by recomputing the trace hash from the stream and checking three predicates: (i) every OP_GATE cross validated the intent

contract, (ii) no executed word lies outside the declared stream (trace-hash match), and (iii) all emitted modalities fall within the contract's epistemic whitelist. Two technical issues arise here:

1. *Control-Flow vs. Data-Flow Verification:* First, when a program contains loops (OP_LOOP) or data-dependent exits (OP_TERMINAL, OP_BYPASS, budget exhaustion), a verifier cannot recompute the dynamic trace hash \mathcal{H}(\text{Trace}) from the static L-word stream alone without knowing the runtime trip counts and branch decisions. Second, checking predicates (i)–(iii) without replaying the data-plane operations (or verifying a cryptographic proof of execution) only establishes control-flow and capability conformance—it does not verify computational/data-flow integrity. Without either deterministic replay between OP_ANCHORCUT checkpoints, trusted hardware attestation, or a succinct cryptographic proof of execution (such as a zk-STARK/SNARK proof over the QnIR transition function inside the \pi field of Equation 10), a faulty or Byzantine node could hash a valid sequence of instruction words in \mathcal{O}(L) while fabricating the data values written to \mathcal{H}(S_{\text{out}}).
2. *Conditioning Direction in R12:* Writing H(\text{Intent} \mid \text{Execution}) = 0 inverts the conditioning of Equation (1). Anti-drift requires that execution be uniquely determined by declared intent and dependencies (H(\text{Execution} \mid \text{Intent}, \mathcal{D}) = 0), whereas H(\text{Intent} \mid \text{Execution}) = 0 states that the high-level intent can be uniquely reconstructed from the low-level execution trace (which fails whenever two distinct intents share a common low-level arithmetic sub-routine via Q(reuse)).

H4: Minor Corrections and Typos

- Page 20, Section 8.2 & Page 23, Table 4: Section 8.2 ("8.2 Design requirements") on page 20 is an empty heading with no body text, while its corresponding table (Table 4: QnIR v0.1 design requirements, each traced to its canon ground) floats three pages later onto page 23 after Section 8.4. Moving Table 4 (or adding an introductory pointer sentence) directly under Section 8.2 will restore the reading order before Section 8.3 references "R1–R14."

- Page 21, Table 5 (OP_FACT cost): The cost of OP_FACT is written as n-1 / bypass: 0. Because n - 1 = -1 at n = 0 (even though n \in \{0, 1, 2\} takes the bypass path), writing \max(0, n-1) avoids negative cost expressions.

- Page 25, Section 8.7 vs. Pages 26–28: On page 25, the formal totality statement is labeled "Requirement 1 (Totality of QnIR v0.1 programs)," whereas the QnIR-specific invited objections box on page 26, Objection 15 on page 27, and QO-8 on page 28 refer to it as "Theorem 1." Unify the environment label and cross-references across pp. 25–28 to either Theorem 1 (Candidate) or Requirement 1.

- Page 25, Section 8.8, Example 1: At PC 1, GATE V.hdr uses the syntax V.hdr even though Section 8.3 defines the program header as distinct from the value array \mathbf{V}; at PC 2, DIV src=V0',src2=V1 uses src= instead of src1= (matching the field name in Section 8.6).

- Page 25, Section 8.8, Example 2: In the paragraph immediately below Example 2, "For n \in \{0, 1, 2\} stage 1 hits the identity table" should read "stage 2 hits the identity table" (per Section 8.7, Stage 1 is the budget gate and Stage 2 is the bypass probe).

H3: [Anticipated Objections, Reviewer Open Questions (QO-1..QO-15), and Canonical Appendices] (Pages: 27-38)

H4: Potential Mistakes and Improvements

H5: 1. Cross-Reference Index for Open Questions (QO-1 … QO-15, Section 10, PDF pp. 29–30 / Document pp. 28–29)

Per the review loop protocol in Sections 10 and 11, detailed technical responses to QO-1 through QO-14 are provided in their respective technical segments above, and QO-15 is answered in Subsections 2 and 3 immediately below:

- QO-1 (Decidability of the Two-Tier SIMEMP Hierarchy): See Segment 1, Point 3.

- QO-2 (Landauer-Based Metering vs. Logical Cost Unit): See Segment 1, Point 2.

- QO-3 (Expressiveness of the Zoom/Remainder Telescope): See Segment 2, Point 2.

- QO-4 (Constructing Arbitrary \mathbb{N} and \mathbb{Q} Beyond Q(9)): See Segment 2, Point 1.

- QO-5 (Confluence, Termination, and Totality of OP_REWRITE): See Segment 2, Point 3.

- QO-6 (Slot-Based State Model vs. Flattened SSA Value Graph): See Segment 4, Point 3.

- QO-7 (Missing Opcodes, Bounded OP_CALL, and Receipt-Typed Exceptions): See Segment 4, Point 4.

- QO-8 (Formal Status of Static Opcode Costs): See Segment 4, Point 2.

- QO-9 (Commitment Width, Collision-Risk Accounting, and Collision Receipts): See Segment 4, Point 7.

- QO-10 (Scaling Causal Index \kappa to Multi-Core and Multi-Node Execution): See Segment 4,

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   Point 6.

- QO-11 (Formal Semantics for Intent Contracts and Typing Parser Error Modes): See Segment 3,
   Point 2.

- QO-12 (Constructive Logic of Qlog and Sequent Rules for Q(patterns)): See Segment 3, Point 3.

- QO-13 (Soundness of the Road-Receipt Trust Model and Audit Rights): See Segment 3, Point 1.

- QO-14 (Satisfiability of the Independence Quorum on Legacy History): See Segment 3, Point 4.

- QO-15 (Fatal Objections, Cosmetic Objections, and Unanticipated Objections): See Subsections 2
   and 3 below.


H5: 2. Evaluation of the 16 Self-Critiques (Section 9, PDF pp. 27–29 / Document pp. 26–28) and
    Answer to QO-15 (Parts 1 & 2)

In response to QO-15, the sixteen anticipated objections in Section 9 stratify as follows:

- Three Fatal Objections (Blocking if Unresolved Before v0.2 Implementation):
     1. Objection 14 (Unconstructed \mathbb{N} Beyond Q(9) and Unspecified \kappa(z)): A numeric
        substrate and virtual machine whose value slots hold q_z, z \in \mathbb{N} and whose
        division/zoom semantics depend on \delta_z(d) = Q(1)_d / \kappa(z), without ever defining
        how integers >9 are constructed or fixing the resolution function \kappa(z), cannot
        canonically encode or hash numbers \ge 10 across nodes. (Resolved via the
        radix-polynomial constructor and \kappa(z) = B^z in QO-4.)
     2. Objection 15 (Soundness of Requirement 1 / "Theorem 1" Static Cost Accounting vs.
        Parameterized & Mid-Instruction Work): In Table 5, several opcodes have operand-dependent
        or exponential costs (OP_POW costs B, OP_FACT costs n-1, OP_FOLD costs m, OP_GP is
        2^n-bounded, OP_DIV and OP_CF run internal searches), and the dispatch loop checks
        W_{\text{rem}} \le 0 *before* execution rather than checking W_{\text{rem}} <
        C_{\text{op}}(\text{operands}). Without parameterized pre-execution cost deduction
        (QO-8), a single instruction can overdraw W_{\text{rem}} mid-execution, invalidating the
        totality proof. (Furthermore, coupling this with Objection 8, OP_REUSE can only evaluate
        the breakeven inequality \Delta W_{\text{compute}} > \Delta W_{\text{lookup}} + \epsilon
        deterministically without heuristic prediction if the historical compute work \Delta
        W_{\text{compute}} is explicitly stored in the committed artifact's Metric Envelope at
        OP_COMMIT time.)
     3. Objection 10 & Objection 7 (Boundary Trust Gaps — Unmitigated Statistical Parser at the
        Qs Entrance and Unverified RoadReceipt Execution Claims): All downstream zero-drift
        guarantees (H(\text{Intent} \mid \text{Execution}) = 0, R12) are vacuous if the formal
        Intent object was hallucinated by an unverified LLM parser from human language (Objection
        10), or if a remote node can sign an arbitrary output hash \mathcal{H}(S_{\text{out}}) in
        a RoadReceipt without a cryptographic execution proof \pi (Objection 7). Both boundary
        gaps must be closed via the round-trip human co-signing gate (QO-11) and zk-STARK /
        interactive bisection proofs (QO-13).

- Three Cosmetic Objections (Resolvable via Editorial Phrasing or Existing Scoping):
     1. Objection 4 (Hilbert's Tenth Problem / DPRM Phrasing): The underlying mathematical
        mechanism (Northcott's theorem on bounded logarithmic height h(u/v) \le H_{\max} combined
        with finite budget exhaustion) is completely sound; uniformizing the prose across the
        corpus to state "finiteness buys guaranteed termination, not mathematical decidability"
        is a straightforward editorial cleanup.
     2. Objection 11 (Chronosa Sheaf-Intelligence Metaphor): Because Chronosa is already
        explicitly isolated as a non-normative VISION PROPOSAL (Sec. 7.2, Appendix D.4) and sits
        outside the Layers 1–3 horizontal execution core, its speculative claims have zero impact
        on the soundness of Qm, Qexpr, or QnIR v0.1.
     3. Objection 12 (Axiom 12 License Lineage Coupling): Preserving an immutable
        provenance/paternity metadata field in artifact envelopes and compiled binary headers is
        standard practice (analogous to SPDX/SBOM provenance attestations) and does not affect
        the technical invariants of the VM as long as runtime execution does not require active
        network telemetry.


H5: 3. Critical Unanticipated Objections Not Foreseen in Section 9 (Answering the Final Part of
    QO-15)

Cross-checking the QnIR v0.1 specification (Section 8) against the rest of the document reveals
four critical architectural and encoding issues that were not anticipated in Objections 1–16
(detailed mechanically in Segment 4, Points 1, 3, 4, 6, and 7):

1. Unanticipated Objection A — Definitive Encoding Contradiction Between the 64-Bit Instruction
   Word Layout (Sec. 8.6) and Higher-Arity / Dual-Output Opcodes in Table 5 (Sec. 8.4):
     While Objection 16 notes that the 64-bit word limits slot counts to \le 256 and aux to \le
    65535, it misses that the two 8-bit source fields (src1, src2), single 8-bit dest field, and
    16-bit aux immediate cannot encode OP_TUPLE (arity 4), OP_FOLD (arity k \ge 3), OP_DIV (dual
    outputs: quotient to dest, remainder to aux while still needing aux for its Diophantine step
    ceiling), or 256-bit commitment hashes in OP_REUSE H(3!).
2. Unanticipated Objection B — Absence of Forward Conditional Branching and Inability to Recover
   from ORDERING_INDETERMINATE_AT_ZOOM in QnIR v0.1:
     While Objection 6 notes the absence of OP_CALL and indirect jumps, it overlooks that QnIR
    v0.1 has no forward conditional branch or conditional select instruction at all (OP_LOOP is
    strictly a backward jump), and that classifying ORDERING_INDETERMINATE_AT_ZOOM as a closed
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    terminal status in Section 2.5 and Appendix B.1 halts the program rather than allowing the
    caller to catch it and invoke OP_ZOOM (z \to z + \Delta z) as required by Section 3.4.
3. Unanticipated Objection C — Conflation of the Static Program Hash Chain (W_i) with the
   Dynamic Execution Trace (\kappa) Under Loops (OP_LOOP):
     Hashing the runtime causal counter \kappa_i into the static program word commitment W_i =
    \text{Hash}(W_{i-1} \parallel \text{word}_i \parallel \kappa_i) (Sec. 8.6) makes W_i
    ill-defined whenever OP_LOOP executes the same static instruction word \text{word}_i
    multiple times at different runtime values of \kappa.
4. Unanticipated Objection D — Unified V/B Slot Index Space Ambiguity and Syntactic vs. Semantic
   Hash Equality (R6):
     Section 8.6's 8-bit unified V/B index space has no defined partition boundary, breaking
    static type expectations when OP_ROOT dynamically promotes an even
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